

LCN Installation

LCN Guidelines

Implementation, Troubleshooting, and Service

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CE Compatible

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About This Document

This publication provides **TPS Network** (TPN) guidelines for use by system hardware planners, installation and checkout personnel, and maintenance personnel. The term TPN replaces the previously used term Local control Network (LCN), except in cases where we are referring to the physical LCN cable. To reduce confusion for experienced users and still evolve to the new terminology, there will be references to both LCN and TPN in this document. This document is designed to help you:

- Correct any problems due to improper system configuration and implementation.
- Correct coax cable and fiber optic link errors.

This publication supports TotalPlant Solution (TPS) Network Release 682 and earlier software releases. TPS is the evolution of TDC 3000X.

This publication supports CE Compliant equipment. Any equipment designated as "CE Compliant" complies with the European Union EMC, Health, and Safety Directives. All equipment shipping into European Union countries after January 1, 1996 requires this type of compliance—denoted by the "CE Mark."

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Binder Title	Binder Number
<i>TPS System Site Planning</i>	System Site Planning	TPS 3020-1
<i>LCN System Installation</i>	LCN Installation	TPS 3025
<i>LCN System Checkout</i>	LCN Installation	TPS 3025
<i>Five/Ten-Slot Module Service</i>	LCN Service-2	TPS2060-2
<i>Dual Node Module Service</i>	LCN Service-2	TPS2060-2
<i>History Module Service</i>	LCN Service-2	TPS 3060-2
<i>Engineer's Reference Manual</i>	Implementation/Startup & Reconfiguration	TPS 3030-2
<i>TPS System Implementation Guide</i>	N/a	N/a
<i>TPS System Administration Guide</i>	N/a	N/a

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE — INTERNAL: Identifies an additional source of information within the bookset.
	CAUTION: Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process. CAUTION: Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation, which, if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING, Risk of electrical shock: Potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.
	ESD HAZARD: Danger of an electro-static discharge to which equipment may be sensitive. Observe precautions for handling electrostatic sensitive devices.
	Protective Earth (PE) terminal: Provided for connection of the protective earth (green or green/yellow) supply system conductor.
	Functional earth terminal: Used for non-safety purposes such as noise immunity improvement. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national local electrical code requirements.

Symbol	Definition
	Earth Ground: Functional earth connection. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.
	Chassis Ground: Identifies a connection to the chassis or frame of the equipment shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.

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1. INTRODUCTION

This section provides an overview of the contents of the manual and lists references.

1.1 Purpose and Scope of This Publication

This guide provides information and references to supporting publications needed to plan, implement, operate, and maintain a TPS Network (TPN). It will assist you in understanding the functional components that make up the **TotalPlant** Solution (TPS) system (Figure 1-1). A TPN is made up of the coaxial and fiber optic cable network (Local Control Network – LCN) that provides the data path for communications between the TPS Network-based control modules and TPS system nodes. That physical cable network will still be referred to as the LCN cable in this document.

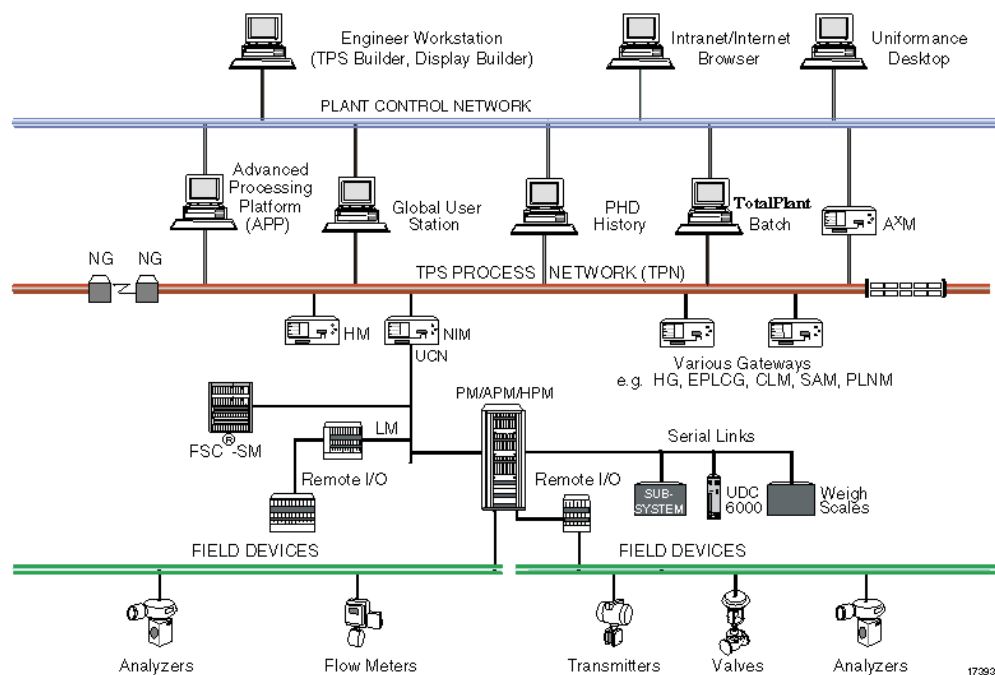


Figure 1-1 — TPS System Overview

1. INTRODUCTION

1.1. Purpose and Scope of This Publication

Several devices (nodes) that connect to the TPN also connect to another network. They are listed below. See Figure 1-1.

Device/Name	Abbreviation
Application Module ^X	A ^X M
Application Processing Platform	APP Node
Global User Station	GUS
Hiway Gateway	HG
Plant Network Module	PLNM
Programmable Logic Controller Gateway	PLCG
Scanner Application Module	SAM
Network Gateway	NG
Network Interface Module	NIM
Universal Station ^X	U ^X S

Your attention is called to these networks only for the purpose of considering them as you plan the TPS system Equipment placement and cable routing.

2. IMPLEMENTATION OF COAXIAL CABLE SEGMENTS

This section provides guidelines and rules to be followed when implementing a Local Control Network (LCN) coaxial cable segment.

2.1 System Requirements

2.1.1 Removable Media Requirements

A Removable Media Disk Drive may be either a floppy drive or a cartridge drive. The minimum Removable Media Disk Drive requirements for all systems are as follows:

- At least one Universal Station with an Engineer's Keyboard must have two Removable Media Disk Drives.
- All other Universal Stations with Engineer's Keyboards must have at least one Removable Media Disk Drive.

For systems without a History Module, the following rules also apply:

- All Operator Consoles must have two Removable Media Disk Drives.
- All Universal Stations with Engineer's Keyboards must have two Removable Media Disk Drives.

For systems with a History Module, the following rule also applies:

- Each console must have two Removable Media Disk Drives. One of these must be on a Universal Station with an Engineer's Keyboard.

Additional Removable Media Disk Drive guidelines:

- Systems requiring fast reload should have the Removable Media Drive — Cartridge Disk option in each console.
- A console on a remote segment (a coaxial cable segment separated from the main coaxial cable segment by a fiber optic link) of the TPN must have at least one US with at least one Removable Media Drive.

2. IMPLEMENTATION OF COAXIAL CABLE SEGMENTS

2.1. System Requirements

2.1.2 System Maximums

- The maximum number of nodes in a TPS system is 64 (software limitation).
- For each segment, the number of nodes plus the number of LCN Extender sets must not exceed 40. An LCN Extender set is a pair of XLCNE2 boards—one for Cable A and one for Cable B. A Dual Node Module counts as one node load towards the 40-node limit in each segment, whether there are one or two nodes in it.
- The number of Application Modules per TPN is not restricted.
- The maximum number of Computer Gateways per TPN is 10.
- The maximum number of Process Networks per TPN is 20. This means a maximum number of 40 physical Process Network nodes per TPN (20 redundant pairs). A Process Network is defined as a Data Hiway, a Universal Control Network, or a PLC Gateway.
- The maximum number of History Modules per TPN is 20.
- The maximum number of Universal Stations in a single console is 10.
- The maximum number of Network Gateway modules per TPN is 10.
- The maximum number of consoles per system is 10.

2.2 Coaxial Cable Hardware Description

NOTE

The XLCNE2 board is a replacement for the LCNE2 board. They are functionally equivalent. The LCNE2 and XLCNE2 cannot be mixed on the same fiber. A XLCNE2 can only talk to another XLCNE2.

TPN modules communicate with each other through interconnections of coaxial cable and fiber optic cable. There are two LCN cables for redundancy, Cable A and Cable B. Each cable must be routed separately and must be marked to indicate that it is Cable A or Cable B. Cable A is marked with yellow and Cable B with green shrink-fit bands at the connectors. Use only Honeywell specified cables.

Each node on the TPN connects to both LCN Cable A and LCN Cable B through connectors called BNC tees. These connectors allow a node to be disconnected from the coaxial cable segment without disrupting communication among the other nodes. The tee connectors are Honeywell part number 51190728-105 (this is not the same part number used for Data Hiway tee connectors).

An LCN can have up to seven coaxial cable segments—a central segment and up to six remote segments or remote nodes. The remote segments are connected to the central segment with LCN Extenders (LCNE2s/XLCNE2s). LCN Extenders use fiber optic cables that provide ground isolation and can run through hazardous areas to interconnect LCN coaxial cable bus segments. Four LCNE2s/XLCNE2s are required to interconnect two LCN bus segments—one at each end of fiber optic Cable A and one at each end of fiber optic Cable B. (Each fiber optic cable, A and B, contains at least two fiber filaments—one for transmits and one for receives). Two LCNE2s and one LCNFL (LCN Fiber Link) are required to connect a remote node to the central segment. An XLCNE2 is used for Cable A and another XLCNE2 is used for Cable B at the central segment. An LCNFL is used at the remote node. It has fiber optic transmitters and receivers for both Cable A and Cable B. Remote nodes and remote segments can be connected only to the central coaxial cable segment.

ATTENTION

The LCNFL (LCN Fiber Link) cannot be used in conjunction with the XLCNE2 (XLCN Extender II). Therefore, it is not possible to connect a remote node to the central segment, when using XLCNE2.

2. IMPLEMENTATION OF COAXIAL CABLE SEGMENTS

2.2. Coaxial Cable Hardware Description

The clock may or may not need to be transmitted between two coaxial cable segments, although it is generally required to guarantee accuracy of alarm events, specifically Sequence of Events (SOE) alarms. If the remote segment requires only the 5 Mbits/s digital clock and the central segment can generate this clock, additional hardware is not required. The digital clock is transmitted on the LCN Cable A and Cable B as a special data message. If the remote segment requires the 12.5 kHz clock, fiber optic clock transmitter and receiver boards must be used (FOC/XMTR and FOC/RCVR) in addition to the LCN Extender boards (LCNE2s). Refer to Section 4 for additional information on fiber optic links. Figure 2-1 illustrates the LCN coaxial cable segment architecture.

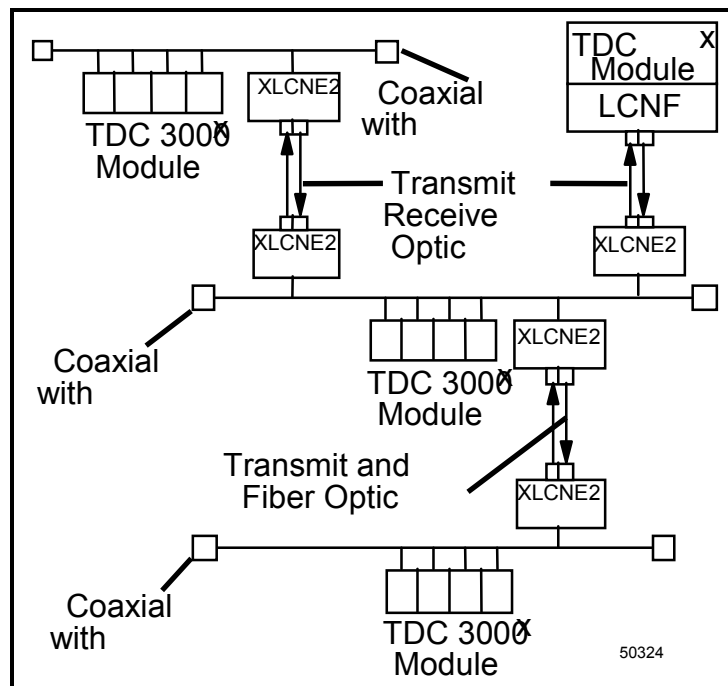


Figure 2-1 — LCN Coaxial Cable Segment Architecture

NOTE

In order to provide LCN redundancy, the system as shown must be duplicated (with the exception of the LCNFL, which includes fiber optic transmitters and receivers for both A and B cables).

Follow these rules as you plan and install your coaxial cable segments:

- A segment can have a maximum of two Clock Source/Repeater (CS/R) boards—one for Cable A and one for Cable B.
- The maximum LCN coaxial cable length per bus segment is 300 meters. The segment cannot have stubs or branches. Use part number 51109181-XXX coaxial cable between consoles and between cabinets and consoles (“XXX” is the cable length in meters). **DO NOT SUBSTITUTE** another type of cable.
- Minimum cable length between nodes is two meters. For adjacent nodes, use Honeywell cable set 51109806-002, which contains two cables marked “Cable A” and “Cable B.”
- The maximum total difference in Cable A and Cable B length between any two modules in the system is 300 meters. The length of each cable is the sum of all the coaxial cable and fiber optic cable sections between the two modules. The reason for this restriction is to limit the difference in communications delays between the two cables.
- Fiber optic cables can be up to 2000 meters (6600 feet) in length.
- In each node, the node address is selected with jumpers on the LCNI I/O or LCNFL transceiver board, or on the K2LCN/K4LCN board, whichever is present. In the case of a remote node that has a K2LCN/K4LCN and an LCNFL, the pinning is done on the LCNFL and all address jumpers should be removed from the K2LCN/K4LCN. The address uses seven bits with odd parity. A jumper in place is “zero” and a jumper removed is a “one.” Refer to 2.4 for additional information about TPN node addresses.
- The nodes' TPN addresses should be assigned in increasing numerical order. For communication efficiency, try not to leave gaps in the address numbers. We recommend that addresses be assigned in accordance with the physical placement of the modules in the control room. For example, start at one end of the LCN cable and assign consecutive addresses until the opposite end of the cable is reached. Repeat for each remote segment. Sometimes it is desirable to leave planned gaps in the node numbers when it is known that nodes will be added later. Refer to 2.4 for additional information about TPN node addresses.
- Terminator resistors are required at each end of both coaxial cables. Use Honeywell part number 30732052-001.
- It is imperative that Cable A be connected only to Cable A interfaces in each of the nodes, and that Cable B be connected only to Cable B interfaces. **Cables cannot be crossed. Do not substitute unauthorized cables.**

2. IMPLEMENTATION OF COAXIAL CABLE SEGMENTS

2.2. Coaxial Cable Hardware Description

NOTE

Crossed cables will work until a communication error occurs, then some nodes or the entire system may isolate and/or fail.

- LCN coaxial cable physical and electrical characteristics are suitable for internal building use only. **The LCN coax cables must not be routed outside the control room building.**
- Although LCNE2s/XLCNE2s, FOC/XMTRs, and FOC/RCVRs can be mounted in open I/O slots of any node type, it is desirable to locate them in nodes that are the least likely to have the power turned off. It is recommended that the order specified in Table 2-1 be used in selecting node types for housing these extender boards.
- The LCNE2s/XLCNE2s, FOC/XMTRs, and FOC/RCVRs for Cable A must be in a different node than the ones for Cable B and must be powered from separate ac power entries. If more space is required, an otherwise empty Dual Node Module with a power supply and KJMP load board (51401594-200) installed in each node should be used to house the extender board sets. The LCNE2s/XLCNE2 and associated FOC/XMTR or FOC/RCVR for Cable A are placed in the I/O section of the two-slot module and the corresponding boards for Cable B are placed in the three-slot node. If FOC/XMTR or FOC/RCVRs are not used, two LCNE2s/XLCNE2s for Cable A can be placed in the two-slot node, and two LCNE2s/XLCNE2s for Cable B can be placed in the three-slot node (the third slot would also be available for an extender board that is part of another extender set).

Table 2-1 — Order for Selecting Node for LCN Extension Components

Preference	Node	Description
1	AMR	Redundant Application Module
2	HG	Hiway Gateway (preferably in a redundant pair)
3	NIM	Network Interface Module (redundant)
4	PLCG	Programmable Logic Controller Gateway (redundant)
5	AM	Application Module
6	A ^X M	Application Module ^X
7	SAM	Scanner Application Module
8	HM	History Module
9	PLNM	Plant Network Module
10	NG	Network Gateway
11	CG	Computer Gateway
12	US	Universal Station
13	U ^X S	Universal Station ^X

2.3 Rules for Placing Modules on the TPN

- Universal Stations (USs) with parity only memory (PMEM boards) may be used only for no critical monitor applications. They must not be used for control.
- There must be at least two USs per console, per area database, to ensure a window to the process in case of failure in the primary US.
- Because of the lower reliability of the fiber optic extension to remote segments, locate redundant HGs and added HGs on the same coaxial cable segment. Also, locate redundant NIMs and redundant PLCGs on the same coaxial cable segment.
- Any coaxial segment with a Process Network interface (HG, NIM, or PLCG) must have at least one Universal Station on that segment (two are recommended).
- If possible, avoid configuring only two nodes on a remote segment due to error scenarios, which might occur when the fiber optic cable is broken, or when LCNE2s/XLCNE2s are powered off.

2.4 TPN Node Addresses

The TPN address for each TPN node is in a range from 1 to 64. The address for each node is established by jumper pins on the CLCN A/B board at the back of LCN module chassis (CLCN A/B boards have the LCN T-connectors on them), or by jumper pins on the LCNFL board at the back of an LCN module chassis, or for a single node at the far end of a fiber optic link. The pinning scheme for an LCNFL board is illustrated on Figure 9-1. It is the same for the CLCN A/B board. For a node implemented in a Dual Node Module, address pinning is accomplished on the K2LCN/K4LCN board, using the same pinning scheme as for an LCNFL or a CLCN A/B. For the special case of a single remote node that has both a K2LCN/K4LCN and an LCNFL, the LCNFL is mounted in I/O slot one (the lowest) behind the K2LCN/K4LCN. In this case, the address pinning should be done on the LCNFL and all address jumpers should be removed from the K2LCN/K4LCN.

USs and UWSs should be assigned the lowest TPN node addresses. Redundant module pairs (HGs, NIMs, PLCGs, and AMs) should be assigned consecutive addresses, one odd and one even. In addition to a node address, HMs are assigned node pair numbers (for more information about node pair numbers, refer to 7.2.1 in the *Engineer's Reference Manual*). In spite of the term “node pair number,” HMs are not configured as redundant pairs of nodes. HMs can have redundant disk drives but such HMs still require only one TPN node address.

2.5 Clock System and Segment Grounding

2.5.1 12.5 kHz Clock Systems

This subsection covers systems that use only the 12.5 kHz clock. These systems that do not include K2LCN/K4LCN kernels (processor boards).

Each coaxial cable segment must have two Clock Source/Repeater (CS/R) boards—one for Cable A and one for Cable B. The Clock Source/Repeater boards transmit the 12.5 kHz clock on each LCN cable and automatically provide the single point LCN shield connection to logic ground for each cable.

ATTENTION

The LCN Clock Source nodes **MUST** reside on the same LCN **Coaxial Cable** segment. If the clock source nodes are **NOT** on the same LCN Coax segment, it may cause system anomalies.

MP-MCSR02 includes two P-MCSR2 (CS/R boards), and associated cables. The two-meter cables connect the CS/Rs to the LCN coaxial cables (Cable A and Cable B).

Each CS/R board must be installed in I/O slot one of a module that includes an EMPU, HMPU, or HPK2. For network integrity, it is desirable to locate them in modules that are least likely to have the power turned off. The order of module preference is shown in Table 2-1.

CS/Rs can be connected with twisted pair wires to a Fiber Optic Clock Transmitter (FOC/XMTR) in order to propagate the clock signal to remote segments over a 2-kilometer maximum length fiber optic cable pair with a matching Fiber Optic Clock Receiver

(FOC/RCVR) and CS/R set on the remote segment. Alternatively, a current loop over 300 meter maximum length twisted pair wires can be used to directly interconnect CS/Rs for propagating the clock signal to remote segments, although this is not recommended due to lower reliability. Current loops cannot be used to transverse hazardous areas. The fiber optic clock approach is the preferred method of clock propagation to remote segments.

Two LCNE2s and one LCNFL or two XLCNE2s are required to connect a single remote node to an LCN coaxial cable segment. The 12.5 kHz clock signal cannot be transmitted to this remote node. The LCNFL takes the place of the LCNI I/O board in the remote node and includes the jumpers for setting the node's address.

ATTENTION

LCNFL is Not compatible with XLCNE2.

Two clock source nodes must be defined in the software Network Configuration File (NCF). These nodes must contain CS/R boards and they must be in the central segment. This allows the central cable segment to maintain synchronization with other remote segments if one remote link fails.

The power supply in each node of the system must be set up by a pinning option for the internal or line frequency clock unless the system is required to run Sequence of Events (SOE) functionality on the NIMs. If SOE functionality is required, the clock master, the clock slave, and the NIMs with SOE requirements must be pinned for use with external clocks, which would normally be provided by the Precision Clock PWA (Printed Wire Assembly) or the Precision Clock Source Repeater PWA. All other nodes are pinned for internal (line frequency) clock. For information on the clock pinning option, refer to subsection 4.3.1.

2.5.2 5 Mb/s Digital Clock Only Systems (All-K2LCN/K4LCN Nodes)

This subsection covers systems that use only K2LCN/K4LCNs as the kernel boards (no EMPU, HMPU, or HPK2 boards). These systems use only the digital mode of clock operation, in which clock synchronization signals are digital frames or messages incorporated in the 5 Mb/s TPN communications. Because remote segments and remote nodes receive the clock through TPN messages, fiber optic clock transmitters and receivers (FOC/XMTRs and FOC/RCVRs) are not required.

Two clock source nodes must be configured in the software Network Configuration File (NCF). You must identify these two clock source nodes in the central coaxial cable segment. They must follow the node selection order of preference as shown in Table 2-1. This allows the central cable segment to maintain synchronization with other remote cable segments if one remote link fails.

Note that systems containing only K2LCN/K4LCN boards do not use CS/R boards.

The power supply of each node must be pinned to use the internal or power line frequency clock. For information on the clock pinning option for the Dual Node Module, refer to *Dual Node Module Service* in the *LCN Service* binder.

2. IMPLEMENTATION OF COAXIAL CABLE SEGMENTS

2.5. Clock System and Segment Grounding

Two LCNE2s and one LCNFL or two XCLNE2s are required to connect a single remote node to an LCN coaxial cable segment. There can be only one K2LCN/K4LCN and one LCNFL in a Dual Node Module. The other node must be left empty. The LCNFL board must be a Revision F or greater to be compatible with the K2LCN/K4LCN. The LCNFL is mounted in I/O slot one (the lowest) behind the K2LCN/K4LCN.

NOTE

A remote node in an all-K2LCN/K4LCN system will receive the clock through TPN communications messages. This allows the remote node to have the sequence-of-events (SOE) function.

Each Dual Node Module requires a CLCNA in I/O Slot 1 of the two-slot node, and a CLCNB in I/O Slot 1 of the three-slot node, except for the case of a remote node using the LCNFL.

ATTENTION

LCNFL is Not compatible with XLCNE2.

If an LCN coaxial cable segment does not have CS/R boards (which is true for an all-K2LCN/K4LCN segment), a grounding wire must be installed on one, and only one, CLCNA board and on one, and only one, CLCNB board. This grounds the shield of Cable A and the shield of Cable B at one point each, to chassis ground. If possible, do not ground Cable A and Cable B in the same module. Refer to *Dual Node Module Service* in the *LCN Service* binder for additional information on TPN network grounding.

The power supply in each node of the system must be set up by a pinning option for the internal or line frequency clock unless the system is required to run Sequence of Events (SOE) functionality on the NIMs. If SOE functionality is required, the clock master, the clock slave, and the NIMs with SOE requirements must be pinned for use with external clocks. For these nodes, the internal Precision Clock Oscillator on the K2LCN/K4LCN will be used as the clock source instead of the power line. All other nodes are pinned for internal (line frequency) clock. For information on the clock pinning option, refer to subsection 4.3.1.

2.5.3 Combined 12.5 kHz Clock and Digital Clock Systems

This subsection covers combined mode systems—systems which include both the 12.5 kHz (Subchannel) and 5 Mbits/s (Digital) clocks. The 12.5 kHz clock is required for nodes which use HMPU, or HPK2 processors, while the 5 Mbits/s digital clock is used by nodes which have K2LCN/K4LCN processors.

Each combined clock mode segment and each 12.5 kHz clock only segment must have two Clock Source/Repeater (CS/R) boards—one for Cable A and one for Cable B. The Clock Source/Repeater boards transmit the 12.5 kHz clock on each LCN cable and automatically provide the single point LCN shield connection to logic ground of the host module for each cable. The CS/Rs must be mounted in non-K2LCN/K4LCN nodes. Digital clock only segments do not use CS/Rs. Grounding for the cable shields in digital clock only segments is done by ground wires at one CLCNA and one CLCNB board as described in the previous subsection (2.5.2).

Each CS/R board must be installed in I/O slot one of a module that includes an HMPU, or HPK2. For network integrity, it is desirable to locate them in nodes that are least likely to have the power turned off. The order of module preference is shown in Table 2-1.

CAUTION

Do not put CS/Rs in I/O Slot 1 of a K2LCN node. Damage could result.
--

In a combined clock mode system, the central segment must contain at least two nodes with HMPU, or HPK2 processors. These are used in conjunction with CSR boards to source the 12.5 kHz clock. If one or more remote segments are all K2LCN/K4LCN, or could become all-K2LCN/K4LCN through upgrades or replacements, then the central segment must contain at least two nodes with K2LCN/K4LCNs. One K2LCN/K4LCN will serve as a translator, receiving the 12.5 kHz clock and generating the 5 Mbits/s digital clock to be used by other K2LCN/K4LCNs in the system. The second K2LCN/K4LCN required in the central system serves as the backup for the translator function, and will take over automatically if the primary translator fails.

Two clock source nodes must be defined in the software network configuration file. These nodes must be nodes which contain CS/R boards and they should be on the central segment. This will allow the central segment to maintain synchronization with other remote segments if one remote link fails.

A minimum of two K2LCN/K4LCN nodes must be on a segment in order to connect it to an all-Digital Clock segment using LCNE2s/XLCNE2s.

FOC/XMTR/FOC/RCVRs or CS/Rs with twisted pair wires must be used to attach an all-12.5 kHz clock segment with LCNE2s/XLCNE2s to a combined clock mode segment or to attach one combined clock mode segment to another combined clock mode segment.

An all-12.5 kHz segment cannot be connected with LCNE2s/XLCNE2s to an all Digital Clock segment. At least one of the segments must have combined clock modes.

A software algorithm automatically configures a K2LCN/K4LCN node to become the Digital Clock source node that translates the 12.5 kHz clock into the Digital Clock. The algorithm also configures backups as needed.

2. IMPLEMENTATION OF COAXIAL CABLE SEGMENTS

2.5. Clock System and Segment Grounding

Adding non-K2LCN/K4LCN nodes to a Digital Clock only system requires that at least two nodes be added to the central segment and that they contain the CS/Rs. This scenario might occur when adding redundant Application Modules to a Digital Clock only system.

Two LCNE2s and one LCNFL are required to connect a single remote node to an LCN coaxial cable segment. The clock signal cannot be transmitted to this node unless it is a K2LCN/K4LCN-based node, in which case the Digital Clock is transmitted through the LCN connection.

ATTENTION

LCNFL is Not compatible with XLCNE2.

There can be only one K2LCN/K4LCN and one LCNFL in a single Dual Node Module. The other node must be left empty. The LCNFL board must Revision F to be compatible with the K2LCN/K4LCN. The LCNFL is mounted in I/O slot one (the lowest) behind the K2LCN/K4LCN.

NOTE

This allows a remote node to have the Digital Clock for sequence-of-events (SOE).

Each Dual Node Module requires a CLCNA board in the two-slot node and a CLCNB board in the three-slot node except for remote nodes using the LCNFL.

The power supply in each node of the system must be set up by a pinning option for the internal or line frequency clock unless the system is required to run Sequence of Events (SOE) functionality on the NIMs. If SOE functionality is required, the clock master and the clock slave must be pinned for use with external clocks, normally provided by Precision Clock Source Repeaters (PCS/Rs). Each NIM with SOE requirements must also be pinned for external clocks. In addition, if the NIM with SOE requirements is a non-K2LCN/K4LCN node, it must have a precision clock source such as a PCS/R. All other nodes are pinned for internal (line frequency) clock. For information on the clock pinning option, refer to subsection 4.3.1

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

This section provides guidelines and suggestions for improving performance and reliability of a TPN that has LCN Extenders and one or more fiber optic links installed.

NOTE

The XLCNE2 board is a replacement for the LCNE2 board. They are functionally equivalent. The LCNE2 and XLCNE2 cannot be mixed on the same fiber. A XLCNE2 can only talk to another XLCNE2.

3.1 General Fiber Optic Link Information

3.1.1 System-Level Characteristics

Figure 3-1 illustrates the maximum configuration of LCN coaxial cable segments and fiber optic cable segments. There can be no more than one central and six remote coaxial cable segments in such a network.

- Each fiber optic link is limited to a maximum length of 2000 meters (6600 feet). The fiber optic link is nonelectrical, so it can be run safely through hazardous areas. Because fiber optic cables use light as the transmission media, they provide complete electrical isolation between the segments at each end of the link.
- The fiber optic transmitters use LEDs rather than lasers as the light source, so there is no danger of eye damage if you accidentally look directly at a transmitter output or into the end of a fiber cable. If you were to do this with a laser system, eye damage could result.

3.1.2 Honeywell-Provided LCN Extender and Clock Packages

- Four LCN Extender (XLCNE2) boards are provided with model number TP-LCNECM(CE0 or TP-LCENNM(non-CE). This package handles one Extender with Cable A and Cable B.
- One Fiber Optic Clock Transmitter (FOC/XMTR) board; one Fiber Optic Clock Receiver (FOC/RCVR) board; and two, two-meter shielded, twisted pair cables are provided with model number MP-MFOC01. The cables connect the two boards to the Clock Source/Repeater (CS/R) boards.
- Two CS/R boards and two two-meter coaxial cables to connect to the LCN coaxial segment are provided with model number MP-MCSR02.

3.2 LCN Extender Hardware Description

3.2.1 LCN Extension Set

An LCN Extension Set is not a module. It is a group of circuit boards whose function is to add remote segments and/or remote nodes to an LCN using fiber optic cable transmission media. To connect a remote segment, an LCN Extender set includes two pairs of XLCNE2 I/O boards—one pair for Cable A and one pair for Cable B. There are several reasons why an LCN Extender might be necessary or desirable:

- To provide control functions in an area which is too far away from the central area to be serviced by coaxial cable.
- To traverse an area where hazardous or explosive conditions exist and electrical signals would be dangerous.
- To traverse an area where environmental factors would prohibit coaxial cable, such as an outdoor or buried cable run.
- To implement a system which requires more than 40 nodes.

LCN Extension Sets allow the extension (expansion) of the LCN as follows:

- There can be a main, or central, coaxial segment and up to six remote segments, each of which can be located up to two km away from the main coaxial segment.
- The maximum length of each coaxial segment is 300 meters.
- For each segment, the number of nodes plus the number of XLCNE2 pairs must not exceed 40. A Dual Node Module counts as one node load towards the 40 node limit in each segment, whether there are one or two nodes in it.
- The maximum number of nodes in a TPS system is 64.

Fiber optic links have these additional advantages:

- A fiber optic cable is an intrinsically safe transmission medium that isolates the LCN from any hazardous area of a plant it passes through.
- Fiber optic cables are nonelectric, and therefore, offer extremely high immunity to electromagnetic interference (EMI) or radio frequency interference (RFI).
- Fiber optic cables allow long distance (up to 2 km) extension of an LCN without having to worry about ground potential difference between the connected sites.
- Using fiber optics, an LCN may be implemented in a star configuration. Such a topology may facilitate rapid diagnosis and improve survivability if a communications problem develops. First, the segments can be isolated by disabling the fiber optic links. Next, the faulty segment can be determined by noting communication statistics on the separate segments (see subsection 6.2). Finally, the “good” segments can be reconnected.

3.2.2 Components of an LCN Extension Set

An LCN Extension Set consists of optional circuit boards that are used to extend the LCN data cables, Cable A and Cable B, to a remote location. Also, when required, additional boards are used to transmit the 12.5 kHz system clock to the remote segment.

A minimum LCN Extension Set consists of four LCN Extender (XLCNE2) boards—one for each end of fiber optic Cable A and one for each end of fiber optic Cable B. (Each fiber optic cable, A and B, contains at least two fiber filaments, including one for transmission of data from the central segment to the remote segment and one for receiving data from the remote segment). This configuration extends the LCN data signals, and the 5 Mbits/s digital clock if present, to the remote segment, but does not extend the 12.5 kHz clock to the remote segment. If the 12.5 kHz clock is required at the remote segment, a Fiber Optic Clock Transmitter (FOC/XMTR) board is required at the main (central) segment, and a Fiber Optic Clock Receiver (FOC/RCVR) board is required at the remote segment; also, two Clock Source/Repeater (CS/R) boards are required at the remote segment to transmit the 12.5 kHz clock onto the coaxial cables (A and B). Figure 3-3 and Figure 3-4 illustrate an LCN Extension configuration which includes both data and 12.5 kHz clock extension.

The boards which comprise the LCN extension family are summarized below.

- **XLCNE2—LCN Extender:** Provides an interface to one end of a fiber optic link that passes data. One XLCNE2 is required for each end of fiber optic cables A and B. See subsection 3.2.2.1 for more detailed information.
- **CS/R—Clock Source/Repeater:** Transmits the 12.5 kHz system clock onto an LCN coaxial cable (Cable A or Cable B). Has optional current loop input and output so that it can receive the clock from an FOC/RCVR board or another CS/R board, or transmit the clock to FOC/XMTR boards and/or another CS/R board. See subsection 3.2.2.4 for more detailed information.
- **FOC/XMTR—Fiber Optic Clock Transmitter:** Receives the 12.5 kHz system clock from the CS/R on through a current loop interface and transmits it for a given cable (A or B) over a fiber optic link to a remote segment. See subsection 3.2.2.2 for more detailed information.
- **FOC/RCVR—Fiber Optic Clock Receiver:** Receives the system clock transmitted by an FOC/XMTR and converts it to current loop format to send to a CS/R. See subsection 3.2.2.3 for more details.

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.2. LCN Extender Hardware Description

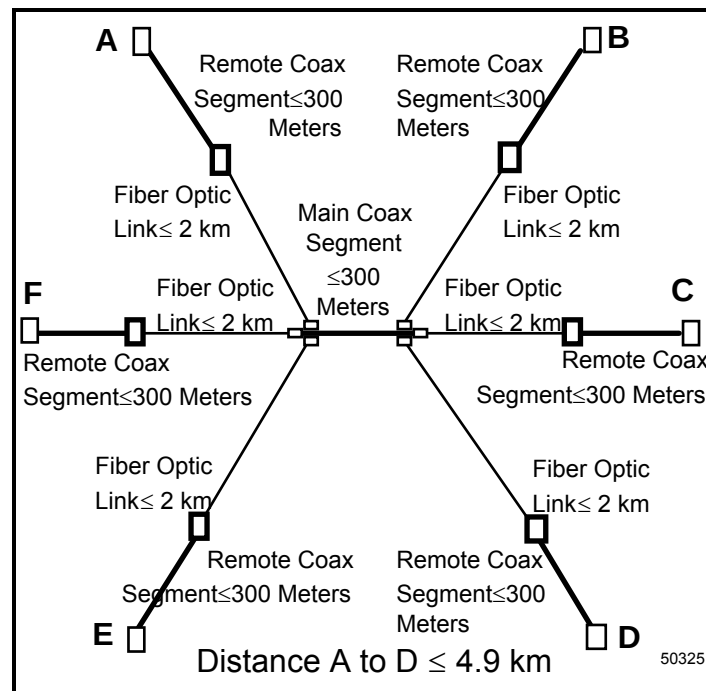


Figure 3-1 — LCN Extender Segment Lengths

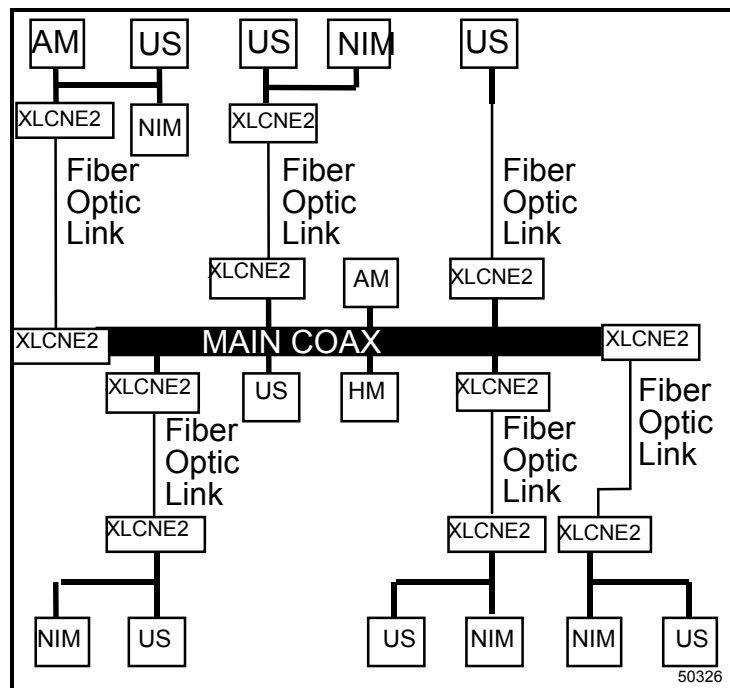


Figure 3-2— LCN Extender Segments

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.2. LCN Extender Hardware Description

Figures 3-3 and Figure 3-4 illustrate an LCN Extension configuration that includes both data and 12.5 kHz clock extensions. Both figures must be used together. The A and B connections designated in Figure 3-3 are respectively connected to A and B in Figure 3-4.

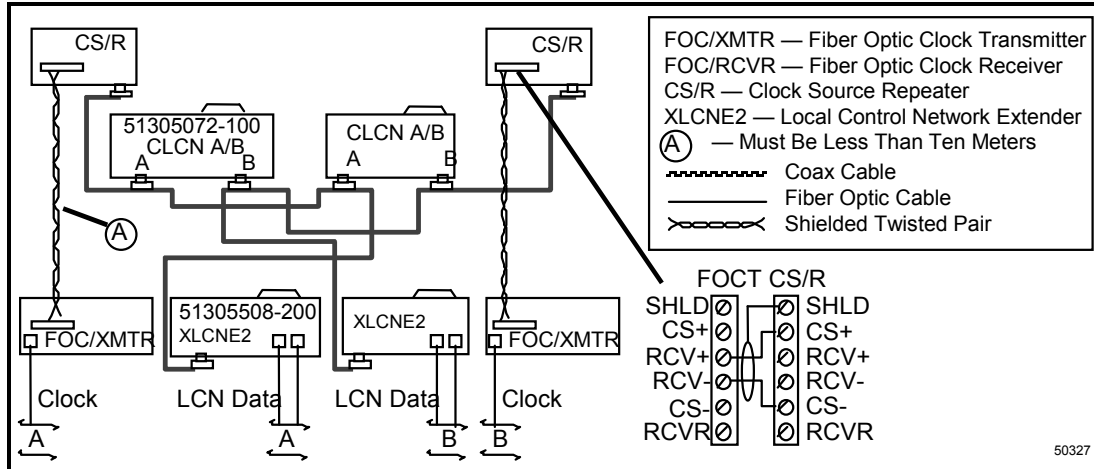


Figure 3-3— LCN Extension Set Connections for One of a Pair of XLCNE2s

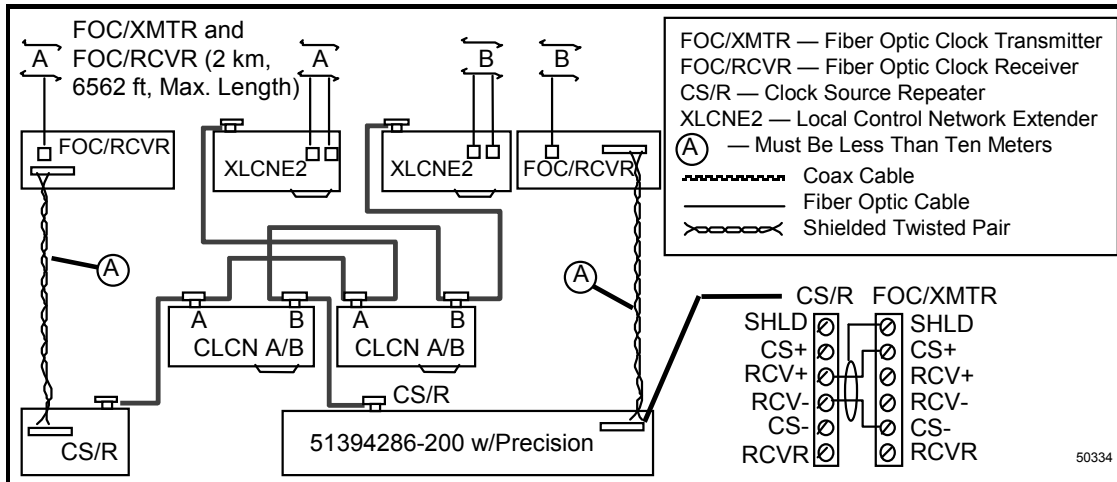


Figure 3-4 — LCN Extension Set Connections for One of a Pair of XLCNE2s

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.2. LCN Extender Hardware Description

LCN Extender (XLCNE2)

The XLCNE2 is used at both ends of a fiber optic link to extend the LCN signals to and from a remote segment. The XLCNE2 includes both a fiber optic transmitter and receiver. This allows data to be sent in both directions between the main coaxial segment and the remote coaxial segment. The XLCNE2-to-XLCNE2 fiber link cannot transmit the 12.5 kHz system clock to the remote segment (refer to subsection 2.5 for clock system details). If this clock is required at the remote segment (such as for the sequence-of-events function), a twisted-pair loop of ≤ 300 m, or a separate fiber link of ≤ 2 km will be required for each LCN cable (A and B). FOC/XMTR and FOC/RCVR boards are used in the implementation of the fiber clock link (see the next two subsections).

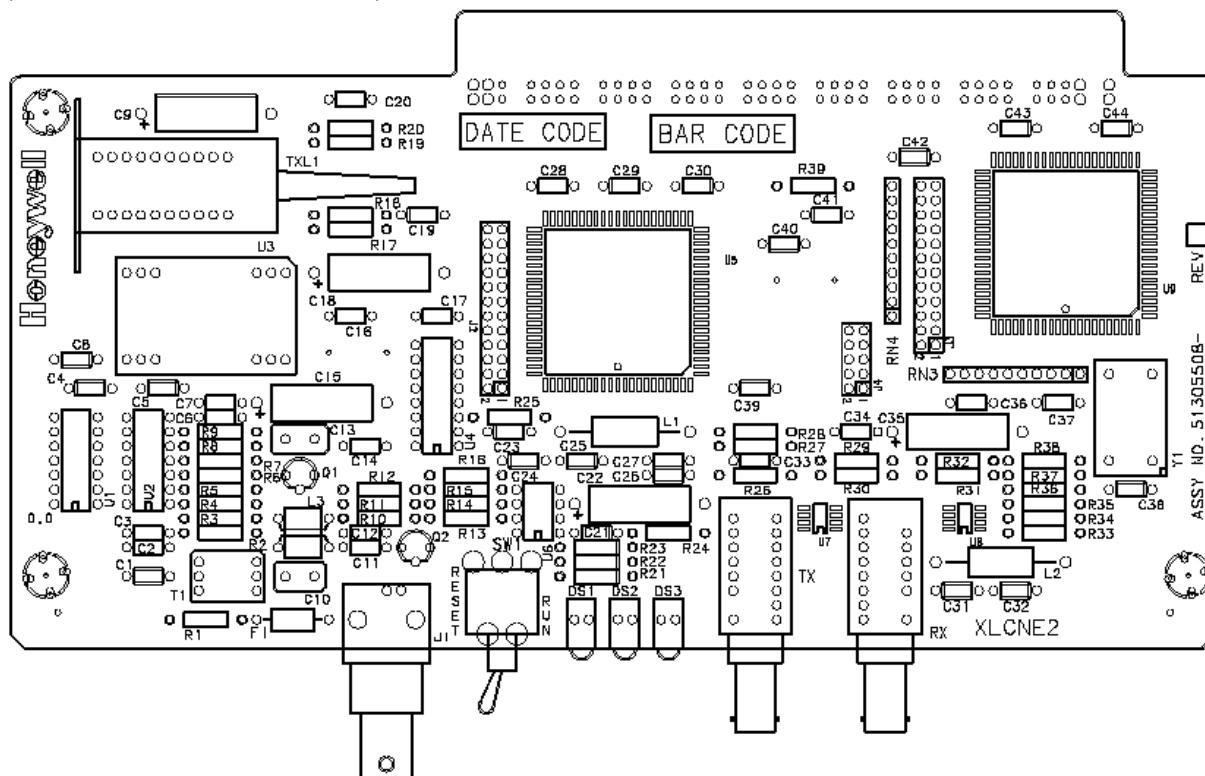


Figure 3-5 — XLCNE2 Board Assembly, 51305508-200

NOTE

The XLCNE2 board, 51305508, replaces the LCNE2 board 51304540. However, an individual fiber optic link segment may not contain XLCNE2 and LCNE2 boards in the same segment.

3.2.2.1 Fiber Optic Clock Transmitter (FOC/XMTR)

FOC/XMTR boards receive 12.5 kHz clock information through a current loop from a CS/R board and transmit it to a remote coaxial segment, through a single fiber that is up to 2 km long, to a FOC/RCVR board. If the remote segment is less than 300 meters away from the central segment, the clock may be transmitted directly with a shielded, twisted-pair current loop from the central CS/R to the remote CS/R, eliminating the need for the FOC/XMTR, FOC/RCVR, and fiber cable components.

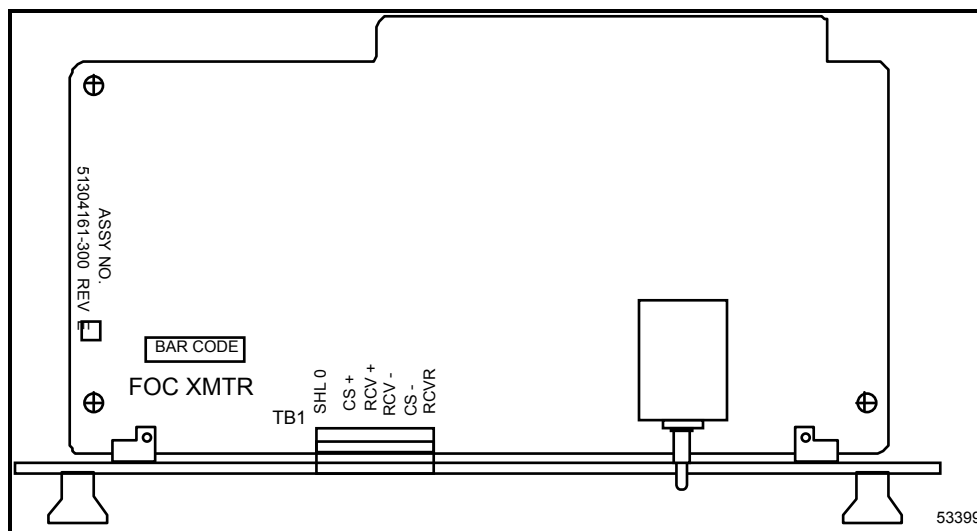


Figure 3-6— FOC/XMTR Assembly

NOTE

The FOC XMTR board, 51304540-300, is a later version of the FOCT board, 51304161-100. They are functionally equivalent and your system may include either or both types.

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.2. LCN Extender Hardware Description

3.2.2.2 Fiber Optic Clock Receiver (FOC/RCVR)

FOC/RCVR boards receive 12.5 kHz clock information from a fiber optic cable (transmitted by a FOC/XMTR board at the other end) and transmit it through a current loop to a CS/R board, which in turn transmits the clock onto an LCN coaxial cable. The current loop may be up to 10 m long. The FOC/RCVR board is used in applications that require transmission of 12.5 kHz clock data from a main segment to a remote segment, over a distance of up to 2 km. If the remote segment is less than 300 meters away from the central segment, the clock may be transmitted directly with a shielded, twisted-pair current loop from the central CS/R to the remote CS/R, eliminating the need for the FOC/XMTR, FOC/RCVR, and fiber cable components.

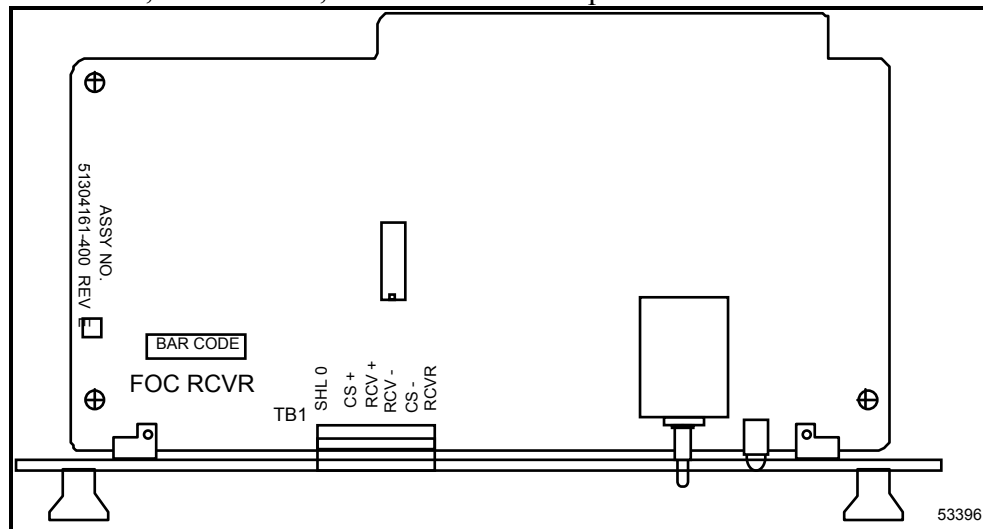


Figure 3-7 — FOC/RCVR

NOTE

The FOC/RCVR board, 51304161-400, is a later version of the FOC/RCVR board, 5130416-200. They are functionally equivalent and your system may include either or both types.

3.2.2.3 Clock Source/Repeater (CS/R)

Each CS/R board must be mounted in Slot 1 of the I/O card cage of a 5-Slot or 10-Slot Module, directly behind an EMPU, HMPU, or HPK2 processor board. This is the only location where the CS/R board can receive the ± 12 Vdc power that it requires.

At the central (main) segment, two CS/R boards (one for LCN Cable A and one for Cable B) each receive the 12.5 kHz clock signal through the I/O connector from the processor.

Each CS/R transmits the clock onto an LCN coaxial cable of the central segment. It also provides a 300 m maximum length current loop output which may be used to drive up to four loads. The loads may be FOC/XMTR boards (used for transmission of the clock over fiber optic cables to remote segments), and/or other CS/R boards. If there are more than four remote segments (up to six are allowed), the first CS/R can include in its loop a second CS/R, which is used as a repeater and can drive the additional required FOC/XMTR or remote CS/R boards. These scenarios are covered in more detail in subsection 3.3.1.

If the remote segment is less than 300 meters away from the central segment, the clock may be transmitted directly with a shielded, twisted-pair current loop from the central CS/R to the remote CS/R, eliminating the need for FOC/XMTR, FOC/RCVR, and fiber cable components.

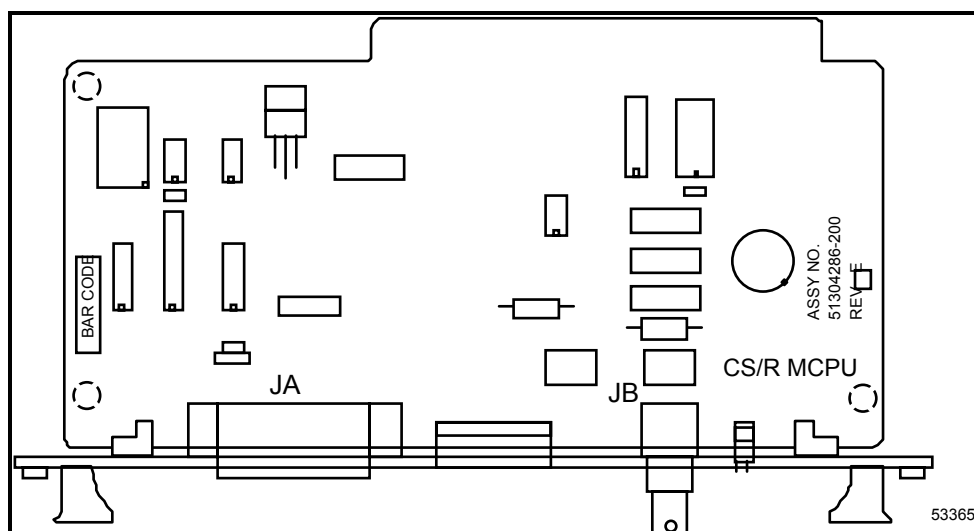


Figure 3-8 — CS/R Board Assembly

NOTE

The CS/R board, 51304266-200, is a later version of the XLCNE2 board, 51304266-100. They are functionally equivalent and your system may include either or both types.

NOTE

In a Dual Node Module, the CLCN A/B would be replaced by a CLCNA and a CLCNB board.

3.3 LCN Extender Hardware Implementation

Section 2 of this publication covers rules and guidelines for the implementation of a TPS system from the standpoint of coaxial cable segments while Section 3 covers similar aspects of fiber optic extender links. This subsection, 3.3, includes some rules and guidelines concerning fiber optic extension links. It is intended to be supplemental to the material in Section 2. In some cases it may overlap material presented in Section 2 and elsewhere. Nevertheless, it is not intended to be a stand-alone reference for extending LCN segments. When planning or expanding a system, it is essential to consider all aspects, including rules and guidelines for systems, coaxial segments, and fiber optic extensions, because all of these elements are interrelated.

3.3.1 LCNE Configuration Rules

Consider the following general information, rules, and guidelines when you configure a TPS system that is to contain one or more fiber optic link(s).

- LCN Extender sets, and fiber optic cables up to eight km (26,246 ft) in length, can be used to interconnect LCN coaxial bus segments. Four LCN Extender boards (XLCNE2) and four fibers are required to extend both Cable A and Cable B to a remote area. Two fibers (one for transmit and one for receive) are used for the Cable A extension and, similarly, two fibers are used for the Cable B extension. An XLCNE2 is used at each end of the A fiber pair and at each end of the B fiber pair. For examples, see Figures 3-18, 3-19, and 3-20.
- The LCN extension with XLCNE2 and fibers takes care of extending the A and B data path (which can include the 5 Mbits/s digital clock) to the remote segment. It does not, however, accomplish transmission of the 12.5 kHz clock to the remote segment. If the remote segment requires the system clock and cannot use the digital clock exclusively (i.e., it contains EMPU, HMPU, or HPK2 processors), then separate means must be provided to transmit the 12.5 kHz clock to the remote segment (see the next two paragraphs). Subsection 2.5 has additional information about the 5 Mbits/s and 12.5 kHz system clocks.
- If a remote segment requires the 12.5 kHz clock and it is located more than 300 m from the central segment, then two additional fibers must be used to transmit the 12.5 kHz clock from the central segment to the remote segment. One fiber carries the clock for Cable A and the other fiber carries the clock for Cable B. Two Fiber Optic Clock Transmitter boards (FOC/XMTR) are used at the central segment. They receive the clock over twisted pair current loops from Clock Source/Repeater boards also located in the central segment. The FOC/XMTR boards transmit the clock over fiber links to Fiber Optic Clock Receiver boards (FOC/RCVR) located

in the remote segment. The FOC/RCVR boards transmit the clock over 10 m maximum length twisted-pair current loops to CS/R boards in the remote segment, which, in turn, transmit the clock onto the LCN coaxial cables. This configuration is illustrated in Figures 3-18, 3-19, and 3-20. The CS/Rs in the central segment are located in nodes which have been configured as clock sources in the software Network Configuration File (NCF).

If a remote segment requires the 12.5 kHz clock and it is located less than 300 m from the central segment, the clock optionally can be transmitted directly from the CS/Rs in the central segment to the CS/Rs in the remote segment over 300 m maximum length twisted pair current loops. This eliminates the need for the FOC/XMTRs, FOC/RCVRs, and clock fibers as described in the previous paragraph; however, with this approach you lose the advantages of fiber optics described in subsection 3.1.1.

- Although XLCNE2s, FOC/XMTRs, and FOC/RCVRs can be mounted in open I/O slots of any node type, it is desirable to locate them in nodes that are the least likely to have the power turned off. Table 2-1 shows the recommended order that should be used in selecting node types for housing these extender boards.
- The XLCNE2s, FOC/XMTRs, and FOC/RCVRs for Cable A must be in a different node than the ones for Cable B and must be powered from separate ac power entries. If more space is required, an otherwise empty Dual Node Module with a power supply and KJMP load board (51401594-200) installed in each node should be used to house the extender board sets. The XLCNE2 and associated FOC/XMTR or FOC/RCVR for Cable A are placed in the I/O section of the two-slot module and the corresponding boards for Cable B are placed in the three-slot node. If FOC/XMTR or FOC/RCVRs are not used, two XLCNE2s for Cable A may be placed in the two-slot node and two XLCNE2s for Cable B may be placed in the three-slot node.
- When LCN extension hardware is used, the configuration of the LCN has increased importance from the standpoint of reliability and failure analysis. This hardware represents points in the system where failure or power loss to a relatively small percentage of the system can potentially isolate and thereby affect a whole segment or group of modules due to disruption of communications. The extension hardware is transparent to the system communications software, and therefore in non-failure conditions the topology of the system may seem unimportant. However, when failure scenarios are considered, it is evident that system layout can dramatically affect diagnosis, recovery, and visibility into processes. The following rules and guidelines are essential to maximize system reliability in the event of LCN Extender failures.
 - The configuration of Removable Media Disk Drives and Engineer's Keyboards is very important. This subject is addressed in detail in subsection 2.1.1.

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.3. LCN Extender Hardware Implementation

- Fiber optic A and B channel fibers should be in separate cables and routed on separate paths from source to destination. Therefore, if one cable is damaged, the other will still be operable. However, for any two nodes in the system, the maximum difference in Cable A and Cable B lengths (including the sum of all coaxial and fiber optic sections) cannot exceed 300 meters, or communication problems may result. Refer to subsection 3.6.1 for additional details.
- The number of coaxial cable bus segments connected in serial, by way of fiber optic cables and XLCNE2s, must not exceed three.
- The LCNE2/XLCNE2 board for LCN Cable A should be installed in a different module than the XLCNE2 board for LCN Cable B. This is also true for FOC/XMTR, FOC/RCVR, and CS/R pairs. For example, one NIM of a redundant pair could have the XLCNE2 for Cable A and its partner NIM should have the XLCNE2 for Cable B. The same principle applies to redundant HG pairs. See Figure 3-2.
- All nodes containing an XLCNE2 should be on an uninterruptible power supply (if available) to ensure that the LCN coaxial cable segments cannot become disconnected because of power outages and thus cause reconnection problems.
- If possible, avoid configuring only two nodes on a remote segment due to error scenarios which might occur if the fiber optic cable is broken or if XLCNE2s have had the power removed.
- Because of the lower reliability of the fiber optic extension to remote segments, locate redundant HGs and added HGs on the same coaxial cable segment. Also, locate redundant NIMs and redundant PLCGs on the same coaxial cable segment.
- To prevent loss of view to the process, two or more Universal Stations or Universal Work Stations in a console should be on the same coaxial cable bus segment. In addition, for plant integrity, we recommend that the NIMs and HGs most often associated with a Universal Station be on the same coaxial cable bus segment.
- The two clock sources configured in the Network Control File must be located on the main coaxial cable bus segment.
- If the software release is 210 or earlier, both fiber optic links (A and B) on the XLCNE2 should not be turned off at the same time and then on again without bringing down the nodes on one side of the XLCNE2. If the nodes are not brought down (power removed or shut down), reconnection problems could occur.
- Unless the Precision Clock Source option is installed, the power supply in each node of the system must be set up by a pinning option for the internal or line frequency clock. If the Precision Clock Source option is installed, the power supplies in the nodes that have the Precision Clock Source installed in them must be pinned for the external clock option, and all other nodes are pinned for internal clock. Figures 3-11 through 3-17 show the clock pinning option for the standard, enhanced, and dual node power supplies, respectively.

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.3. LCN Extender Hardware Implementation

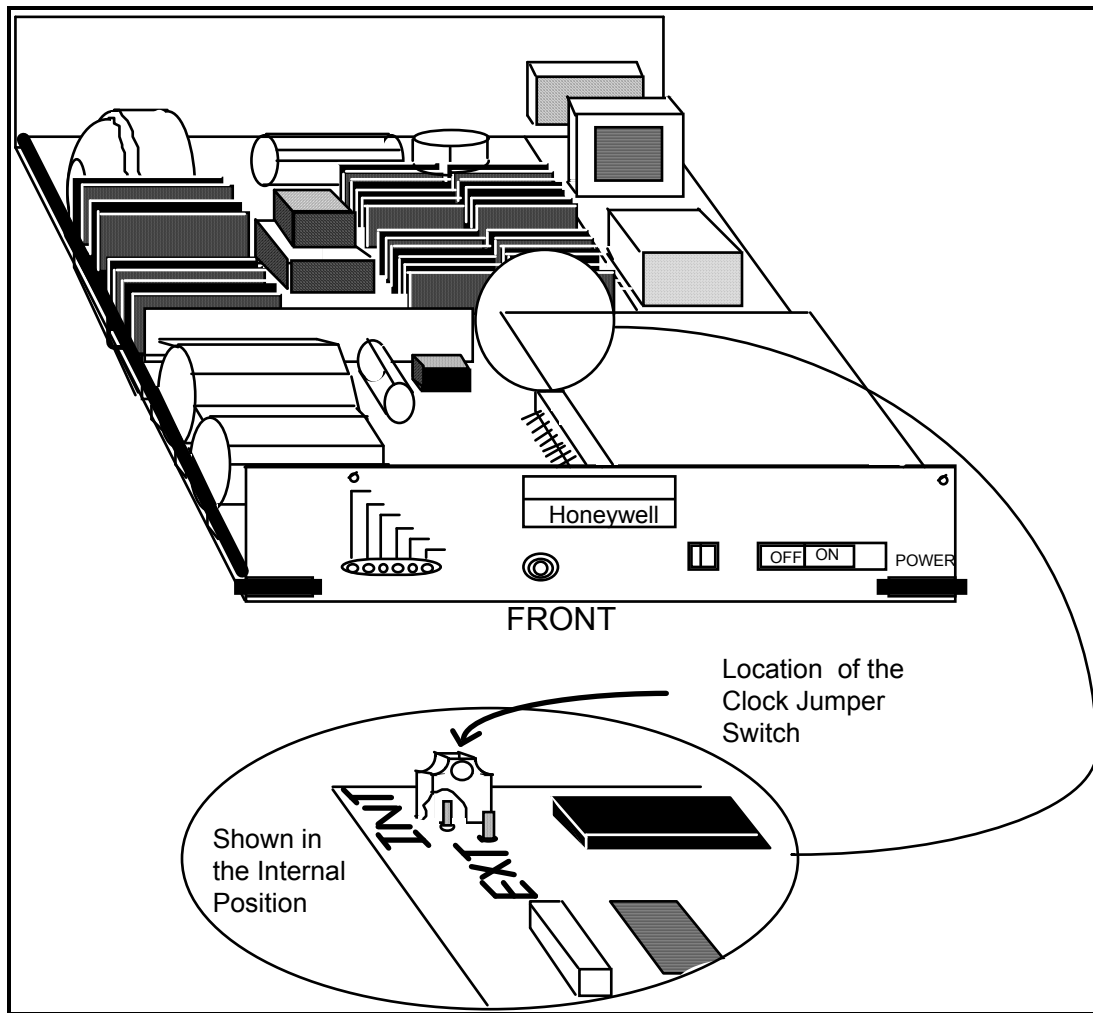


Figure 3-9 — Standard Power Supply Showing Location of Clock Jumper

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.3. LCN Extender Hardware Implementation

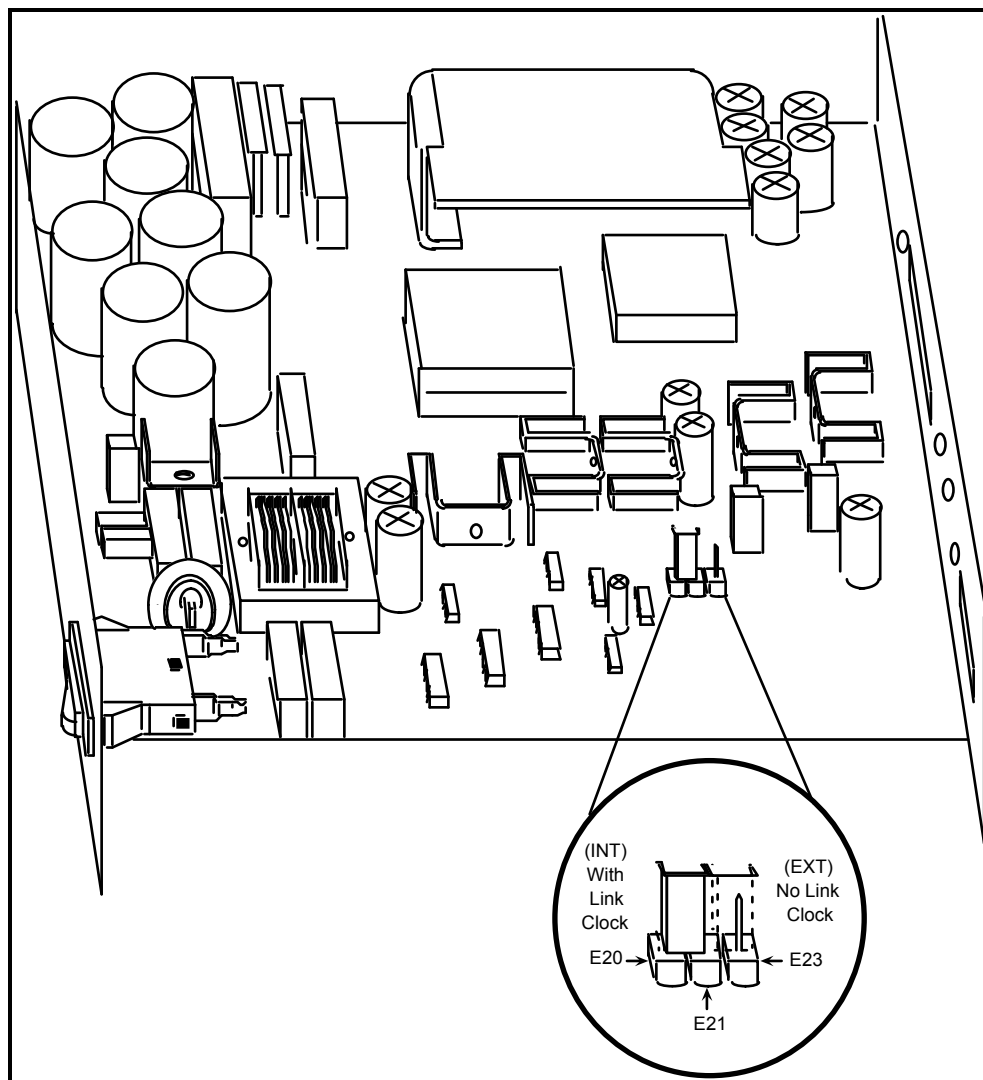


Figure 3-10 — Enhanced Power Supply Showing Location of Clock Jumper

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.3. LCN Extender Hardware Implementation

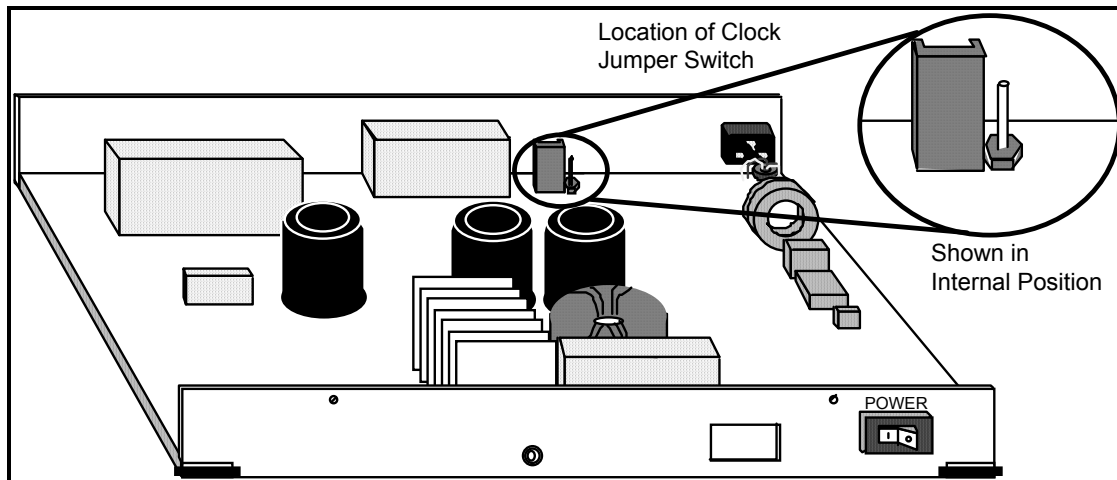


Figure 3-11 — CE Compliant Five-Slot Power Supply Showing Location of Clock Jumper

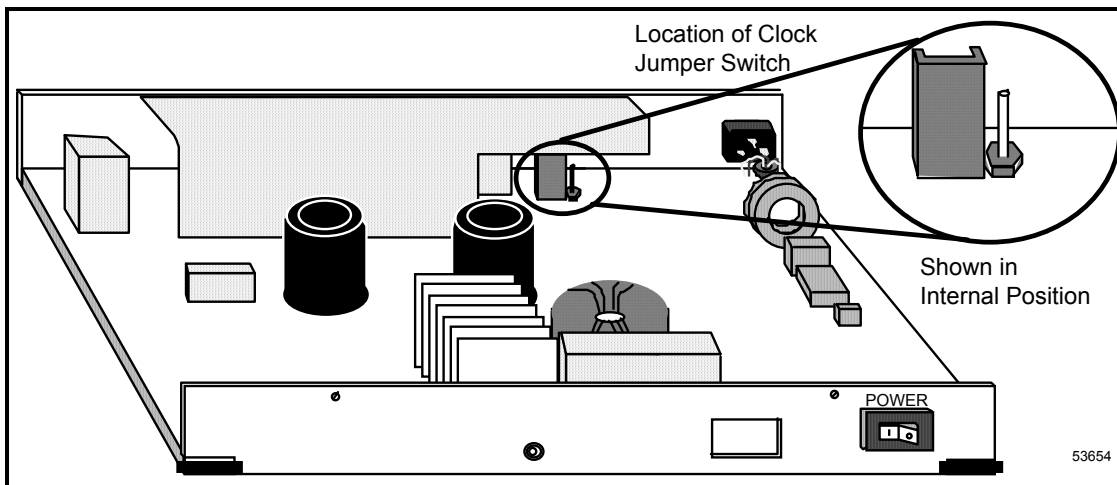


Figure 3-12 — CE Compliant Ten-Slot Power Supply Showing Location of Clock Jumper

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.3. LCN Extender Hardware Implementation

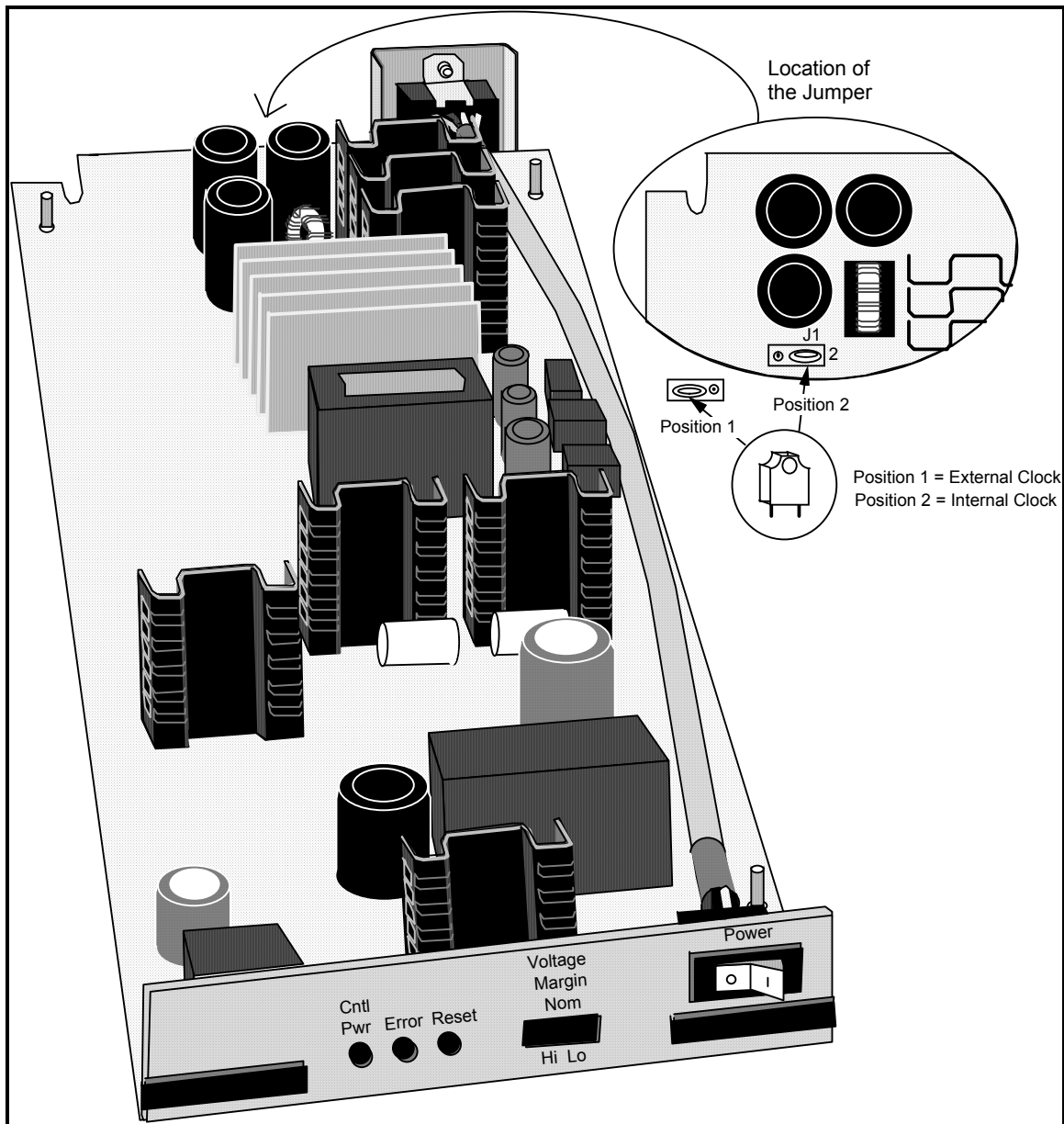


Figure 3-13 — Dual Node Power Supply Showing Location of Clock Jumper

There are two versions of the CE Compliant Dual Node power Supply. Match your power supply to either Figure 3-14 or Figure 3-15.

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.3. LCN Extender Hardware Implementation

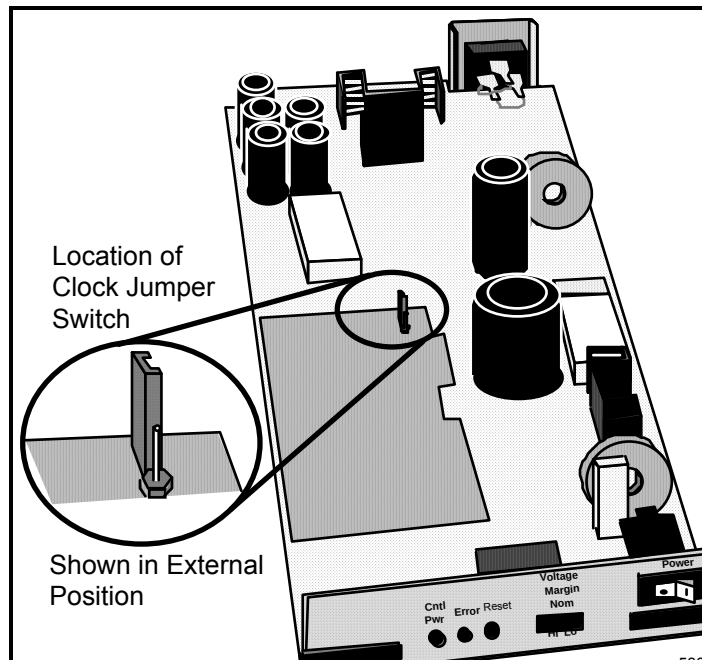


Figure 3-14 — CE Compliant Dual Node Power Supply Showing Location of Clock Jumper

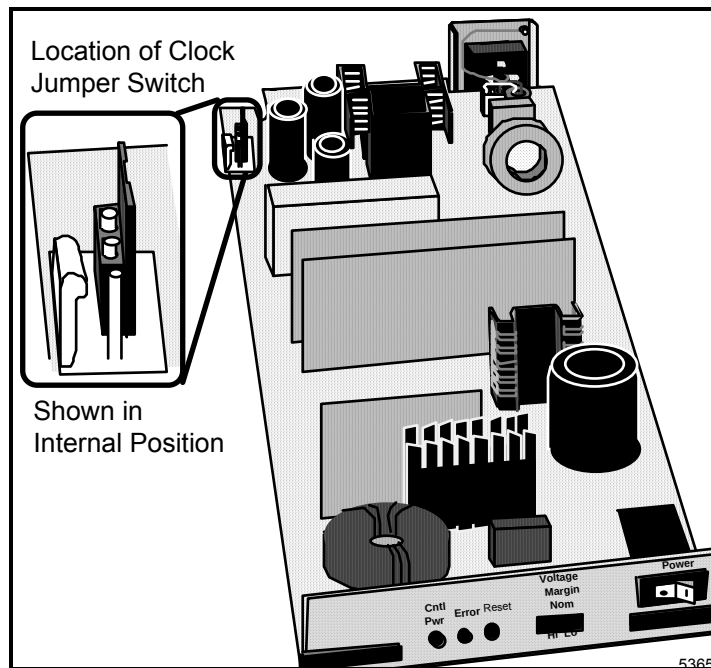


Figure 3-15 — CE Compliant Dual Node Power Supply Showing Location of Clock Jumper

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.3. LCN Extender Hardware Implementation

3.3.2 LCN Extension Implementation Examples

This subsection includes examples of the use of LCN extension hardware, including the use of XLCNE2s to extend the LCN data circuits, the use of an LCNFL in a single remote node, and the use of FOC/XMTRs, FOC/RCVRs, and CS/Rs to extend the 12.5 kHz system clock to a remote segment. It illustrates the optional use of a twisted pair current loop in place of the FOC/XMTRs, FOC/RCVRs, and clock fibers when the distance between the segments is 300 meters or less. Also shown in detail are the fiber and twisted pair interconnections among the various extension components for different configurations.

Figures 3-18 and 3-19 show a central or main segment and the remote segment split between the two figures. They show the interconnections of the two coaxial cables (Cable A and Cable B) in the both the central and the remote segments. The example includes a fiber optic clock link using FOC/XMTRs and FOC/RCVRs for transmission of the 12.5 kHz system clock from the central segment to the remote segment. At the central segment, note the twisted pair cables which carry the clock from the CS/R boards to the FOC/XMTR clock transmitter boards, and at the remote segment, note the twisted pair cables which carry the clock from the FOC/RCVR clock receiver boards to the remote CS/Rs. The CS/Rs in the remote segment then transmit the clock onto the A and B coaxial cables.

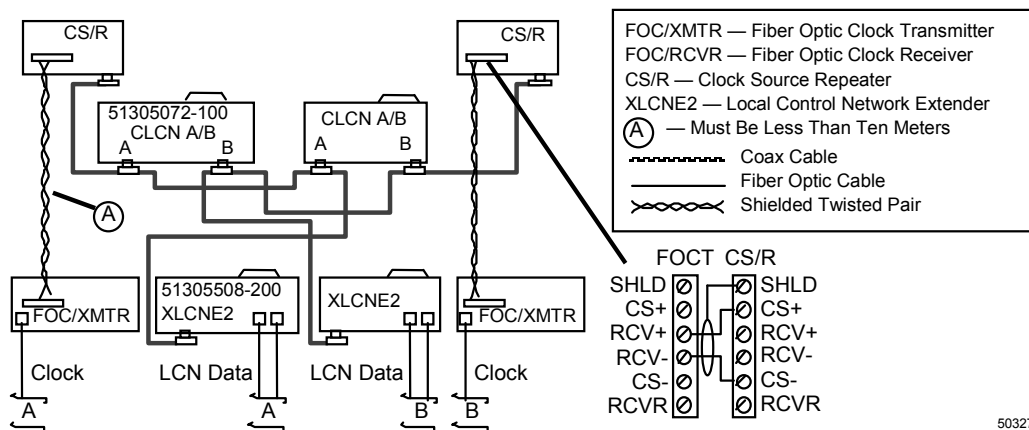


Figure 3-16 — LCN Extension Set Connections for One of a Pair of XLCNE2s

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.3. LCN Extender Hardware Implementation

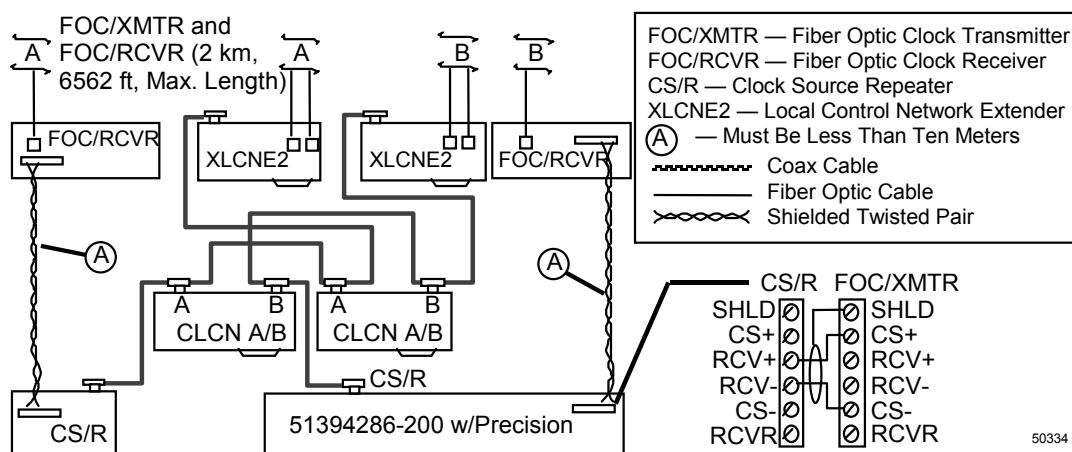


Figure 3-17 — LCN Extension Set Connections for One of a Pair of XLCNE2s

Figure 3-18 illustrates pictorially the use of LCN Extender components to connect a remote segment to a central segment. This example also includes the FOC/XMTRs, FOC/RCVRs, and fibers required to transmit the 12.5 kHz system clock to the remote segment. In this illustration, the extension components (XLCNE2s, FOC/XMTRs, FOC/RCVRs, and CS/Rs) associated with Cable A are located in a separate module from the components associated with Cable B. This separation occurs in both the central segment and the remote segment. As indicated in the figure, this placement of the Cable A components in a separate module from the Cable B components minimizes the possibility of total communications loss in the event of a power off condition or power supply failure at one module. Although this example has the CS/R, FOC/XMTR or FOC/RCVR, and XLCNE2 for each coaxial cable section located in the same module, this is not required. It is desirable for reliability that all of the components for at least one cable (A or B), including fiber interface components, be on an uninterruptible power source. It is also desirable that the fibers associated with each cable be routed separately, and that spare fibers be included, as is covered in more detail in subsection 3.6.1.

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.3. LCN Extender Hardware Implementation

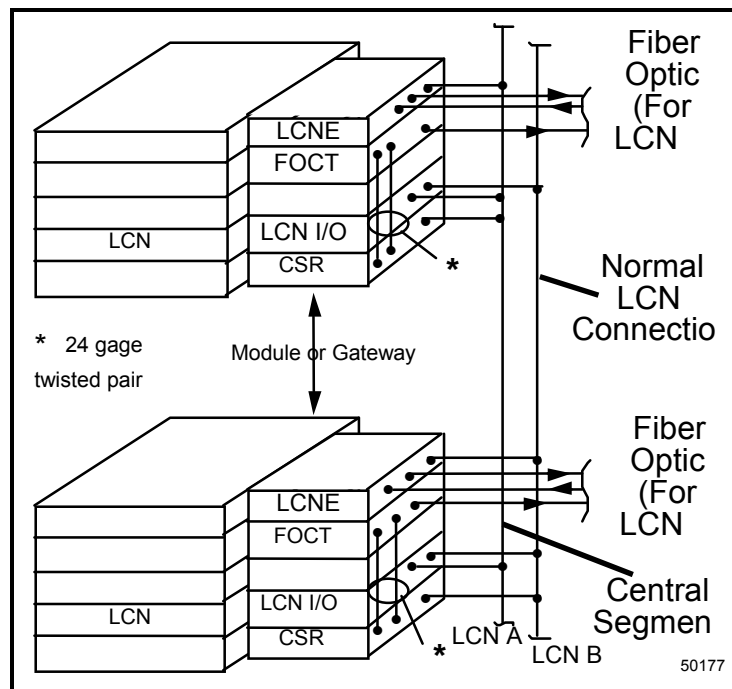


Figure 3-18 — Connection Example for Cable A and B

Coax Segment Beta S 300

CS/ FOC/RCV XLCNE2

Coax Alpha S300 Meter

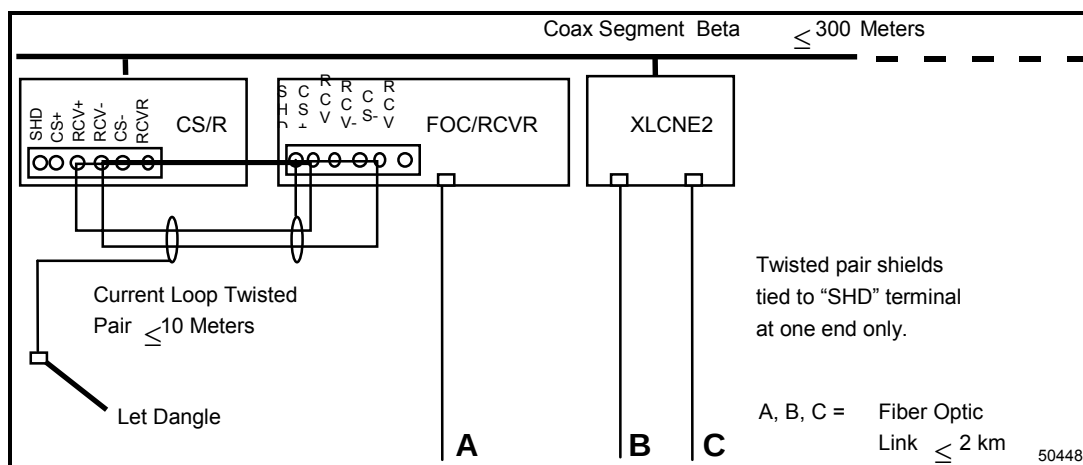
FOC/XMT CS/ FOC/XMT XLCNE2

CS/ FOC/RCV XLCNE2

Coax Segment S 300

50447

Figure 3-20 shows a central segment “Alpha” that has three remote segments, “Beta,” “Delta,” and “Gamma.” Remote segment “Delta” has the 12.5 kHz clock transmitted to it with fiber optic components as in the previous example. However, remote segments “Beta” and “Gamma” have the clock transmitted to them by current loop. This figure shows in detail the connections for the current loop twisted pairs and shields. Note that the current loop from the CS/R in the central segment includes as loads the FOC/XMTR (for transmitting the clock to segment “Delta”) as well as the CS/Rs for segments “Beta” and “Gamma.” The total length of this current loop (which includes the sections marked “X,” “Y,” and “Z” in Figure 3-28 and 3-29) must be no greater than 300 meters.



3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.3. LCN Extender Hardware Implementation

Figure 3-20 — LCN Extender Fiber Optic Cabling Current Loop Connections

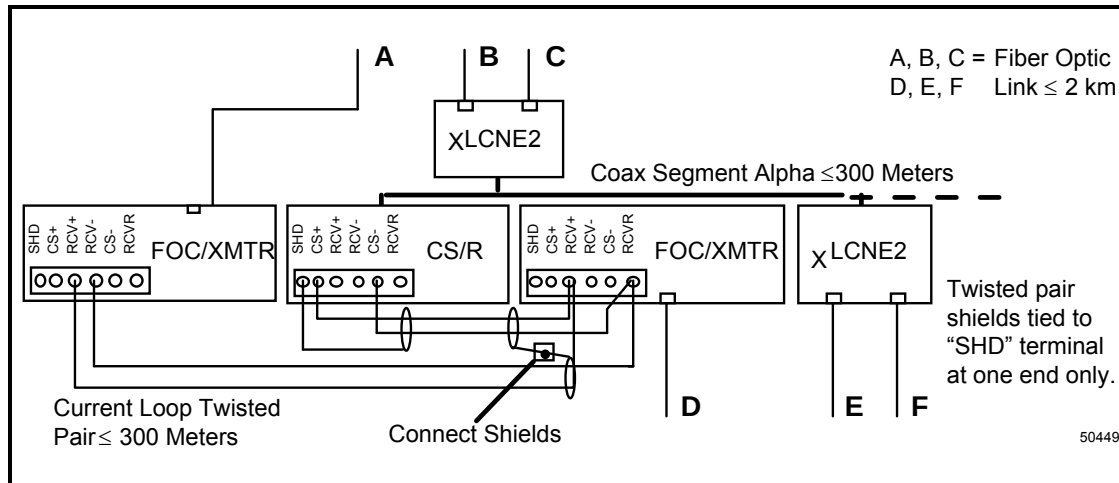


Figure 3-21 — LCN Extender Fiber Optic Cabling Current Loop Connections

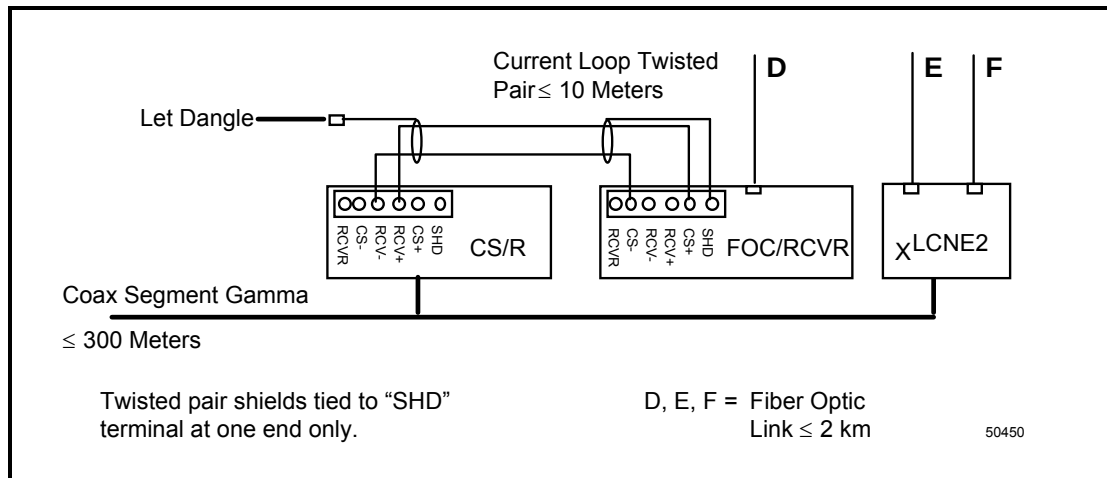


Figure 3-22 — LCN Extender Fiber Optic Cabling Current Loop Connections

Figure 3-23 is an overview that illustrates this function, showing a CS/R driving three FOC/XMTRs and another CS/R, which in turn drives three additional FOC/XMTRs. The clock can be retransmitted only twice in this manner, or excessive distortion of the signal may result. The maximum length of each current loop is 300 meters, giving an overall maximum length of 900 meters, assuming the loop is retransmitted twice.

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.3. LCN Extender Hardware Implementation

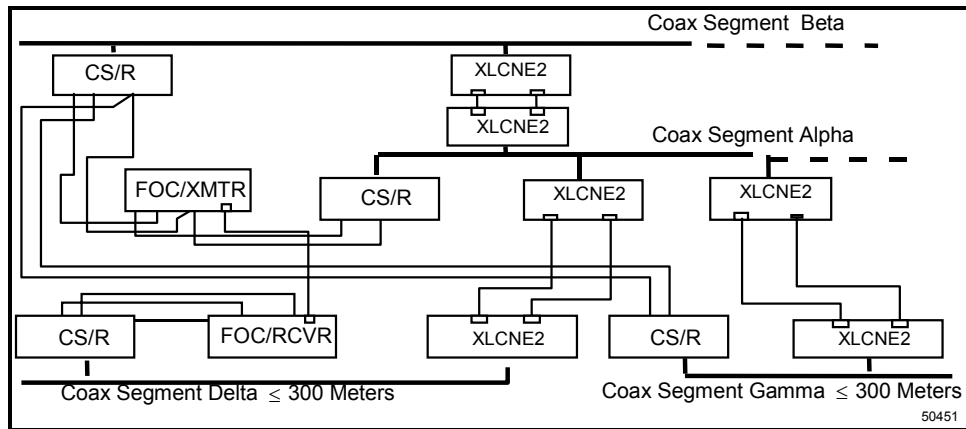


Figure 3-23 — Multiple Current Loop Connections with Retransmitted Clock

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.3. LCN Extender Hardware Implementation

Figure 3-24 and 3-25 illustrate the detailed wiring of the current loops used in Figure 3-23.

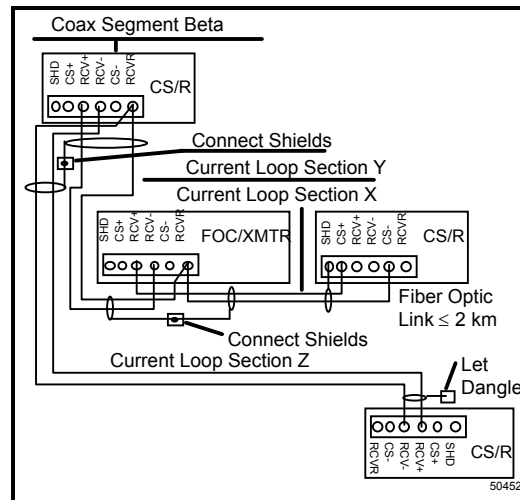


Figure 3-24 — Wiring for Multiple Current Loop Connections

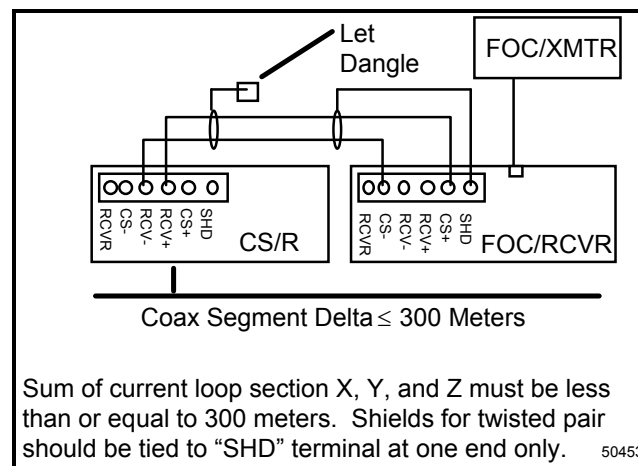


Figure 3-25 — Wiring for Multiple Current Loop Connections with Retransmitted Clock

The current loop output of a CS/R can drive up to four loads, which can be either FOC/XMTRs or other CS/R inputs. If more than four loads are required (i.e., there are more than four remote segments which require the 12.5 kHz clock), a CS/R can be used to repeat, or retransmit, the clock.

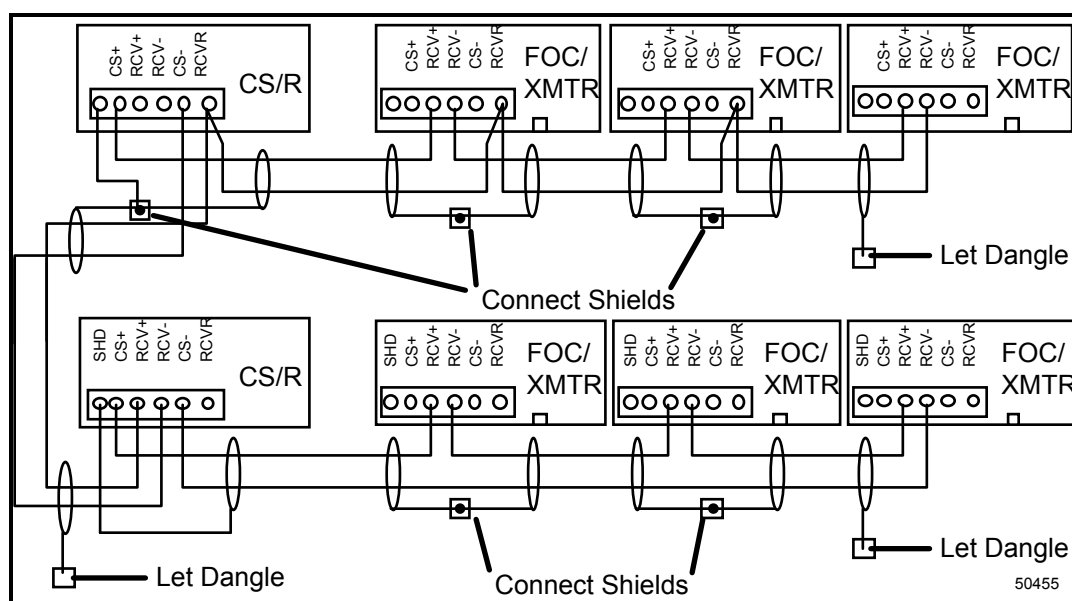


Figure 3-26 — Multiple Current Loop Connections with Retransmitted Clock

NOTE

1. A CS/R current loop can be repeated only twice.
2. This set of boards and connections is for Cable A and must be repeated for Cable B.

3.4 Fiber Optic Cable Specifications

There are four Honeywell raw fiber optic cable specifications. They describe both indoor and outdoor versions of fiber optic cable in both the 100 micron and 62.5 micron fiber core sizes. In addition, there are three Honeywell fiber optic cable assembly drawings, each with associated marketing model numbers. “Micron” is another term for “micrometer,” a unit of measure, one thousandth of a millimeter, or one millionth of a meter. The symbol “ μm ” is often used to denote micron or micrometer. The symbol “nm” means nanometer (one billionth of a meter) and is used as a unit of measurement of wavelength of light.

3.4.1 100 Micron and 62.5 Micron Fiber

100 micron was the fiber core size originally specified for the **TotalPlant** Solution (TPS) system. To date, most TPS systems that have fiber installed use 100 micron fiber. In the meantime, however, the industry has moved toward 62.5 micron fiber as a de facto standard for multimode fiber. Hastening this process is the preference for 62.5 micron fiber in the recently completed ANSI Fiber Distributed Data Interface (FDDI) specification. As a result of this shift, 62.5 micron fiber is now more readily available at a lower cost than 100 micron fiber. Recognizing this industry trend, Honeywell has added 62.5 micron fiber as an option in implementing TPS LCN fiber optic installations. As a matter of fact, Honeywell now recommends that all new installations use 62.5 micron fiber.

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.4. Fiber Optic Cable Specifications

Now that two different sizes of fiber are allowed in TPS systems, there exists the possibility of inadvertently mixing fiber sizes in a fiber link. This is generally undesirable. There is no additional loss when light goes from the smaller fiber to the larger one, but a 4.1 dB loss is suffered when going from larger to smaller. Since fiber cables are run in pairs, one for transmit and one for receive, light will be going in opposite directions in the individual fibers of the pair. Adding a different fiber size to both cables at one end of a link (such as when transitioning from outdoor cable to indoor cable) means that at least one cable will suffer at least a 4.1 dB loss. Therefore, to be safe, DO NOT mix fiber sizes in a link.

NOTE

1. XLCNE2 fiber optic extensions, using 62.5 micron fiber with a maximum attenuation of 0.9 db/km can be made up to 8 kilometers maximum.
2. XLCNE2 fiber optic extensions, using 100 micron fiber, can be made up to 1 kilometer. Refer to upgrade kit model numbers TP-ZEC100 and TP-ZEN100.
3. XLCNE2 fiber optic extensions, using 62.5 micron fiber with a maximum attenuation of 2 db/km, can be made up to 2.25 kilometers maximum. Refer to upgrade kit model numbers TP-ZEC625 and TP-ZEN625.

3.4.2 Procurement Policies

3.4.2.1 Outdoor Fiber Optic Cable

In general, Honeywell does not wish to supply or install outdoor fiber optic cables. Honeywell relies on the cable installation expertise of cable vendors and installation contractors to perform the cable installation. Honeywell also does not wish to restrict the purchase of outdoor fiber optic cable to a particular vendor. Honeywell recognizes that vendors may be able to supply better service in some parts of the world than in others, thus making it desirable to have a choice worldwide. Also, the installation conditions at various project sites may call for widely differing types of cable construction. For these reasons, the outdoor cable specifications were written rather loosely with respect to physical construction details and mechanical parameters. The actual glass fiber itself is completely specified to ensure proper operation of the fiber link.

If the customer requires, Honeywell will contract with cable vendors and installation contractors for the customer to purchase cable, oversee, and guarantee a proper installation. However, if the customer procures his fiber optic cable directly from the supplier, and arranges his own installation, the cable supplier and/or installing contractor must certify to the customer that his cable fully meets or exceeds the applicable Honeywell cable specification. Honeywell will freely supply our outdoor cable specifications to our customers for this purpose.

3.4.2.2 Indoor Fiber Optic Cable

In contrast, the indoor cable specifications completely specify both the mechanical construction details of the cable, and the important parameters of the glass fiber. The recommended method of procurement is direct purchase from Honeywell as a finished cable assembly, including connectors, under the applicable Honeywell model number. See subsection 3.4.5 for a description of the cable assemblies that are available, and the corresponding Honeywell model numbers.

3.4.3 Indoor Grade Cable Specifications

Figure 3-27 illustrates the construction of indoor fiber optic cable as specified by Honeywell. Both 100 micron and 62.5 micron indoor cables share this basic construction, differing only in the fiber size.

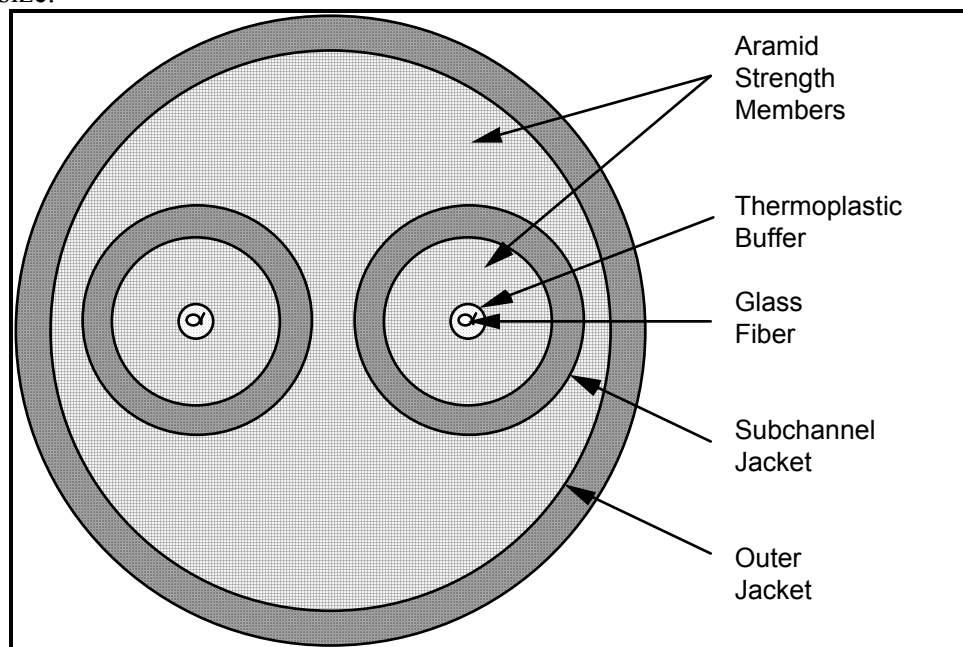


Figure 3-27 — Indoor Tight Buffered Cable

This type of cable is known as Tight Buffered cable because the optical fibers are tightly held by the cable fillers. Unmatched coefficients of expansion between the cable materials and the glass fiber can subject the fibers to significant microbending losses if the cable is exposed to temperature extremes. For this reason, tight buffered cable is limited to indoor use. The advantage indoors is good physical protection for the fiber while maintaining the cable flexibility required for routing the cable inside electronic cabinets, under floors, etc. The cable contains two, four, or six subunits. Each subunit protects a single fiber and can have independent connectors. The standard indoor cable assemblies are duplex (two fibers). Finished cable assemblies with four or six fibers are available on special order. There are two cable jacket options available for the indoor cable. The National Fire Protection Association (NFPA) publishes the National Electrical Code (NEC) to establish fire safety standards for premises wiring. Honeywell specifies jacketing material conforming to either NEC optical cable rating OFNR (Optical Fiber, Nonconducting, for Riser applications), or OFNP (Optical Fiber, Nonconducting, for Plenum applications). The standard cable assemblies, which can be purchased from the Honeywell price book by model number, are OFNR rated cables. Should OFNP rated cables be required, they can be special ordered through Honeywell Purchasing. OFNP rated cables are required only when routing indoor fiber runs through air handling chambers.

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.4. Fiber Optic Cable Specifications

3.4.3.1 Key Parameters, Indoor Grade Cable, 62.5 Micron (Drawing Number 51191597)

The important parameters of the 62.5 micron indoor grade glass fiber specified by Honeywell are as follows:

Core diameter:	62.5 ± 3 microns
Cladding diameter:	125 ± 2 microns
Bandwidth @ 1300 nm:	500 MHz-km min.
Attenuation @ 1300 nm:	.9 dB/km max.

3.4.3.2 Key Parameters, Indoor Grade Cable, 100 Micron (Drawing Number 51190697)

The important parameters of the 100 micron indoor grade glass fiber specified by Honeywell are as follows:

Core diameter:	100 ± 4 microns
Cladding diameter:	140 ± 6 microns
Bandwidth @ 850 nm:	100 MHz-km min.
Attenuation @ 850 nm:	6.0 dB/km max.

3.4.3.3 Mechanical Properties, 62.5 Micron Indoor Grade Cable

Table 3-1 includes the mechanical properties of indoor grade cable supplied by Honeywell. “Subchannel” refers to an individual jacketed fiber, which is part of a multi-fiber cable (see Figure 3-27). “N” stands for “Newton,” a metric unit of force equal to approximately 0.225 lb.

Table 3-1 — Mechanical Properties of 62.5 µm Indoor Grade Cable

Characteristic	Two Fiber Cable	Four Fiber Cable	Six Fiber Cable
Tensile Load, Short Term, N (LBF)	900 (200)	1000 (225)	1000 (225)
Tensile Load, Log Term, N (LBF)	300 (68)	400 (90)	350 (79)
Minimum Bend Radius, Loaded, CM (IN)	9.0 (3.5)	10.2 (4.0)	11.9 (4.7)
Minimum Bend Radius, Installed, CM (IN)	5.9 (2.3)	6.8 (2.7)	7.9 (3.1)
Nominal Outer Diameter, MM (IN)	5.9 (0.3)	6.8 (0.3)	7.9 (0.3)
Nominal Weight KG/KB (LB/KFT)	34 (23)	46 (31)	65 (44)

3.4.4 Outdoor Grade Cable Specifications

Honeywell specifies a cable construction known as Loose Tube for its outdoor grade fiber optic cable. An illustration of this construction is shown in Figure 3-28.

This construction is characterized by loose fitting, gel filled tubes into which the fibers are placed. The fibers are actually longer than the tubes, so that when thermal expansion lengthens the buffer tubes, the glass fibers (which have a lower coefficient of expansion) are never subject to tensile stress.

The cable is given buckle resistance typically by a glass reinforced plastic (GRP) rod through the center of the cable. An aramid wrap around the buffer tubes provides tensile strength.

This type of cable is significantly stiffer than tight buffered cable, and the jacketing material does not meet NEC requirements. These factors make it unsuitable for indoor use. The loose buffer design's advantage is being able to take environmental extremes without suffering any significant optical performance degradation.

Figure 3-28 shows a typical construction for an aerial/duct cable. A direct burial cable is similar, except that it would probably have two additional layers: a steel armor tape layer for rodent protection, covered by an additional polyethylene jacket. Vendors also may have double armoring options available. Again, exact construction details will vary from vendor to vendor, but the basic loose tube concept remains the same.

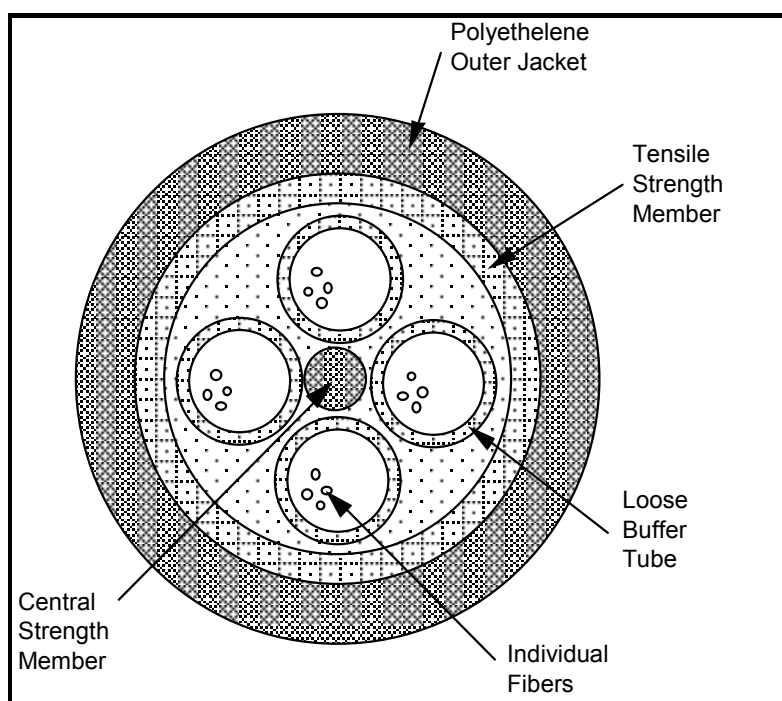


Figure 3-28 — Outdoor Loose Tube Cable

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.4. Fiber Optic Cable Specifications

The National Electrical Code ratings do not apply to outdoor cable because it does not fall into the category of premises wiring. The NEC does mention that the maximum length of unrated outdoor cable that is permitted inside a building is 50 feet. This allows enough length to bring the cable to a splice/interconnect enclosure where the transition to indoor cable can be made.

3.4.4.1 Key Parameters, Outdoor Grade Cable, 62.5 Micron (Drawing Number 51190919)

The important parameters of the 62.5 micron outdoor grade glass fiber specified by Honeywell are identical to the parameters specified for indoor cable:

Core diameter:	62.5 ± 3 microns
Cladding diameter:	125 ± 2 microns
Bandwidth @ 1300 nm:	500 MHz-km min.
Attenuation @ 1300 nm:	0.9 dB/km max.

3.4.4.2 Mechanical Properties of Outdoor Cable

As discussed in subsection 3.4.2, Honeywell does not recommend specific cable for outdoor use. When a cable is selected, refer to the vendor's specifications for that cable to obtain applicable mechanical properties.

3.4.5 Fiber Optic Cable Assemblies

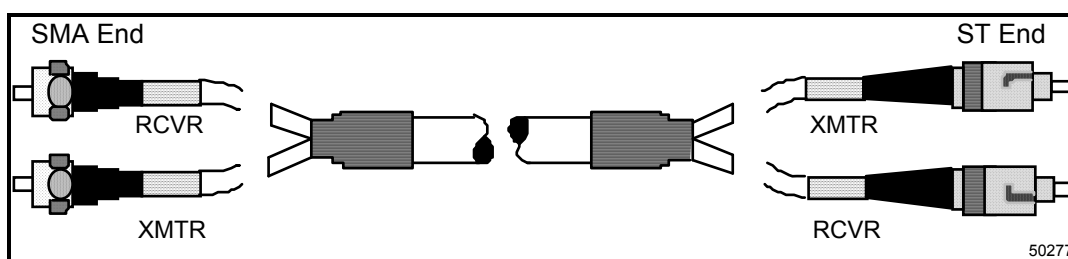
This subsection discusses the duplex indoor fiber optic cable assemblies with preinstalled connectors that are available from Honeywell. Table 3-2 lists the all of the standard assemblies and the paragraphs that follow the table discuss each type.

Table 3-2 — Standard Duplex Indoor Cable Assemblies

Model Number	Length (Meters)	Core Diameter (µm)	Description
P-KFB02	2	62.5	ST to ST
P-KFB05	5		
P-KFB10	10		
P-KFB20	20		
P-KFB50	50		

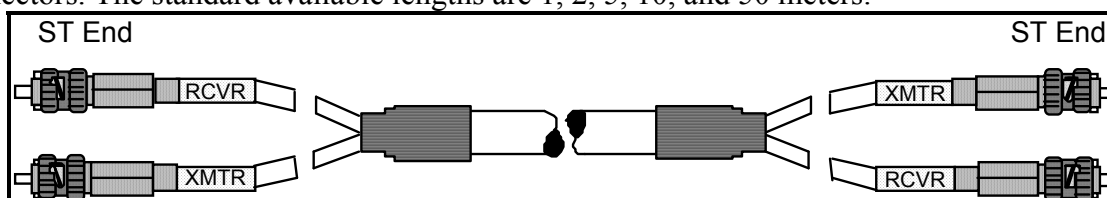
3.4.5.1 Model Number P-KFHxx — 62.5 Micron SMA/ST

This is a hybrid cable assembly in the sense that different connector types are used on each end. Just as 62.5 micron fiber has displaced 100 micron fiber as a de facto industry standard, the ST connector has largely displaced the SMA style connector. TPN equipment still uses SMA active device mounts, so the SMA connectors must still be used at the LCN end. This cable is provided for connection to a fiber optic interconnect panel which typically would be implemented using a bank of ST-style connector receptacles. See subsection 3.6.3.2 for a discussion of interconnect panels.

Figure 3-29 — P-KFHxx 62.5 μ m SMA to ST Cable Assembly

3.4.5.2 Model Number P-KFBxx — 62.5 Micron ST/ST

This duplex cable assembly is provided with ST connectors at both ends. XLCNE2 uses ST connectors. The standard available lengths are 1, 2, 5, 10, and 50 meters.

Figure 3-30 — P-KFBxx 62.5 μ m ST to ST Cable Assembly

3.4.6 Cable Connectors

Honeywell recommends the fiber optic connectors shown in Table 3-3 for use on indoor tight-buffered fiber optic cable only, because outdoor cable does not permit the small bend radii needed in a cabinet or equipment rack and is always spliced to a short length of indoor cable for connection purposes.

Table 3-3 — Cable Connectors

Supplier	Supplier's Part Number	Type	Power Loss (dB)	Fiber (μ m)
Amphenol Fiber Optic Products 1925 Ohio Street Lisle, IL 60532 708-810-5800	953-101-5010	ST - Ceramic Ferrule	0.13 dB median	62.5

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.5. Calculating the Link Power Budget

Special tools and procedures are required for installing connectors. The cable must be stripped, cut, finished, and attached to the connector. Typical tools and materials for performing these operations are shown in Table 3-4.

Table 3-4 — Tools and Materials

Supplier	Supplier's Part Number	Type	Use
Amphenol Fiber Optic Products 1925 A. Ohio Street Lisle, Ill 60532 Phone: 630-960-1010 Fax: 630-810-5640 sales@amphenol-fiberoptics.com http://www.amphenol-fiberoptics.com	927-100-2053	Add-on Kit	953 Series ST Add-on Kit
Amphenol Fiber Optic Products 1925 A. Ohio Street Lisle, Ill 60532 Phone: 630-960-1010 Fax: 630-810-5640 sales@amphenol-fiberoptics.com http://www.amphenol-fiberoptics.com	927-100-5000	Basic Termination Kit	Basic Termination Kit with multiport over (3 ST/FC ports and 3 SC/SMA ports)
Amphenol Fiber Optic Products 1925 A. Ohio Street Lisle, Ill 60532 Phone: 630-960-1010 Fax: 630-810-5640 sales@amphenol-fiberoptics.com http://www.amphenol-fiberoptics.com	927-100-5041	Designer Kit	ST Designer Kit includes Basic Termination Kit 927-100-5000, 953 Series ST Add-on Kit 927-100-2053 and samples for 6 practice terminations.

3.5 Calculating the Link Power Budget

In planning the installation of a fiber optic link, one of the most important tasks is the calculation of the power budget. The power budget determines the link distance achievable, and indicates how much power margin exists. It depends on a number of factors, such as fiber attenuation, number of connector joints, and the number and type of splices in the link. The XLCNE2 transmitter module is guaranteed to couple 10 μ W (-20 dBm) of optical power into 62.5 micron fiber when new. The receiver modules are guaranteed to operate correctly (with a bit-error-rate better than 1 in 10^9) with only .8 μ W (-31 dBm) of received power.

3.5.1 Calculating the Power Budget for 62.5 Micron Fiber

Table 3-5 details the calculation of a typical link's power budget using 62.5 micron fiber:

Table 3-5 — Power Budget Calculation for 62.5 Micron Fiber

Minimum power coupled into fiber:	10 μ W = -20.0 dBm
Minus minimum power receiver needs:	(-) .8 μ W = (-) -31.0 dBm
Total power budget:	11.0 dB
Losses:	
Connector loss (2 * 1.0 dB):	1.0 dB
Splice loss (2 * 0.25 dB):	0.5 dB
Safety margin:	2.3 dB
Budget remaining for loss in cable:	7.2 dB
Divide by:	
Specified attenuation per km:	$\div$ 0.9 dB
Achievable link distance:	8.0 km

Starting with the minimum power guaranteed to be coupled into 62.5 micron fiber, 10 μ W or -20 dBm, subtract from that figure the minimum power level guaranteed to be properly decoded by the receiver, 0.8 μ W or -31.0 dBm. This calculation gives a total power budget of 11.0 dB. From this 11.0 dB, certain losses must be subtracted. The first shown is the loss of two connector-to-connector butt splices. Typically, two such connections are made in the link at interconnect boxes at the interface between the outdoor and the indoor cable—one at each end of the link.

Similarly, each indoor-to-outdoor cable interface may require a fiber-to-fiber splice, so two splice losses are subtracted from the budget. Lastly, a 2.3 dB safety factor is subtracted. A safety factor of 2.3 dB is minimal, and it is strongly recommended that if a larger safety factor can be arranged (either by shortening the required link distance, or by using fiber with a better attenuation figure), by all means do so. The larger the safety margin, the more reliable the link will be.

After all the above loss factors have been subtracted, the 7.2 dB that remains is the power left to be expended in cable loss. It is this number, when divided by the attenuation figure of the fiber, that determines the achievable link distance. As can be seen above, using the attenuation figure specified in Honeywell drawing number 51191598, a typical link distance of slightly more than 8 km can be achieved.

NOTE

It is important that the fiber optic cable be positioned and supported such that no stress is placed on the fiber optic cable's mechanical connection at the board. Any stress on the fiber optic connections will increase the signal loss at the connection reducing the available operating distance.

3.6 Cable Plant Implementation

3.6.1 Fiber Count and Redundant Routing

When extending the TPS LCN, two fibers are needed for each coaxial cable channel—a transmit and a receive fiber. Thus, to extend both A and B channels of the redundant LCN, four fibers are required. In addition, if clock synchronization is required across the fiber link, two more fibers are required to carry the clock signal to the remote end (one each for A and B channels), bringing the total to six fibers.

However, Honeywell does not recommend that both A and B channels be carried in the same cable. To maintain the inherent reliability of redundant channels, Honeywell recommends that the A and B channel fibers be carried in separate cables on separate routes. For example, the A cable may be buried on one side of a service road, and the B cable on the other side; or bury one cable and hang the other aerially. Only one basic rule needs to be observed when routing the cables along separate paths: the maximum distance between any two nodes in the system as measured on the A cable must not differ from the distance measured along the B cable by more than 300 meters. Note that the distances referred to above include the sum of all fiber optic and coaxial sections of cable.

Figure 3-31 is an example of a poor configuration. Segment Alpha is a central coaxial segment, with two remote coaxial segments, Beta and Gamma. There is no problem with communications between the Alpha segment and either of the two remote segments. Cable A is 200 meters shorter than Cable B between Alpha and Beta. Cable A is also 200 meters shorter than Cable B between Alpha and Gamma. 200 meters is within the 300 meter differential limit. **However, Cable A is a total of 400 meters shorter than Cable B between segments Beta and Gamma. Unreliable communication would result.**

Keep in mind that coaxial cable length differential also plays a part in this. In the example, we assumed that the coaxial A and B channels in each segment were equal. If they are not, they would also contribute to the total differential calculation. Remember that the criteria is 300 meters maximum differential between any two nodes.

Due to the expense of installing outdoor cable runs, it is generally recommended that outdoor cable runs contain 33% to 100% more fibers than is immediately required. The incremental cost of providing extra fibers in the cable is small compared with the cost of installing another cable if more data communications capacity is required in the future. For example, consider a single remote LCN extension requiring remote clock synchronization. If the recommendation for routing the A and B cables on separate paths is being followed, two cables of at least three fibers each would be required. To provide the extra capacity for future contingencies, the recommended fiber count per cable would be four to six (cables are generally available only in even fiber counts).

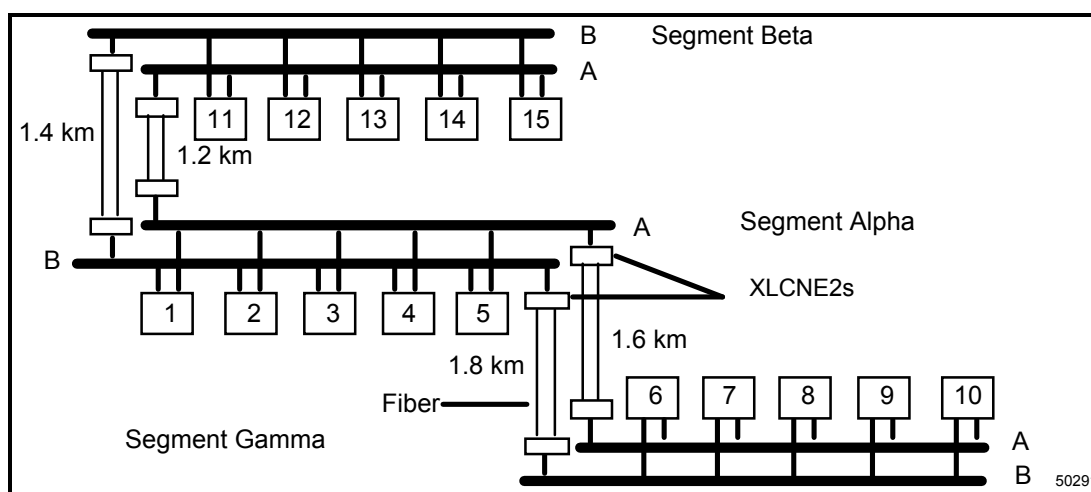


Figure 3-31 — Example of Incorrect Routing Configuration of Redundant Fiber Cables

3.6.2 Major Classes of Cable Installation

Aerial, underground duct, and direct cable burial are the three most popular ways to install outdoor cable runs. The method of choice for a particular site depends on any number of factors. Each has its own pluses and minuses.

3.6.2.1 Aerial

Aerial cable installations are subject to the greatest environmental stresses of the major classes of outdoor cable installation. An aerial cable is subject to wind loading, ice loading, extremes of temperature variation, and even gunshots.

There are two basic kinds of aerial cable available. One is designed to be lashed to a messenger wire as it cannot support itself. The other is self-supporting. Of the self-supporting aerial cables, there are two main subclasses: One contains the tensile stress bearing member within the normal cable sheath; the other type, which is capable of longer unsupported runs, is the "figure-8" type of cable, where the cable contains its own built-in messenger wire from which hangs the fiber-carrying capsule.

3.6.2.2 Underground Duct

This type of installation buries a plastic tube or tubes into which a cable or several cables can be pulled. A recommended fill ratio is 50% by cross sectional area. The cable is usually of virtually the same construction as a lashed aerial cable. Under the ground it is well protected from environmental stress. It is recommended that the duct be buried as deeply as possible—definitely below the frost line. Frost heaving can crack the duct. The relatively large diameter of the duct provides rodent protection. Rodents can't chew on it if they can't get their mouths around it. If empty inner ducts were provided within a larger duct, then new cables could be pulled in the future without too much trouble. Pulling new cable into an already occupied inner duct, however, is not recommended.

3.6.2.3 Direct Burial

This type of installation buries the cable directly into the earth. As with the underground duct, bury this cable as deeply as possible. Make sure to document and mark where the cable is buried for future reference. An armor sheath is required on this cable for rodent protection. Keep in mind, though, that the metal content of the cable may make it subject to lightning strikes. Also, the armor must be grounded at each end where it enters the building.

3.6.2.4 Cable with Preinstalled Connectors

In certain situations, it may be desirable to preinstall connectors to the cable at the factory. This may be the case where the skilled labor required to perform splicing and installing of connectors will not be available at the installation site. However, preinstallation of connectors makes pulling the cable through duct or conduit very difficult, so in duct situations, it may not be feasible. If handled carefully, it should not be as much of a problem in aerial or direct burial situations. Still, the risk of damaging the terminations during installation is substantial. Therefore, the general preference is to splice or preconnect the cable in the field once the cable is installed.

3.6.3 The Transition from Outdoor to Indoor Cable

It is necessary to use both indoor and outdoor fiber optic cable types when implementing a fiber link between two buildings. The indoor cable has the flexibility to allow routing under floors and into equipment cabinets, while the outdoor grade has superior performance over temperature and environmental extremes.

3.6.3.1 Splicing

Perhaps the most straight-forward method of transitioning from outdoor to indoor cable types is to use a splice, housed in an in-line splice enclosure. Figure 3-32 shows an example of using in-line splices and splice enclosures.

Since this method does not involve any connector-to-connector couplings, it suffers the least power budget penalty. The drawback, though, is lack of flexibility. The in-line splice enclosures have provision for only one cable in, and one cable out. If extra dark fibers were supplied in the outdoor cable, and it becomes necessary to use them in the future, the in-line splice enclosure should probably be replaced with a wall-mounted interconnect panel. Multiple indoor cable runs cannot be fanned out from the simple in-line splice enclosure. Of course, if it is considered highly unlikely that future expansion needs to be accommodated, this is a very cost-effective method of transitioning.

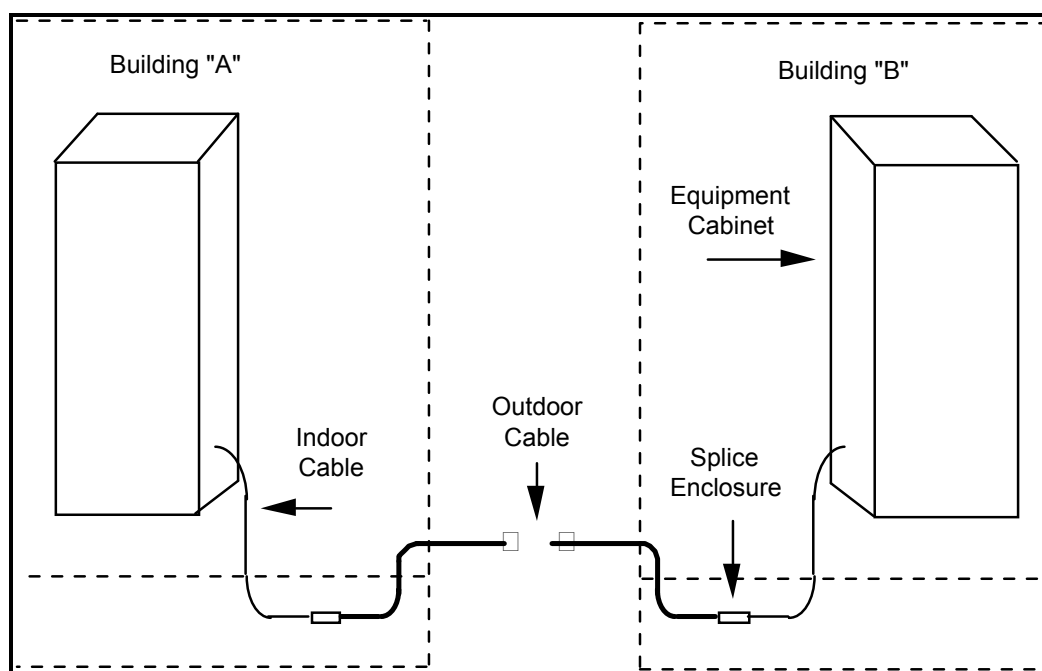


Figure 3-32 — Indoor to Outdoor Cable Transition Using In-Line Splices

As far as the fiber splice itself is concerned, there are two basic methods used to join the fibers: fusion splicing and mechanical splices.

Fusion splicing is accomplished by bringing the two fiber ends together in a precision micropositioning fixture, then hitting the joint with a precisely timed electric arc to melt the fiber ends together. Fusion splicing is the most reliable method, and when done properly, results in less than .2 dB of power loss in the link. Fusion splicing's biggest drawback is expense. The machine is quite expensive, as is the technical talent. The cost to get a trained individual with his fusion splicer to a project site won't be cheap.

In recent years, fiber optics vendors have begun to produce mechanical splices with excellent insertion loss performance. The low insertion loss of fusion splicing can even be approached. The newest mechanical splices can be mated and remated if the first try is not good enough. The mechanical splice aligns the two fibers to be joined in a V-groove, or a precision capillary. The joint between the two fibers is filled with an index-matching gel to minimize back reflection and maximize forward coupling. Mechanical splices typically exhibit insertion losses less than .5 dB, or better if tuning is employed. Since these splices can be made with a minimum of training, and the alignment jig is relatively inexpensive, the mechanical splice is quite cost effective in most cases.

NOTE

Representative splicing kits and splice enclosures are available from several suppliers, one of which is Corning Cable Systems (formerly Siecor Corporation): 1-800-743-2671.

3. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR XLCNE2

3.6. Cable Plant Implementation

3.6.3.2 Interconnect Panels

The use of interconnect (or patch) panels in the fiber optic cable plant is strongly encouraged. Their use allows the quick and easy future use of extra dark fibers. Also, cables from several remote buildings may be brought to a single interconnect box to provide a central point of fiber channel management for plant-wide data communications.

This method can make use of fanout tubing to avoid the need to perform any splicing. Fanout tubing is essentially a tight buffer subunit without any fiber. A fanout tube is slipped over each of the individual fibers in the loose tube cable after the sheath and buffer tubes have been stripped off. Cable vendors supply "break out" or "fan-out" kits, which consist of fanout tubing and a strain-relieving boot. These kits allow the installation of connectors to the individual fibers. This technique can jacket only about 3 to 15 feet of bare fiber, so it is really limited to use inside an interconnect panel.

Each fiber in the outdoor cable (now jacketed by the breakout kit) is terminated and plugged into the back of a bulkhead-mounted barrel connector. These connectors will generally be the ST style of connector. On the other side, the connection to LCN end equipment is made by means of the hybrid duplex cable assembly discussed in subsection 3.4.5.4. Unused receptacles should be capped off to prevent the ingress of contaminants.

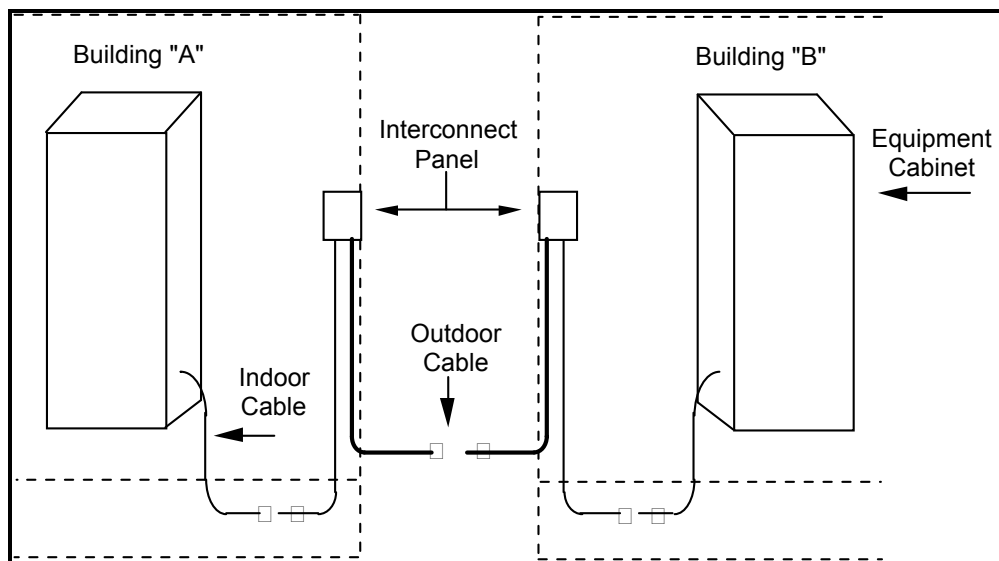


Figure 3-33 — Indoor to Outdoor Cable Transitions Using Interconnect Panels

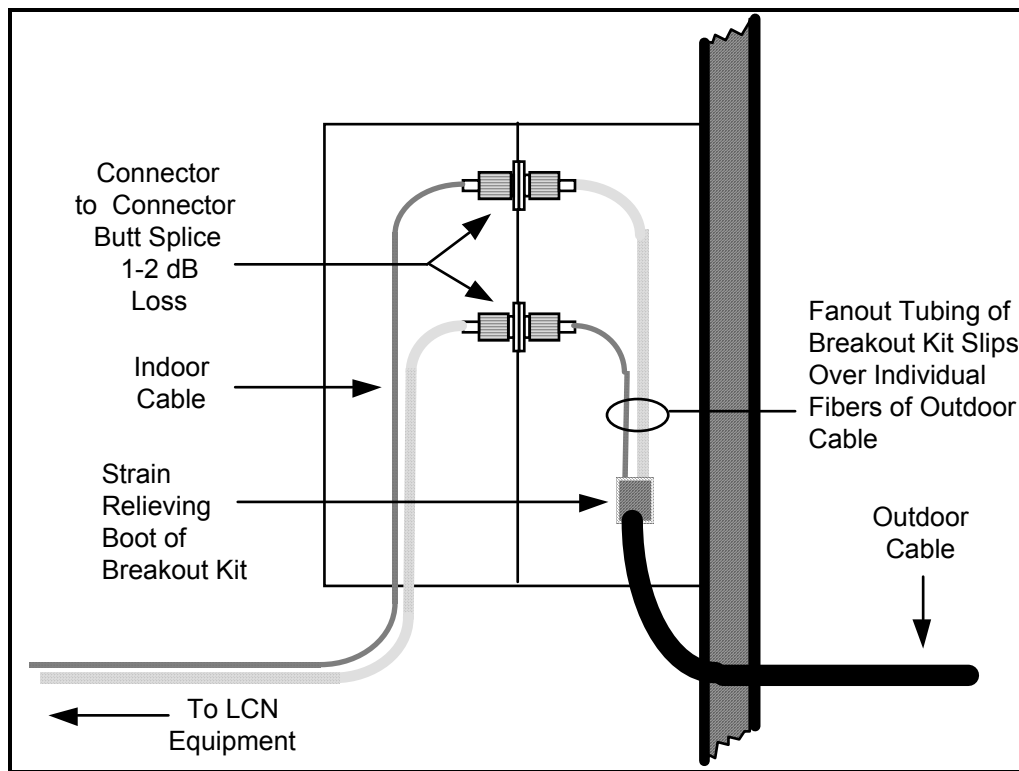


Figure 3-34 — Interconnect Panel Schematic Detail

NOTE

Representative splicing kits and splice enclosures are available from several suppliers, one of which is Corning Cable Systems (formerly Siecor Corporation): 1-800-743-2671.

3.6.4 Qualifying the Link

Once the cable plant is installed, a final optical check should be made. For multimode fiber installations, an optical time domain reflectometer (OTDR) is not necessary. An end-to-end attenuation check plus a single-ended minimum receive power check is sufficient. Assuming that an XLCNE2 is available as an optical power source, only a hand-held optical power meter and a short length of patch cable is required.

These tests are very simple to perform. First, the end-to-end attenuation check:

When the fiber link was planned, the link power budget should have been calculated (refer to subsection 3.5) in order to get an idea of where the optical losses would be taken, and how much safety margin would remain. The end-to-end attenuation check verifies the planned link attenuation against the actual installation.

First, measure the output of an XLCNE2 fiber optic transmitter by connecting a short length of patch cable to a hand-held optical power meter. Use a patch cable of the same fiber diameter as the fiber used in the fiber link to be tested. Record the transmitter's power output in dBm. Now connect the XLCNE2 transmitter to one end of the fiber optic link under test. At the other end, measure the optical power output in dBm. The difference between the power output recorded from the patch cable measurement and the power output measured over the link under test is the end-to-end attenuation of the link. (The error introduced because the patch cable wasn't also used in the second measurement is negligible because the length of the patch cable is so short.) The end-to-end attenuation thus measured should substantially agree with the planned end-to-end attenuation.

Now, the minimum receive power check can be made: Pay attention to the actual value of the power measurement at the far end of the link under test as opposed to the difference from a reference. Recall from the discussion on power budget (subsection 3.5) that the XLCNE2 fiber optic receiver requires a minimum of $.8 \mu\text{W}$ (-31.0 dBm) of power to operate. Recall also, that it is desirable to leave at least 2.3 dB as a general safety margin. Therefore, if the XLCNE2 is new, the power measured at the far end of the link should be a minimum of ($-31.0 \text{ dBm} + 2.3 \text{ dB} =$) -28.7 dBm .

If both of these tests return satisfactory results, the link can be commissioned with confidence.

NOTE

It is important that the fiber optic cable be positioned and supported such that no stress is placed on the fiber optic cable's mechanical connection at the board. Any stress on the fiber optic connections will increase the signal loss at the connection reducing the available operating distance

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

This section provides guidelines and suggestions for improving performance and reliability of a TPN that has LCN Extenders and one or more fiber optic links installed.

NOTE

The XLCNE2 board is a replacement for the LCNE2 board. They are functionally equivalent. The LCNE2 and XLCNE2 cannot be mixed on the same fiber. A XLCNE2 can only talk to another XLCNE2.

4.1 General Fiber Optic Link Information

4.1.1 System-Level Characteristics

Figure 4-1 illustrates the maximum configuration of LCN coaxial cable segments and fiber optic cable segments. There can be no more than one central and six remote coaxial cable segments in such a network.

- Each fiber optic link is limited to a maximum length of 2000 meters (6600 feet). The fiber optic link is nonelectrical, so it can be run safely through hazardous areas. Because fiber optic cables use light as the transmission media, they provide complete electrical isolation between the segments at each end of the link.
- The fiber optic transmitters use LEDs rather than lasers as the light source, so there is no danger of eye damage if you accidentally look directly at a transmitter output or into the end of a fiber cable. If you were to do this with a laser system, eye damage could result.

4.1.2 Honeywell-Provided LCN Extender and Clock Packages

- Four LCN Extender (LCNE2) boards are provided with model number MP-MEXT01. This package handles one Extender with Cable A and Cable B.
- Two LCNE2 boards and one LCN Fiber Link (LCNFL) board are provided with model number MP-MEFL01. This handles the extension of the LCN to a single-node remote segment.
- One Fiber Optic Clock Transmitter (FOC/XMTR) board; one Fiber Optic Clock Receiver (FOC/RCVR) board; and two, two-meter shielded, twisted pair cables are provided with model number MP-MFOC01. The cables connect the two boards to the Clock Source/Repeater (CS/R) boards.
- Two CS/R boards and two two-meter coaxial cables to connect to the LCN coaxial segment are provided with model number MP-MCSR02.

4.2 LCN Extender Hardware Description

4.2.1 LCN Extension Set

An LCN Extension Set is not a module. It is a group of circuit boards whose function is to add remote segments and/or remote nodes to an LCN using fiber optic cable transmission media. To connect a remote segment, an LCN Extender set includes two pairs of LCNE2 I/O boards—one pair for Cable A and one pair for Cable B. For the case of a single remote node, the LCN Extender set includes two LCNE2 I/O boards and one LCNFL board (the next two subsections will cover these configurations in more detail). There are several reasons why an LCN Extender might be necessary or desirable:

- To provide control functions in an area that is too far away from the central area to be serviced by coaxial cable.
- To traverse an area where hazardous or explosive conditions exist and electrical signals would be dangerous.
- To traverse an area where environmental factors would prohibit coaxial cable, such as an outdoor or buried cable run.
- To implement a system that requires more than 40 nodes.

LCN Extension Sets allow the extension (expansion) of the LCN as follows:

- There can be a main, or central, coaxial segment and up to six remote segments, each of which can be located up to two km away from the main coaxial segment.
- The maximum length of each coaxial segment is 300 meters.
- For each segment, the number of nodes plus the number of LCNE2 pairs must not exceed 40. A Dual Node Module counts as one node load towards the 40 node limit in each segment, whether there are one or two nodes in it.
- The maximum number of nodes in a TPS system is 64.

Fiber optic links have these additional advantages:

- A fiber optic cable is an intrinsically safe transmission medium that isolates the LCN from any hazardous area of a plant it passes through.
- Fiber optic cables are nonelectric, and therefore, offer extremely high immunity to electromagnetic interference (EMI) or radio frequency interference (RFI).
- Fiber optic cables allow long distance (up to 2 km) extension of an LCN without having to worry about ground potential difference between the connected sites.
- Using fiber optics, an LCN may be implemented in a star configuration. Such a topology may facilitate rapid diagnosis and improve survivability if a communications problem develops. First, the segments can be isolated by disabling the fiber optic links. Next, the faulty segment can be determined by noting communication statistics on the separate segments (see subsection 6.2). Finally, the “good” segments can be reconnected.

4.2.2 Components of an LCN Extension Set

An LCN Extension Set consists of optional circuit boards that are used to extend the LCN data cables, Cable A and Cable B, to a remote location. Also, when required, additional boards are used to transmit the 12.5 kHz system clock to the remote segment.

A minimum LCN Extension Set consists of four LCN Extender (LCNE2) boards—one for each end of fiber optic Cable A and one for each end of fiber optic Cable B. (Each fiber optic cable, A and B, contains at least two fiber filaments, including one for transmission of data from the central segment to the remote segment and one for receiving data from the remote segment). This configuration extends the LCN data signals, and the 5 Mbits/s digital clock if present, to the remote segment, but does not extend the 12.5 kHz clock to the remote segment. If the 12.5 kHz clock is required at the remote segment, a Fiber Optic Clock Transmitter (FOC/XMTR) board is required at the main (central) segment, and a Fiber Optic Clock Receiver (FOC/RCVR) board is required at the remote segment; also, two Clock Source/Repeater (CS/R) boards are required at the remote segment to transmit the 12.5 kHz clock onto the coaxial cables (A and B). Figure 4-3 and Figure 4-4 illustrate an LCN Extension configuration which includes both data and 12.5 kHz clock extension.

To add a single remote node to the central segment, a slightly different configuration is used. Two LCNE2 boards are used at the central segment, and an LCN Fiber Link board (LCNFL) is used at the remote node. The LCNFL has fiber optic transmitters and receivers for both Cable A and Cable B. Figure 4-10 illustrates this configuration. A remote node connected in this manner cannot receive the 12.5 kHz clock. However, if it is implemented with a K2LCN/K4LCN kernel (processor), it can receive the 5 Mbits/s digital clock, assuming there is a source or translator for this clock in the central segment. Subsection 2.5 in this document contains additional information on implementing the 12.5 kHz and 5 Mbits/s system clocks. The boards which comprise the LCN extension family are summarized below.

- **LCNE2—LCN Extender:** Provides an interface to one end of a fiber optic link that passes data. One LCNE2 is required for each end of fiber optic cables A and B. See subsection 4.2.2.1 for more detailed information.
- **CS/R—Clock Source/Repeater:** Transmits the 12.5 kHz system clock onto an LCN coaxial cable (Cable A or Cable B). Has optional current loop input and output so that it can receive the clock from an FOC/RCVR board or another CS/R board, or transmit the clock to FOC/XMTR boards and/or another CS/R board. See subsection 4.2.2.4 for more detailed information.
- **FOC/XMTR—Fiber Optic Clock Transmitter:** Receives the 12.5 kHz system clock from the CS/R on through a current loop interface and transmits it for a given cable (A or B) over a fiber optic link to a remote segment. See subsection 4.2.2.2 for more detailed information.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.2. LCN Extender Hardware Description

- **FOC/RCVR—Fiber Optic Clock Receiver:** Receives the system clock transmitted by an FOC/XMTR and converts it to current loop format to send to a CS/R. See subsection 4.2.2.3 for more details.
- **LCNFL—LCN Fiber Link:** Connects the node on a 1-node remote segment to both A and B fiber optic links. This board replaces the LCNI I/O board or the CLCNA and CLCNB boards. The LCNFL has a pinning option to set the node address. The 12.5 kHz system clock is not available to this remote node. See 3.2.2.5 for more details.

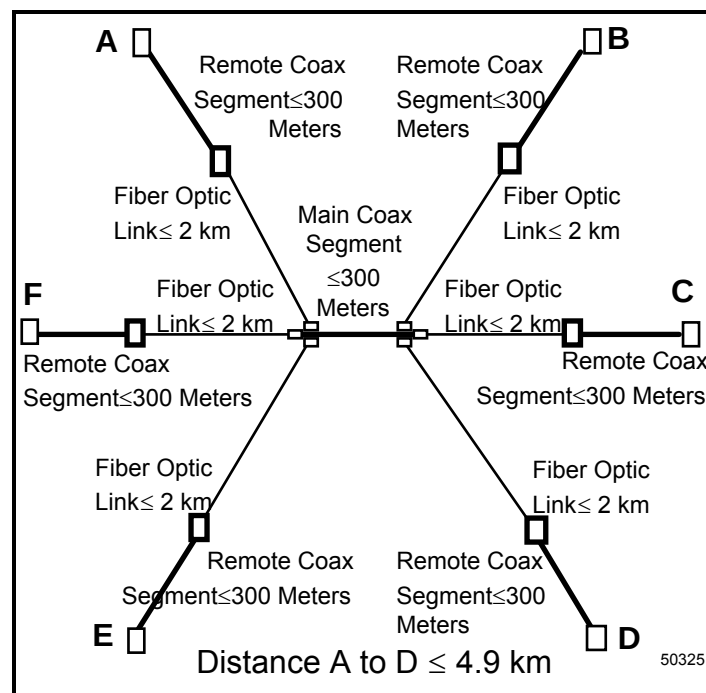


Figure 4-1 — LCN Extender Segment Lengths

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.2. LCN Extender Hardware Description

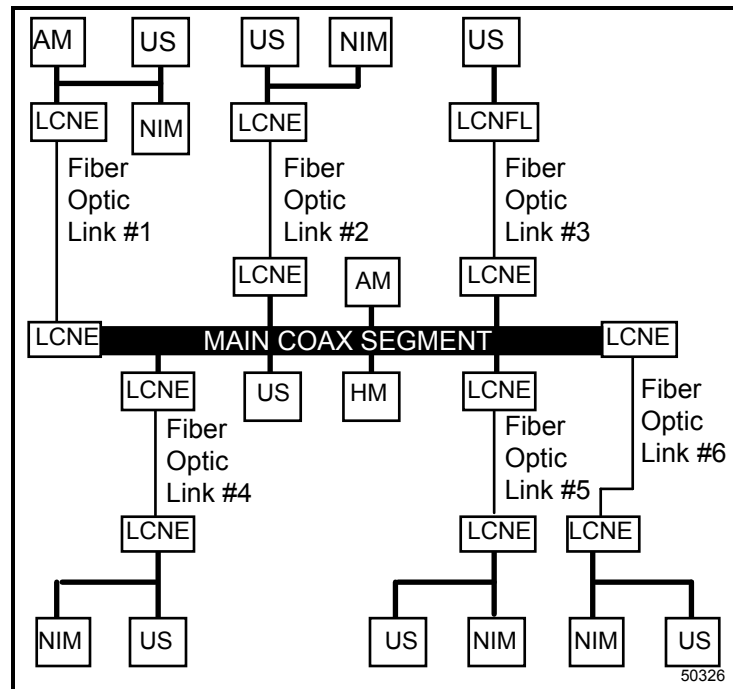


Figure 4-2— LCN Extender Segments

Figures 4-3 and Figure 4-4 illustrate an LCN Extension configuration that includes both data and 12.5 kHz clock extensions. Both figures must be used together. The A and B connections designated in Figure 4-3 are respectively connected to A and B in Figure 4-4.

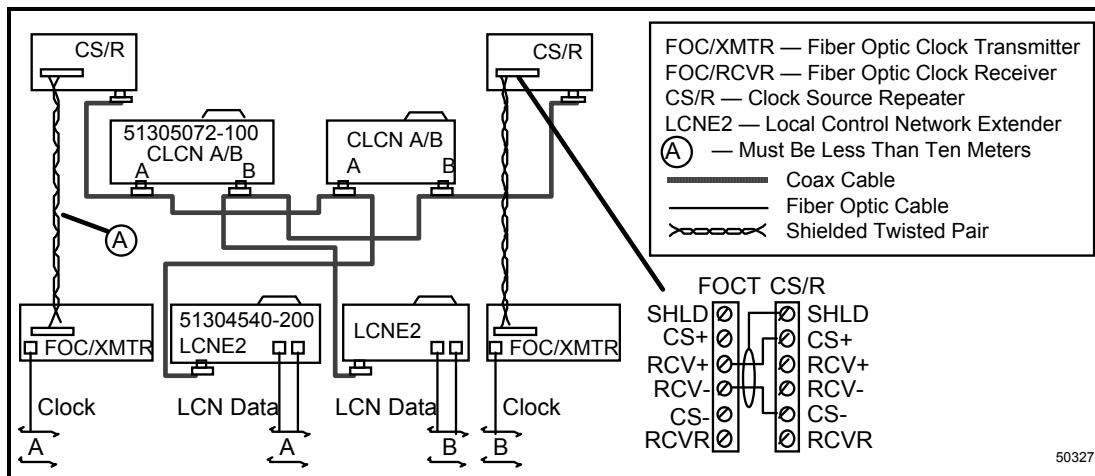


Figure 4-3— LCN Extension Set Connections for One of a Pair of LCNE2s

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.2. LCN Extender Hardware Description

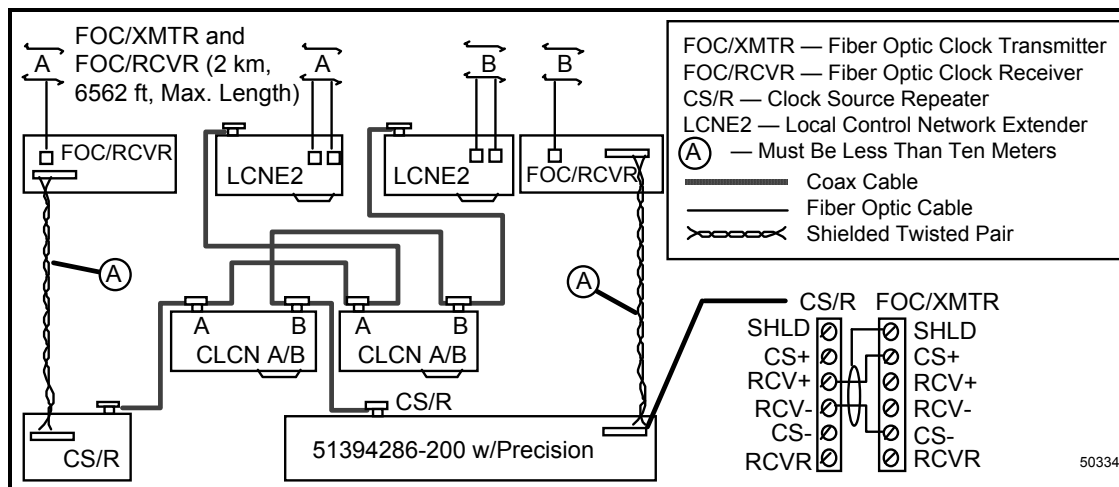


Figure 4-4 — LCN Extension Set Connections for One of a Pair of LCNE2s

4.2.2.1 LCN Extender (LCNE2)

The LCNE2 is used at both ends of a fiber optic link to extend the LCN signals to and from a remote segment. The LCNE2 includes both a fiber optic transmitter and receiver. This allows data to be sent in both directions between the main coaxial segment and the remote coaxial segment. The LCNE2-to-LCNE2 fiber link cannot transmit the 12.5 kHz system clock to the remote segment (refer to subsection 2.5 for clock system details). If this clock is required at the remote segment (such as for the sequence-of-events function), a twisted-pair loop of ≤ 300 m, or a separate fiber link of ≤ 2 km will be required for each LCN cable (A and B). FOC/XMTR and FOC/RCVR boards are used in the implementation of the fiber clock link (see the next two subsections).

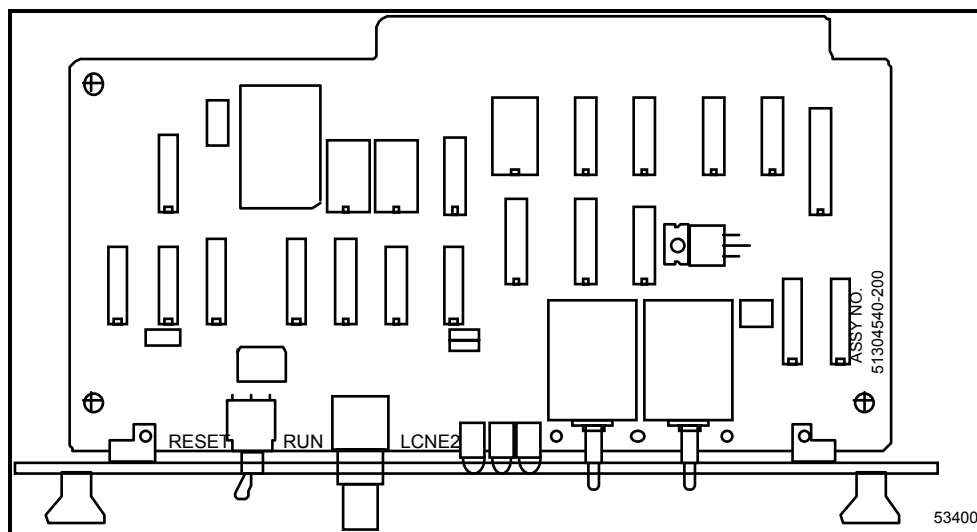


Figure 4-5 — LCNE2 Board Assembly

NOTE

The LCNE2 board, 51304540-100, is a later version of the LCNE2 board, 51109881-100. They are functionally equivalent and your system may include either or both types.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.2. LCN Extender Hardware Description

4.2.2.2 Fiber Optic Clock Transmitter (FOC/XMTR)

FOC/XMTR boards receive 12.5 kHz clock information through a current loop from a CS/R board and transmit it to a remote coaxial segment, through a single fiber that is up to 2 km long, to a FOC/RCVR board. If the remote segment is less than 300 meters away from the central segment, the clock may be transmitted directly with a shielded, twisted-pair current loop from the central CS/R to the remote CS/R, eliminating the need for the FOC/XMTR, FOC/RCVR, and fiber cable components.

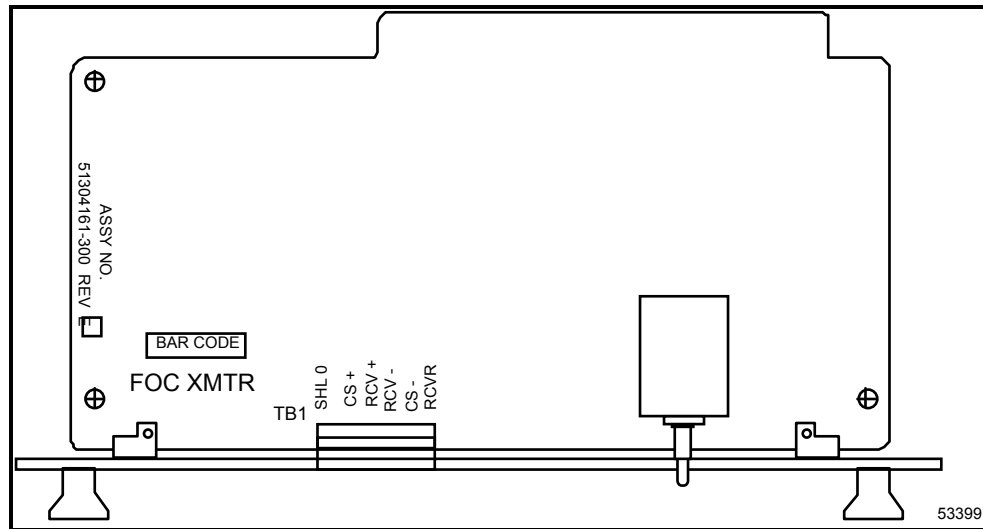


Figure 4-6— FOC/XMTR Assembly

NOTE

The FOC XMTR board, 51304540-300, is a later version of the FOCT board, 51304161-100. They are functionally equivalent and your system may include either or both types.

4.2.2.3 Fiber Optic Clock Receiver (FOC/RCVR)

FOC/RCVR boards receive 12.5 kHz clock information from a fiber optic cable (transmitted by a FOC/XMTR board at the other end) and transmit it through a current loop to a CS/R board, which in turn transmits the clock onto an LCN coaxial cable. The current loop may be up to 10 m long. The FOC/RCVR board is used in applications that require transmission of 12.5 kHz clock data from a main segment to a remote segment, over a distance of up to 2 km. If the remote segment is less than 300 meters away from the central segment, the clock may be transmitted directly with a shielded, twisted-pair current loop from the central CS/R to the remote CS/R, eliminating the need for the FOC/XMTR, FOC/RCVR, and fiber cable components.

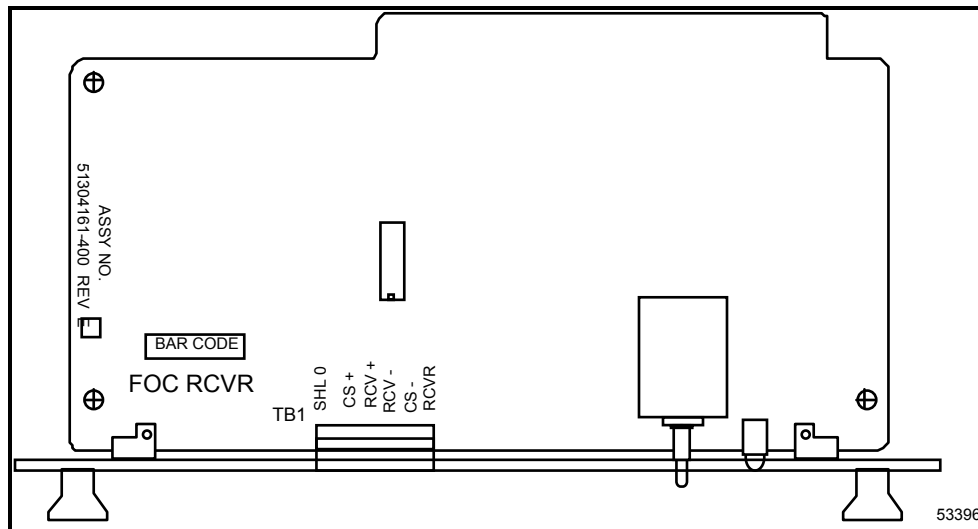


Figure 4-7 — FOC/RCVR

NOTE

The FOC/RCVR board, 51304161-400, is a later version of the FOC/RCVR board, 5130416-200. They are functionally equivalent and your system may include either or both types.

4.2.2.4 Clock Source/Repeater (CS/R)

Each CS/R board must be mounted in Slot 1 of the I/O card cage of a 5-Slot or 10-Slot Module, directly behind an EMPU, HMPU, or HPK2 processor board. This is the only location where the CS/R board can receive the ± 12 Vdc power that it requires. At the central (main) segment, two CS/R boards (one for LCN Cable A and one for Cable B) each receive the 12.5 kHz clock signal through the I/O connector from the processor.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.2. LCN Extender Hardware Description

Each CS/R transmits the clock onto an LCN coaxial cable of the central segment. It also provides a 300 m maximum length current loop output which may be used to drive up to four loads. The loads may be FOC/XMTR boards (used for transmission of the clock over fiber optic cables to remote segments), and/or other CS/R boards. If there are more than four remote segments (up to six are allowed), the first CS/R can include in its loop a second CS/R, which is used as a repeater and can drive the additional required FOC/XMTR or remote CS/R boards. These scenarios are covered in more detail in subsection 4.3.1.

If the remote segment is less than 300 meters away from the central segment, the clock may be transmitted directly with a shielded, twisted-pair current loop from the central CS/R to the remote CS/R, eliminating the need for FOC/XMTR, FOC/RCVR, and fiber cable components.

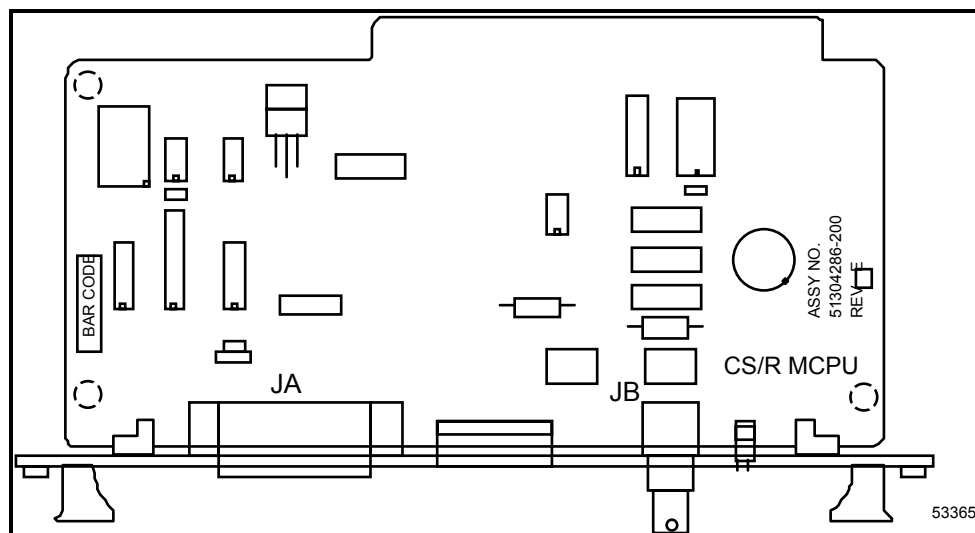


Figure 4-8 — CS/R Board Assembly

NOTE

The CS/R board, 51304266-200, is a later version of the LCNE2 board, 51304266-100. They are functionally equivalent and your system may include either or both types.

4.2.2.5 LCN Fiber Link (LCNFL)

The LCNFL is used when adding a single remote node to the central segment. Two LCNE2 boards are used at the central segment, and an LCN Fiber Link board (LCNFL) is used at the remote node. The LCNFL has fiber optic transmitters and receivers for both Cable A and Cable B. Figure 4-10 illustrates this configuration. The LCNFL replaces the LCNI I/O board or the CLCNA and CLCNB boards in the remote node. The LCNFL has a pinning option to set the node address. The LCNFL does not connect to LCN coaxial connectors; therefore, its application is limited to a single remote node. A remote node connected in this manner cannot receive the 12.5 kHz clock. However, if it is implemented with a K2LCN/K4LCN kernel (processor), it can receive the 5 Mbits/s digital clock, assuming there is a source or translator for this clock in the central segment. Subsection 2.5 in this document contains additional information on implementing the 12.5 kHz and 5 Mbits/s system clocks.

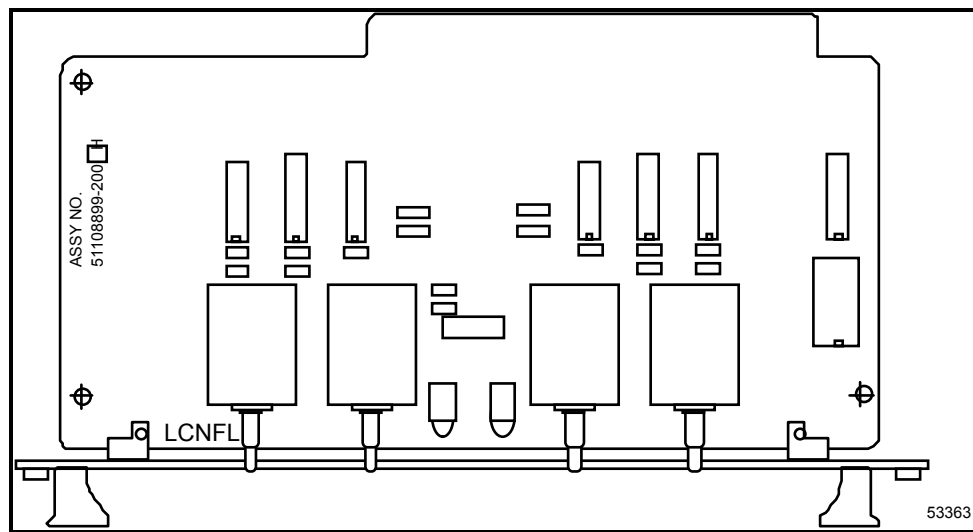


Figure 4-9— LCNFL Assembly

NOTE

The LCNFL board, 51308899-200, is a later version of the LCNE2 board, 51308899-100. They are functionally equivalent and your system may include either or both types.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.3. LCN Extender Hardware Implementation

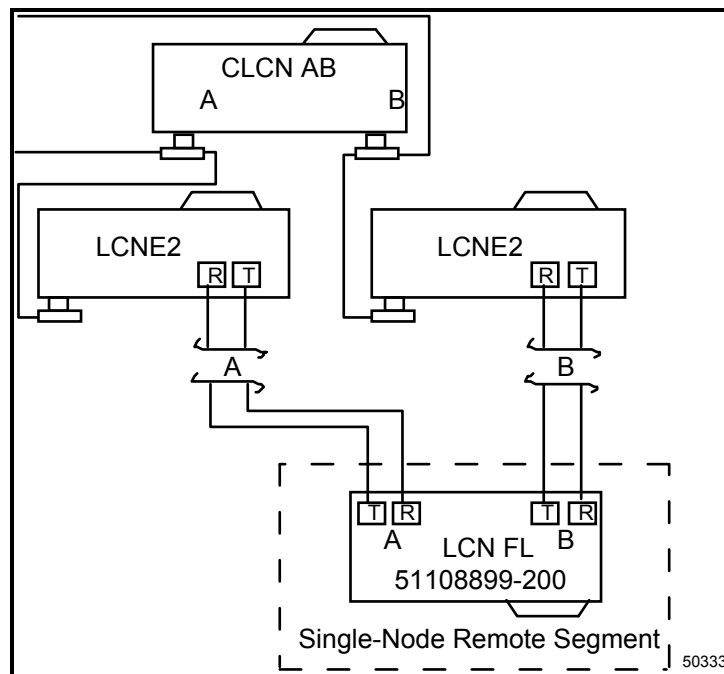


Figure 4-10 — LCN Extension to a Single Node — without 12.5 kHz Clock

NOTE

In a Dual Node Module, the CLCN A/B would be replaced by a CLCNA and a CLCNB board.

4.3 LCN Extender Hardware Implementation

Section 2 of this publication covers rules and guidelines for the implementation of a TPS system from the standpoint of coaxial cable segments while Section 4 covers similar aspects of fiber optic extender links. This subsection, 3.3, includes some rules and guidelines concerning fiber optic extension links. It is intended to be supplemental to the material in Section 2. In some cases it may overlap material presented in Section 2 and elsewhere. Nevertheless, it is not intended to be a stand-alone reference for extending LCN segments. When planning or expanding a system, it is essential to consider all aspects, including rules and guidelines for systems, coaxial segments, and fiber optic extensions, because all of these elements are interrelated.

4.3.1 LCNE Configuration Rules

Consider the following general information, rules, and guidelines when you configure a TPS system that is to contain one or more fiber optic link(s).

- LCN Extender sets, and fiber optic cables up to two km (6560 ft) in length, can be used to interconnect LCN coaxial bus segments. Four LCN Extender boards (LCNE2) and four fibers are required to extend both Cable A and Cable B to a remote area. Two fibers (one for transmit and one for receive) are used for the Cable A extension and, similarly, two fibers are used for the Cable B extension. An LCNE2 is used at each end of the A fiber pair and at each end of the B fiber pair. For examples, see Figures 3-18, 3-19, and 3-20.
- The LCN extension with LCNE2 and fibers takes care of extending the A and B data path (which can include the 5 Mbits/s digital clock) to the remote segment. It does not, however, accomplish transmission of the 12.5 kHz clock to the remote segment. If the remote segment requires the system clock and cannot use the digital clock exclusively (i.e., it contains EMPU, HMPU, or HPK2 processors), then separate means must be provided to transmit the 12.5 kHz clock to the remote segment (see the next two paragraphs). Subsection 2.5 has additional information about the 5 Mbits/s and 12.5 kHz system clocks.
- If a remote segment requires the 12.5 kHz clock and it is located more than 300 m from the central segment, then two additional fibers must be used to transmit the 12.5 kHz clock from the central segment to the remote segment. One fiber carries the clock for Cable A and the other fiber carries the clock for Cable B. Two Fiber Optic Clock Transmitter boards (FOC/XMTR) are used at the central segment. They receive the clock over twisted pair current loops from Clock Source/Repeater boards also located in the central segment. The FOC/XMTR boards transmit the clock over fiber links to Fiber Optic Clock Receiver boards (FOC/RCVR) located in the remote segment. The FOC/RCVR boards transmit the clock over 10 m maximum length twisted-pair current loops to CS/R boards in the remote segment, which, in turn, transmit the clock onto the LCN coaxial cables. This configuration is illustrated in Figures 3-18, 3-19, and 3-20. The CS/Rs in the central segment are located in nodes which have been configured as clock sources in the software Network Configuration File (NCF).

If a remote segment requires the 12.5 kHz clock and it is located less than 300 m from the central segment, the clock optionally can be transmitted directly from the CS/Rs in the central segment to the CS/Rs in the remote segment over 300 m maximum length twisted pair current loops. This eliminates the need for the FOC/XMTRs, FOC/RCVRs, and clock fibers as described in the previous paragraph; however, with this approach you lose the advantages of fiber optics described in subsection 4.1.1.

- The LCN Fiber Link (LCNFL) allows a single remote node to be connected with fiber optics to the central LCN coaxial cable bus segment. At the central segment, two LCNE2s and two pair of fibers are used exactly as described above for connecting a remote segment to the central segment. A single remote node, however, does not require the LCN A and B coaxial cables normally used to interconnect the nodes of a multi-node segment. Therefore, coaxial cable interface

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.3. LCN Extender Hardware Implementation

boards (CLCNA and CLCNB or CLCNA and CLCNB boards, and CS/R boards) are not required at the remote node. The LCNFL, which may be used instead, has fiber optic transmitters and receivers for both Cable A and Cable B. It replaces the LCNI I/O board or the CLCNA and CLCNB boards. It is installed in slot one (the lowest) in the I/O card cage of the remote node. The LCNFL has a pinning option to set the node address. If the remote node uses a K2LCN/K4LCN (which would normally be pinned for the node address), the addressing is done on the LCNFL and all address jumpers are removed from the K2LCN/K4LCN. LCNFL address pinning is illustrated in Figure 9-1. A remote node connected in this manner cannot receive the 12.5 kHz clock. However, if it is implemented with a K2LCN/K4LCN kernel (processor), it can receive the 5 Mbits/s digital clock, assuming there is a source or translator for this clock in the central segment. Subsection 2.5 in this document contains additional information on implementing the 12.5 kHz and 5 Mbits/s system clocks.

- Figures 3-18, 3-19, and 3-25 illustrate the application of the LCNFL.
- Although LCNE2s, FOC/XMTRs, and FOC/RCVRs can be mounted in open I/O slots of any node type, it is desirable to locate them in nodes that are the least likely to have the power turned off. Table 2-1 shows the recommended order that should be used in selecting node types for housing these extender boards.
- The LCNE2s, FOC/XMTRs, and FOC/RCVRs for Cable A must be in a different node than the ones for Cable B and must be powered from separate ac power entries. If more space is required, an otherwise empty Dual Node Module with a power supply and KJMP load board (51401594-200) installed in each node should be used to house the extender board sets. The LCNE2 and associated FOC/XMTR or FOC/RCVR for Cable A are placed in the I/O section of the two-slot module and the corresponding boards for Cable B are placed in the three-slot node. If FOC/XMTR or FOC/RCVRs are not used, two LCNE2s for Cable A may be placed in the two-slot node and two LCNE2s for Cable B may be placed in the three-slot node.
- When LCN extension hardware is used, the configuration of the LCN has increased importance from the standpoint of reliability and failure analysis. This hardware represents points in the system where failure or power loss to a relatively small percentage of the system can potentially isolate and thereby affect a whole segment or group of modules due to disruption of communications. The extension hardware is transparent to the system communications software, and therefore in non-failure conditions the topology of the system may seem unimportant. However, when failure scenarios are considered, it is evident that system layout can dramatically affect diagnosis, recovery, and visibility into processes. The following rules and guidelines are essential to maximize system reliability in the event of LCN Extender failures.
 - The configuration of Removable Media Disk Drives and Engineer's Keyboards is very important. This subject is addressed in detail in subsection 2.1.1.
 - Fiber optic A and B channel fibers should be in separate cables and routed on separate paths from source to destination. Therefore, if one cable is damaged,

the other will still be operable. However, for any two nodes in the system, the maximum difference in Cable A and Cable B lengths (including the sum of all coaxial and fiber optic sections) cannot exceed 300 meters, or communication problems may result. Refer to subsection 4.6.1 for additional details.

- The number of coaxial cable bus segments connected in serial, by way of fiber optic cables and LCNE2s, must not exceed three.
- The LCNE2 board for LCN Cable A should be installed in a different module than the LCNE2 board for LCN Cable B. This is also true for FOC/XMTR, FOC/RCVR, and CS/R pairs. For example, one NIM of a redundant pair could have the LCNE2 for Cable A and its partner NIM should have the LCNE2 for Cable B. The same principle applies to redundant HG pairs. See Figure 4-2.
- All nodes containing an LCNE2 should be on an uninterruptible power supply (if available) to ensure that the LCN coaxial cable segments cannot become disconnected because of power outages and thus cause reconnection problems.
- If possible, avoid configuring only two nodes on a remote segment due to error scenarios which might occur if the fiber optic cable is broken or if LCNE2s have had the power removed.
- Because of the lower reliability of the fiber optic extension to remote segments, locate redundant HGs and added HGs on the same coaxial cable segment. Also, locate redundant NIMs and redundant PLCGs on the same coaxial cable segment.
- To prevent loss of view to the process, two or more Universal Stations or Universal Work Stations in a console should be on the same coaxial cable bus segment. In addition, for plant integrity, we recommend that the NIMs and HGs most often associated with a Universal Station be on the same coaxial cable bus segment.
- The two clock sources configured in the Network Control File must be located on the main coaxial cable bus segment.
- If the software release is 210 or earlier, both fiber optic links (A and B) on the LCNE2 should not be turned off at the same time and then on again without bringing down the nodes on one side of the LCNE2. If the nodes are not brought down (power removed or shut down), reconnection problems could occur.
- Unless the Precision Clock Source option is installed, the power supply in each node of the system must be set up by a pinning option for the internal or line frequency clock. If the Precision Clock Source option is installed, the power supplies in the nodes that have the Precision Clock Source installed in them must be pinned for the external clock option, and all other nodes are pinned for internal clock. Figures 3-11 through 3-17 show the clock pinning option for the standard, enhanced, and dual node power supplies, respectively.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.3. LCN Extender Hardware Implementation

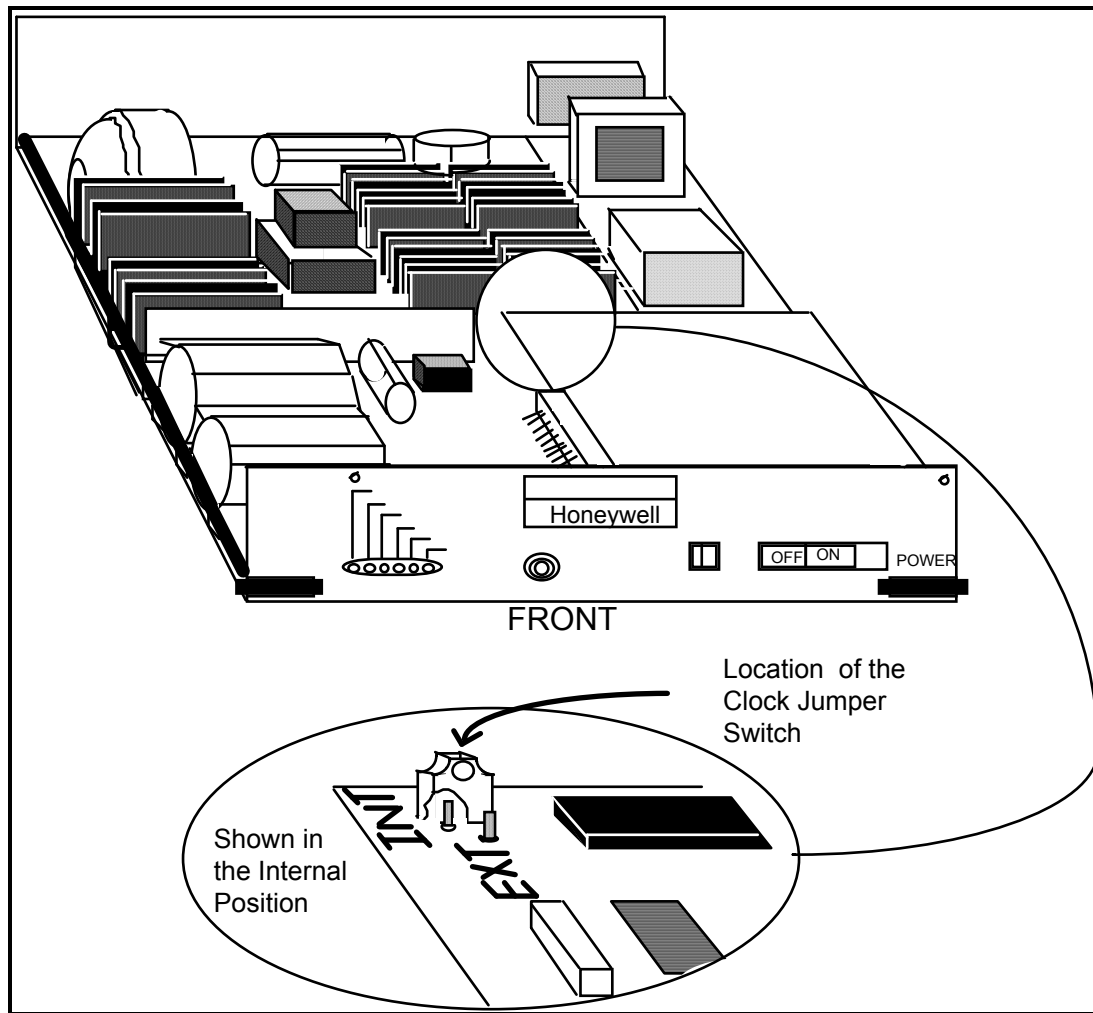


Figure 4-11 — Standard Power Supply Showing Location of Clock Jumper

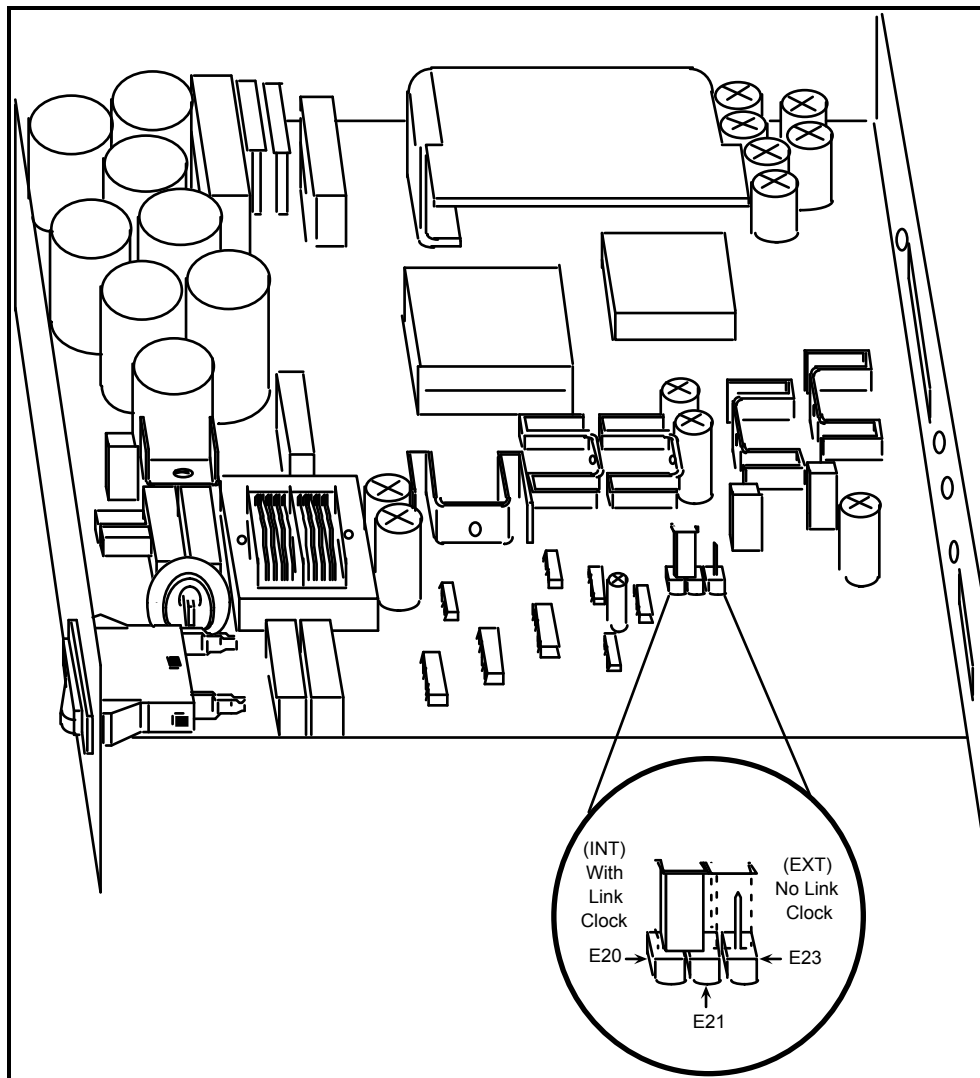


Figure 4-12 — Enhanced Power Supply Showing Location of Clock Jumper

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.3. LCN Extender Hardware Implementation

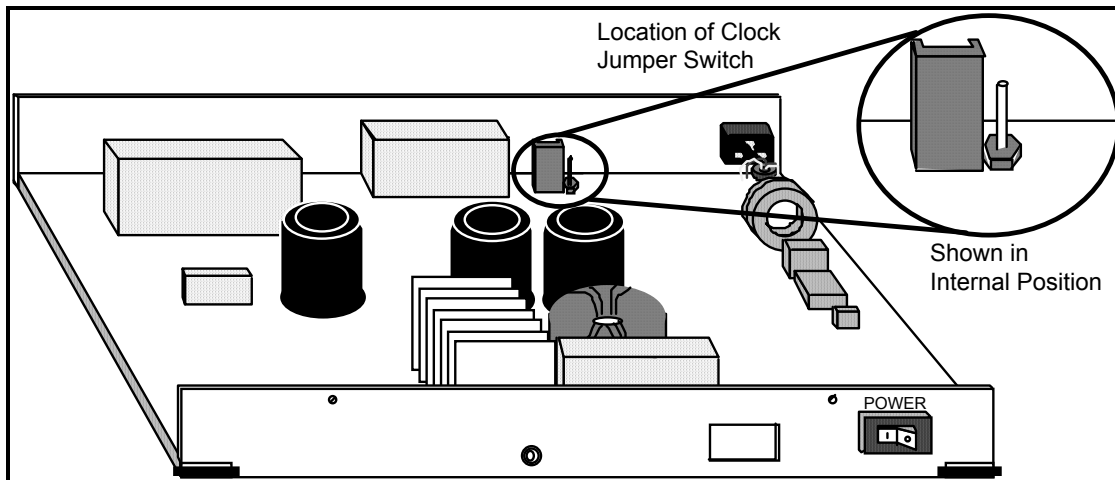


Figure 4-13 — CE Compliant Five-Slot Power Supply Showing Location of Clock Jumper

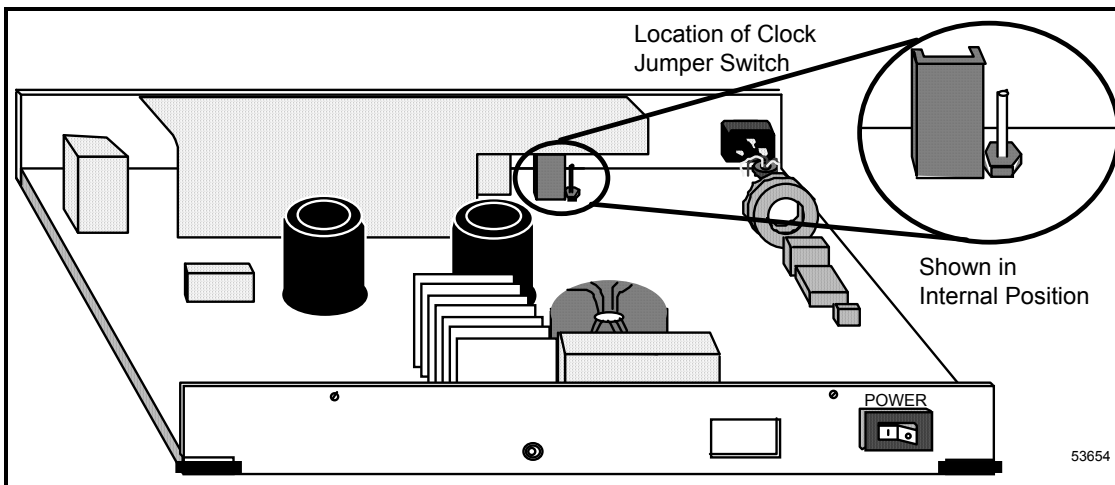


Figure 4-14 — CE Compliant Ten-Slot Power Supply Showing Location of Clock Jumper

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.3. LCN Extender Hardware Implementation

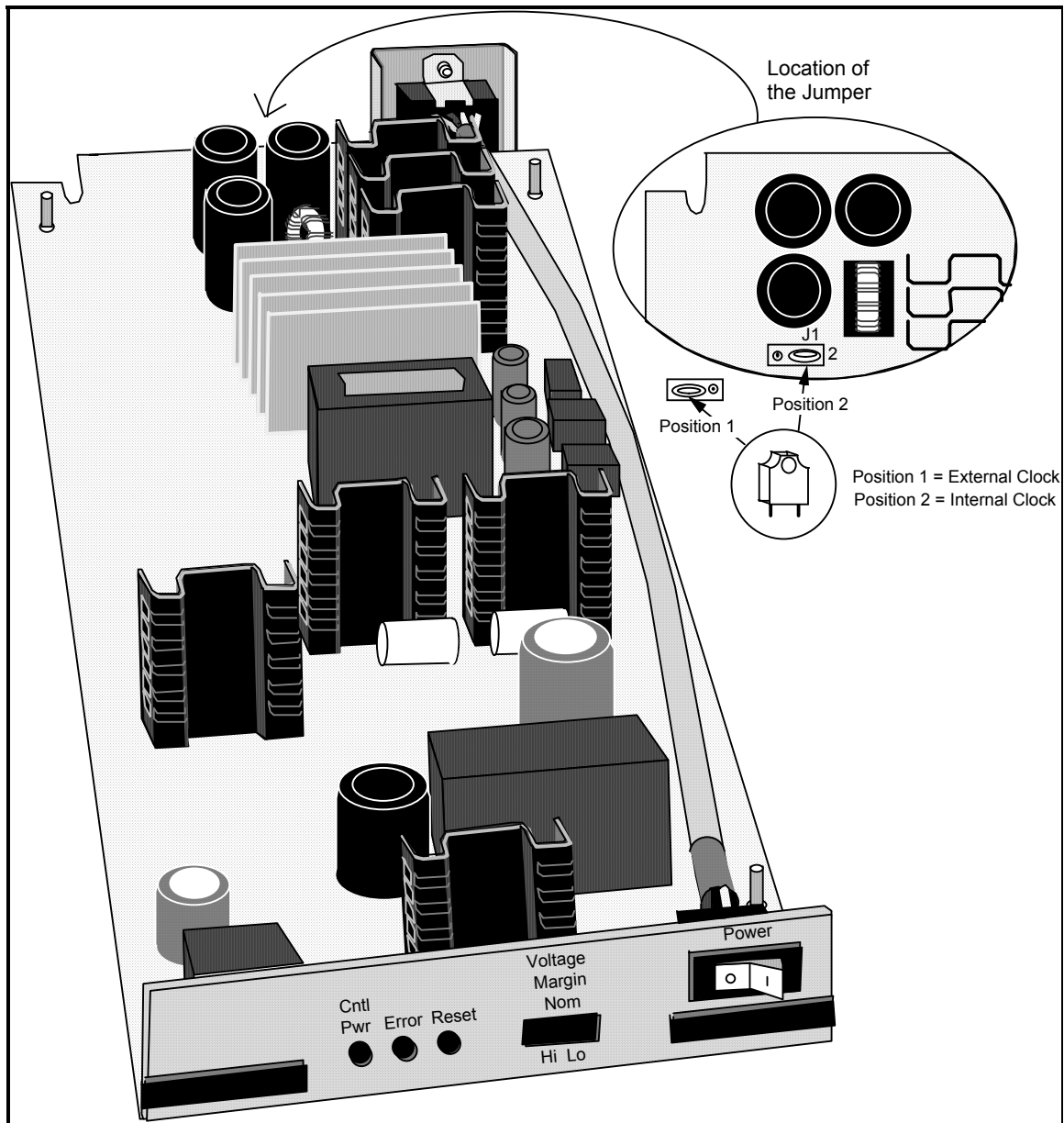


Figure 4-15 — Dual Node Power Supply Showing Location of Clock Jumper

There are two versions of the CE Compliant Dual Node power Supply. Match your power supply to either Figure 4-16 or Figure 4-17.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.3. LCN Extender Hardware Implementation

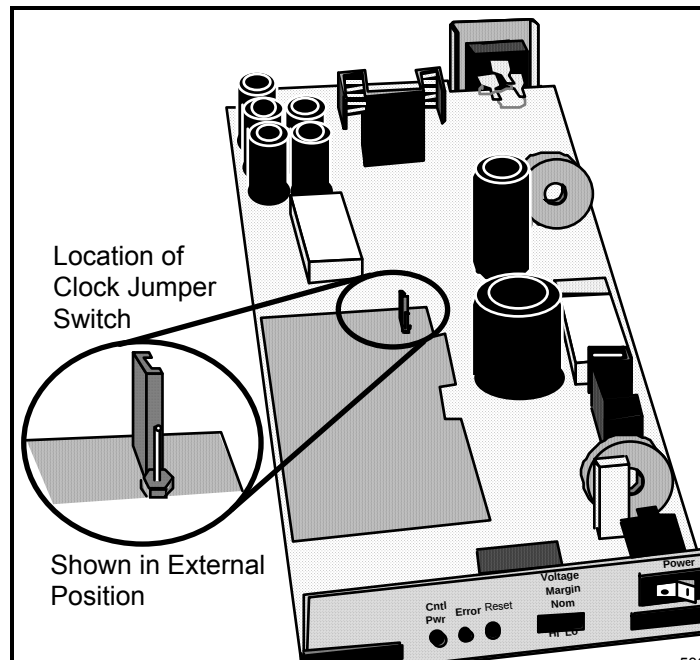


Figure 4-16 — CE Compliant Dual Node Power Supply Showing Location of Clock Jumper

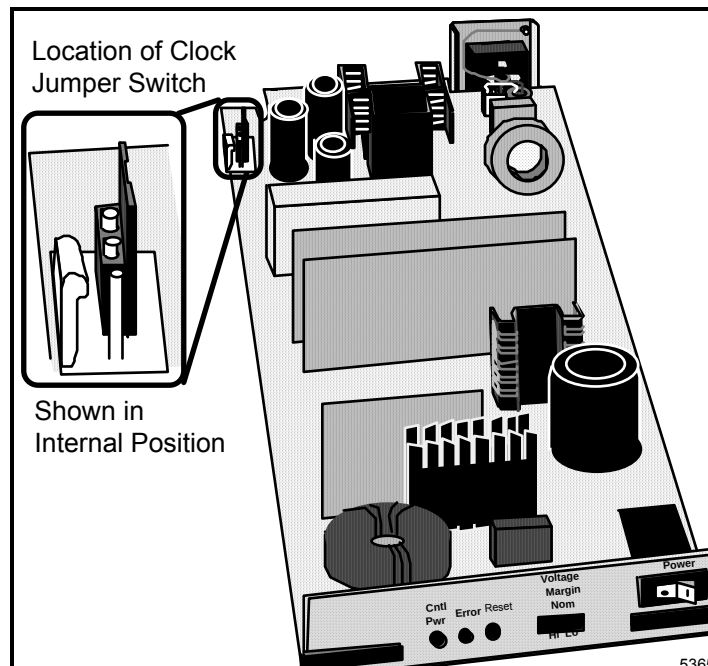


Figure 4-17 — CE Compliant Dual Node Power Supply Showing Location of Clock Jumper

4.3.2 LCN Extension Implementation Examples

This subsection includes examples of the use of LCN extension hardware, including the use of LCNE2s to extend the LCN data circuits, the use of an LCNFL in a single remote node, and the use of FOC/XMTRs, FOC/RCVRs, and CS/Rs to extend the 12.5 kHz system clock to a remote segment. It illustrates the optional use of a twisted pair current loop in place of the FOC/XMTRs, FOC/RCVRs, and clock fibers when the distance between the segments is 300 meters or less. Also shown in detail are the fiber and twisted pair interconnections among the various extension components for different configurations.

Figures 3-18 and 3-19 show a central or main segment and the remote segment split between the two figures. They show the interconnections of the two coaxial cables (Cable A and Cable B) in the both the central and the remote segments. The example includes a fiber optic clock link using FOC/XMTRs and FOC/RCVRs for transmission of the 12.5 kHz system clock from the central segment to the remote segment. At the central segment, note the twisted pair cables which carry the clock from the CS/R boards to the FOC/XMTR clock transmitter boards, and at the remote segment, note the twisted pair cables which carry the clock from the FOC/RCVR clock receiver boards to the remote CS/Rs. The CS/Rs in the remote segment then transmit the clock onto the A and B coaxial cables.

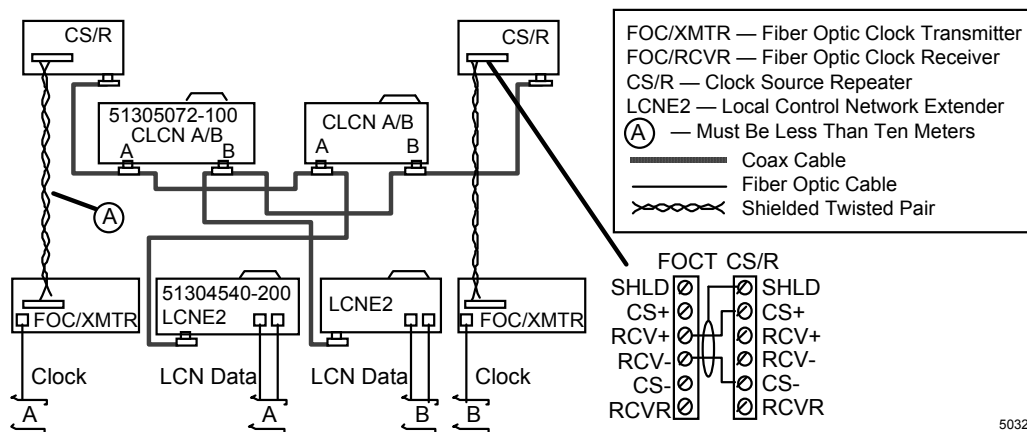


Figure 4-18 — LCN Extension Set Connections for One of a Pair of LCNE2s

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.3. LCN Extender Hardware Implementation

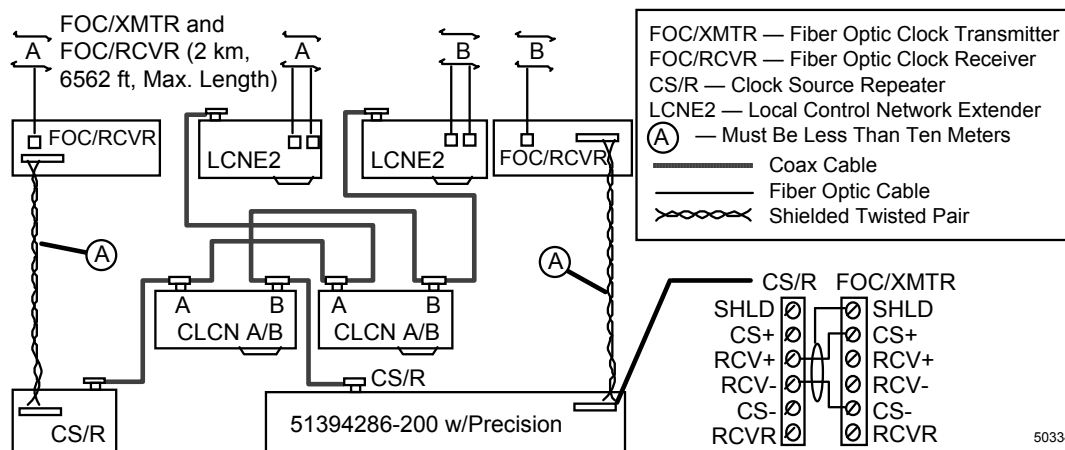


Figure 4-19 — LCN Extension Set Connections for One of a Pair of LCNE2s

Figure 4-20 illustrates pictorially the use of LCN Extender components to connect a remote segment to a central segment. This example also includes the FOC/XMTRs, FOC/RCVRs, and fibers required to transmit the 12.5 kHz system clock to the remote segment. In this illustration, the extension components (LCNE2s, FOC/XMTRs, FOC/RCVRs, and CS/Rs) associated with Cable A are located in a separate module from the components associated with Cable B. This separation occurs in both the central segment and the remote segment. As indicated in the figure, this placement of the Cable A components in a separate module from the Cable B components minimizes the possibility of total communications loss in the event of a power off condition or power supply failure at one module. Although this example has the CS/R, FOC/XMTR or FOC/RCVR, and LCNE2 for each coaxial cable section located in the same module, this is not required. It is desirable for reliability that all of the components for at least one cable (A or B), including fiber interface components, be on an uninterruptible power source. It is also desirable that the fibers associated with each cable be routed separately, and that spare fibers be included, as is covered in more detail in subsection 4.6.1.

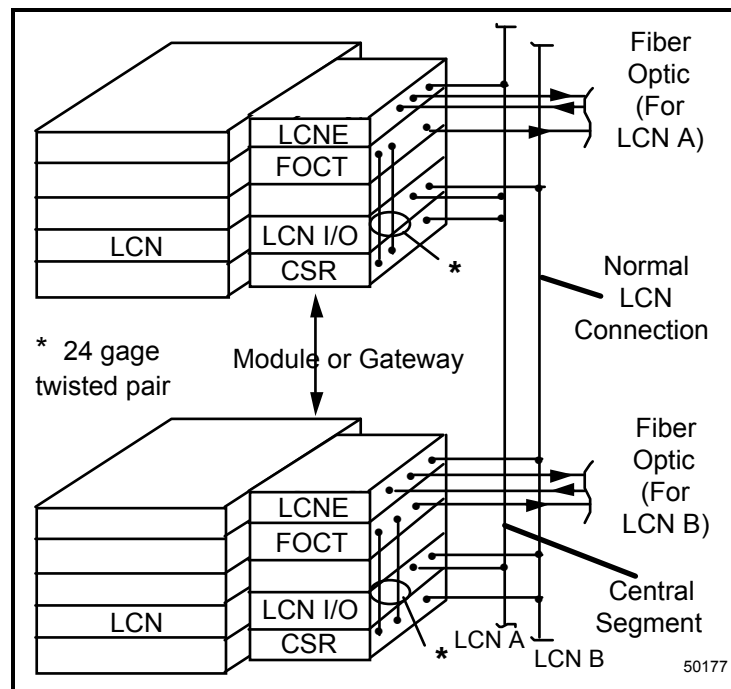


Figure 4-20 — Connection Example for Cable A and B

A single remote node can be thought of as a remote segment that has only one node. Therefore, the coaxial cable links that normally interconnect nodes on a segment are not required. Figures 3-21 and 3-22 are pictorial illustrations of the implementation of a single remote node using the LCNFL. Also refer to Figures 3-10, 3-18 and 3-19 which show single remote nodes. Figure 4-21 shows a remote node in a 5-Slot Module. The 12.5 kHz system clock is not transmitted to this node. Figure 4-22 shows a remote node in a Dual Node Module. The upper (2-slot) node in the Dual Node Module must be empty. The 5 Mbits/s Digital Clock is transmitted to this node if the central segment has a clock source or translator (see subsection 2.5).

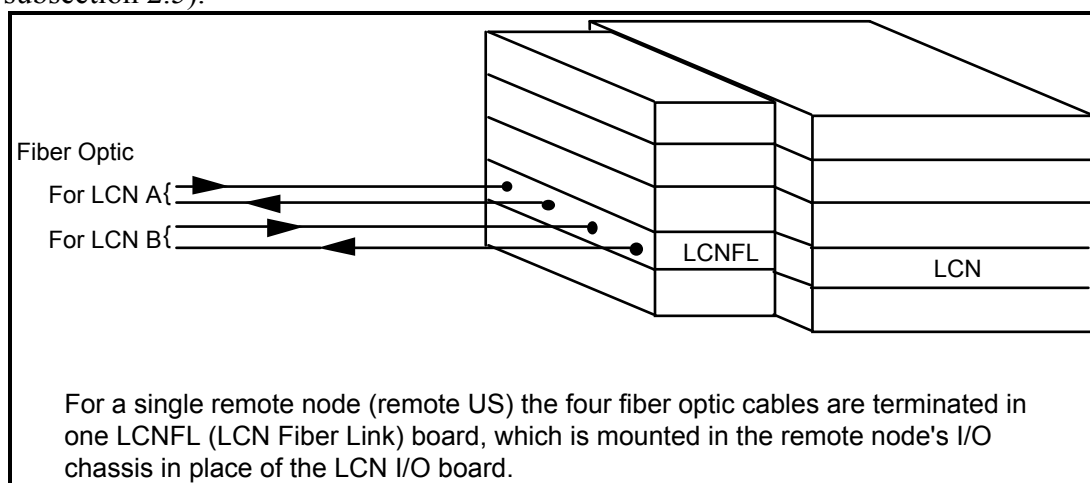


Figure 4-21 — Single Remote Node in a 5-Slot Module

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.3. LCN Extender Hardware Implementation

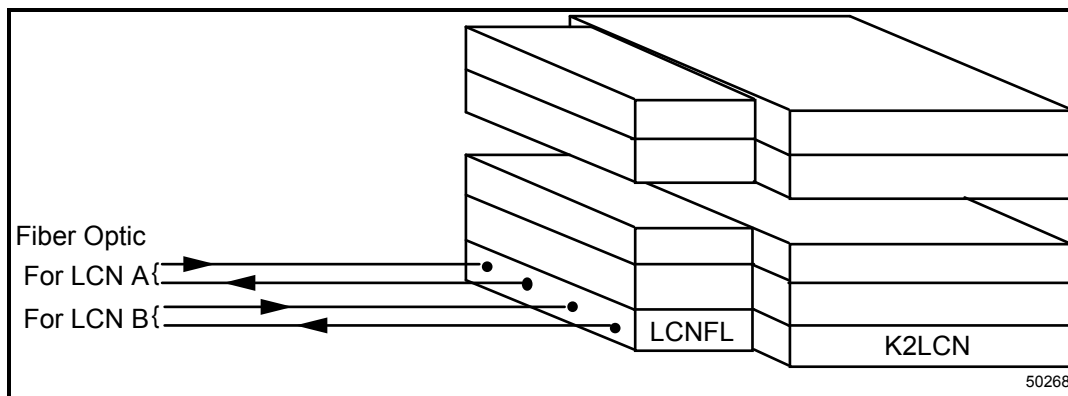


Figure 4-22 — Single Remote Node in Dual Node Module Package

Figure 4-23 illustrates an **overview** of a central segment “Alpha” with two remote cable segments, “Beta” and “Gamma.” Both remote cable segments receive the 12.5 kHz system clock through fiber optic links.

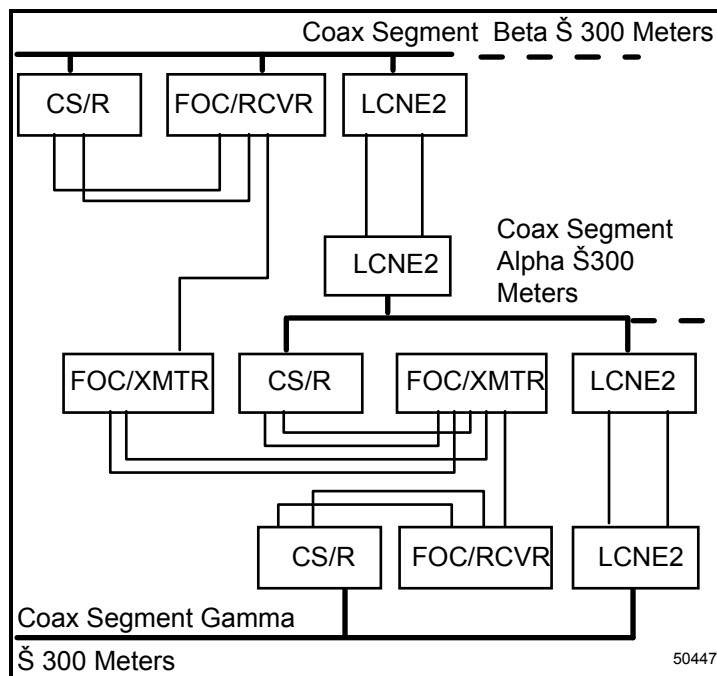


Figure 4-23 — LCN Extender Fiber Optic Cabling with Current Loop Connections

Figure 4-24 shows a central segment “Alpha” that has three remote segments, “Beta,” “Delta,” and “Gamma.” Remote segment “Delta” has the 12.5 kHz clock transmitted to it with fiber optic components as in the previous example. However, remote segments “Beta” and “Gamma” have the clock transmitted to them by current loop. This figure shows in detail the connections for the current loop twisted pairs and shields. Note that the current loop from the CS/R in the central segment includes as loads the FOC/XMTR (for transmitting the clock to segment “Delta”) as well as the CS/Rs for segments “Beta” and “Gamma.” The total length of this current loop (which includes the sections marked “X,” “Y,” and “Z” in Figure 3-28 and 3-29) must be no greater than 300 meters.

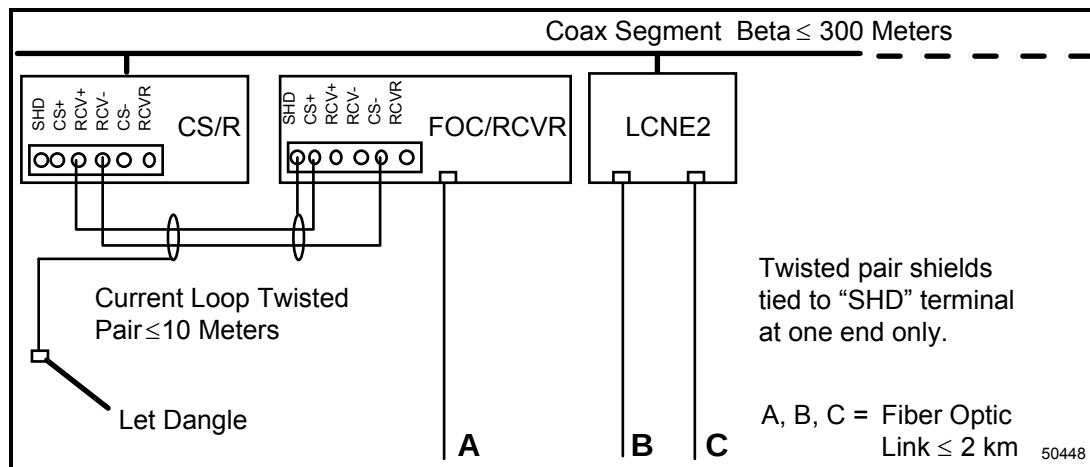


Figure 4-24 — LCN Extender Fiber Optic Cabling Current Loop Connections

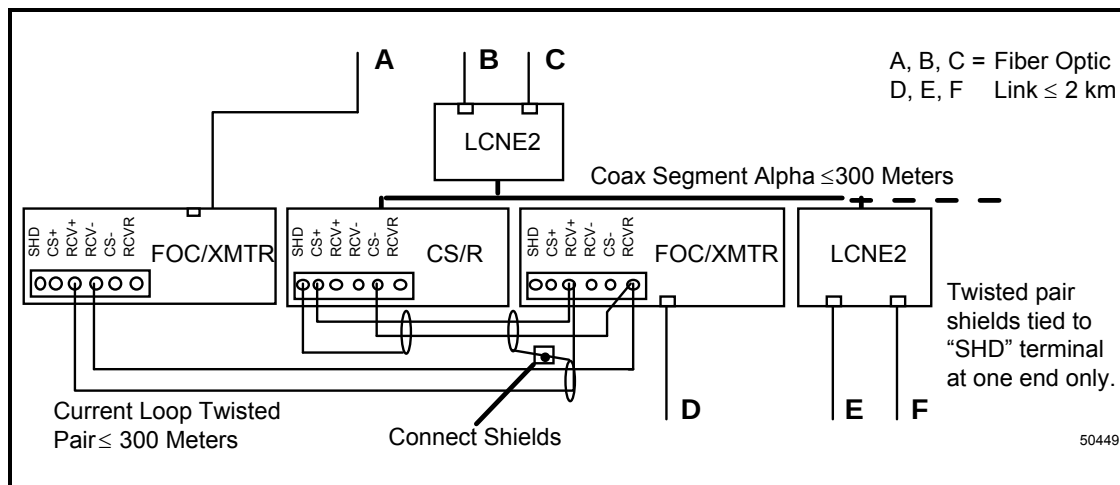


Figure 4-25 — LCN Extender Fiber Optic Cabling Current Loop Connections

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.3. LCN Extender Hardware Implementation

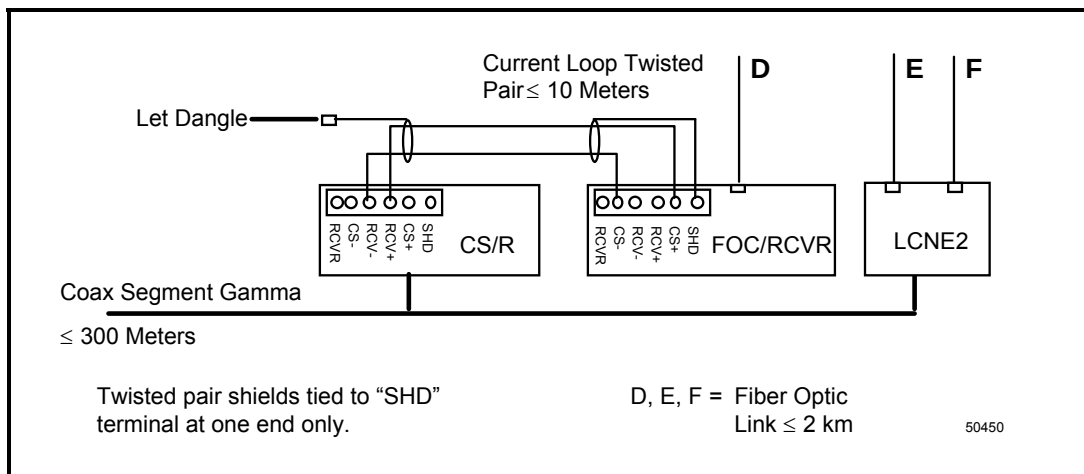


Figure 4-26 — LCN Extender Fiber Optic Cabling Current Loop Connections

Figure 4-27 is an overview that illustrates this function, showing a CS/R driving three FOC/XMTR boards and another CS/R, which in turn drives three additional FOC/XMTRs. The clock can be retransmitted only twice in this manner, or excessive distortion of the signal may result. The maximum length of each current loop is 300 meters, giving an overall maximum length of 900 meters, assuming the loop is retransmitted twice.

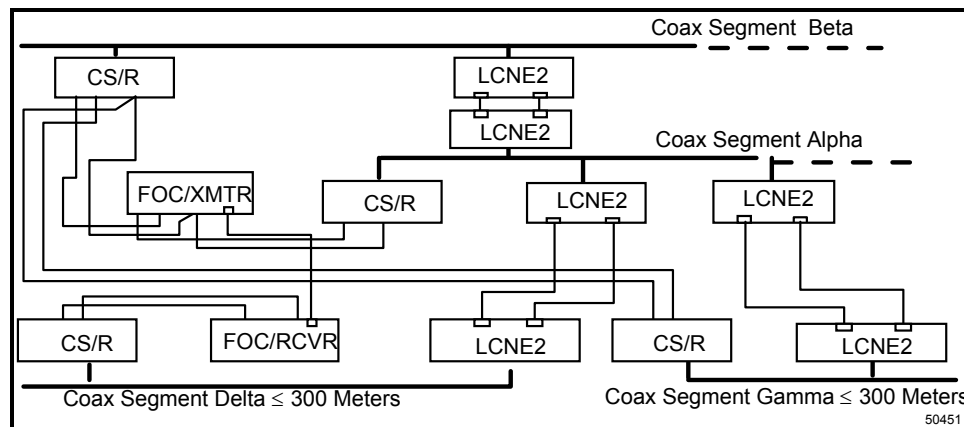


Figure 4-27 — Multiple Current Loop Connections with Retransmitted Clock

Figure 4-28 and 4-29 illustrate the detailed wiring of the current loops used in Figure 4-27.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.3. LCN Extender Hardware Implementation

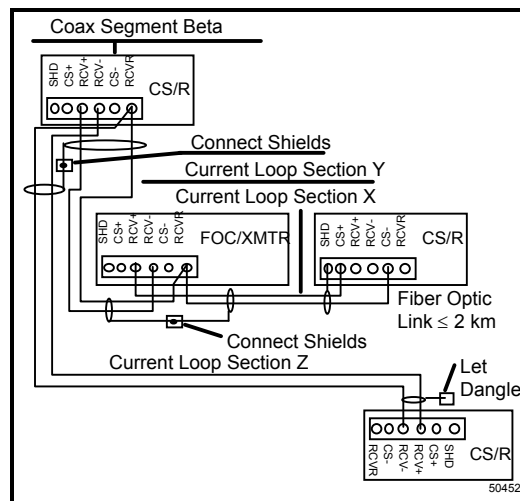


Figure 4-28 — Wiring for Multiple Current Loop Connections

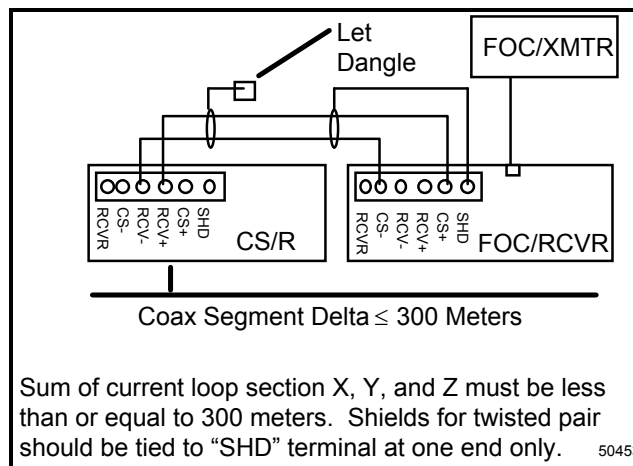


Figure 4-29 — Wiring for Multiple Current Loop Connections with Retransmitted Clock

The current loop output of a CS/R can drive up to four loads, which can be either FOC/XMTRs or other CS/R inputs. If more than four loads are required (i.e., there are more than four remote segments which require the 12.5 kHz clock), a CS/R can be used to repeat, or retransmit, the clock.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.4. Fiber Optic Cable Specifications

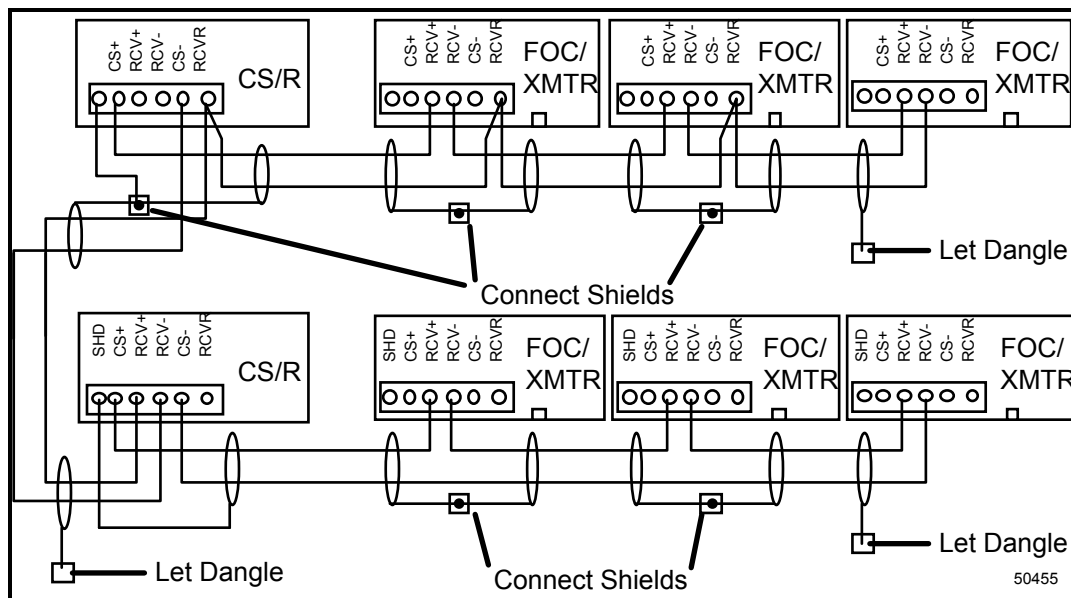


Figure 4-30 — Multiple Current Loop Connections with Retransmitted Clock

NOTE

1. A CS/R current loop can be repeated only twice.
2. This set of boards and connections is for Cable A and must be repeated for Cable B.

4.4 Fiber Optic Cable Specifications

There are four Honeywell raw fiber optic cable specifications. They describe both indoor and outdoor versions of fiber optic cable in both the 100 micron and 62.5 micron fiber core sizes. In addition, there are three Honeywell fiber optic cable assembly drawings, each with associated marketing model numbers. “Micron” is another term for “micrometer,” a unit of measure, one thousandth of a millimeter, or one millionth of a meter. The symbol “ μm ” is often used to denote micron or micrometer. The symbol “nm” means nanometer (one billionth of a meter) and is used as a unit of measurement of wavelength of light.

4.4.1 100 Micron and 62.5 Micron Fiber

100 micron was the fiber core size originally specified for the **TotalPlant** Solution (TPS) system. To date, most TPS systems that have fiber installed use 100 micron fiber. In the meantime, however, the industry has moved toward 62.5 micron fiber as a de facto standard for multimode fiber. Hastening this process is the preference for 62.5 micron fiber in the recently completed ANSI Fiber Distributed Data Interface (FDDI) specification. As a result of this shift, 62.5 micron fiber is now more readily available at a lower cost than 100 micron fiber. Recognizing this industry trend, Honeywell has added 62.5 micron fiber as an option in implementing TPS LCN fiber optic installations. As a matter of fact, Honeywell now recommends that all new installations use 62.5 micron fiber.

Now that two different sizes of fiber are allowed in TPS systems, there exists the possibility of inadvertently mixing fiber sizes in a fiber link. This is generally undesirable. There is no additional loss when light goes from the smaller fiber to the larger one, but a 4.1 dB loss is suffered when going from larger to smaller. Since fiber cables are run in pairs, one for transmit and one for receive, light will be going in opposite directions in the individual fibers of the pair. Adding a different fiber size to both cables at one end of a link (such as when transitioning from outdoor cable to indoor cable) means that at least one cable will suffer at least a 4.1 dB loss. Therefore, to be safe, DO NOT mix fiber sizes in a link.

4.4.2 Procurement Policies

4.4.2.1 Outdoor Fiber Optic Cable

In general, Honeywell does not wish to supply or install outdoor fiber optic cables. Honeywell relies on the cable installation expertise of cable vendors and installation contractors to perform the cable installation. Honeywell also does not wish to restrict the purchase of outdoor fiber optic cable to a particular vendor. Honeywell recognizes that vendors may be able to supply better service in some parts of the world than in others, thus making it desirable to have a choice worldwide. Also, the installation conditions at various project sites may call for widely differing types of cable construction. For these reasons, the outdoor cable specifications were written rather loosely with respect to physical construction details and mechanical parameters. The actual glass fiber itself is completely specified to ensure proper operation of the fiber link.

If the customer requires, Honeywell will contract with cable vendors and installation contractors for the customer to purchase cable, oversee, and guarantee a proper installation. However, if the customer procures his fiber optic cable directly from the supplier, and arranges his own installation, the cable supplier and/or installing contractor must certify to the customer that his cable fully meets or exceeds the applicable Honeywell cable specification. Honeywell will freely supply our outdoor cable specifications to our customers for this purpose.

4.4.2.2 Indoor Fiber Optic Cable

In contrast, the indoor cable specifications completely specify both the mechanical construction details of the cable, and the important parameters of the glass fiber. The recommended method of procurement is direct purchase from Honeywell as a finished cable assembly, including connectors, under the applicable Honeywell model number. See subsection 4.4.5 for a description of the cable assemblies that are available, and the corresponding Honeywell model numbers.

4.4.3 Indoor Grade Cable Specifications

Figure 4-31 illustrates the construction of indoor fiber optic cable as specified by Honeywell. Both 100 micron and 62.5 micron indoor cables share this basic construction, differing only in the fiber size.

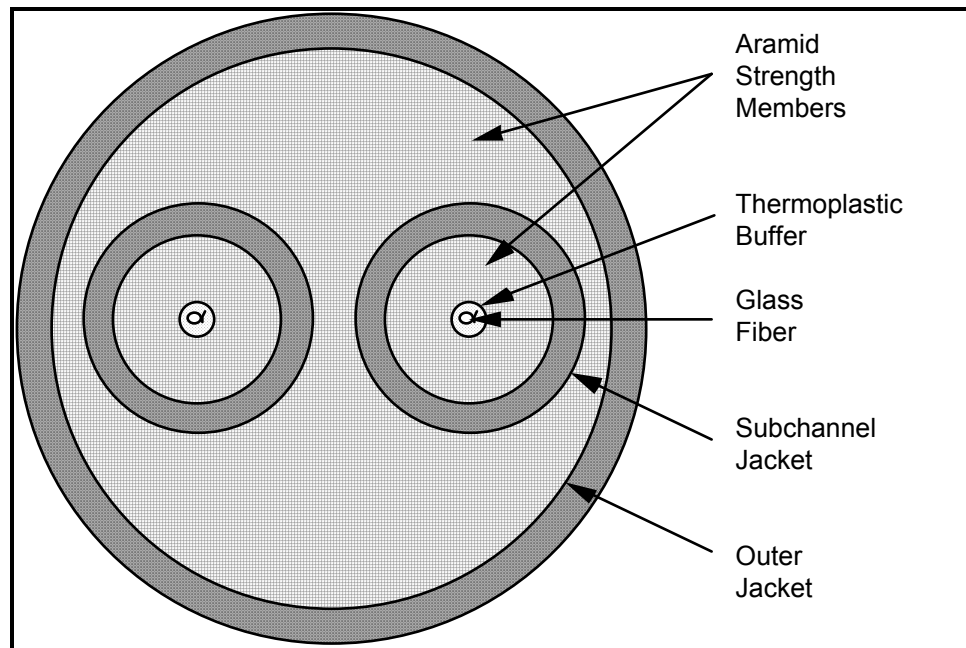


Figure 4-31 — Indoor Tight Buffered Cable

This type of cable is known as Tight Buffered cable because the optical fibers are tightly held by the cable fillers. Unmatched coefficients of expansion between the cable materials and the glass fiber can subject the fibers to significant microbending losses if the cable is exposed to temperature extremes. For this reason, tight buffered cable is limited to indoor use. The advantage indoors is good physical protection for the fiber while maintaining the cable flexibility required for routing the cable inside electronic cabinets, under floors, etc.

The cable contains two, four, or six subunits. Each subunit protects a single fiber and can have independent connectors. The standard indoor cable assemblies are duplex (two fibers). Finished cable assemblies with four or six fibers are available on special order.

There are two cable jacket options available for the indoor cable. The National Fire Protection Association (NFPA) publishes the National Electrical Code (NEC) to establish fire safety standards for premises wiring. Honeywell specifies jacketing material conforming to either NEC optical cable rating OFNR (Optical Fiber, Nonconducting, for Riser applications), or OFNP (Optical Fiber, Nonconducting, for Plenum applications). The standard cable assemblies, which can be purchased from the Honeywell price book by model number, are OFNR rated cables. Should OFNP rated cables be required, they can be special ordered through Honeywell Purchasing. OFNP rated cables are required only when routing indoor fiber runs through air handling chambers.

4.4.3.1 Key Parameters, Indoor Grade Cable, 62.5 Micron (Drawing Number 51190918)

The important parameters of the 62.5 micron indoor grade glass fiber specified by Honeywell are as follows:

Core diameter:	62.5 ± 3 microns
Cladding diameter:	125 ± 2 microns
Bandwidth @ 850 nm:	160 MHz-km min.
Attenuation @ 850 nm:	4.0 dB/km max.

4.4.3.2 Key Parameters, Indoor Grade Cable, 100 Micron (Drawing Number 51190697)

The important parameters of the 100 micron indoor grade glass fiber specified by Honeywell are as follows:

Core diameter:	100 ± 4 microns
Cladding diameter:	140 ± 6 microns
Bandwidth @ 850 nm:	100 MHz-km min.
Attenuation @ 850 nm:	6.0 dB/km max.

4.4.3.3 Mechanical Properties, 62.5 and 100 Micron Indoor Grade Cable

Table 4-1 includes the mechanical properties of indoor grade cable supplied by Honeywell. “Subchannel” refers to an individual jacketed fiber, which is part of a multi-fiber cable (see Figure 4-31). “N” stands for “Newton,” a metric unit of force equal to approximately 0.225 lb.

Table 4-1 — Mechanical Properties of 62.5 and 100 µm Indoor Grade Cable

Characteristic	Subchannel	2 Fiber Cable	4 Fiber Cable	6 Fiber Cable
Tensile Load, Short Term at Installation	300 N (67 lb)	650 N (146 lb)	1300 N (292 lb)	2000 N (450 lb)
Tensile Load, Long Term	50 N (11 lb)	100 N (22.5 lb)	200 N (45 lb)	300 N (67.5 lb)
Minimum Bend Radius, Short Term at Installation (For Installation at Maximum Tensile Load)	5 cm (2.0 in)	17 cm (6.7 in)	22 cm (8.7 in)	26 cm (10.2 in)
Minimum Bend Radius, Long Term (Unloaded for Expected Life)	3 cm (1.2 in)	13.0 cm (5.1 in)	16.5 cm (6.5 in)	20.0 cm (7.9 in)
Maximum Diameter of Jacketed Cable Assembly	N/A	9.5 mm (0.374 in)	11.0 mm (0.433 in)	12.5 mm (0.492 in)
Maximum Cable Weight	N/A	60 kg/km (40 lb/kft)	90 kg/km (60.5 lb/kft)	120 kg/km (80.7 lb/kft)

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.4. Fiber Optic Cable Specifications

4.4.4 Outdoor Grade Cable Specifications

Honeywell specifies a cable construction known as Loose Tube for its outdoor grade fiber optic cable. An illustration of this construction is shown in Figure 4-32.

This construction is characterized by loose fitting, gel filled tubes into which the fibers are placed. The fibers are actually longer than the tubes, so that when thermal expansion lengthens the buffer tubes, the glass fibers (which have a lower coefficient of expansion) are never subject to tensile stress.

The cable is given buckle resistance typically by a glass reinforced plastic (GRP) rod through the center of the cable. An aramid wrap around the buffer tubes provides tensile strength.

This type of cable is significantly stiffer than tight buffered cable, and the jacketing material does not meet NEC requirements. These factors make it unsuitable for indoor use. The loose buffer design's advantage is being able to take environmental extremes without suffering any significant optical performance degradation.

Figure 4-32 shows a typical construction for an aerial/duct cable. A direct burial cable is similar, except that it would probably have two additional layers: a steel armor tape layer for rodent protection, covered by an additional polyethylene jacket. Vendors also may have double armoring options available. Again, exact construction details will vary from vendor to vendor, but the basic loose tube concept remains the same.

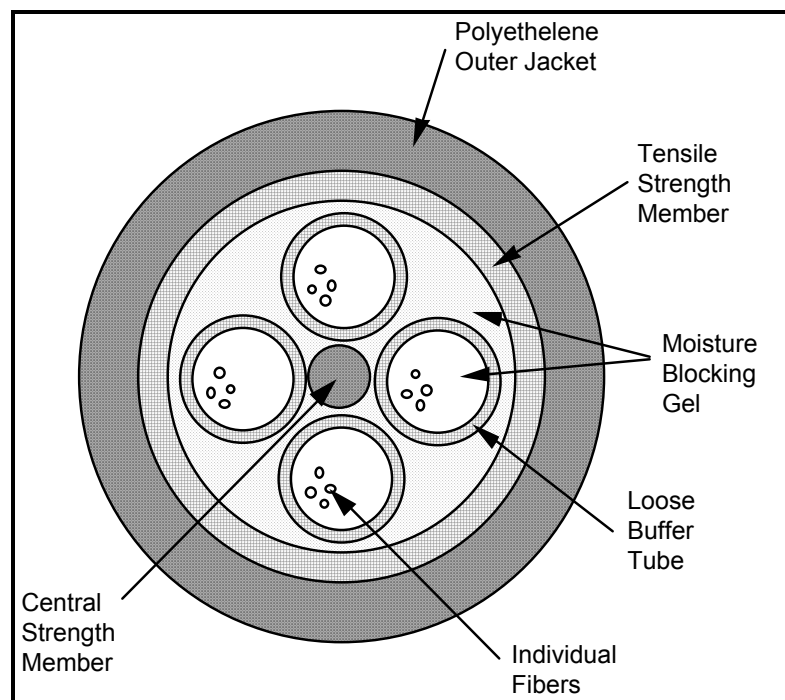


Figure 4-32 — Outdoor Loose Tube Cable

The National Electrical Code ratings do not apply to outdoor cable because it does not fall into the category of premises wiring. The NEC does mention that the maximum length of unrated outdoor cable that is permitted inside a building is 50 feet. This allows enough length to bring the cable to a splice/interconnect enclosure where the transition to indoor cable can be made.

4.4.4.1 Key Parameters, Outdoor Grade Cable, 62.5 Micron (Drawing Number 51190919)

The important parameters of the 62.5 micron outdoor grade glass fiber specified by Honeywell are identical to the parameters specified for indoor cable:

Core diameter:	62.5 ± 3 microns
Cladding diameter:	125 ± 2 microns
Bandwidth @ 850 nm:	160 MHz-km min.
Attenuation @ 850 nm:	4.0 dB/km max.

4.4.4.2 Key Parameters, Outdoor Grade Cable, 100 Micron (Drawing Number 51190698)

The important parameters of the 100 micron indoor grade glass fiber specified by Honeywell are identical to the parameters specified for indoor cable:

Core diameter:	100 ± 4 microns
Cladding diameter:	140 ± 6 microns
Bandwidth @ 850 nm:	100 MHz-km min.
Attenuation @ 850 nm:	6.0 dB/km max.

4.4.4.3 Mechanical Properties of Outdoor Cable

As discussed in subsection 4.4.2, Honeywell does not recommend specific cable for outdoor use. When a cable is selected, refer to the vendor's specifications for that cable to obtain applicable mechanical properties.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.4. Fiber Optic Cable Specifications

4.4.5 Fiber Optic Cable Assemblies

This subsection discusses the duplex indoor fiber optic cable assemblies with preinstalled connectors that are available from Honeywell. Table 4-2 lists the all of the standard assemblies and the paragraphs that follow the table discuss each type.

Table 4-2 — Standard Duplex Indoor Cable Assemblies

Model Number	Length (Meters)	Core Diameter (μm)	Description
C-KFT02	2	100	SMA 905 to SMA 905
C-KFT04	4		
C-KFT06	6		
C-KFT10	10		
C-KFT25	25		
C-KFT50	50		
C-KFN02	2	100	SMA 905 to Pigtail
C-KFN04	4		
C-KFN06	6		
C-KFN10	10		
C-KFN25	25		
C-KFN50	50		
P-KFA02	2	62.5	SMA 905 to SMA 905
P-KFA05	5		
P-KFA10	10		
P-KFA20	20		
P-KFA50	50		
P-KFH02	2	62.5	SMA 905 to ST
P-KFH05	5		
P-KFH10	10		
P-KFH20	20		
P-KFH50	50		
P-KFB02	2	62.5	ST to ST
P-KFB05	5		
P-KFB10	10		
P-KFB20	20		
P-KFB50	50		

In all the following cable assembly model numbers, replace the trailing "xx" with two numeric digits specifying a standard length. For example, if a 25-meter long 100 micron SMA to SMA cable assembly was desired, the correct model number would be C-KFT25.

4.4.5.1 Model Number C-KFTxx — 100 Micron SMA/SMA

This duplex cable assembly is provided with SMA 905 connectors on both ends. Standard available lengths are 2, 4, 6, 10, 25, and 50 meters.

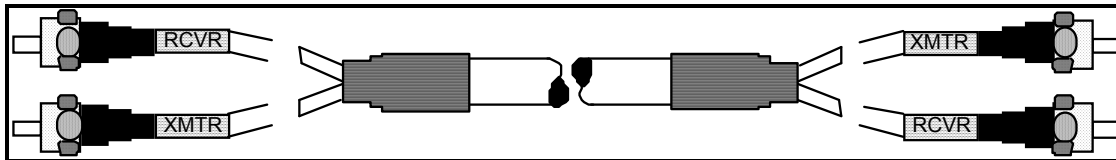


Figure 4-33 — SMA to SMA Cable Assembly

Originally, the SMA 906 connector was specified for this cable assembly. A Delrin alignment sleeve needs to be installed on the tip of this connector to assure accurate alignment in an active device mount (such as the fiber optic transmitter or receiver). Often, the sleeve would fall off and get lost. If used without the alignment sleeve, excess power loss at the connector would result. Therefore the switchover was made to the SMA 905 connector, which does not require the alignment sleeve. The 906 series connector can continue to be used as long as the alignment sleeve is in place. Figure 4-34 compares the difference between these two connector types.

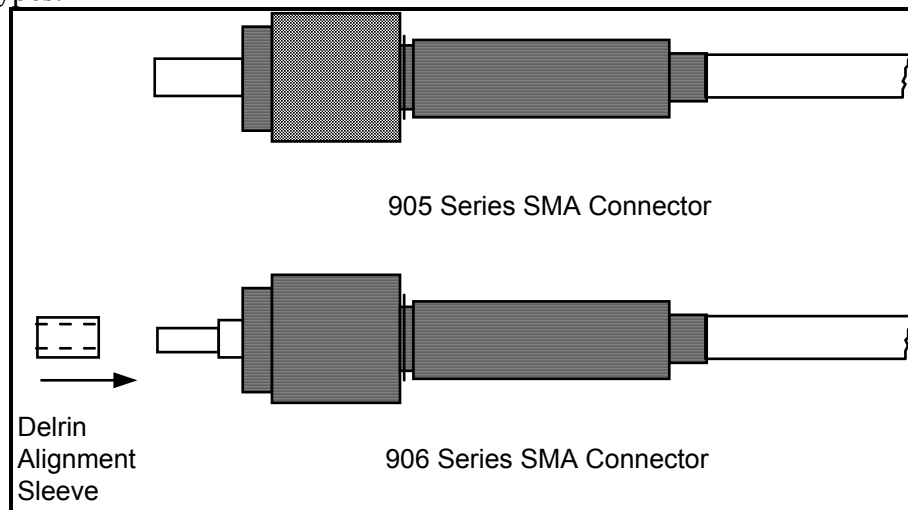


Figure 4-34 — Comparison of SMA 905 and SMA 906 Connectors

4.4.5.2 Model Number C-KFNxx — 100 Micron SMA/Pigtail

This duplex cable assembly is very similar to C-KFTxx, except that it is provided with SMA 905 connectors on one end only. The other end is left unterminated for direct splicing to the outdoor cable run. Typically, this cable is used where the outdoor cable is brought indoors to a splice enclosure. The splice is made to the indoor SMA/Pigtail cable, which is then run to the equipment. Standard available lengths are 2, 4, 6, 10, 25, and 50 meters.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.4. Fiber Optic Cable Specifications

4.4.5.3 Model Number P-KFAxx — 62.5 Micron SMA/SMA

This cable assembly is also very similar to C-KFTxx, the difference being, in this case, that 62.5 micron fiber is used. It has preinstalled SMA 905 series connectors on both ends. Standard lengths are 1, 2, 5, 10, 20, and 50 meters.

4.4.5.4 Model Number P-KFHxx — 62.5 Micron SMA/ST

This is a hybrid cable assembly in the sense that different connector types are used on each end. Just as 62.5 micron fiber has displaced 100 micron fiber as a de facto industry standard, the ST connector has largely displaced the SMA style connector. TPN equipment still uses SMA active device mounts, so the SMA connectors must still be used at the LCN end. This cable is provided for connection to a fiber optic interconnect panel which typically would be implemented using a bank of ST-style connector receptacles. See subsection 4.6.3.2 for a discussion of interconnect panels.

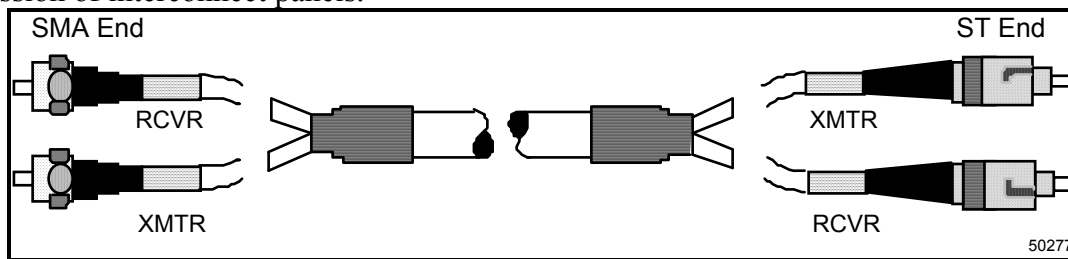


Figure 4-35 — P-KFHxx 62.5 μm SMA to ST Cable Assembly

4.4.5.5 Model Number P-KFBxx — 62.5 Micron ST/ST

This fiber optic cable assembly contains ST connectors at both ends. Standard TPN equipment does not use ST connectors but special long distance versions of the LCNE2 have been developed which do. Also, the PM I/O Link Extender used ST connectors. These cable assemblies may also be used in interconnect panels as patch cables. Refer to subsection 4.6.3.2 for a discussion of interconnect panels. Figure 4-36 illustrates the model P-KFBxx cable.

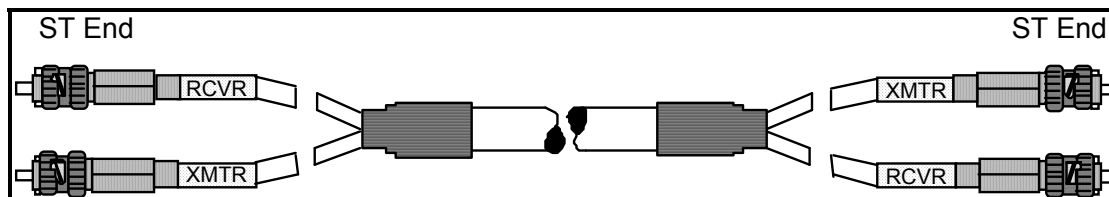


Figure 4-36 — P-KFBxx 62.5 μm ST to ST Cable Assembly

4.4.6 Cable Connectors

Honeywell recommends the fiber optic connectors shown in Table 4-3 for use on indoor tight-buffered fiber optic cable only, because outdoor cable does not permit the small bend radii needed in a cabinet or equipment rack and is always spliced to a short length of indoor cable for connection purposes. The SMA connectors are used to connect to TPN equipment and the ST connectors typically are used connect to interconnect panels (subsection 4.6.3.2).

Table 4-3 — Cable Connectors

Supplier	Supplier's Part Number	Type	Power Loss (dB)	Fiber (μm)
Amphenol Fiber Optic Products 1925 Ohio Street Lisle, IL 60532 708-810-5800	905-400-5007	SMA - Ceramic Ferrule	<0.8 dB typical	100
	905-405-5007	SMA - Ceramic Ferrule	<0.8 dB typical	62.5
	953-101-5010	ST - Ceramic Ferrule	0.13 dB median	62.5

Special tools and procedures are required for installing connectors. The cable must be stripped, cut, finished, and attached to the connector. Typical tools and materials for performing these operations are shown in Table 4-4.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.4. Fiber Optic Cable Specifications

Table 4-4 — Tools and Materials

Supplier	Supplier's Part Number	Type	Use
Siecor Corporation 489 Siecor Park Hickory, NC 28603	100299-01 (handle) 100311-01 (hook blade)	Utility knife	Stripping—to slit outer jacket of cable.
	100300-01	Side cutters (diagonal cutting)	Stripping—to cut off the central member (core).
	100107-01	Buffer jacket cutter	Stripping to remove the buffer jacket
	100345-01	Blade pack	
	100294-01	Scissors	Stripping—to cut off the peeled-back outer jacket, protective yarn, and tape.
	100304-01	Sheath ripper	Used to cut helically-wrapped binders when stripping cable.
Amphenol Fiber Optic Products 1925 Ohio Street Lisle, IL 60532 708-810-5800	927-100-5017	SMARt Termination Kit	Complete kit to install Amphenol SMARt series fiber optic connectors

4.5 Calculating the Link Power Budget

In planning the installation of a fiber optic link, one of the most important tasks is the calculation of the power budget. The power budget determines the link distance achievable, and indicates how much power margin exists. It depends on a number of factors, such as fiber attenuation, number of connector joints, and the number and type of splices in the link. To date, all TPN product line fiber optic equipment uses the same fiber optic transmitter and receiver modules. The transmitter module is guaranteed to couple 100 μW (-10 dBm) of optical power into 100 micron fiber when new. The smaller diameter of 62.5 micron fiber allows only 39 μW (-14.1 dBm) to be coupled in (4.1 dB less than 100 micron fiber). The receiver modules are guaranteed to operate correctly (with a bit-error-rate better than 1 in 10^9) with only 1.5 μW (-28.2 dBm) of received power.

4.5.1 Calculating the Power Budget for 100 Micron Fiber

Table 4-5 details the calculation of a typical link's power budget using 100 micron fiber:

Table 4-5 — Power Budget Calculation for 100 Micron Fiber

Minimum power coupled into fiber:	100 μW = -10.0 dBm
Minus minimum power receiver needs:	(-) 1.5 μW = (-) -28.2 dBm
Total power budget:	18.2 dB
Losses:	
Connector loss (2 * 1.0 dB):	2.0 dB
Splice loss (2 * 0.25 dB):	.5 dB
Safety margin:	3.0 dB
Budget remaining for loss in cable:	12.7 dB
Divide by:	
Specified attenuation per km:	$\div$ 6.0 dB
Achievable link distance:	2.2 km

Starting with the minimum power guaranteed to be coupled into 100 micron fiber, 100 μW or -10 dBm, subtract from that figure the minimum power level guaranteed to be properly decoded by the receiver, 1.5 μW or -28.2 dBm. This calculation gives a total power budget of 18.2 dB. From this 18.2 dB, certain losses must be subtracted. The first shown is the loss of two connector-to-connector butt splices. Typically, two such connections are made in the link at interconnect boxes at the interface between the outdoor and the indoor cable—one at each end of the link.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.5. Calculating the Link Power Budget

Similarly, each indoor-to-outdoor cable interface may require a fiber-to-fiber splice, so two splice losses are subtracted from the budget. Lastly, a 3 dB safety factor is subtracted. A safety factor of 3 dB is minimal, and it is strongly recommended that if a larger safety factor can be arranged (either by shortening the required link distance, or by using fiber with a better attenuation figure), by all means do so. The larger the safety margin, the more reliable the link will be.

After all the above loss factors have been subtracted, the 12.7 dB that remains is the power left to be expended in cable loss. It is this number, when divided by the attenuation figure of the fiber, that determines the achievable link distance. As can be seen above, using the attenuation figure specified in Honeywell drawing number 51190698, a typical link distance of slightly more than 2 km can be achieved.

4.5.2 Calculating the Power Budget for 62.5 Micron Fiber

Table 4-6 shows a similar analysis for 62.5 micron fiber:

Table 4-6 — Power Budget Calculation for 62.5 Micron Fiber

Minimum power coupled into fiber:	39 μ W = -14.1 dBm
Minus minimum power receiver needs:	(-) 1.5 μ W = (-) -28.2 dBm
Total power budget:	14.1 dB
Losses:	
Connector loss (2 * 1.0 dB):	2.0 dB
Splice loss (2 * 0.25 dB):	.5 dB
Safety margin:	3.0 dB
Budget remaining for loss in cable:	8.6 dB
Divide by:	
Specified attenuation per km:	÷ 4.0 dB
Achievable link distance:	2.2 km

The first difference to notice in this table is that less power is coupled into 62.5 micron fiber. Since the cross-sectional area of the fiber is less than that of 100 micron fiber, less light gets in. This reduction in input power is about 4.1 dB. So instead of an initial budget of 18.2 dB, the total budget is only 14.1 dB.

The same loss factors are subtracted as they were for the 100 micron example shown earlier, leaving 8.6 dB to be expended in the cable. The 4.1 dB lower level of coupled power is completely made up by the lower attenuation figure of 62.5 micron fiber. Virtually the same distance can be achieved using either the specified 62.5 micron fiber or the specified 100 micron fiber.

4.6 Cable Plant Implementation

4.6.1 Fiber Count and Redundant Routing

When extending the TPS LCN, two fibers are needed for each coaxial cable channel—a transmit and a receive fiber. Thus, to extend both A and B channels of the redundant LCN, four fibers are required. In addition, if clock synchronization is required across the fiber link, two more fibers are required to carry the clock signal to the remote end (one each for A and B channels), bringing the total to six fibers.

However, Honeywell does not recommend that both A and B channels be carried in the same cable. To maintain the inherent reliability of redundant channels, Honeywell recommends that the A and B channel fibers be carried in separate cables on separate routes. For example, the A cable may be buried on one side of a service road, and the B cable on the other side; or bury one cable and hang the other aurally. Only one basic rule needs to be observed when routing the cables along separate paths: the maximum distance between any two nodes in the system as measured on the A cable must not differ from the distance measured along the B cable by more than 300 meters. Note that the distances referred to above include the sum of all fiber optic and coaxial sections of cable.

Figure 4-37 is an example of a poor configuration. Segment Alpha is a central coaxial segment, with two remote coaxial segments, Beta and Gamma. There is no problem with communications between the Alpha segment and either of the two remote segments. Cable A is 200 meters shorter than Cable B between Alpha and Beta. Cable A is also 200 meters shorter than Cable B between Alpha and Gamma. 200 meters is within the 300 meter differential limit. **However, Cable A is a total of 400 meters shorter than Cable B between segments Beta and Gamma. Unreliable communication would result.**

Keep in mind that coaxial cable length differential also plays a part in this. In the example, we assumed that the coaxial A and B channels in each segment were equal. If they are not, they would also contribute to the total differential calculation. Remember that the criteria is 300 meters maximum differential between any two nodes.

Due to the expense of installing outdoor cable runs, it is generally recommended that outdoor cable runs contain 33% to 100% more fibers than is immediately required. The incremental cost of providing extra fibers in the cable is small compared with the cost of installing another cable if more data communications capacity is required in the future. For example, consider a single remote LCN extension requiring remote clock synchronization. If the recommendation for routing the A and B cables on separate paths is being followed, two cables of at least three fibers each would be required. To provide the extra capacity for future contingencies, the recommended fiber count per cable would be four to six (cables are generally available only in even fiber counts).

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.6. Cable Plant Implementation

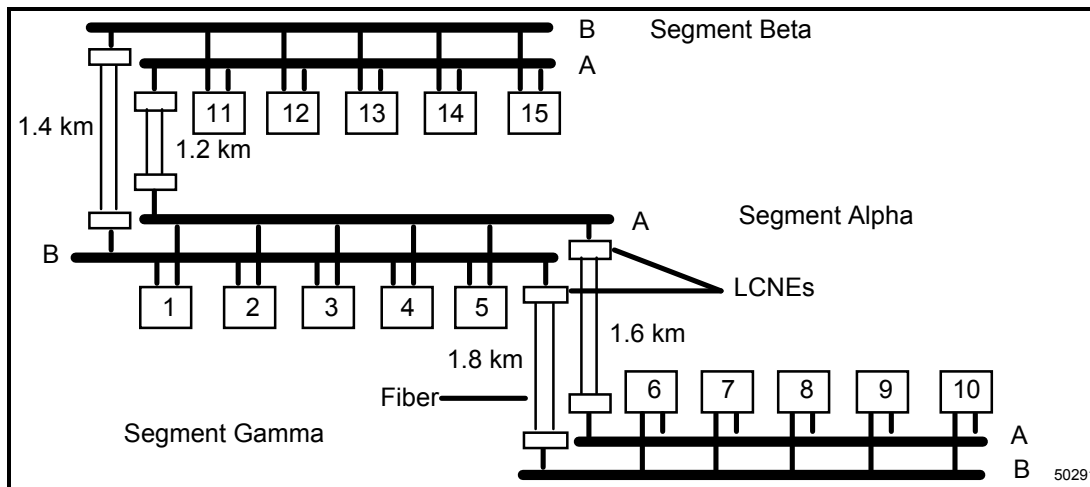


Figure 4-37 — Example of Incorrect Routing Configuration of Redundant Fiber Cables

4.6.2 Major Classes of Cable Installation

Aerial, underground duct, and direct cable burial are the three most popular ways to install outdoor cable runs. The method of choice for a particular site depends on any number of factors. Each has its own pluses and minuses.

4.6.2.1 Aerial

Aerial cable installations are subject to the greatest environmental stresses of the major classes of outdoor cable installation. An aerial cable is subject to wind loading, ice loading, extremes of temperature variation, and even gunshots.

There are two basic kinds of aerial cable available. One is designed to be lashed to a messenger wire as it cannot support itself. The other is self-supporting. Of the self-supporting aerial cables, there are two main subclasses: One contains the tensile stress bearing member within the normal cable sheath; the other type, which is capable of longer unsupported runs, is the "figure-8" type of cable, where the cable contains its own built-in messenger wire from which hangs the fiber-carrying capsule.

4.6.2.2 Underground Duct

This type of installation buries a plastic tube or tubes into which a cable or several cables can be pulled. A recommended fill ratio is 50% by cross sectional area. The cable is usually of virtually the same construction as a lashed aerial cable. Under the ground it is well protected from environmental stress. It is recommended that the duct be buried as deeply as possible—definitely below the frost line. Frost heaving can crack the duct. The relatively large diameter of the duct provides rodent protection. Rodents can't chew on it if they can't get their mouths around it. If empty innerducts were provided within a larger duct, then new cables could be pulled in the future without too much trouble. Pulling new cable into an already occupied inner duct, however, is not recommended.

4.6.2.3 Direct Burial

This type of installation buries the cable directly into the earth. As with the underground duct, bury this cable as deeply as possible. Make sure to document and mark where the cable is buried for future reference. An armor sheath is required on this cable for rodent protection. Keep in mind, though, that the metal content of the cable may make it subject to lightning strikes. Also, the armor must be grounded at each end where it enters the building.

4.6.2.4 Cable with Preinstalled Connectors

In certain situations, it may be desirable to preinstall connectors to the cable at the factory. This may be the case where the skilled labor required to perform splicing and installing of connectors will not be available at the installation site. However, preinstallation of connectors makes pulling the cable through duct or conduit very difficult, so in duct situations, it may not be feasible. If handled carefully, it should not be as much of a problem in aerial or direct burial situations. Still, the risk of damaging the terminations during installation is substantial. Therefore, the general preference is to splice or preconnect the cable in the field once the cable is installed.

4.6.3 The Transition from Outdoor to Indoor Cable

It is necessary to use both indoor and outdoor fiber optic cable types when implementing a fiber link between two buildings. The indoor cable has the flexibility to allow routing under floors and into equipment cabinets, while the outdoor grade has superior performance over temperature and environmental extremes.

4.6.3.1 Splicing

Perhaps the most straight-forward method of transitioning from outdoor to indoor cable types is to use a splice, housed in an in-line splice enclosure. Figure 4-38 shows an example of using in-line splices and splice enclosures.

Since this method does not involve any connector-to-connector couplings, it suffers the least power budget penalty. The drawback, though, is lack of flexibility. The in-line splice enclosures have provision for only one cable in, and one cable out. If extra dark fibers were supplied in the outdoor cable, and it becomes necessary to use them in the future, the in-line splice enclosure should probably be replaced with a wall-mounted interconnect panel.

Multiple indoor cable runs cannot be fanned out from the simple in-line splice enclosure. Of course, if it is considered highly unlikely that future expansion needs to be accommodated, this is a very cost-effective method of transitioning.

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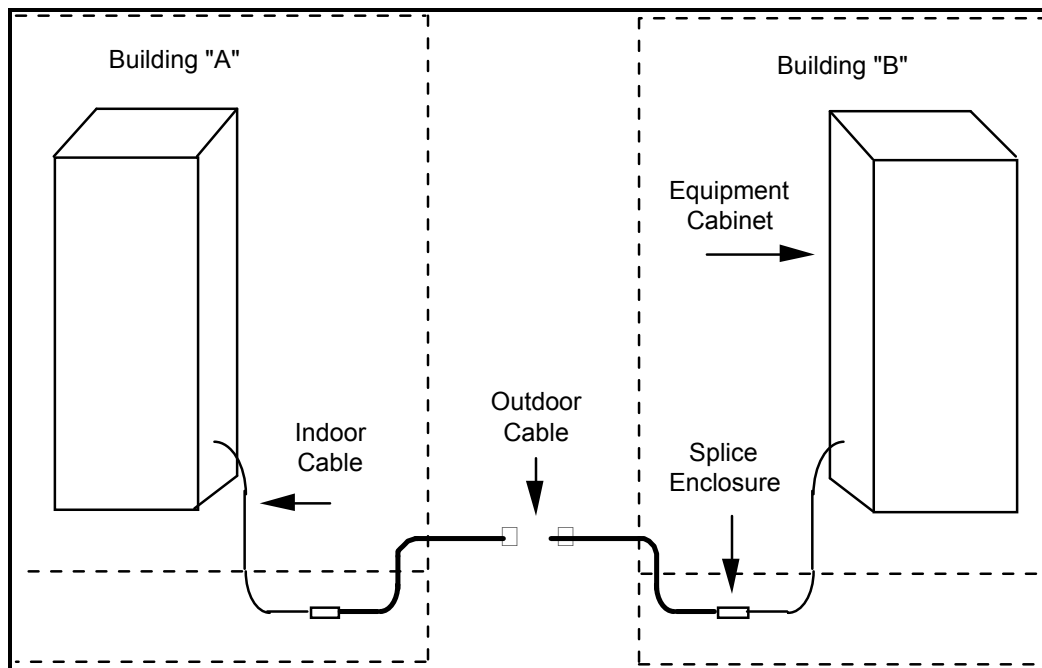


Figure 4-38 — Indoor to Outdoor Cable Transition Using In-Line Splices

As far as the fiber splice itself is concerned, there are two basic methods used to join the fibers: fusion splicing and mechanical splices.

Fusion splicing is accomplished by bringing the two fiber ends together in a precision micropositioning fixture, then hitting the joint with a precisely timed electric arc to melt the fiber ends together. Fusion splicing is the most reliable method, and when done properly, results in less than .2 dB of power loss in the link. Fusion splicing's biggest drawback is expense. The machine is quite expensive, as is the technical talent. The cost to get a trained individual with his fusion splicer to a project site won't be cheap.

In recent years, fiber optics vendors have begun to produce mechanical splices with excellent insertion loss performance. The low insertion loss of fusion splicing can even be approached. The newest mechanical splices can be mated and remated if the first try is not good enough. The mechanical splice aligns the two fibers to be joined in a V-groove, or a precision capillary. The joint between the two fibers is filled with an index-matching gel to minimize back reflection and maximize forward coupling. Mechanical splices typically exhibit insertion losses less than .5 dB, or better if tuning is employed. Since these splices can be made with a minimum of training, and the alignment jig is relatively inexpensive, the mechanical splice is quite cost effective in most cases.

Table 4-7 lists some typical or representative mechanical splice hardware that is commercially available. The Siecor "SeeSplice" is so named because it incorporates a unique fiber alignment method that enables the craftsperson to actually see the fibers during splicing, thereby ensuring successful mating and accurate alignment. The Siecor "CamSplice" incorporates the same alignment as the "SeeSplice," but features a cam locking mechanism, which requires no adhesive. Also, one part fits all fiber coating diameters.

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Table 4-7 — Representative Splicing Kits

Supplier	Part Number	Description
Siecor Corporation 489 Siecor Park Hickory, NC 28603 704-327-5000	95-000-01	SeeSplice for 250 µm to 250 µm coated fiber
	95-000-02	SeeSplice for 250 µm to 900 µm coated fiber
	95-000-03	SeeSplice for 900 µm to 900 µm coated fiber
	95-000-04	CamSplice
GTE Products Corporation 2401 Reach Road Williamsport, PA 17701	7713-0006-000	Emergency Restoration Kit (contains precision cleaving tool for splicing—produces splices that require no grinding or polishing). Used to strip and splice cable. Splices have a low power loss (0.25 dB typical). Complete instructions are included.

Table 4-8 — Representative Splice Enclosures

Supplier	Supplier's Part Number	Item	Use	Comments
Siecor Corporation 489 Siecor Park Hickory, NC 28603 704-327-5000	QSK-010	Quick-Splice (Trademark) Kit	Protect fusion splices. Indoor and/or outdoor.	12 fibers
	QSK-011	Quick-Splice (Trademark) Kit	Protect fusion splices. Indoor use only.	12 fibers
	QSK-012	Quick-Splice (Trademark) Kit	Protect mechanical splice. Indoor and/or outdoor.	6 fibers, holder designed for SeeSplice or CamSplice.
	QSK-013	Quick-Splice Kit (Trademark)	Protect mechanical splice. Indoor use only.	6 fibers, holder designed for SeeSplice or CamSplice.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.6. Cable Plant Implementation

4.6.3.2 Interconnect Panels

The use of interconnect (or patch) panels in the fiber optic cable plant is strongly encouraged. Their use allows the quick and easy future use of extra dark fibers. Also, cables from several remote buildings may be brought to a single interconnect box to provide a central point of fiber channel management for plant-wide data communications.

This method can make use of fanout tubing to avoid the need to perform any splicing. Fanout tubing is essentially a tight buffer subunit without any fiber. A fanout tube is slipped over each of the individual fibers in the loose tube cable after the sheath and buffer tubes have been stripped off. Cable vendors supply "break out" or "fan-out" kits, which consist of fanout tubing and a strain-relieving boot. These kits allow the installation of connectors to the individual fibers.

This technique can jacket only about 3 to 15 feet of bare fiber, so it is really limited to use inside an interconnect panel.

Each fiber in the outdoor cable (now jacketed by the breakout kit) is terminated and plugged into the back of a bulkhead-mounted barrel connector. These connectors will generally be the ST style of connector. On the other side, the connection to LCN end equipment is made by means of the hybrid duplex cable assembly discussed in subsection 4.4.5.4. Unused receptacles should be capped off to prevent the ingress of contaminants.

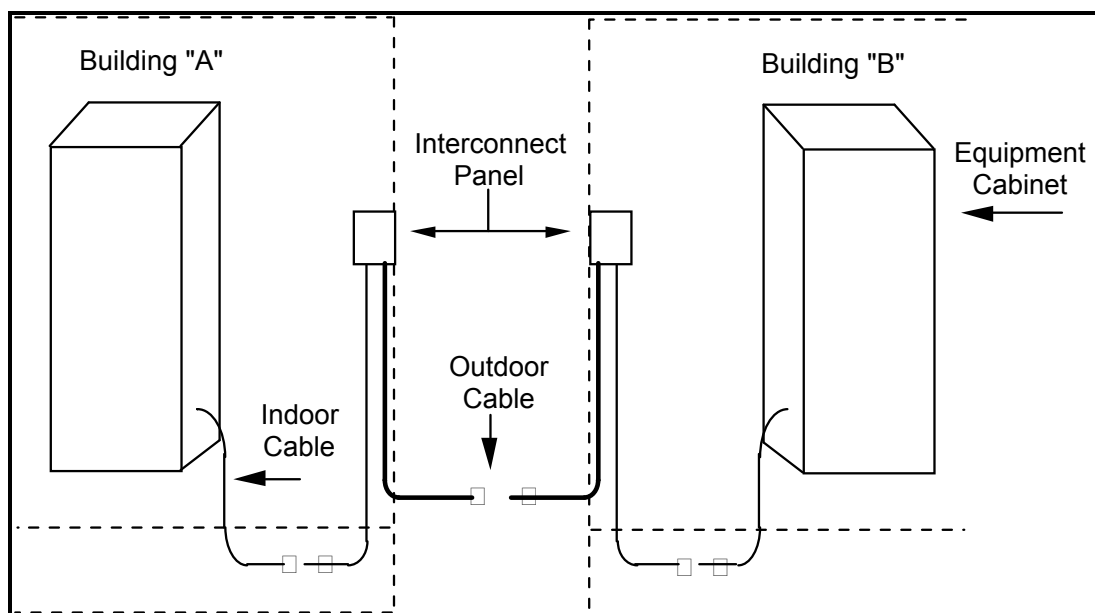


Figure 4-39 — Indoor to Outdoor Cable Transitions Using Interconnect Panels

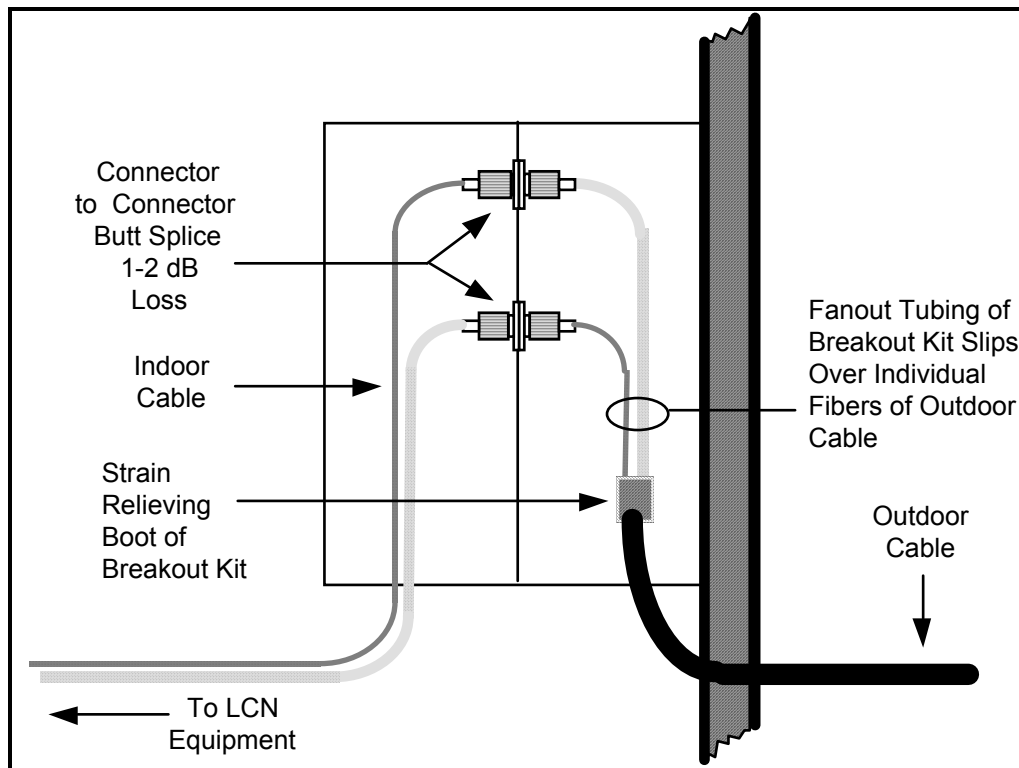


Figure 4-40 — Interconnect Panel Schematic Detail

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.6. Cable Plant Implementation

Table 4-9 — Representative Interconnect Panels

Supplier	Supplier's Part Number	Description	Comments
Siecor Corporation 489 Siecor Park Hickory, NC 28603 704-327-5000	WIC-001	Wall-Mountable Interconnect Panel (12 fiber capacity)	Compact wall-mount unit. Contact Siecor for WIC options, including connector panel modules.
	WIC-024	Wall-Mountable Interconnect Panel (24 fiber capacity)	
	FDC-003	Fiber Distribution Center (24 fiber capacity)	Modular units for rack-mount or wall-mount. Contact Siecor for FDC options, which include connector panel modules and splice trays.
	FDC-002	Fiber Distribution Center (72 fiber capacity)	
	FDC-001	Fiber Distribution Center (144 fiber capacity)	

4.6.4 Qualifying the Link

Once the cable plant is installed, a final optical check should be made. For multimode fiber installations, an optical time domain reflectometer (OTDR) is not necessary. An end-to-end attenuation check plus a single-ended minimum receive power check is sufficient. Assuming that an LCNE2 is available as an optical power source, only a hand-held optical power meter and a short length of patch cable is required.

These tests are very simple to perform. First, the end-to-end attenuation check:

When the fiber link was planned, the link power budget should have been calculated (refer to subsection 4.5) in order to get an idea of where the optical losses would be taken, and how much safety margin would remain. The end-to-end attenuation check verifies the planned link attenuation against the actual installation.

First, measure the output of an LCNE2 fiber optic transmitter by connecting a short length of patch cable to a hand-held optical power meter. Use a patch cable of the same fiber diameter as the fiber used in the fiber link to be tested. Record the transmitter's power output in dBm. Now connect the LCNE2 transmitter to one end of the fiber optic link under test. At the other end, measure the optical power output in dBm. The difference between the power output recorded from the patch cable measurement and the power output measured over the link under test is the end-to-end attenuation of the link. (The error introduced because the patch cable wasn't also used in the second measurement is negligible because the length of the patch cable is so short.) The end-to-end attenuation thus measured should substantially agree with the planned end-to-end attenuation.

Now, the minimum receive power check can be made: Pay attention to the actual value of the power measurement at the far end of the link under test as opposed to the difference from a reference. Recall from the discussion on power budget (subsection 4.5) that the LCNE2 fiber optic receiver requires a minimum of $1.5 \mu\text{W}$ (-28.2 dBm) of power to operate. Recall also, that it is desirable to leave at least 3 dB as a general safety margin. Therefore, if the LCNE2 is new, the power measured at the far end of the link should be a minimum of $(-28.2 \text{ dBm} + 3 \text{ dB} =) -25.2 \text{ dBm}$.

If both of these tests return satisfactory results, the link can be commissioned with confidence.

4. IMPLEMENTATION OF LCN FIBER OPTIC EXTENDERS FOR LCNE2

4.6. Cable Plant Implementation

5. IMPLEMENTATION OF UCN FIBER OPTIC EXTENDERS

This section provides a description of a Universal Control Network fiber optic link Extender (UCNE) and UCNE mounting kit installation procedures.

5.1 General UCNE Information

5.1.1 Description

The Universal Control Network two-kilometer fiber optic Extender (UCNE) is used to:

- Increase the end-to-end distance of a single UCN without resorting to special, wide-diameter coaxial trunk cables.
- Increase the UCN signal quality at the ends of the UCN that would otherwise be connected by long coaxial trunk cables.
- Implement certain UCN topologies, such as a “star configuration,” that is not possible by using coaxial trunk cables.
- Provide ground isolation, when necessary, between UCN trunk segments.

Each UCNE provides coax-to-fiber optic and fiber optic-to-coax cable conversion for the UCN signals.

UCNEs are installed in pairs at each end of a redundant fiber optic link, that is a UCNE-A and UCNE-B at one end of the fiber optic link and a UCNE-A and UCNE-B at the other end of the fiber optic link. Therefore, a total of four UCNEs are required to implement one redundant fiber optic link.

The UCNE, at each end of the fiber optic link, never connects directly to the UCN coaxial trunk, but instead connects to the drop port of a UCN tap on that coaxial segment.

Standard coaxial UCN drop cables (RG6-type) are used to connect the UCNEs to the UCN tap drop ports.

5. IMPLEMENTATION OF UCN FIBER OPTIC EXTENDERS

5.1. General UCNE Information

5.1.2 Universal Control Network Topologies

There are two classes of UCN topology when using UCN fiber optic Extenders (UCNEs) to link coaxial UCN trunk segments.

- Serial
- Star

Serial Topology

The simplest “serial” UCN topology consists of two UCN coaxial trunk segments that are interconnected by a single fiber optic link. Figure 5-1 illustrates the “one series link” configuration.

Another serial UCN topology consists of three UCN coaxial trunk segments that are interconnected by two fiber optic links. Figure 5-2 illustrates the “two series link” configuration.

Each UCN redundant fiber optic link requires two UCNEs (UCNE-A and UCNE-B) at each end of the link, or a total of four UCNEs. Two fiber optic links, as illustrated in Figure 5-2, would require eight UCNEs.

The maximum number of series-connected fiber optic links that are allowed in a physical UCN is two. The use of three or more fiber optic links in a serial topology is prohibited because the delay imposed by the additional UCNE(s) and fiber optic cable could violate the UCN’s slot time setting and cause UCN communication errors.

Figure 5-1 illustrates a typical serial UCN topology with two UCN coaxial segments and one fiber optic link. Only network “A” is shown. Network “B” is not shown.

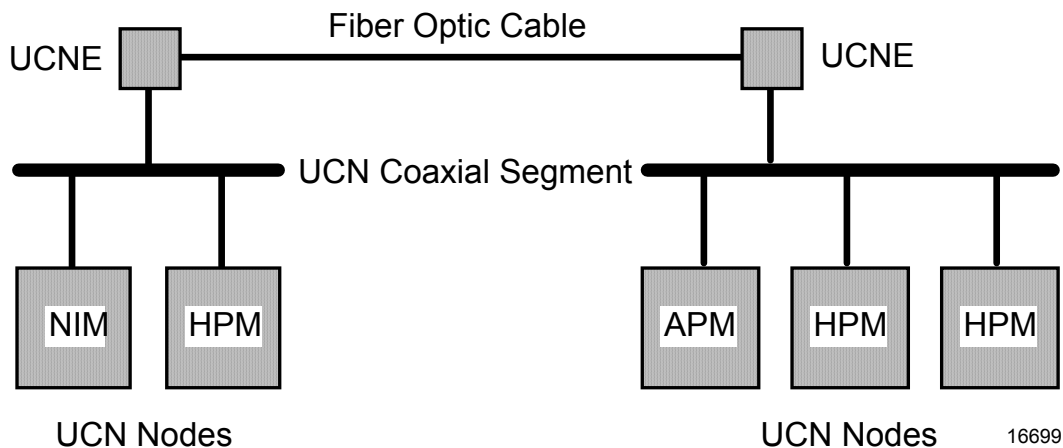


Figure 5-1 — Typical Serial UCN Topology – Two Coaxial Segments

Figure 5-2 illustrates a typical serial UCN topology with three UCN coaxial segments and two fiber optic links. Only network “A” is shown. Network “B” is not shown.

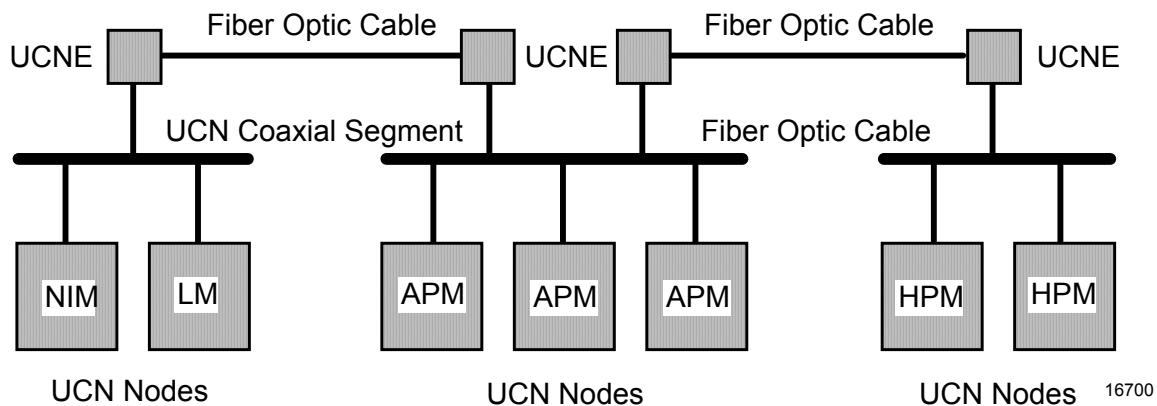


Figure 5-2 — Typical Serial UCN Topology – Three Coaxial Segments

Star Topology

A “star” UCN topology consists of a central UCN coaxial cable segment with peripheral UCN coaxial cable segments connected to the central UCN coaxial cable segment through UCNEs. Figure 5-3 illustrates a typical star UCN “star” configuration example.

UCNEs must not be used with passive star couplers because passive couplers have excessive optical power loss. Instead, star topology is implemented by using multiple drop ports on a central UCN coaxial segment to branch out through multiple UCN fiber optic links to other UCN coaxial segments. An example of their use is a central control room that contains the NIM(s) with multiple radial UCN fiber optic links (using UCNEs) to remote UCN coaxial segments that accommodate other UCN nodes. Figure 5-3 illustrates four radial fiber optic links to four remote sites.

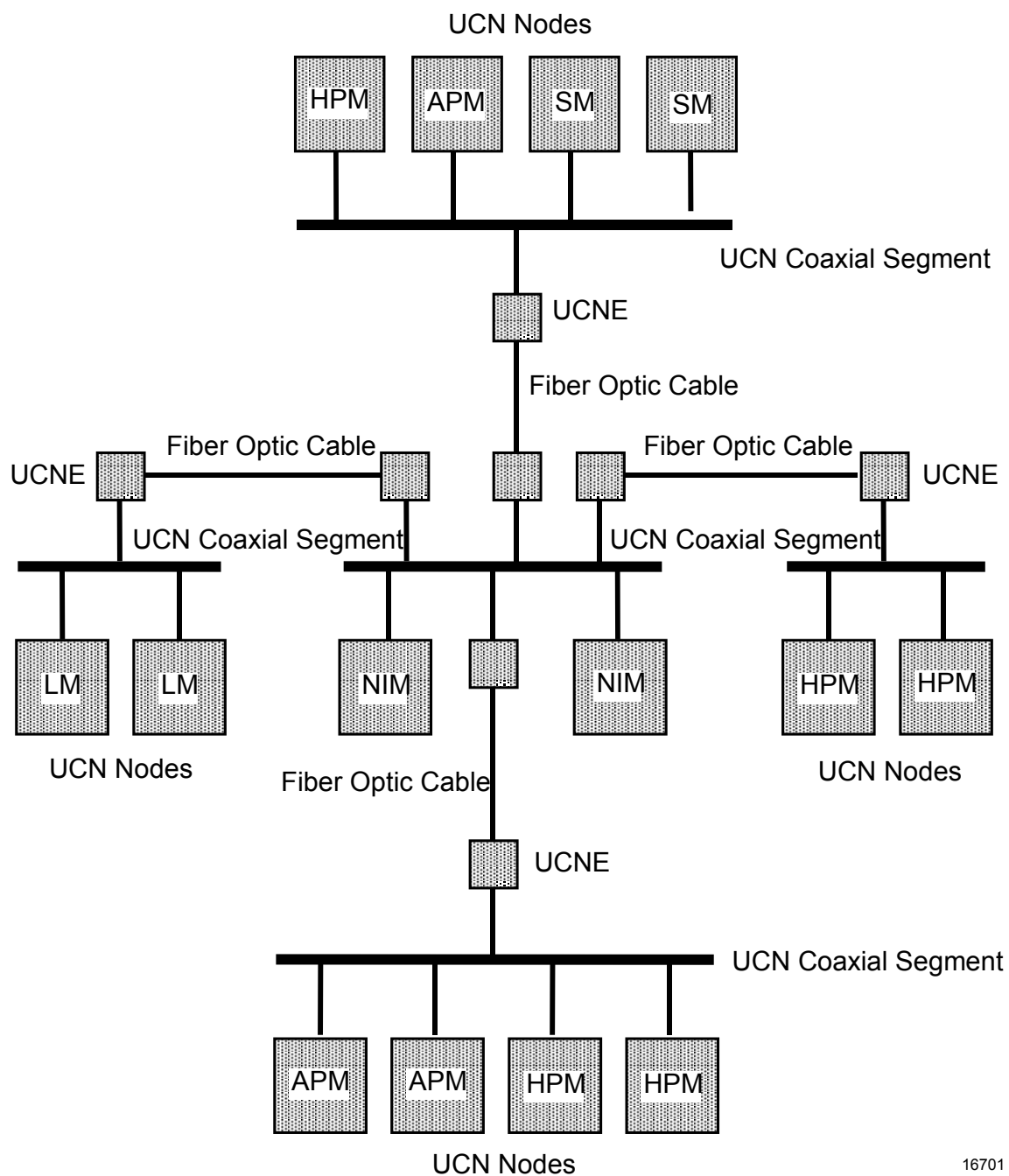
Only one of the redundant UCN cables, A or B, is shown in the figure for simplification.

It is preferable that the NIM(s) is at the center of the “star,” but it is not a requirement, just a recommendation, to minimize the “loss of view” in the unlikely event that a redundant fiber optic link fails completely.

Figure 5-3 illustrates a typical star UCN topology with five UCN coaxial segments. Only network “A” is shown. Network “B” is not shown.

5. IMPLEMENTATION OF UCN FIBER OPTIC EXTENDERS

5.1. General UCNE Information



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Figure 5-3 — Typical Star UCN Topology

5.2 UCNE Configuration Rules

Regardless of the UCN topology, the following configuration requirements must be met.

1. The total length of the UCN (network A or B) between any two nodes, including all series-connected coaxial trunk cable segments and fiber optic links, but not including drop cable lengths, must not exceed 2 kilometers. Since any two UCN nodes can potentially communicate with each other, the restriction is determined by the two nodes that are farthest apart from each other along either UCN Cable A or UCN Cable B and separated by a combination of UCN coaxial trunk cable and UCN fiber optic cable. Violation of this rule could result in UCN errors that are due to communication delays that exceed the UCN slot time. UCN slot time is the maximum time that any UCN node waits for a response from any other UCN node on the network and is affected by cable and UCNE delays.
2. Related to rule 1 above, the difference between the total length of UCN network A and UCN network B must be no greater than 500 meters. That is, the UCN A cable and UCN B cable communication paths between the two farthest apart UCN nodes must not be different by more than 500 meters. Violation of this rule could result in UCN errors that are due to communication delays that exceed the UCN slot time.
3. Related to rule 1 above, the length of fiber optic cable that connects any two UCNEs must not be longer than 2 kilometers, but there is no minimum length requirement. Violation of this rule could result in UCN communications errors because the optical signal loss budget would be exceeded for the interconnected UCNEs (see rule 4 that follows).
4. Related to rule 3 above, the optical signal loss budget for a fiber optic cable connection between two interconnected UCNEs is 10 dB maximum and must not be exceeded. This includes up to 2 kilometers of fiber optic cable that is specified in Honeywell drawing 51190918 that has a maximum loss of 4 dB/km, two ST-type connectors (one at each end of the fiber optic cable) with a maximum loss of 1 dB per connector, and two cable splices with a maximum loss of 0.5 dB per splice. Violation of this rule could result in UCN communications errors because the optical signal loss budget could be exceeded for the interconnected UCNEs.
5. The UCN coaxial cable configuration rules found in other sections of the manual also apply for each separate coaxial trunk segment of the network.

5.3 UCNE Model Numbers

The UCNE model numbers that support the maximum 2-kilometer fiber optic cable distance are:

- MU-FOED02A pair of 18-36 Vdc powered UCNEs
- MU-FOEA02A pair of 85-264 Vac, 47-63 Hz powered UCNEs

The dc-powered UCNE is used where a PM, APM, or HPM Power System's 24 Vdc is available. Figure 5-4 is an illustration of a dc-powered UCNE.

The ac-powered UCNE is typically used at UCN locations where a PM, APM, or HPM Power System's 24 Vdc is not available, such as in cabinets or consoles that house a NIM, LM, or SM. However, the ac-powered UCNE should be powered by UPS-backed ac power if it is expected to continue supporting UCN communications during an ac line power outage. Figure 5-5 is an illustration of an ac-powered UCNE.

UCNEs must be installed independently from each other such that a failed UCNE in a redundant pair (UCNE-A and UCNE-B) can be replaced while the other UCNE continues operating. This is accomplished using the standard mounting kits that are described in subsection 5.6.

Figure 5-4 illustrates the front panel of a dc-powered UCNE.

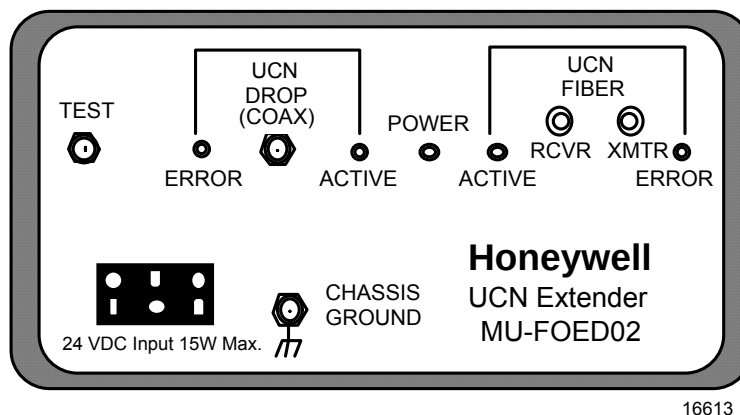
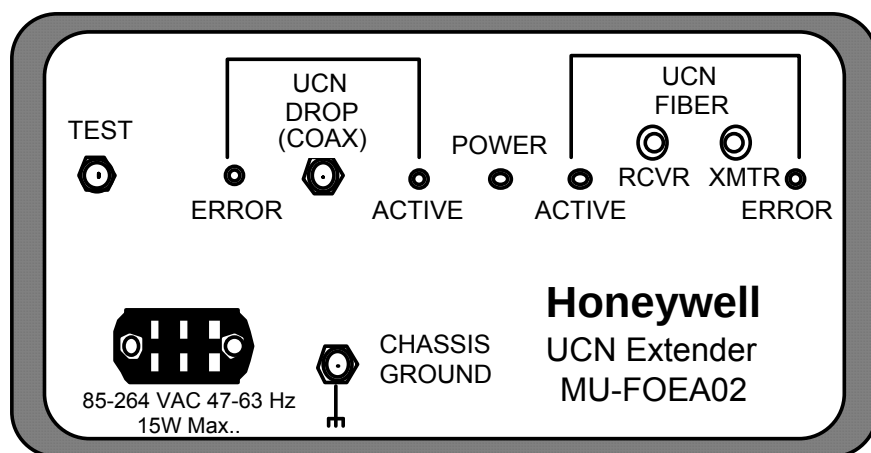


Figure 5-4 — DC-Powered UCNE

Figure 5-5 illustrates the front panel of an ac-powered UCNE.



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Figure 5-5 — AC-Powered UCNE

5.4 UCNE Mounting Kits

5.4.1 LCN Cabinets and Consoles

The standard mounting location for a redundant pair of UCNEs in LCN cabinets or consoles is in the UCNE enclosures that are included in the following mounting kits.

MP-FOMK02	LCN Cabinet 120 Vac UCNE Mounting Kit
MP-FOMK03	LCN Cabinet 240 Vac UCNE Mounting Kit
MP-FOMK04	Classic Console 120 Vac UCNE Mounting Kit
MP-FOMK05	Classic Console 240 Vac UCNE Mounting Kit
MP-FOMK06-100	Z-Console Upper Equipment Pod UCNE Mounting Kit
MP-FOMK06-200	Z-Console Lower Equipment Pod UCNE Mounting Kit

The UCNE enclosures can also be installed in LM and SM cabinets. The model MP-FOMK02 and MP-FOMK03 mounting kits are recommended for that purpose.

The model MP-FOMK06 mounting kits do not include a pair of replacement leg covers which are required when adding a UCNE enclosure to Z-console furniture. They must be ordered separately. Two factors must be considered before ordering, the height of the console furniture (high, medium, or low) and whether the UCNE enclosure's equipment pod will be the only (upper) pod or a second (lower) pod.

For each of the kits, a redundant pair of ac-powered UCNEs (model MU-FOEA02), which must be ordered separately, are mounted side by side in a metal enclosure (module). Viewing the rear of the enclosure, the UCNE on the right side of the enclosure is connected to the UCN-A cable and the UCNE on the left side of the enclosure is connected to the UCN-B cable.

Labels on the enclosure identify the UCN-A and UCN-B mounting locations. The labels are color-coded orange and brown, respectively, to agree with the standard UCN redundant cable color-coding scheme.

5. IMPLEMENTATION OF UCN FIBER OPTIC EXTENDERS

5.4. UCNE Mounting Kits

Each mounting kit includes two ac power cables to power the pair of ac-powered UCNEs that must be installed in the enclosure. If redundant ac power sources are available, it is recommended that each UCNE in the redundant pair be connected to a different ac power source to maintain operation of at least one UCNE if a power source should fail. An Uninterruptible Power System (UPS) is recommended to maintain ac power to the UCNEs during ac line outages.

Table 5-1 is the list of parts in the model MP-FOMK02 mounting kit. The two UCNEs that are mounted in the UCNE enclosure must be ordered separately by model MU-FOEA02.

Table 5-1 — Model MP-FOMK02 Parts List

Description	Part Number	Quantity
UCNE Enclosure	51403223-100	1
AC Power Cable, 120 Vac	51305409-500	2
UCNE Enclosure Front Panel	51305413-100	1
M5 Thread Forming Screw	51195168-616	4
M5 Black Hex-Head Machine Screw	51108888-616	1
Grounding Serrated Washer	51109931-100	2
Black Hex-Head Machine Screw	51108888-625	1
Spacer	51108572-100	1
M5 Serrated Washer	51109933-003	4

Table 5-2 is the list of parts in the model MP-FOMK03 mounting kit. The two UCNEs that are mounted in the UCNE enclosure must be ordered separately by model MU-FOEA02.

Table 5-2 — Model MP-FOMK03 Parts List

Description	Part Number	Quantity
UCNE Enclosure	51403223-100	1
AC Power Cable, 240 Vac	51305410-500	2
UCNE Enclosure Front Panel	51305413-100	1
M5 Thread Forming Screw	51195168-616	4
M5 Black Hex-Head Machine Screw	51108888-616	1
Grounding Serrated Washer	51109931-100	2
Black Hex-Head Machine Screw	51108888-625	1
Spacer	51108572-100	1
M5 Serrated Washer	51109933-003	4

Table 5-3 is the list of parts in the model MP-FOMK04 mounting kit. The two UCNEs that are mounted in the UCNE enclosure must be ordered separately by model MU-FOEA02.

Table 5-3 — Model MP-FOMK04 Parts List

Description	Part Number	Quantity
UCNE Enclosure	51403223-100	1
AC Power Cable, 120 Vac	51305409-500	2
UCNE Enclosure Front Panel	51305413-100	1
M5 Thread Forming Screw	51195168-616	4
Grounding Serrated Washer	51109933-003	4

Table 5-4 is the list of parts in the model MP-FOMK05 mounting kit. The two UCNEs that are mounted in the UCNE enclosure must be ordered separately by model MU-FOEA02.

Table 5-4 — Model MP-FOMK05 Parts List

Description	Part Number	Quantity
UCNE Enclosure	51403223-100	1
AC Power Cable, 240 Vac	51305410-500	2
UCNE Enclosure Front Panel	51305413-100	1
M5 Thread Forming Screw	51195168-616	4
Grounding Serrated Washer	51109933-003	4

Table 5-5 is the list of parts in the model MP-FOMK06-100 mounting kit. The two UCNEs that are mounted in the UCNE enclosure must be ordered separately by model MU-FOEA02.

Table 5-5 — Model MP-FOMK06-100 Parts List

Description	Part Number	Quantity
UCNE Enclosure	51403223-100	1
AC Power Cable	51305450-100	2
Module Enclosure Front Cover	51403089-300	1
Module Enclosure Assembly	51403117-100	1
Module Enclosure Mounting Bracket	51402745-100	1
Module Enclosure Mounting Bracket	51402745-200	1
Module Enclosure Front Cover Mounting Bracket	51308039-100	2
Module Enclosure Rear Cover Assembly	51403152-100	1
M5 Pan-Head Machine Screw	51108385-610	4
M5 Thread Forming Screw	51195168-616	4
M6 Flat-Head Screw	51108881-716	4
Air Filter	51308003-100	1

5. IMPLEMENTATION OF UCN FIBER OPTIC EXTENDERS

5.4. UCNE Mounting Kits

Table 5-6 is the list of parts in the model MP-FOMK06-200 mounting kit. The two UCNEs that are mounted in the UCNE enclosure must be ordered separately by model MU-FOEA02.

Table 5-6 — Model MP-FOMK06-200 Parts List

Description	Part Number	Quantity
UCNE Enclosure	51403223-100	1
AC Power Cable	51305450-100	2
Module Enclosure Front Cover	51403090-300	1
Module Enclosure Assembly	51403117-100	1
Module Enclosure Mounting Bracket	51402745-100	1
Module Enclosure Mounting Bracket	51402745-200	1
Module Enclosure Front Cover Mounting Bracket	51308039-100	2
Module Enclosure Rear Cover Assembly	51403152-100	1
M5 Pan-Head Machine Screw	51108385-610	4
M5 Thread Forming Screw	51195168-616	4
M6 Flat-Head Screw	51108881-716	4
Air Filter	51308003-100	1

A pair of UCN drop cables of appropriate length (model MU-NKDxxx where xxx = length in meters) is required to connect the pair of UCNEs to a pair of UCN taps. The drop cables are not supplied with the UCNE mounting kit and must be ordered separately.

If sufficient UCN tap drop ports are not available when expanding a UCN with UCNEs, then either the number of ports per tap must be increased, or the number of taps in the existing UCN coaxial segments (A and B) must be increased. UCN taps can be ordered by model number as follows.

MU-NTAP02 Two 2-drop taps with mounting/grounding hardware

MU-NTAP04 Two 4-drop taps with mounting/grounding hardware

MU-NTAP08 Two 8-drop taps with mounting/grounding hardware

5.4.2 PM, APM, and HPM Cabinets

NOTE

The information that follows is supplemental to the LCN cabinet and console information. Detailed PM, APM, and HPM cabinet information can be found in the *Universal Control Network Planning* and *Universal Control Network Installation* manuals.

The standard mounting location for a redundant pair of UCNEs is in model MU-FOMK01 mounting kit hardware. The kit's enclosure is installed in a Left or Right 7-slot card file position in the card file side of a PM, APM, or HPM cabinet. Typically, the kit is installed after first mounting all the other PM, APM, or HPM card files that connect to the UCN, and in a cabinet that contains UCN taps because UCNEs connect to the UCN, through drop cables to the UCN taps.

The model MU-FOMK01 mounting kit provides for vertical stacking of up to four UCNEs. A redundant pair of dc-powered UCNEs (model MU-FOED02) will occupy vertical slots 1 and 3.

The model MU-FOMK11 mounting kit provides the hardware to accommodate a second redundant pair of dc-powered UCNEs for use in vertical slots 2 and 4 for an optional second UCN fiber optic link. The second pair of UCNEs must be ordered separately by ordering another model MU-FOED02.

Each of the four vertical mounting slots includes a slide-out mounting tray that attaches to the bottom of a UCNE by using four screws that are provided in the mounting kit.

Mounting positions 1 and 2 are identified by orange-colored "UCN-A" labels, and mounting positions 3 and 4 are identified by brown-colored "UCN-B" labels.

Mounting kits model MU-FOMK01 and MU-FOMK11 include two 5-meter power cables (51201397-002) to provide power to the dc-powered UCNEs. If longer power cables are needed, they must be ordered separately.

A pair of UCN drop cables of appropriate length (model MU-NKDxxx where xxx = length in meters) is required to connect a pair of UCNEs to a pair of UCN taps. The drop cables are not supplied with the UCNE mounting kits and must be ordered separately.

If sufficient UCN tap drop ports are not available when expanding a UCN with UCNEs, then either the number of ports per tap must be increased, or the number of taps in the existing UCN coaxial segments (A and B) must be increased. UCN taps can be ordered by model number as follows.

MU-NTAP02	Two 2-drop taps with mounting/grounding hardware
MU-NTAP04	Two 4-drop taps with mounting/grounding hardware
MU-NTAP08	Two 8-drop taps with mounting/grounding hardware

5.5 GROUNDING

The chassis ground stud at the front of the UCNE must be connected to Safety Ground, similar to the grounding of a UCN tap.

A #10 AWG ground wire is provided in all the UCNE mounting kits to connect the UCNE's front panel ground stud to ground on the metal enclosure in which the UCNEs are installed. The eventual connection to Safety Ground depends on proper grounding of the metal enclosure. As required when installing a UCN tap, the installer must confirm that there is electrical continuity between the UCNE's front panel grounding stud and the cabinet Safety Ground stud after the installation is complete. The Safety Ground stud is typically located at the bottom rear of the cabinet.

5.6 Fiber Optic Cable

The multi-node fiber optic cables recommended for use with a UCNE are the same as the 62.5-micron fiber optic cables that are recommended for the Honeywell I/O Link Extender in the PM, APM, or HPM. Honeywell drawings 51190918 (indoor) and 51190919 (outdoor) define the cables. The cable has 4 dB/km maximum optical attenuation according to the signal loss budget requirements of subsection 5.2.

Other fiber optic cable types, such as the 100-micron fiber optic cable, have not been tested with the UCNE by Honeywell and are not recommended. The UCNE is not compatible with single-mode fiber optic cable.

UCN indoor fiber optic cable assemblies should be fabricated using the information in Honeywell drawing 51195498. The RCVR and XMTR fibers at each end of the cable should be labeled as such to agree with the connector labeling on the UCNEs.

A UCN fiber optic cable link consists of two transmit-receive pairs of optic fibers (UCN-A and UCN-B). If possible, cables that have high installation costs, such as cables routed in conduit or buried in the ground should have spare optic fibers.

Refer to the *Universal Control Network Planning* and *Local Control Network Planning* manuals for fiber optic cable information on the following subjects.

- Cable routing
- Cable burial
- Cable aerial lashing
- Cable vertical installations
- Indoor cable requirements
- Loose buffered cable
- Optic fiber losses
- Optic fiber signal loss budget calculation
- Cable installation
- Allowable indoor cable bend radius
- Cable construction selection
- Cable splices and connections
- Power level measurement

5.7 UCNE Installation

5.7.1 Preinstallation Considerations

This subsection provides the procedures for the installation of a UCNE mounting kit in an LCN cabinet or console, or in an LM or SM cabinet. It does not cover the installation of UCNE mounting kits in a PM, APM, or HPM cabinet. That information is found in the *Universal Control Network Installation* manual.

Before beginning a UCNE installation, be sure that you are adequately prepared. Consider the following.

- Are the required number of unused UCN drop tap ports available on each existing UCN coaxial segment, A and B? One port on the A and B segments is needed for each UCN Extender (two UCNEs will be used). The port must be available on the taps that are positioned the same distance from other taps.
- If the required number of usable UCN drop tap ports are not available, have additional taps been ordered to replace existing taps and expand the number of unused ports? If replacement of existing taps is not appropriate, have additional taps been ordered so that they can be added to the trunk cables? When a tap is added to the trunk cable, an additional coaxial cable of the appropriate length is required and cable's loading factor must be recalculated. Refer to the *Local Control Network Planning* manual.
- Have the required number of UCN trunk cables (RG11-type) and drop taps been ordered to support the coaxial cable segment(s) that will be added to the UCN?
- Have the required number of drop cables (RG6-type) been ordered to connect the UCNEs to the taps and support the coaxial cable segment(s) that will be added to the UCN?
- Are a sufficient number of ac outlets of the proper voltage available (120 or 240 Vac) to accommodate the UCNEs power cables (a minimum of two are required)? If the number of unused outlets is inadequate, additional outlets must be installed, either in the cabinet or external to the cabinet where the UCNEs will be installed.
- Has a contractor been contacted to fabricate and install the fiber optic cables?

5.7.2 Installation Procedure Overview

Proper installation and modification, if necessary, of the UCN coaxial cable segments must be verified before beginning the installation of the UCNE hardware and the fiber optic link(s).

Use the rules and procedures in Section 2 and Appendix C of the *Universal Control Network Installation* manual to fabricate, install, and modify, if necessary, the UCN coaxial cable segments. Relative information can also be found in the *Universal Control Network Planning* manual.

Use the information and procedures in Section 5 of the *Universal Control Network Installation* manual to test and verify the coaxial cable segments.

The UCNE enclosure is the same physical size as a Five-Slot or Dual Node Module and is mounted at the front-side of an LCN, LM, or SM cabinet. The two UCNEs are mounted side by side on a sliding tray in the enclosure that faces the rear of the enclosure. Each UCNE is secured to the tray by four screws. The tray, which slides toward the front of the enclosure, is secured in place by a single screw. A panel covers the front of the enclosure when it is installed in an LCN, LM, or SM cabinet, or a Classic console.

Figure 5-6 provides pictorial views of the UCNE enclosure, and Figure 5-7 provides pictorial views of the slide-tray with the two UCNEs mounted on the tray.

The UCNE enclosure is the same physical size as a Five-Slot or Dual Node Module and is mounted at the rear side of a Classic or Ergonomic console. The two UCNEs that are mounted side by side on a sliding tray in the enclosure face the front of the cabinet.

The model MP-FOMK02 through MP-FOMK06 UCNE mounting kits provide a UCNE enclosure and the required installation hardware for a pair of UCNEs. Two UCNEs are provided in the model MU-FOEA02 kit. Optionally, the model MU-FOED02 kit can be installed if reliable redundant 24 Vdc power sources are available.

If additional UCNEs are required, each pair of UCNEs requires another UCNE mounting kit. Figure 5-6 provides pictorial views of the UCNE enclosure.

5. IMPLEMENTATION OF UCN FIBER OPTIC EXTENDERS

5.7. UCNE Installation

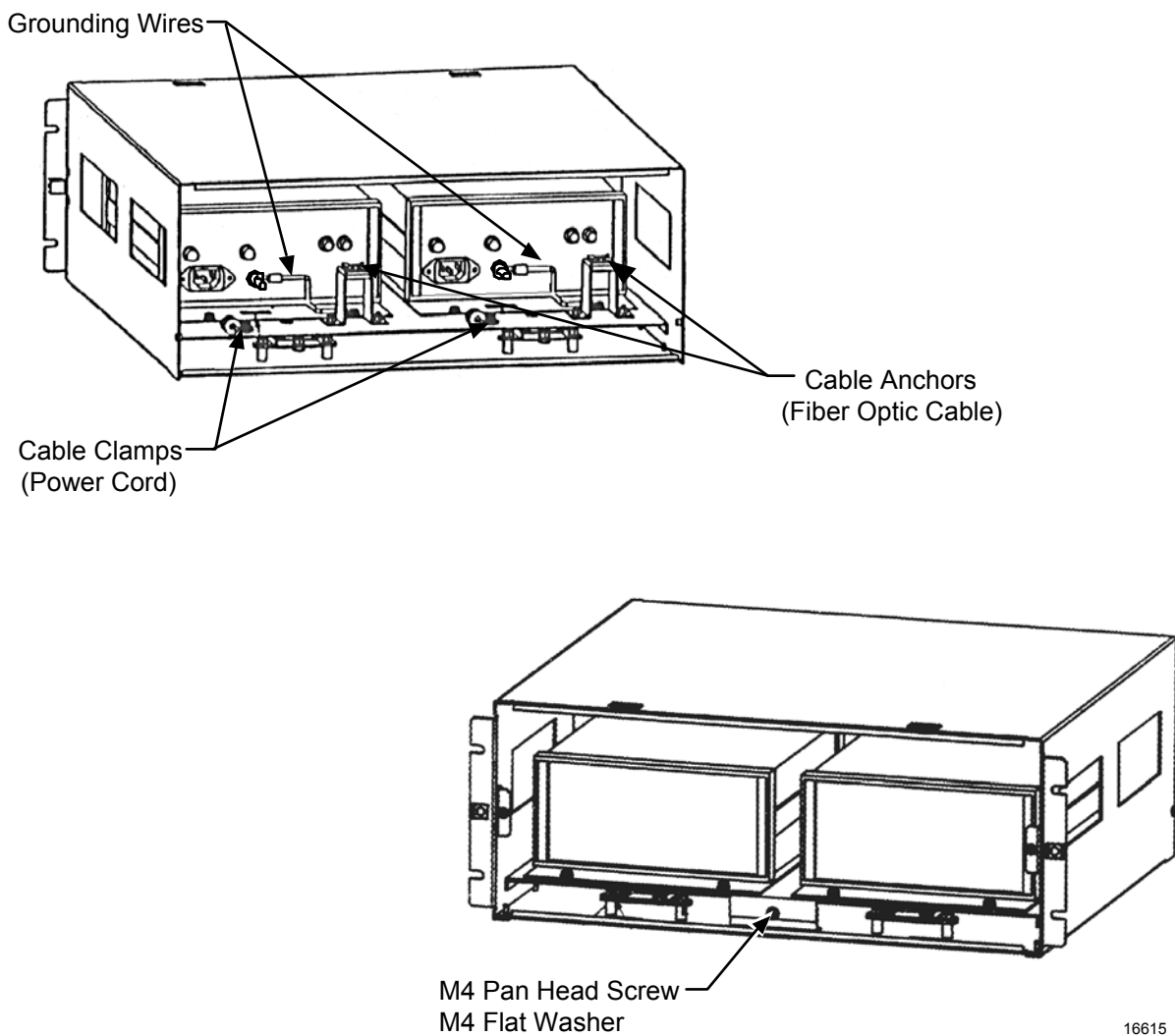
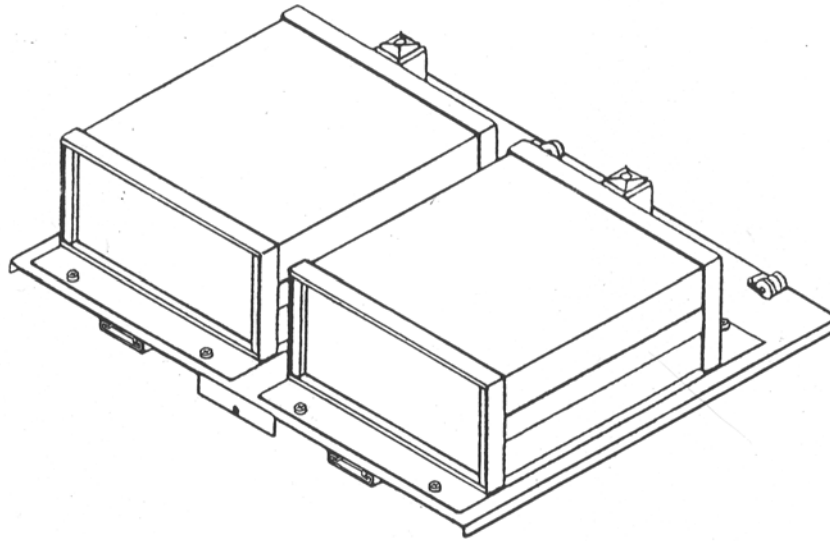


Figure 5-6 — UCNE Enclosure Pictorial Views

Figure 5-7 provides a pictorial view of the UCNE enclosure's slide-tray.



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Figure 5-7 — UCNE Enclosure Slide-Tray**5.7.3 LCN, LM, and SM Cabinet Installation Procedure**

Use the procedure in Table 5-7 to install a UCNE enclosure and pair of UCNEs in an LCN, LM, or SM cabinet.

Table 5-7 — LCN, LM, and SM Cabinet Installation Procedure

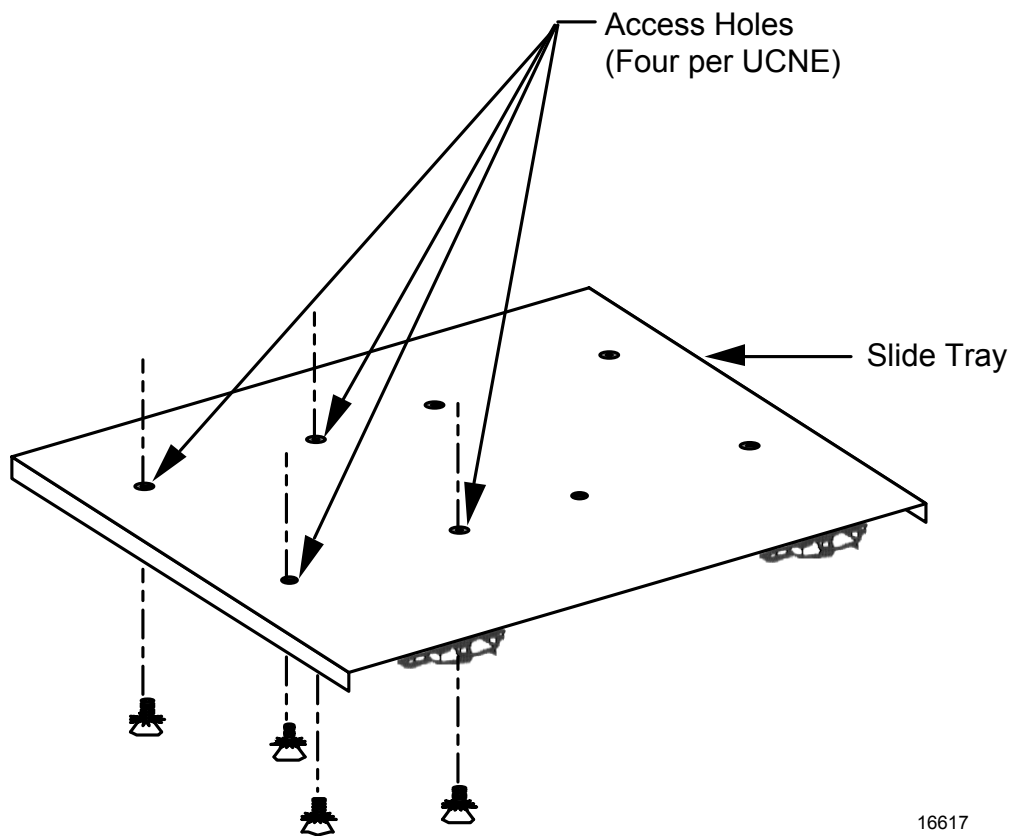
Step	Action
1	Because the UCNE enclosure (51403223-100) may be mounted in a lower module position in the cabinet where it would be difficult to have access to the bottom of the enclosure when installing the UCNEs, it is recommended that you mount the UCNEs on the enclosure's slide-tray before mounting the enclosure in the cabinet. If the UCNE enclosure will not be mounted in a lower module position in the cabinet and access to the bottom of the slide-tray is not a problem, the enclosure can be mounted in the cabinet before installing the UCNEs. Modify the procedure to accommodate the change.
2	Place the UCNE enclosure in an upright position on a suitable work surface.
3	Loosen and remove the machine screw and flat washer at the front of the enclosure that secures the slide-tray in place in the enclosure. Refer to Figure 5-6.
4	Fully extend the slide-tray outward until stopped by the tray's detents. You must fully extend the tray to be able to release the detents and return the tray to its normal position in the enclosure.
5	Attach a UCNE to each UCNE mounting plate with four M4 flat-head screws (51108386-510) and M4 external-tooth lock washers (51108393-005). The screws and washers are included with the UCNE enclosure (51403223-100). The holes in the plates can be accessed through enlarged holes in the bottom of the slide tray. Refer to Figure 5-8. When mounting the UCNEs, be sure the front of the UCNEs face the rear of the enclosure.

5. IMPLEMENTATION OF UCN FIBER OPTIC EXTENDERS

5.7. UCNE Installation

Step	Action
6	Release and return the slide-tray to its normal position in the enclosure by pressing on the two detent release buttons beneath the tray.
7	Secure the slide-tray in place with the machine screw and flat washer that were previously removed.
8	At the rear of the enclosure, attach the existing grounding wire (51201807-100) for each UCNE to the Chassis Ground terminal at the front of the UCNE.
9	Decide on an available module position in the cabinet in which to mount the UCNE enclosure. Use four M5 serrated washers (51109933-003) and M5 Black thread-forming screws (51195168-616) to attach the enclosure to the front cabinet rails. Use an M5 black hex-head machine screw (51108888-625) with an M5 serrated washer (51109931-100) and spacer (51108572-100), and an M5 black hex-head machine screw (51108888-616) with an M5 serrated washer (51109931-100) to attach the enclosure to the rear cabinet rails as illustrated in Figure 5-9.
10	Install the front panel (51305413-100) on the enclosure with its two captive screws. See Figure 5-9.
11	Connect a power cable (51305409-500) for 120 Vac operation, 51305410-500 for 240 Vac operation) between an ac outlet and the power connector at the front of each UCNE. Use the cable clamp in front of the UCNE's power connector to secure the power cable. See Figure 5-7.
12	Repeat Steps 2 through 11 if more than one UCNE enclosure is being installed.
13	Installation of the UCNE enclosure(s) is complete. Use the procedures in Tables 4-10 and 4-11 to connect the UCNEs to the appropriate UCN coaxial segment and install the fiber optic cable(s) between the UCNEs.

Figure 5-8 provides pictorial view of the UCNE enclosure slide-tray access holes.



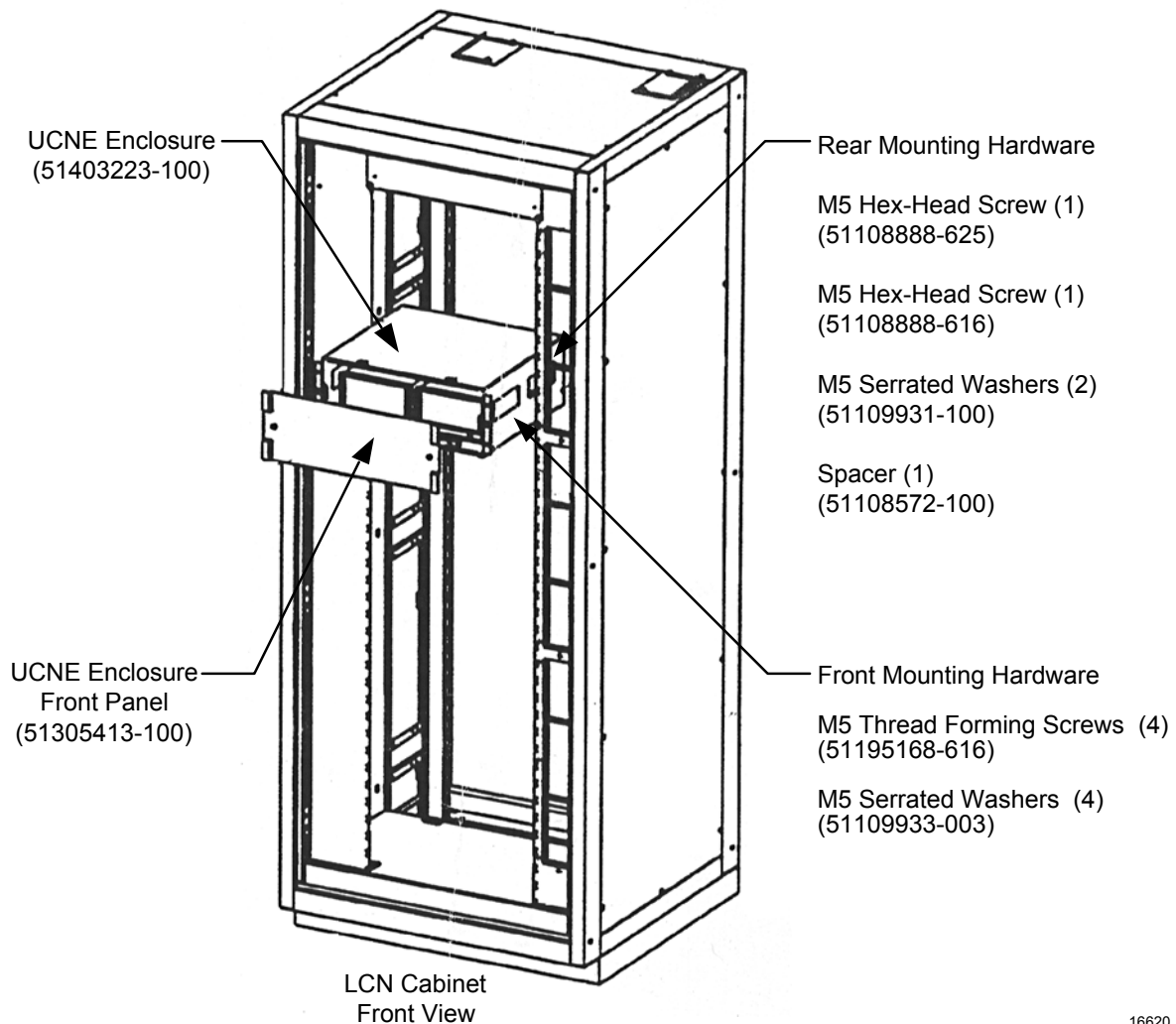
16617

Figure 5-8 — UCNE Enclosure Slide-Tray Access Holes

5. IMPLEMENTATION OF UCN FIBER OPTIC EXTENDERS

5.7. UCNE Installation

Figure 5-9 illustrates the method of mounting a UCNE enclosure in an LCN cabinet.



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Figure 5-9 — UCNE Enclosure LCN Cabinet Mounting Pictorial View

5.7.4 Classic Console Furniture Installation Procedure

Use the procedure in Table 5-8 to install a UCNE enclosure and pair of UCNEs in Classic console furniture.

Table 5-8 — Classic Console Furniture Installation Procedure

Step	Action
1	Because the UCNE enclosure (51403223-100) will be mounted in the lower portion of the console furniture where it is difficult to have access to the bottom of the enclosure when installing the UCNEs, it is recommended that you mount the UCNEs on the enclosure's slide-tray before mounting the enclosure in the cabinet. If you feel that you will have access to the bottom of the slide-tray after the UCNE enclosure is mounted in the console furniture, the enclosure can be mounted in the cabinet before installing the UCNEs. Modify the procedure to accommodate the change.
2	Place the UCNE enclosure in an upright position on a suitable work surface.
3	Loosen and remove the machine screw and flat washer at the front of the enclosure that secures the slide-tray in place in the enclosure. Refer to Figure 5-6.
4	Fully extend the slide-tray outward until stopped by the tray's detents. You must fully extend the tray to be able to release the detents and return the tray to its normal position in the enclosure.
5	Attach a UCNE to each UCNE mounting plate with four M4 flat-head screws (51108386-510) and M4 external-tooth lock washers (51108393-005). The screws and washers are included with the UCNE enclosure (51403223-100). The holes in the plates can be accessed through enlarged holes in the bottom of the slide tray. Refer to Figure 5-8. When mounting the UCNEs, be sure the front of the UCNEs face the rear of the enclosure.
6	Release and return the slide-tray to its normal position in the enclosure by pressing on the two detent release buttons beneath the tray.
7	Secure the slide-tray in place with the M4 pan-head machine screw and flat washer that were previously removed.
8	At the rear of the enclosure, attach the existing grounding wire (51201807-100) for each UCNE to the Chassis Ground terminal at the front of the UCNE.
9	Decide on an available module position in the console furniture in which to mount the UCNE enclosure. Use four M5 serrated washers (51109933-003) and M5 thread-forming screws (51195168-616) to attach the enclosure to the rear furniture rails as illustrated in Figure 5-10.
10	Install the front panel (51305413-100) on the enclosure with its two captive screws. Refer to Figure 5-10.

5. IMPLEMENTATION OF UCN FIBER OPTIC EXTENDERS

5.7. UCNE Installation

Table 5-8 — Classic Console Furniture Installation Procedure

Step	Action
11	Connect a power cable (51305409-500 for 120 Vac operation, 51305410-500 for 240 Vac operation) between an ac outlet and the power connector at the front of each UCNE. Use the cable clamp in front of the UCNE's power connector to secure the power cable. Refer to Figure 5-7.
12	Installation of the UCNE enclosure(s) is complete. Use the procedures in Tables 4-10 and 4-11 to connect the UCNEs to the appropriate UCN coaxial segment and install the fiber optic cable(s) between the UCNEs.

Figure 5-10 illustrates the method of mounting a UCNE enclosure in Classic console furniture.

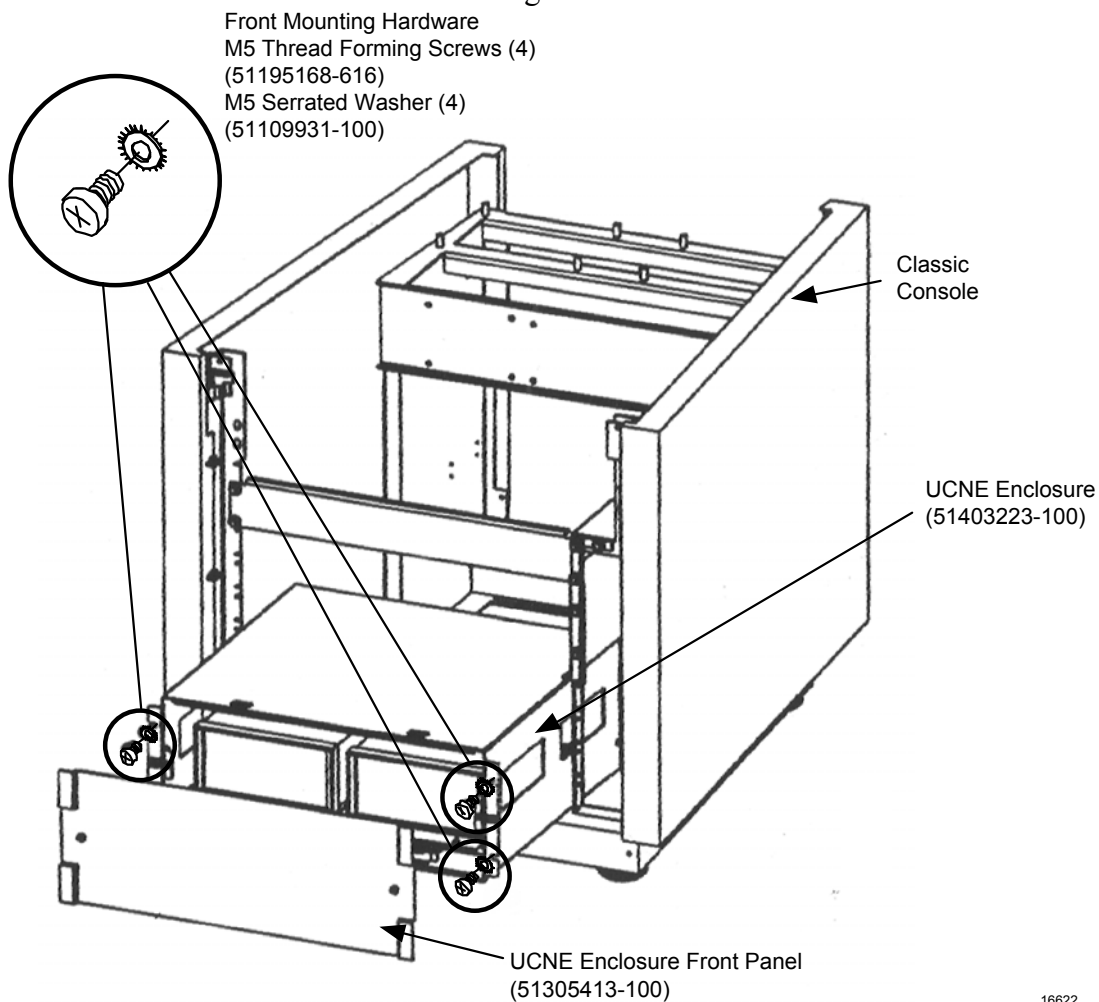


Figure 5-10 — UCNE Enclosure Classic Console Furniture Pictorial View

5.7.5 Ergonomic Console Furniture Installation Procedure

Use the procedure in Table 5-9 to install a UCNE enclosure and pair of UCNEs in Ergonomic console furniture.

Table 5-9 — Ergonomic Console Furniture Installation Procedure

Step	Action
1	Because the UCNE enclosure (51403223-100) could be mounted in the lower portion of the console furniture where it is difficult to have access to the bottom of the enclosure when installing the UCNEs, it is recommended that you mount the UCNEs on the enclosure's slide-tray before mounting the enclosure in the cabinet. If you feel that you will have access to the bottom of the slide-tray after the UCNE enclosure is mounted in the console furniture, the enclosure can be mounted in the cabinet before installing the UCNEs. Modify the procedure to accommodate the change.
2	Place the UCNE enclosure in an upright position on a suitable work surface.
3	Loosen and remove the machine screw and flat washer at the front of the enclosure that secures the slide-tray in place in the enclosure. Refer to Figure 5-6.
4	Fully extend the slide-tray outward until stopped by the tray's detents. You must fully extend the tray to be able to release the detents and return the tray to its normal position in the enclosure.
5	Attach a UCNE to each UCNE mounting plate with four M4 flat-head screws (51108386-510) and M4 external-tooth lock washers (51108393-005). The screws and washers are included with the UCNE enclosure (51403223-100). The holes in the plates can be accessed through enlarged holes in the bottom of the slide tray. Refer to Figure 5-8. When mounting the UCNEs, be sure the front of the UCNEs face the rear of the enclosure.
6	Release and return the slide-tray to its normal position in the enclosure by pressing on the two detent release buttons beneath the tray.
7	Secure the slide-tray in place with the machine screw and flat washer that was previously removed.
8	At the rear of the enclosure, attach the existing grounding wire (51201807-100) for each UCNE to the Chassis Ground terminal at the front of the UCNE.
9	Detach and remove the leg covers at each side of the console furniture by rotating each of their four fasteners or screws counter clockwise. Refer to Figure 5-11 to identify the leg covers.
10	The UCNE enclosure will reside in either an upper or lower equipment pod below the keyboard. The equipment pod being installed is composed of left and right module mounting brackets (51402745-100 and 51402745-200, respectively), the UCNE enclosure (51403223-100), a module enclosure assembly (51403117-100), a rear module enclosure cover (51403152-100), and a front module enclosure cover [51403089-100 (upper pod) or 51403090-100 (lower pod)] as illustrated in Figure 5-11. The assembly breakdown of a lower equipment pod is shown in the figure.
11	Use four flat-head machine screws (51108881-716) to attach the module mounting brackets (51402745-100 and 51402745-200) to the furniture legs as illustrated in Figure 5-11.

5. IMPLEMENTATION OF UCN FIBER OPTIC EXTENDERS

5.7. UCNE Installation

Table 5-9 — Ergonomic Console Furniture Installation Procedure

Step	Action
12	Use four M5 pan-head screws (51108385-610) to attach the UCNE enclosure (51403223-100) to the module mounting brackets as illustrated in Figure 5-11.
13	Slide the module enclosure assembly (51403117-100) over the UCNE enclosure as illustrated in Figure 5-11. Tabs on the cover will secure it in place.
14	Install the module enclosure's rear cover assembly (51403152-100) on the module rear cover as illustrated in Figure 5-11 and secure it with its two fasteners.
15	Use four M5 thread forming screws (51195168-616) to mount the two front module cover mounting brackets (51308039-100) on the front of furniture legs as illustrated in Figure 5-11.
16	Connect the ac power cables (51305450-400) between the furniture's ac power distribution module and the power connector at the front of each UCNE. Use the cable clamp in front of the UCNE's power connector to secure the power cable. Refer to Figure 5-7. Dress the cables inside the cabinet down the right-leg channel.
17	Installation of the UCNE enclosure is complete except for installing the module enclosure's front cover [51403089-300 (upper pod) or 51403090-300 (lower pod)] and a pair of leg covers as illustrated in Figure 5-11. Install the covers after using the procedures in Tables 4-11 and 4-12 to connect the UCNEs to the appropriate UCN coaxial segment and install the fiber optic cable(s) between the UCNEs. Install a pair of leg covers that are appropriate for the height of the console furniture and the number of equipment pods. There are three console furniture heights to be considered and one or two equipment pods may be present. Use the following information to select the appropriate set of leg covers. <u>Low Height</u> <u>Medium Height</u> <u>High Height</u> Single Pod 51197048-100/101 51197049-100/101 51197050-100/101 Dual Pod 51197048-200/201 51197049-200/201 51197050-200/201

Figure 5-11 illustrates the method of mounting a UCNE enclosure in Ergonomic console furniture.

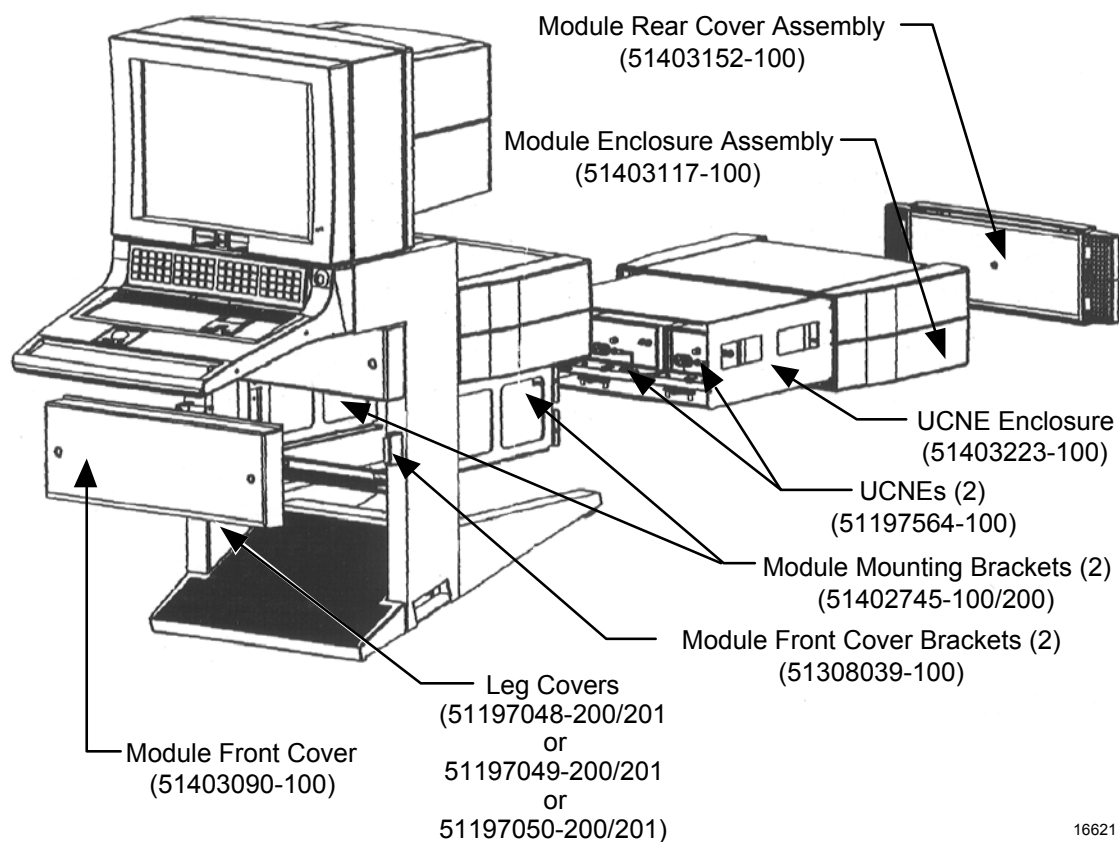


Figure 5-11 — UCNE Enclosure Ergonomic Console Furniture Pictorial View

5. IMPLEMENTATION OF UCN FIBER OPTIC EXTENDERS

5.7. UCNE Installation

5.7.6 UCN Cable Installation Procedure

Use the procedure in Table 5-10 to connect the UCNEs to the UCN drop tap ports on the appropriate UCN coaxial trunk cable segments. The procedure assumes that new UCN coaxial trunk cable segments have been properly fabricated and tested, and the existing cable segments have been properly modified, if necessary, and tested by using information that is provided in Sections 2 and 4 in the *Universal Control Installation* manual and the *Universal Control Network Planning* manual.

Unused ports of drop taps that are strategically located on the cable segments must be available. The Cable A (UCN-A) and Cable B (UCN-B) taps that the UCNEs connect to must be the same cable distance from other taps in their cable segment.

Table 5-10 — UCNE Coaxial Cable Connection Procedure

Step	Action
1	Connect a drop cable (RG6-type) between the UCN DROP (COAX) connector at the front of each UCNE and the appropriate UCN coaxial cable segment drop tap port. Observe the correct color code. Cable A has an orange-colored label and Cable B has a brown-colored label. UCNEs in positions 1 (UCN-A) and 3 (UCN-B), and 2 (UCN-A) and 4 (UCN-B) in the UCNE enclosure must be paired for consistency.
2	Tighten the drop cable connectors to 25 inch/pounds by using the torque wrench that is included in the model MU-NKTQ01 kit.
3	Secure and dress the cables in the cabinet.

5.7.7 Fiber Optic Cable Installation Procedure

Use the procedure in Table 5-11 to install UCNE fiber optic cable. It is highly recommended that an experienced contractor fabricate, install, and test the fiber optic cables after installation. The cables must meet the signal loss budget requirement. Reference information can be found in the *Universal Control Network Planning* manual.

Table 5-11 — UCNE Fiber Optic Cable Connection Procedure

Step	Action
1	Install the fiber optic cables between the two paired UCNEs. The UCNE enclosure may be resident in an LCN cabinet or console, in an LM or SM cabinet, or in a PM, APM, or HPM cabinet. The <i>Local Control Network Installation</i> manual details the installation of a UCNE enclosure in an LCN cabinet or console, or an LM or SM cabinet.
2	Be sure the fiber optic cable connections are proper. The UCNE's Receive (RCVR) connector must be connected to the Transmit (XMTR) connector at the other end for both Cable A and Cable B pairs. Figure 5-12 illustrates the interconnections.
3	Dress the cables and secure them to cable anchors in front of the UCNEs. Refer to Figures 4-6 and 4-7.

Figure 5-12 illustrates the UCNE's fiber optic interconnections.

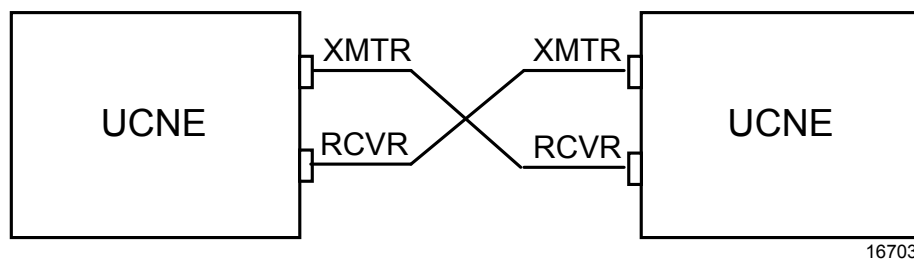


Figure 5-12 — UCNE Fiber Optic Cable Interconnections

5. IMPLEMENTATION OF UCN FIBER OPTIC EXTENDERS

5.7. UCNE Installation

6. TROUBLESHOOTING

This section provides guidelines on troubleshooting problems on coaxial and fiber optic segments of the LCN.

6.1 System Level Information

At the system level there are displays that provide useful status information, including the status of the network configuration file (NCF) and the same status of the LCN, with guidelines for the use of the NCF and LCN information.

6.1.1 System, System Status, and Console Status Displays

NOTE

For PSDP Parameter COMM information see Section 22 of *Engineer's Reference Manual*, and Appendix C of this manual.

The following displays are available in both the Universal Station's Engineering Personality and the Operator Personality:

- System Menu
- System Status display
- Console Status display

In the Engineering Personality, these displays can be called up by selecting the appropriate picks (targets) on the Engineering Main Menu. Once one of these displays is on the screen, either of the other two displays can be called up by pressing the appropriate key on the Operator's Keyboard.

In the Operator's Personality, these displays are called up by pressing the appropriate key on the Operator's Keyboard. Instructions for their use are provided in the *Process Operations Manual* in the *Process Operations* binder.

There are four versions of the System Menu—they are simulated on Figure 6-1. In USs and UWSs without the Printed Summaries option, the System Menu (Engineer's Personality and Operator's Personality) appears as shown on the right-justified screens in Figure 6-1. In USs and UWSs with the Printed Summaries option, the System Menu (Engineer's Personality and Operator's Personality) appears as shown on the left-justified screens in Figure 6-1.

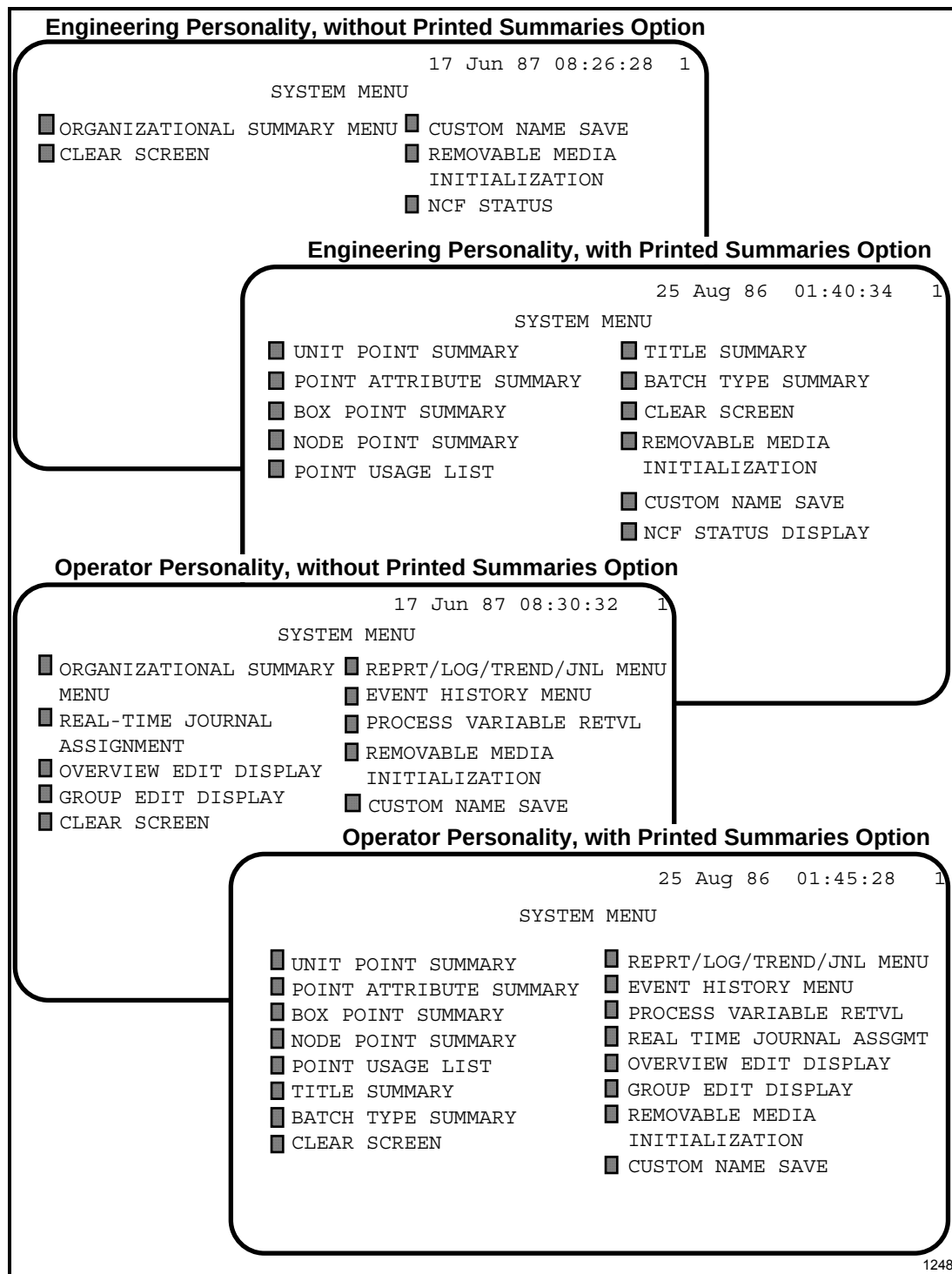


Figure 6-1 — System Menus

Other displays that are important to system engineers and maintenance personnel are available in either the Operator's personality or the Engineer's Personality. These include

- NCF Status display
- LCN Overview Display
- Console Status display
- Gateways Status display
- Node (AM, CG, HM) status displays

In the remainder of this section, we'll discuss these displays and provide guidance for using the information on them.

6.1.2 NCF Status Display

To call up the NCF Status display, on the Engineering Main Menu select the **SYSTEM MENU** target, then select **NCF STATUS**; or in the Operator Personality, press **SCHEM**, then key in **NCF_STS1** (be sure to include the underbar), and press **<ENTER>**. The NCF Status display appears (simulated on Figure 6-2).

Boxes appear in the NCF Status display to represent each of the possible 64 nodes on an (extended) LCN. Figure 6-2 was taken from a system that has five nodes: Node numbers 35, 36, 38, 40, and 43. Only nodes that are running on the LCN are represented by colored blocks with a node number. A failed node, even though connected to the LCN, won't appear. The node numbers begin in the upper-left box (node 1) and continue down the first row (to node 6), up the second row (to node 12), back down the third row (to node 18), and so on.

If all nodes are running on the same version of the NCF (same time stamp) and no on-line reconfiguration NCF (no installation unit) has been installed, all nodes present on the LCN are represented by a green box with a node number, indicating that the NCFs on all nodes are synchronized.

If one or more on-line NCF installation units have been installed (**<F2>** pressed in network configuration), nodes that will be affected by the NCF changes are indicated by a color other than green. As indicated on Figure 6-2, white fill depicts that for this node there will be no impact if the node is not restarted with the new NCF. Blue indicates moderate impact, yellow indicates high impact, and red indicates severe impact.

You will need to consider the type(s) of installation unit(s) that have been installed, the advice that is given on the on-line reconfiguration displays, and what you know about your network and system to determine whether and when any or all of the "out of synch" nodes should be shutdown and restarted. No changes made in the NCF are in effect on a node until that node is shutdown (through the **SHUTDOWN** pick on the node status display) and reloaded.

Instructions for on-line reconfiguration are provided in the *Network Data Entry* manual, in the *Implementation/Startup & Reconfiguration — 1* binder.

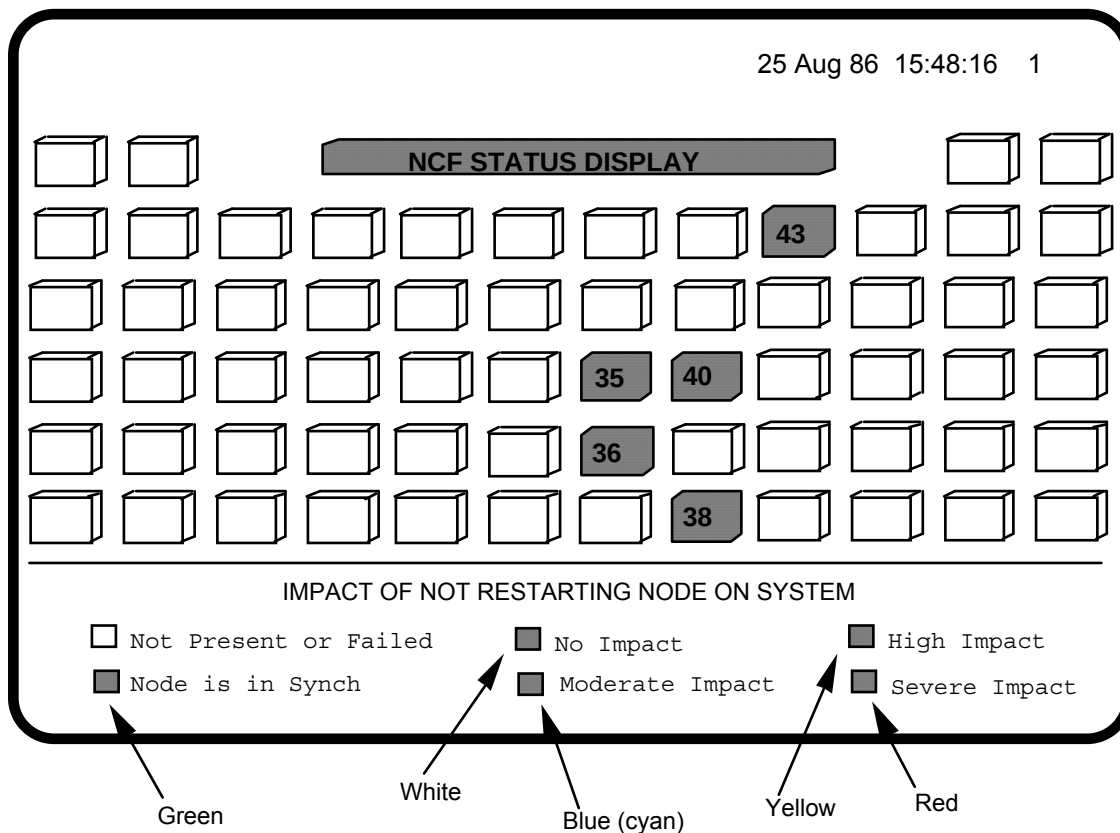


Figure 6-2 — NCF Status Display

6.1.3 LCN Cable Operation and Related Displays

6.1.3.1 Normal LCN Operation

In normal operation, each node on the LCN transmits on both LCN cables (A and B) and listens on the active cable. So that developing problems on a cable don't go undetected and unresolved for a long time, the active cable is switched (swapped) once per minute. You can observe this cable swapping on the System Status display (see Figure 6-3).

6.1.3.2 Cable Marginal Status

The cable status is indicated as OK or SUSPECT depending upon the number of LCN communication errors. With R681, an intermediate status, MRGNAL is added to indicate the cable status.

The cable master node monitors the cable status. The marginal status functionality enables the cable master node to perform the following activities:

- Monitor the errors on the cable and/or node, and
- Set an intermediate MRGNAL status before declaring as SUSPECT

The cable status changes to MRGNAL, when the number of LCN messages/communication errors received by the cable master node $\geq$ the threshold limit. If errors are less than the threshold limit, the cable status switches back to OK.

Cable Threshold parameter

The cable threshold parameter \$CBLMTP is used for specifying the threshold limit for the cable marginal status. The threshold value configured in the cable threshold parameter of Cable master is propagated to the US/GUS/EST nodes.

A value > 0 specified for the cable threshold PSDP parameter activates the marginal status functionality for the cable. This value does not affect the marginal status calculation for the node.



ATTENTION

- A value = 0 does not activate the marginal status functionality.
 - The recommended value for the cable threshold parameter is 1 or 2.
 - The cable status changes to SUSPECT, when the number of errors ≥ 3 .
- You cannot configure the limit for the SUSPECT state.
-

Marginal state indication

The marginal status for the cable is indicated as MRGNAL in yellow and can be viewed in the System Status display.

Events generated

A system status error (event type 60) is generated when the cable state changes from

- OK to MRGNAL state
- MRGNAL to SUSPECT state

A system status error (event type 64) is generated when the cable marginal state is acknowledged.

Cable switching

The marginal state indication does not affect the cable switching mechanism. The cable in MRGNAL state can become the active cable.

6.1.3.3 Node Auxiliary Marginal Status

The auxiliary state for the LCN nodes changes to marginal when the number of LCN communication errors on any or both the cable is greater than the threshold limit. The status changes to a:mrg, b:mrg or ab:mrg depending upon the errors the cables encounter.

The auxiliary status of a node changes to marginal on the following conditions:

- If the number of errors $\geq$ the threshold limit for 1 minute or
- If the number of errors $\geq$ the threshold limit for 15 minutes or
- On both the above conditions

Node Threshold parameters

The node threshold parameters \$NDM1TP and \$NDM15TP are used for specifying the threshold limit for 1 minute and 15 minutes respectively. The threshold values configured in the node threshold parameters of cable master are propagated to all the nodes.

A value > 0 specified for the node threshold PSDP parameter activates the node marginal status functionality for the node. The node threshold value does not affect the marginal state calculation for the cable.



ATTENTION

- A value = 0 specified for the node threshold parameters \$NDM1TP and \$NDM15TP does not activate the node marginal status functionality.
 - The recommended values for the node threshold PSDP parameters are 2 and 3 for 1 minute and 15 minutes respectively.
-

Auxiliary state indication

The marginal state indication for the LCN nodes in the network can be viewed in the following displays:

Display name	Applicable nodes
System Status display	AM/HG/CG/HM/NIM/ESVT/APP/NG
Console Status display	US/GUS/EST

NOTE

The marginal state indication in both the displays blink depending on the option selected in the Area Annunciation policy. If the annunciation option for a particular node is set to **No**, then the marginal state indication for the node does not blink in the System Status display. The indication is displayed as a text.

The following table describes the auxiliary marginal status for a node:

Node Status	Description
A:mrg	Indicates the node status when the number of errors on Cable A of that node is greater than the threshold limit as specified by the PSDP parameter.
B:mrg	Indicates the node status when the number of errors on Cable B of that node is greater than the threshold limit as specified by the PSDP parameter.
AB:mrg	Indicates the node status when the number of errors on both the Cable A and Cable B of that node is greater than the threshold limit as specified by the PSDP parameter.

The following indications can be observed in the displays when a node is in the marginal state:

- ‘A:mrg’ or ‘B:mrg’ or ‘AB:mrg’ appears in yellow in the second line of the individual node box with other auxiliary state indications.
- System Status LED blinks for AM/HG/CG/HM/NIM/ESVT nodes.
- Console Status LED blinks for US/GUS/EST nodes.
- Hooter is ON.
- The **maint** in individual node box of the System Status display and **Yes** in the Console Status display blinks depending on the option selected in the Area annunciation policy.
- When the Aux Info target is selected, **SHOW MRGNAL** target on the System Status display appears in yellow.

Return to OK

The auxiliary state of the node changes to OK, when the number of errors on both the cables is less than the threshold limits.

6.1.3.4 Configuring and propagating threshold value

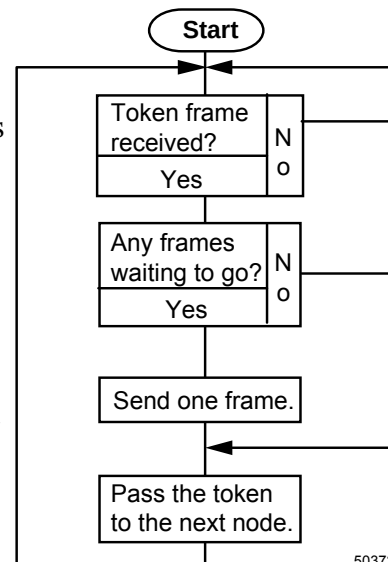
The threshold parameters are available in all the LCN nodes. The default value for these parameters is zero. The user with an ENG access can change the threshold parameter values.

When the current cable master node fails, the other node becomes the active cable master node. If the threshold values are propagated to all the other nodes, the new cable master node uses the propagated threshold values for calculating the marginal state. If the threshold values are not propagated to all the other nodes, the new cable master node uses the default threshold values configured in that node.

When a node is re-loaded, the cable threshold parameter value is obtained from the old cable master node. After re-loading, the cable marginal state functionality continues to be the same. The Cable/Node Marginal Setting display is used for assigning the cable threshold parameter value. The cable marginal state functionality is activated in the next cable switching cycle.

When a node is re-loaded, the threshold values are copied from the cable master node. The node marginal state functionality is activated when one of the threshold parameter value >0 .

A token-passing technique is used to control access to the network by the nodes. A token frame is passed among the nodes to determine which one has access to the network. The LCN is a broadcast type of local area network. All nodes "hear" all transmissions, but they accept only information intended for them. When a node has the token but has nothing to transmit, it passes the token to the next node with a higher node number. When the highest-numbered node receives the token but has nothing to transmit, it passes the token to the lowest numbered node. Thus, the operation is as if the nodes pass the token around a ring.



An operator at the US (with the configured keylock access) can force a swap from an active cable to the other cable by selecting the inactive cable's pick and pressing <ENTER>. He or she might do this to determine if repair to a suspected cable has been successful. If one of the cables is completely inoperative (this might be caused by a break, a disconnected cable, or a transmitter or receiver failure), the LCN can continue to operate because all nodes transmit on both cables and a node that "hears" nothing on one cable for a predetermined period switches to listen on the other cable whether it is active or not. Should the A and B cables be crossed, a failure on one cable appears to be a failure on both cables, and communication between some nodes may be lost.

Every node on the LCN expects to hear from every other node at least once every 30 seconds. Any node that does not receive at least one message from any other node within two minutes will report a watchdog error message against that node. This message is recorded in the System Error Journal and printed on the Real-Time Journal. If no additional messages are received within the next minute, or if two or more other nodes also do not receive any messages from the non-communicative node, then that node becomes **ISOLATED** from each node that did not hear from it. An **ISOLATED** node is treated similarly to a failed node: only power on or qualified messages received from it are accepted (when communication becomes possible), and any other message communication with that node is suspended. Nodes are **ISOLATED** to ensure that the high level of data integrity between the nodes is not compromised when communication is lost, and to avoid the overhead of communication retries on every message.

A **COMM** parameter is provided in every node to accumulate information about LCN operation, as seen by that node. **COMM** and the information it contains can be seen through the LCN Cable Diagnostics displays. These displays are described under subsection 6.2.2.

6.1.3.5 Suspect Cable Detection

There is always one node on the LCN that is identified as the master communication node. The master node is the running Universal Station with the lowest node number. The master node controls:

- the automatic, periodic switching between cables A and B, as well as the declaration of the status of a cable as **SUSPECT** or **GOOD**.
- operator-commanded cable switches. A cable is declared **SUSPECT** when the master node determines that the network cannot continue to operate on that cable. This occurs:
 - every time the master node detects a media dead error on the cable (see Table 6-1),
 - and the third time a switch to a cable causes a forced return to the opposite cable before the first cable has been in use more than one minute. Such a forced return occurs when one or more nodes receives an indication of a single error from the LCNI hardware. The effect is that a cable becomes **SUSPECT** after the master node tries unsuccessfully to switch to it three times.
- the Crossed Cable Count display.

When a cable is declared **SUSPECT**, the other cable is selected and used for ten minutes. Then the master node tries the **SUSPECT** cable again. If errors are still encountered and a return is forced to the good cable, the good cable is selected for the next ten-minute interval. If both cables are declared **SUSPECT**, the network “bounces” back and forth between the cables each time any node detects a fault. If the network can stay on a **SUSPECT** cable for ten minutes without any forced swaps to the other cable, the first cable is declared **GOOD**.

Cases exist where temporarily there may be no master or more than one master. For example, if the master node has power turned off, the network does not detect this for two minutes. There is no master during this time. Duplicate masters can occur if multiple cable faults are severe enough that USs cannot communicate with each other (perhaps due to reflections), and multiple USs may believe that they are the lowest numbered running US. If later they receive a message from a lower-numbered US (even one that is **ISOLATED**), they will drop their status of master.

6.1.3.6 Suspect Cable Annunciation

When the master node declares a cable as SUSPECT and switches to the other cable, it notifies all USs on the network and the Journal Manager on the System HM. This causes SUSPECT to show as the status of the suspected cable at all USs that receive the notification. It may take up to 30 seconds for a US to display the SUSPECT. It is possible for ongoing communication problems to cause a US to miss the notification, but at least one US does show the SUSPECT status, because only a US can be the master node. The master (lowest running node number) US should be used to validate the correct cable status.

When a US receives the SUSPECT notification, it:

- changes the CABLE A or CABLE B pick to blinking, backlighted yellow,
- displays SUSPECT in yellow for the CABLE A or CABLE B status on the LCN Overview display (subsection 6.1.3.6),
- prints “CABLE A (OR B) SUSPECTED” message in the Real Time Journal, if the journal is directed to a printer, and it
- stores the “CABLE SUSPECTED” message in the System Status Journal on an HM.

6.1.3.7 Cases Where a Cable may not Show SUSPECT Status

An LCN cable is not declared SUSPECT if the software does not receive any error indication from the hardware nor if the master node does not see enough forced cable swaps. These are the cases where this can happen:

- A double cable fault (a fault in both cables at the same time) causes only one error: Disconnecting both cables from a node may cause only one media-dead error and may not cause communication errors between nodes on the same side of the cable break. However, both portions of the LCN reestablish their own ring and successfully pass tokens on both the A and B cables.

Communication across the break is discontinued, so nodes on both sides of the break declare nodes on the other side of the break as FAILED (R210M1 and before) or ISOLATED. This can take up to two minutes.

- Low error rate: Errors could occur at such a low rate that forced cable swaps do not occur often enough to declare a cable as suspect, even though the error rate may be above the average error rate.
- Turning off or resetting LCN extenders on both cables: This may not cause enough noise on the LCN to cause a cable to be declared SUSPECT. Both separated portions of the LCN establish their own ring and successfully pass tokens on both cables. Even if the cables are declared SUSPECT, after a maximum of 20 minutes, USs on both sides of the LCN extenders show both cables as GOOD. Typically, only one cable will show SUSPECT.

Communication across the break is not possible, so nodes on both sides of the break declare nodes on the other side as FAILED (R210M1 and before) or ISOLATED. This can take up to two minutes.

- Removing both tee connectors from a node at the same time: If the node does not have the token at the time the tees are removed, the hardware is still able to pass the token to another node without reporting an error to the software.

In this case, the node with the tee connectors removed is declared **FAILED** (R210M1 and before) or **ISOLATED**. This can take up to two minutes.

- Cables A and B crossed: No hardware error is detected when the cables are crossed (Cable A and Cable B on the wrong connectors) at a single node because the messages are sent on both cables and receiving nodes receive the messages on whichever cable is active. However the crossed cable display will show an "out of sync" count.
- Cable swap caused by two errors on the same token pass: If on the same token pass, two nodes detect a cable error that causes a swap to the other cable, the master node still sees messages on the cable selected by the master node. Because of the message pattern on the network, this is not likely to happen.
- A single error when A and B cables are crossed: If the cables are crossed (Cable A and Cable B on the wrong connectors) at a single node, and one node detects an error the network is forced back to the cable that the master has already selected. This can cause the wrong cable to be declared **SUSPECT** or it can force the network to form two token-passing rings with no errors detected.

6.1.3.8 LCN Functions on the System Status Display

Figure 6-3 shows the System Status display. The **CABLE A** and **CABLE B** picks on this display indicate which LCN cable is currently active, and they can be used to request that the inactive cable become active. To do so, select the pick for the inactive cable and press <ENTER>. LCNs can become disconnected intentionally, if proper procedures are followed; and they can become disconnected unintentionally, for example if a cable or fiber is broken, or power to an LCN Extender board is lost. The procedures for reconnection are described in Section 8 of this manual.

6. TROUBLESHOOTING

6.1. System Level Information

In most cases, disconnecting only one cable will allow the LCN to continue operating, but to avoid possibly serious problems, you must follow the procedures in Section 8 for reconnecting LCN cables.

NOTE

Information related to LCN trouble diagnosis is provided in the remainder of this section. Additional information for testing and repair of LCN cables or any of the LCN interface hardware is in *Maintenance Test Operations* in the *LCN Service* binder.

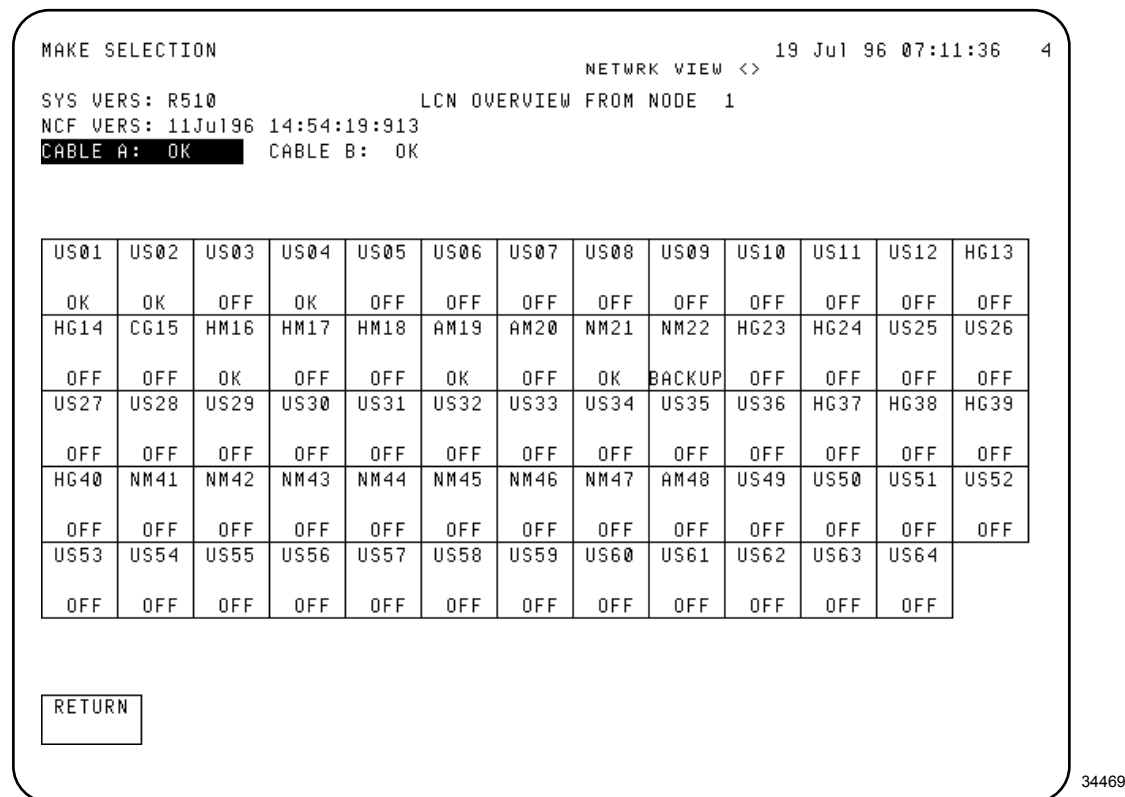


Figure 6-3 — System Status Display

6.1.3.9 LCN Overview Display

To call up the LCN Overview display, use the System Status display to call up the Console Status display, Gateways Status display, or node status display for the node type. Then, select the node whose view of the LCN you wish to see, and select LCN OVERVIEW.

Figure 6-4 is an example of the LCN Overview display that shows the LCN status as seen by node 40. The CABLE A STS and CABLE B STS fields indicate which cable is currently active. Up to 64 blocks define this node's view of the status of each of the other nodes on the LCN; in node number order, left to right, top to bottom. The status indications are the same as those that appear on the node status displays, and include OK, ENGR, UNIVERSL, BACKUP, OFF, PWR_ON, QUALIFY, READY, FAIL, ISOLATED, etc. ENGR and UNIVERSL are used instead of OK by Uss with Engineer and Universal personalities, respectively. ISOLATED indicates that this node sees the other node as isolated. The other node is on an LCN portion that is separated from this node (for more information on isolated cable portions and recovery, refer to Section 8). This node cannot communicate with any node whose status is ISOLATED. In the case of Figure 5-4, if you were to see the LCN Overview display for node 13, it would show the status of node 40 as ISOLATED.

LCN cable problems can cause some nodes to appear to one node to be isolated but not isolated to another node.

6.1.3.10 Node States on the LCN Overview Display

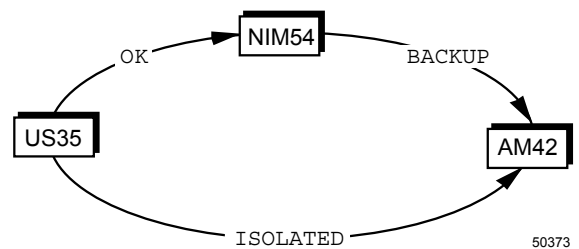
US35 in this sketch sees AM42 as ISOLATED.

It sees NIM54 as OK. AM 42 is running as a BACKUP to a redundant AM partner. When US35

is showing the LCN Overview display for NIM54,

US35 cannot communicate with AM42, so it cannot determine AM42's redundant or auxiliary node states, therefore the LCN Overview display for

NIM54 on US35 shows AM42 as OK*.



If a node is isolated from a US and that US is asked to display the LCN Overview display for another node whose view of the network is different from that of the US, all the nodes the US sees as isolated have their status indicator marked with *. This indicates that the exact state for the node cannot be determined. The indicated status is correct for all node states except for auxiliary states and derived running states. These are the states that are approximated on the LCN Overview display for nodes that the US sees as ISOLATED:

6. TROUBLESHOOTING

6.1. System Level Information

<u>Actual Node State</u>	<u>Node Type</u>	<u>State Shown</u>
UNIVERSL	US	OK*
ENGR	US	OK*
BACKUP	HG, AM, NIM	OK*
UPGRADE	HG	OK*
WARN	AM, HM, CG	OK*
SEVERE	AM, HM, CG	OK*

When a US is showing the LCN Overview display for another node, and that node becomes isolated from the US, the node states for all of the nodes on the display disappear. This indicates that the US can no longer communicate with the other node, so that node's view of the LCN is unknown. If a user calls up another display and then uses the PRIOR DISP key to bring back the LCN Overview display, the node states are all indicated as bad values.

The LCN OVERVIEW pick near the bottom of the LCN Overview display is used to call up the LCN Overview display for other nodes. To do so, select the pick, then in the port that appears, key in the node number and press <ENTER>. The INIT LCN RECONNECT pick is used for reconnecting separated (isolated) portions of an LCN, as described under 7.7.

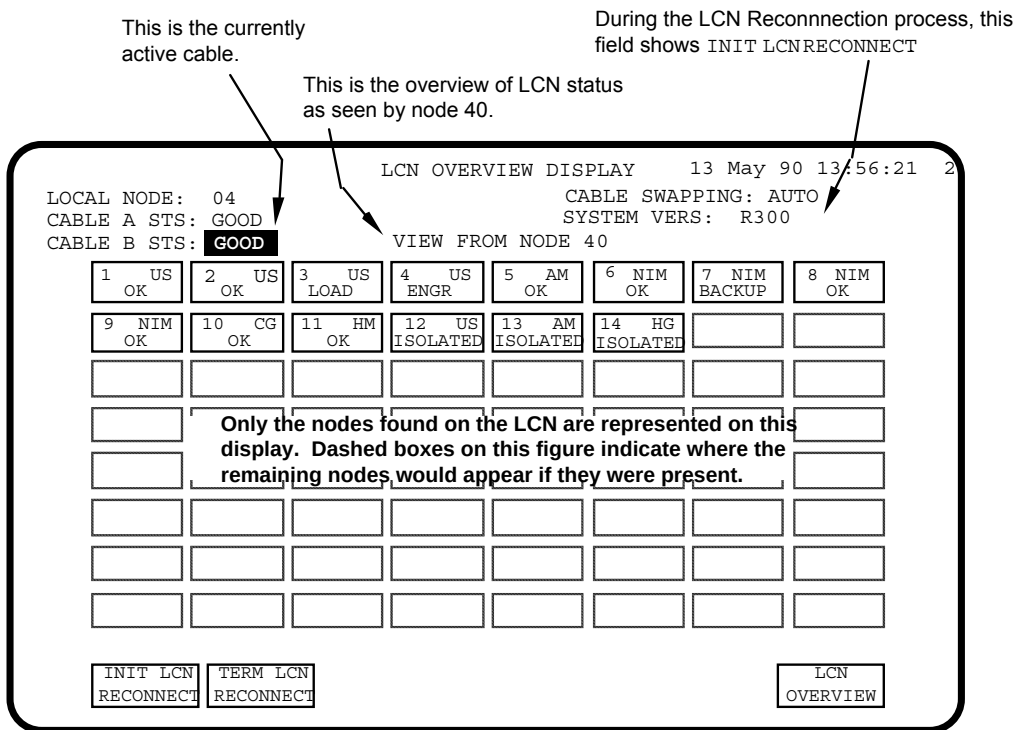


Figure 6-4 — LCN Overview Display

6.2 LCN Cable Diagnosis

NOTE

If the master node has a K4LCN or K2LCN board with firmware revision (LFW) of P or greater, the \$LCNMENUE “Crossed Cable Information” display will provide a count of single cable transmissions received by each node. If an LCN cable is crossed, the display will show a smaller number for any node that does not receive the single cable transmissions.

All LCN nodes must have:

Release 420 loaded

The master nodes must have an K4LCN or K2LCN board with LFW revision P or greater.

6.2.1 Guidelines for LCN Trouble Diagnosis

The primary sources of LCN diagnostic information are the statistics provided on the LCN Cable Diagnostic displays (described in subsection 6.2.2), the information in the System Error Journal (described in subsection 6.1.3.1), and information provided by the LCN Cable Diagnostic Annunciator program (R300 and beyond).

The specified LCN average error rate is 1 error in every 10^9 bits. If the 5-megabits-per-second LCN were carrying data 100% of the time, an error could be expected every 3.3 minutes. In most systems, the error rate is actually far better than 1 in 10^9 and the LCN load is much less than 100%, but errors do occur. Five or six random data errors (framing errors, frame check sequence errors, Manchester decoder errors, oversize frames) in 24 hours should not be considered excessive. A noticeable change in the error rate, however, should be investigated, even if the error rate is still low. For example, a change from 5 to 15 errors per week is cause for further investigation.

Five or six random data errors (framing errors, frame check sequence errors, Manchester decoder errors, oversize frames) in 24 hours should not be considered excessive. A noticeable change in the error rate, however, should be investigated, even if the error rate is still low. For example, a change from 5 to 15 errors per week is cause for further investigation.

Be aware that a single error can be reported by several nodes when a data frame using a logical address is corrupted. Careful study of the available diagnostic information will determine when this is the case.

If an LCN Extender is used, the error rate can be higher. Each fiber optic cable segment and each coax bus segment has an independent bit error rate and these bit error rates are additive. Additionally, when power is applied to or removed from an LCN Extender, its fiber optic transmitter can send a burst of light to the associated receiver that causes an error on the coax bus. These errors should be ignored because they are expected and the system is designed to tolerate such events. Note that these errors are avoided if the RUN/RESET switch on the extender board at each end of the fiber optic cable is put in the RESET position before applying power or turning power off.

A topology diagram—a drawing that shows all of the nodes on the LCN, all of the LCN coax cables, terminators extenders, and fiber optic cables, with the physical order of each of these devices and cables; and the node numbers—can be very helpful in analyzing error information to determine the most probable source of the errors. See Appendix B.

When you believe an LCN communication problem exists, study the information from the diagnostic displays and the System Error Journal to learn as much about the problem as possible. Classify all errors as to type and which cable is involved. Determine which nodes are involved—both transmitting and receiving nodes. Refer to Table 6-1 in this manual to determine the significance of the secondary error codes in the error journal entries.

It may be possible to decide on some corrective actions based solely on the information obtained from the statistics and a topology diagram of the network. If the network contains LCN Extenders and the errors always involve messages that go over a fiber optic link and a specific cable is always involved, a pair of LCN Extenders has been indicated and can be replaced.

If the pattern of errors indicates that a few nodes that are close together on coax cable cannot reliably communicate, excessive reflections on the cable may be the problem. This can result from faulty cables or bad connections. Unless a fault can be visually spotted or manually felt, a Time Domain Reflectometer (TDR) may be required to find the bad cable. Use of a TDR is recommended at appropriate points under subsection 6.2.3.1.

For release R320 and beyond, there are 4 PSDP (Program Status Data Point) parameters and a set of schematics to display the clock statistics on each node. These statistics are useful because a cable error is often indicated by clock errors since the clock is transmitted over the LCN cables. In addition, these schematics can be used to show crossed A and B cables if customers are using Clock Source/Repeater (CS/R) boards. This is done by disconnecting a cable from a CS/R that is acting (configured) as a clock source. If the cables are crossed, clock errors will show up on the wrong cable.

6.2.2 LCN Cable Statistics

Processor Status Data Point (PSDP) parameter COMM provides a large array with statistical information about the LCN performance, as perceived by the each node on the LCN. You can read the values in this parameter through a series of 17 Honeywell-provided LCN Cable Diagnostic displays. These displays are provided in directory DIA1 on cartridge &CR6 or on floppy diskette DIA1 in Media Kit MP-MKUS10 or MP-MKUS20. This parameter may also be used directly to develop custom displays. See Appendix C for additional information.

In a few cases the statistical information on the LCN Cable Diagnostic displays may not be meaningful to customer personnel, but this information can be invaluable to Honeywell personnel who help to analyze difficult situations. If you contact our Solution Support Center's Technical Assistance Center (TAC) for help in diagnosing LCN problems, they may ask you for information from one or more of these displays.

Release 430 Crossed Cable Test—Operation of this schematic requires:

- Release 430 software must be installed and running on each physical node on the LCN.
- The LCN System must be free of other errors.
- The Cable Master Node must have installed a K4LCN or K2LCN board with **Hardware Revision R** and **LCN Firmware Revision P**. The Cable master Node is the Universal Station with the lowest LCN node address running either the Operator, Engineer or Universal personalities. There is a minimum requirement of the one K2LCN board in the Cable Master Node (US01, Node #01).

The schematic maintains a count for node #1. This count is always 1 count greater than the count the other nodes have.

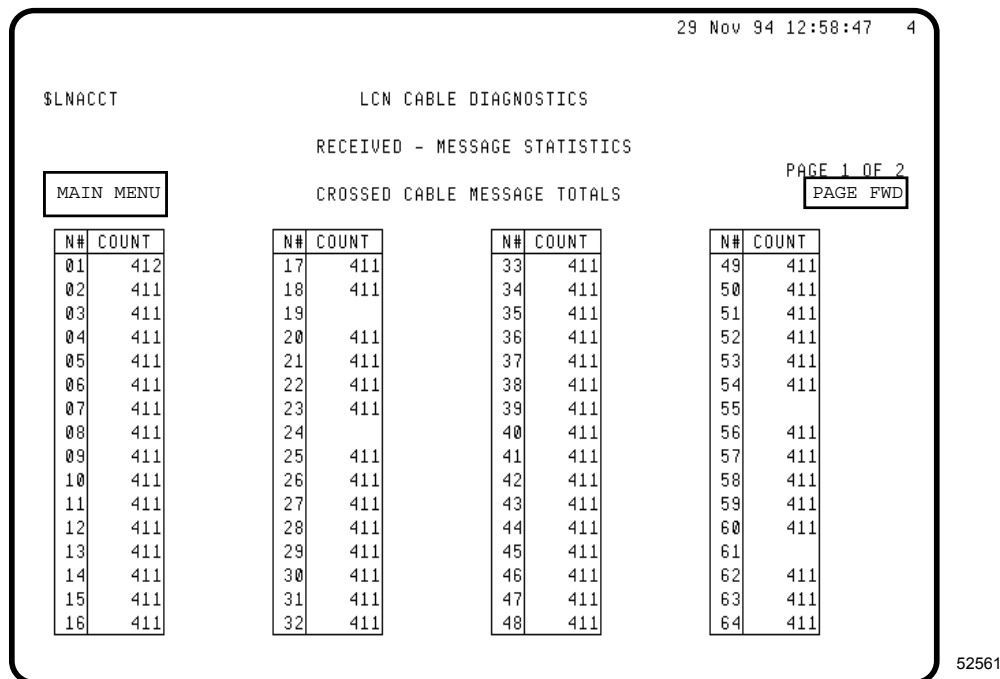


Figure 6-5 — Crossed Cable Message

A message with the count is sent to all nodes on the active cable at one-minute intervals. The nodes put the count in a parameter that any node can read. The schematic displays each nodes count in the LCN Cable Diagnostics display.

- The nodes on the uncrossed cable all display the same count—they are in sync with each other.
- Nodes with crossed cables will display a different count.

(Exception—there is a narrow window where, for one minute, a node with crossed cables can occasionally display the same count.)

To detect crossed cables scan the display for any node, except the LCN master node, that reports a different count than the other nodes.

6.2.2.1 Error Statistics

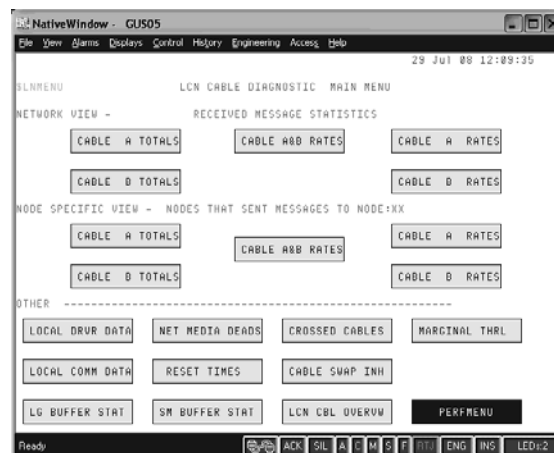
With R681, the additional LCN communication error statistics are collected and represented as PSDP parameters. These errors are monitored for an early indication of the cable status and the auxiliary marginal status of the nodes. The threshold limits are specified in the PSDP parameters.

The following error statistic PSDP parameters are monitored to indicate the auxiliary marginal status of the node:

- Frame Check Sequence Errors
- Framing Errors
- Oversize Errors
- Overrun Errors
- Manchester Errors
- Ring Collapses (common for Cable A and Cable B)

6.2.2.2 How to Install and Display the LCN Cable Statistics

The pathname for the volume or directory with the LCN Cable Diagnostic displays must be listed as one of the schematic pathnames in your Pathname Catalog in the Area Data Base. When it is, you can press the SCHEM key, key in the name of the LCN Diagnostics Menu, \$LNMENU, and press <ENTER>. This menu appears. All of the other LCN Cable Diagnostic displays can be selected from this menu.



If you have a console or a US dedicated to engineering and servicing functions, you might consider configuring a button to call up this menu. For more information, refer to *Button Configuration Form Instructions* and to *Button Configuration Data Entry*. Both of these are in the *Implementation/Engineering Operations — 1* binder.

If you have room in an existing directory on an HM that is already listed in your pathname catalog (the Schematic Titles display in the Operator Personality's Organizational Summary shows these pathnames), you can copy the display-abstract object files from the DIA1 directory on the &CR6 cartridge or the DIA1 floppy to the existing directory. These displays require approximately 410 sectors = 102.5 kb. To copy them in the Engineering Personality, use the Utilities copy command as in this example (for more information, refer to subsection 5.3 in *Utilities Operation in the Implementation/Engineering Operations — 1 binder*):

```
CP $F1>DIA1>*.DO NET>dir>= -D
```

If no directory with adequate space is available, you can add the pathname for the DIA1 directory on cartridge &CR6, or the DIA1 floppy, to the pathname catalog for an area; and do an area change at a US with the Operator Personality to pick up the new pathname. To do so, use these steps:

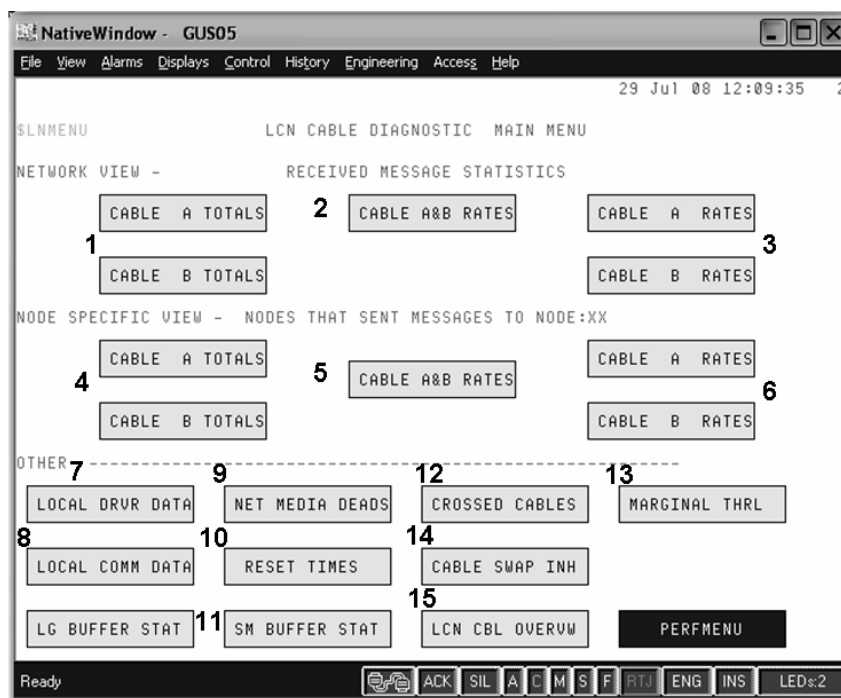
1. At a US with the Engineering Personality and the Engineering Main Menu on the screen, select BUILDER COMMANDS, on the Command Display select SELECT AREA, select the pick for the area served by the US that will show the LCN Diagnostic displays, and press <ENTER>. This tells the Data Entity Builder which area data base is to be changed.
2. On the Command Display, select RECONSTITUTE. New ports appear.
3. In the ENTITY NAME port, key in \$OABSTR and press <ENTER>. After a few seconds, the reconstituted Parameter Entry display for your Pathname Catalog appears.
4. In a spare Schematic Pathname port, key in the pathname for your DIA1 directory on cartridge &CR6 or DIA1 floppy, and press <ENTER>. This is the an example of the pathname;
\$F3>DIA1
5. Hold <CTL> and press <F12> to load the Pathname Catalog into the workfile.
6. Press COMMND. The Command display appears.
7. Select INSTALL AREA and press <ENTER>. This installs the modified area database in directory &Dnn (nn is the area number).
8. At a US running the Operator Personality, on the Console Status display, select AREA CHANGE, then select the target for the area whose data base you just modified, and follow the prompts to complete the area change.
9. With the floppy or cartridge that contains volume or directory DIA1 mounted in the drive named in step 4, press the SCHEM key, key in the LCN Diagnostic Menu name, &LNMENU, and press <ENTER>. The menu appears, and from it you can select any of the sixteen other displays that show the LCN cable statistics for all nodes on the LCN.

6.2.2.3 How to Use the LCN Cable Diagnostic Displays

These displays show the "live" values in the PSDP COMM parameters in all of the nodes on your LCN. To analyze the values displayed, you should consult a topology map of your LCN. You can use this map and the information on the displays to visualize where LCN problems may be. In some cases, you may need assistance from Honeywell's TAC in using the information on these displays to diagnose LCN problems. Examples of topology maps are provided in Appendix B.

The statistical information on these displays is meaningful only if the person analyzing it knows when accumulation of the data began. The values in the COMM parameter accumulate from the time the node is started or from the time the counters were last reset. The RESET COUNTERS target on the Main Menu display, simulated below, calls up a display that you can use to reset the COMM values for any node. This display also shows the date and time the COMM values for each node were last reset. Each of the displays with information for a specific node includes a COUNTERS LAST RESET AT target that shows the last counter reset time for the node and can be selected to reset the COMM values for the node. The key switch must be in the ENG position to reset the COMM counters. A date or time change causes all statistics to be reset in all nodes.

A simulation of each of the LCN Cable Diagnostic displays is provided on this and the following pages. Information on the content of each display is provided with the actions of the targets (picks) on these displays. Each of the targets on the LCN Cable Diagnostics Main Menu selects one of the statistics displays. Numbers were added to this simulated menu display to relate each of the display descriptions that follow. To find the display called for by the main menu targets, look for the corresponding number on the other display images.



6. TROUBLESHOOTING

6.2. LCN Cable Diagnosis

For each node on the LCN this display shows the number of messages received by the node and the number of messages with errors. To show in the space available the very large values that can accumulate, the values appear as a decimal mantissa and exponent. For example, 3931e01 = 3931 x 101, and 16939e02 = 16939 x 102 = 1,693,900. To see the same display for the opposite cable (B in this example), select the CABLE B target. To see a similar display, but with error rates per 10,000 messages, select the RATES target.

1

08 May 90 13:32:11 4

\$LNATOTR LCN CABLE DIAGNOSTICS

CABLE A TOTALS

RECEIVED MESSAGE STATISTICS

MAIN MENU CABLE B RATES

N#	ERRORS	MSGS	N#	ERRORS	MSGS	N#	ERRORS	MSGS	N#	ERRORS	MSGS
01			17			33	0e00	3931e01	49	0e00	19252e01
02	0e00	4554e02	18			34	0e00	3366e01	50	0e00	22408e01
03			19			35	1e00	19792e01	51		
04			20			36	0e00	20579e011	52	3e00	16939e02
05			21			37	0e00	058e01	53	0e00	4559e02
06			22			38	3e00	19578e01	54		
07			23			39	0e00	7329e01	55		
08			24			40	0e00	9518e01	56		
09			25			41	0e00	18299e026	57		
10			26			42	0e00	06e01	58		
11			27			43	0e00	21781e01	59	0e00	5410e02
12			28			44	0e00	20185e01	60		
13			29			45	0e00	21321e01	61		
14			30			46	2e00	5206e01	62	0e00	22296e02
15			31			47	0e00	6033e01	63	0e00	4044e02
16			32			48			64		

For each node on the LCN this display shows the error rate per 10,000 messages received for both the A and the B LCN cables. In this example, the error rate since each node was started or since the COMM parameter values for each node were last reset is either zero or so low that there is no significant error rate. Because all errors are recovered by retransmitting the message, the error rate is the only relevant factor. If a log is maintained for error rates, a trend can be seen, telling of damage or deterioration.

2

08 May 90 13:33:28 4

\$LNABRTR LCN CABLE DIAGNOSTICS

RECEIVED-MESSAGE STATISTICS

ERRORS PER 10,000 MESSAGES

MAIN MENU

N#	CBL-A	CBL-B	N#	CBL-A	CBL-B	N#	CBL-A	CBL-B	N#	CBL-A	CBL-B
01			17			33	0	0	49	0	0
02	0	0	18			34	0	0	50	0	0
03			19			35	0	0	51		
04			20			36	0	0	52	0	0
05			21			37	0	0	53	0	0
06			22			38	0	0	54		
07			23			39	0	0	55		
08			24			40	0	0	56		
09			25			41	0	0	57		
10			26			42	0	0	58		
11			27			43	0	0	59	0	0
12			28			44	0	0	60		
13			29			45	0	0	61		
14			30			46	0	0	62	0	0
15			31			47	0	0	63	0	0
16			32			48			64		

For each node on the LCN, this display shows the number of errors in messages received by the node and the error-rate per 10,000 received messages. To see the same display for the opposite (B in this example) cable, select the CABLE B target. To see a similar display, but with the total number of messages received and total number of errors detected by each node, select the TOTALS target.

08 May 90 13:34:55 4

LCN CABLE DIAGNOSTICS

CABLE B RATES

RECEIVED MESSAGE STATISTICS

3

SLNBRATR

MAIN MENU

CABLE A

TOTALS

N#	ERRORS	ER/10K	N#	ERRORS	ER/10K	N#	ERRORS	ER/10K	N#	ERRORS	ER/10K
01			17			33			49		
02			18			34			50		
03			19			35			51		
04			20			36			52		
05			21			37			53		
06			22			38			54		
07			23			39			55		
08			24			40			56		
09			25			41			57		
10			26			42			58		
11			27			43			59		
12			28			44			60		
13			29			45			61		
14			30			46			62		
15			31			47			63		
16			32			48			64		

This display shows for the specific node indicated in the heading, the number of messages received from each of the other nodes on the LCN and the number of errors detected in those messages. In the MSGS column, to show in the space available the very large values that can accumulate, the values appear as a decimal mantissa and exponent. For example 2324e2 = 2324 x 102 = 232,400. To see the same display for the opposite cable (A in this example), select the CABLE A target. To see a similar display, but with error rates per 10,000 messages, select the RATES target. To see this display for another node, select the SELECT NODE NUMBER target, key in the node number, and press <ENTER>.

08 May 90 13:36:02 4

LCN CABLE DIAGNOSTICS

CABLE A TOTALS

NODES THAT SENT MESSAGES TO NODE: 33

4

SLNATOTS

MAIN MENU

CABLE B

RATES

N#	ERRORS	MSGS	N#	ERRORS	MSGS	N#	ERRORS	MSGS	N#	ERRORS	MSGS
01	0	0e00	17	0	0e00	33	0	0e00	49	0	312e00
02	0	310e00	18	0	0e00	34	0	28466e00	50	0	311e00
03	0	0e00	19	0	0e00	35	0	317e00	51	0	0e00
04	0	0e00	20	0	0e00	36	0	329e00	52	0	3400e00
05	0	0e00	21	0	0e00	37	0	544e00	53	0	311e00
06	0	0e00	22	0	0e00	38	0	337e00	54	0	0e00
07	0	0e00	23	0	0e00	39	0	376e00	55	0	0e00
08	0	0e00	24	0	0e00	40	0	421e00	56	0	0e00
09	0	0e00	25	0	0e00	41	0	275e00	57	0	0e00
10	0	0e00	26	0	0e00	42	0	313e00	58	0	0e00
11	0	0e00	27	0	0e00	43	0	310e00	59	0	310e00
12	0	0e00	28	0	0e00	44	0	312e00	60	0	311e00
13	0	170e00	29	0	0e00	45	0	311e00	61	0	0e00
14	0	32e00	30	0	0e00	46	0	310e00	62	0	310e00
15	0	546e00	31	0	0e00	47	0	324e00	63	0	308e00
16	0	339e00	32	0	0e00	48	0	0e00	64	0	0e00

COUNTERS LAST RESET AT 5/08/89 8:30:57

SE NODE NUMBER

The new values appear very quickly, so watch the node number in the heading to be sure which node's data is present. The COUNTERS LAST RESET target indicates the time the values were last reset for this node and can be selected to reset the COMM values for this node. Occasionally, values may appear for the node that does not exist. This occurs when a message header is corrupted so that the node number is altered.

When this occurs, the number of errors equals the number of messages. This display also can be used to determine if a given node can receive messages from all other nodes on the network (including ISOLATED nodes). This is useful to determine if cables have been fixed after a break.

6. TROUBLESHOOTING
6.2. LCN Cable Diagnosis

This display shows for the specific node indicated in the heading, the error rate per 10,000 messages detected in messages received from each of the other nodes on the LCN. The error rates for both A and B LCN cables are shown. To see this display for another node select the SELECT NODE NUMBER target, key in the node number, and press <ENTER>. The new values appear very quickly, so watch the node number in the heading to be sure which node's data is present. The COUNTERS LAST RESET target indicates the time the values were last reset for this node and can be selected to reset the COMM values for this node.

5

08 May 90 13:38:45 4

SINABRTS

LCN CABLE DIAGNOSTICS

NODES THAT SENT MESSAGES TO NODE: 15

ERRORS PER 10,000 MESSAGES

MAIN MENU

N#	CBL-A	CBL-B	N#	CBL-A	CBL-B	N#	CBL-A	CBL-B	N#	CBL-A	CBL-B
01	0	0	17	0	0	33	0	0	49	0	0
02	0	0	18	0	0	34	0	0	50	0	0
03	0	0	19	0	0	35	0	0	51	0	0
04	0	0	20	0	0	36	0	0	52	0	0
05	0	0	21	0	0	37	0	0	53	0	0
06	0	0	22	0	0	38	0	0	54	0	0
07	0	0	23	0	0	39	0	0	55	0	0
08	0	0	24	0	0	40	0	0	56	0	0
09	0	0	25	0	0	41	0	0	57	0	0
10	0	0	26	0	0	42	0	0	58	0	0
11	0	0	27	0	0	43	0	0	59	0	0
12	0	0	28	0	0	44	0	0	60	0	0
13	0	0	29	0	0	45	0	0	61	0	0
14	0	0	30	0	0	46	0	0	62	0	0
15	0	0	31	0	0	47	0	0	63	0	0
16	0	0	32	0	0	48	0	0	64	0	0

COUNTERS LAST RESET AT 8:30:57

SELECT

NODE NUMBER

This display shows for the specific node indicated in the heading, the total number of errors and the error rate per 10,000 messages detected in messages received from each of the other nodes on the LCN. To see a similar display, but with total errors and total messages, select the TOTALS target. To see the same display for the opposite cable (B in the example), select the CABLE B target. To see this display for another node select the SELECT NODE NUMBER target, key in the node number, and press <ENTER>. The new values appear very quickly, so watch the node number in the heading to be sure which node's data is present. The COUNTERS LAST RESET target indicates the time the values were last reset for this node and can be selected to reset the COMM values for this node.

08 May 90 13:37:20 4

\$LNARATS 6 LCN CABLE DIAGNOSTICS

CABLE A RATES

NODES THAT SENT MESSAGES TO NODE: 15

MAIN MENU			CABLE B			TOTALS		
N#	ERRORS	ERR/10K	N#	ERRORS	ERR/10K	N#	ERRORS	ERR/10K
01	0	0	17	0	0	33	0	0
02	0	0	18	0	0	34	0	0
03	0	0	19	0	0	35	0	0
04	0	0	20	0	0	36	0	0
05	0	0	21	0	0	37	0	0
06	0	0	22	0	0	38	0	0
07	0	0	23	0	0	39	0	0
08	0	0	24	0	0	40	0	0
09	0	0	25	0	0	41	0	0
10	0	0	26	0	0	42	0	0
11	0	0	27	0	0	43	0	0
12	0	0	28	0	0	44	0	0
13	0	0	29	0	0	45	0	0
14	0	0	30	0	0	46	0	0
15	0	0	31	0	0	47	0	0
16	0	0	32	0	0	48	0	0

COUNTERS LAST RESET AT 5/08/89 8:30:57

SELECT NODE NUMBER

This display shows data from the LCN driver software for the node indicated in the display heading. To see this display for another node select the SELECT NODE NUMBER target, key in the node number, and press <ENTER>. The new values appear very quickly, so watch the node number in the heading to be sure which node's data is present. The COUNTERS LAST RESET target indicates the time the values were last reset for this node and can be selected to reset the COMM values for this node. The items on this display and their values are described in the following paragraphs.

08 May 90 13:40:01 4

\$LNLCLDV 7 LCN CABLE DIAGNOSTICS

LOCAL DRIVER DATA

NODE: 33

ITEM	VALUE	ITEM	VALUE
COMM Status	On Net	Indicated Primary	A
Driver Errors	0	Cable After MD	B
Cable Master	Slave	Selected Cable	B
Primary Cable	A	Swap Ticks Left	N/A
Large Input Buffers	12	Buf Shortage - Large	0
Small Input Buffers	70	Buf Shortage - Small	0
Output Pending Chain	0		
Output Packets Lost	0		
Messages Transmitted	8969 e00		
Transmit Errors	0		
Receive IOCB Errors	0	Failed State Time	05/08/89 08:30:57
LCN Recovery Retries	0		

COUNTERS LAST RESET AT 5/08/89 8:30:57

SELECT NODE NUMBER

COMM Status—LCN Driver Status for this node values are: Initializing, Recovering, Initialized, On Network, Entering Ring, Failed, and Waiting.

Driver Errors—Secondary error code for this node in a range from 0 to 31. These codes are defined in Table 6-1.

Cable Master—Indicates if this node is the Master.

Primary Cable—The cable this node sees as the primary for receiving messages: A = Cable A is the primary cable and B is the backup; B = Cable B is the primary cable and A is the backup.

Large Input Buffers—The current number of large message input buffers for this node.

Small Input Buffers—The current number of the small message input buffers for this node.

Output Pending Chain—The current number pending output messages from this node.

Output Packets Lost—The number of packets thrown away by this node due to off-network status.

Message Transmitted—The number of messages transmitted by this node.

Transmit Errors—Hardware detected transmit error.

Receive IOCB Errors—The number of messages received by this node with an error in their I/O control block.

LCN Recovery Retries—The number of times this node attempted to recover from an abnormal LCN error (corrupted messages and "media dead" events are considered normal and do not result in attempts to recover). Range is 0 through 5.

Indicated Primary—The cable the LCNI in this node last reported as selected.

Cable After MD—The cable (A or B) the LCNI in this node reported after a media-dead interrupt.

Selected Cable—The cable (A or B) most recently selected by the software in this node.

Swap Ticks Left—On the master node only, indicates the number of 50 ms time ticks remaining until the master node selects the other LCN cable.

Buffer Shortage — Large—The number of times messages that were lost because large-message buffers were not available.

Buffer Shortage — Small—The number of times messages that were lost because small-message buffers were not available.

Failed State Time—The date and time when the LCN driver in this node last entered a failed state.

Table 6-1 — LCN Driver Errors

Secondary (SECD) error Code	Error Definition
001	<u>Definition:</u> Error encountered during a module bus data transfer <u>Probable cause:</u> LCNI hardware error or software error
002	<u>Not Used</u>
003	<u>Not Used</u>
004	<u>Definition:</u> Integrity failure <u>Probable cause:</u> LCNI hardware error
005	<u>Definition:</u> Illegal command received by LCNI <u>Probable cause:</u> Software error
006	<u>Definition:</u> CCB format not valid <u>Probable cause:</u> Software error
007	<u>Definition:</u> IOCB format not valid <u>Probable cause:</u> Software error
008	<u>Definition:</u> Illogical software condition <u>Probable cause:</u> Software error
009	<u>Definition:</u> Self-test or board error, red LED on LCNI is on <u>Probable cause:</u> LCNI hardware error
010	<u>Not Used</u>
011	<u>Definition:</u> Invalid IOCB code <u>Probable cause:</u> DRAM corruption by software error or hardware malfunction
012	<u>Not Used</u>
013	<u>Not Used</u>
014	<u>Not Used</u>
015	<u>Not Used</u>
016	<u>Definition:</u> Token received but when ready to transmit, LCN not silent <u>Probable cause:</u> Noise on cable or misoperation of some other LCNI. Possible duplicate LCNI addresses.
017	<u>Definition:</u> In background testing a parity error was detected in the LCN address connections on the transceiver board, or there is no match between the value in local RAM and what was read.

6. TROUBLESHOOTING

6.2. LCN Cable Diagnosis

Table 6-1 — LCN Driver Errors

Secondary (SECD) error Code	Error Definition
	<u>Probable cause:</u> LCNI hardware error
018	<u>Definition:</u> LCNI command time out <u>Probable cause:</u> LCNI hardware error
019	<u>Definition:</u> Frame-check sequence error detected while receiving data. <u>Probable cause:</u> LCNI hardware error or cable problems
020	<u>Definition:</u> Framing error—received frame didn't have an even number of bytes. <u>Probable cause:</u> LCNI hardware error or cable problems
021	<u>Definition:</u> LCNI receiver overrun—LCNI received data faster than it could be transferred to memory. <u>Probable cause:</u> LCNI not installed in slot 2 or LCNI hardware error
022	<u>Definition:</u> Underrun during LCNI transmission—LCNI can't get data from memory fast enough to keep up with transmission rate. <u>Probable cause:</u> LCNI not installed in slot 2 or LCNI hardware error
023	<u>Definition:</u> Received an oversized frame. <u>Probable cause:</u> Software error. It could also be caused by hardware error in detecting end-of-frame delimiter before maximum frame time.
024	<u>Definition:</u> Manchester decoder error—LCNI could not decode incoming data. <u>Probable cause:</u> LCNI hardware error or poor signal quality on cable.
025	<u>Definition:</u> Alone in the ring—LCNI received the token or tried to initialize the token-passing ring and was unable to pass the token to another LCNI. <u>Probable cause:</u> LCNI hardware error, all other nodes disconnected or without power, or cable failure.
026	<u>Definition:</u> Media dead—LCNI has not received a token pass addressed to itself or any other address for approx. 18 ms, so the token-passing ring has collapsed. This error is also reported if the LCNI has received 4352 token-pass frames addressed to other nodes since it last received a token-pass frame addressed to itself—the LCN can't receive data passed by its predecessor in the token-passing ring. <u>Probable cause:</u> Power was removed from the node that held the token, noise on the cable, or cable fault.
027	<u>Definition:</u> LCNI in nonautomatic-cable-switchover mode <u>Probable cause:</u> LCNI hardware error
028	<u>Not Used</u>
029	<u>Not Used</u>
030	<u>Definition:</u> Swapped to Cable B because of errors on Cable A. Reported only by the master node (normally, the node with the lowest address).

Table 6-1 — LCN Driver Errors

Secondary (SECD) error Code	Error Definition
	<u>Probable cause:</u> Occurrence of errors 019 through 024, above
031	<u>Definition:</u> Swapped to Cable A because of errors on Cable B. Reported only by the master node (normally, the node with the lowest address). <u>Probable cause:</u> Occurrence of errors 019 through 024, above

For release R400 and beyond, LCN Driver Errors can also be printed on the Real Time Journal. To direct system error messages (including LCN Driver Errors) to the printer, select the REAL TIME JOURNAL ASSIGNMENTS target under the System Menu. Select SYSTEM ERROR MESSAGES. If the indicated status is SUSPEND, touch the ACTIVATE target, and then touch the CHANGE PRINTER target and enter the desired printer number. LCN Driver Errors are reported at a maximum rate of one every 30 seconds to avoid filling the log with repetitious errors. A sample error is as follows:

14:59:31 US 35 LCN DRIVER 000 SLOT 002 CMD REG 0200 PRIM -000005 SECD 120 8000 0000 0000 16

The fields of interest in this example are the time (14:59:31), the node type and number (US 35), the error type (LCN DRIVER), the primary error code (-000005, which indicates a physical error), and the secondary error code (120). The last two digits of the secondary error code (20 in the above example), are the code shown in Table 6-1 (20 = Framing Error for the example). The first digit of the secondary error code (1 in the example), is encoded to indicate the cable on which the error occurred (0 = Cable A, 1 = Cable B). Error codes 30 and 31, the last two shown in Table 6-1, are not used when LCN Driver Errors are reported on the Real Time Journal.

The following display shows statistics for communication errors detected by the node indicated in the display heading. The TOTAL ERRORS and TOTAL MESSAGES values are shown in this form: 4051e01. This represents $4051 \times 10^1 = 40,510$.

6. TROUBLESHOOTING
6.2. LCN Cable Diagnosis

To see this display for another node select the SELECT NODE NUMBER target, key in the node number, and press <ENTER>. The new values appear very quickly, so watch the node number in the heading to be sure which node's data is present. The COUNTERS LAST RESET target indicates the time the values were last reset for this node and can be selected to reset the COMM values for this node. The items on this display are described in the following paragraphs.

08 May 90 13:41:14 4		
\$LNLCM	8	
LCN CABLE DIAGNOSTICS		
LOCAL COMMUNICATION ERRORS		
NODE: 33		
MAIN MENU		
MESSAGE ERROR TYPE	CABLE "A"	CABLE "B"
FCS	0	0
FRAMING	0	0
OVERSIZE	0	0
OVERRUN	0	0
MANCHESTER	0	0
TOTAL ERRORS	0 e00	0 e00
TOTAL MESSAGES	4051 e01	3957 e01
ERRORS PER 10,000	0	0
OTHER ERRORS	CABLE "A"	CABLE "B"
ALONE-IN-RING	0	0
MEDIA DEAD	0	0
STOLEN TOKEN	0	0
COUNTERS LAST RESET AT	SELECT	
5/08/89 8:30:57	NODE NUMBER	

- FCS—Frame Checksum Error count for this node.
- FRAMING—Number of framing errors detected by this node.
- OVERSIZE—Number of messages received by this node that were larger than the allocated buffer size.
- OVERRUN—Number of times this node's LCNI could not gain direct-memory access in time.
- MANCHESTER—Number of Manchester errors detected by this node.
- ALONE-IN-RING—The number of alone-in-ring events detected by this node's LCNI.
- MEDIA DEAD—The number of media-dead events detected by this node's LCNI.
- STOLEN TOKEN—The number of times the LCNI was ready to transmit and detected a signal on one or both cables (possibly noise or a valid transmission by another node).

This display shows the number of media-dead events detected by the LCNI in each node on the LCN.

08 May 90 13:42:12 4

\$LNDDEADS

9

LCN CABLE DIAGNOSTICS

NUMBER OF TIMES "MEDIA DEAD" DETECTED

MAIN MENU

N#	CBL-A	CBL-B	N#	CBL-A	CBL-B	N#	CBL-A	CBL-B	N#	CBL-A	CBL-B
01			17			33	0	0	49	2	4
02	2	4	18			34	0	0	50	3	7
03			19			35	2	4	51		
04			20			36	2	4	52	2	8
05			21			37	2	4	53	2	4
06			22			38	2	4	54		
07			23			39	2	4	55		
08			24			40	2	4	56		
09			25			41	2	4	57		
10			26			42	2	4	58		
11			27			43	2	4	59	0	0
12			28			44	4	2	60		
13			29			45	2	4	61		
14			30			46	4	8	62	4	3
15			31			47	3	8	63	4	3
16			32			48			64		

For each node on the LCN, this display shows the date and time that the COMM values were last reset. To reset the COMM values for a node, select its date-and-time line and press <ENTER> (the key switch must be in the ENG position).

08 May 90 13:43:13 4

\$LNRESET

10

LCN CABLE DIAGNOSTICS

COUNTER RESET TIMES

MAIN MENU

N#	RESET TIME	N#	RESET TIME	N#	RESET TIME	N#	RESET TIME
01		17		33	5/05 8:30:57	49	5/05 13:32:52
02	5/05 13:37:36	18		34	1/01 0:16:28	50	5/05 13:32:46
03		19		35	5/05 13:09:13	51	
04		20		36	5/05 12:13:26	52	5/05 13:32:41
05		21		37	5/05 12:13:13	53	5/05 12:18:01
06		22		38	5/05 13:09:14	54	
07		23		39	5/05 12:21:08	55	
08		24		40	5/05 12:24:42	56	
09		25		41	5/05 12:25:56	57	
10		26		42	5/05 12:24:33	58	
11		27		43	5/05 13:31:43	59	5/05 16:25:14
12		28		44	5/05 12:47:21	60	
13		29		45	5/05 12:21:38	61	
14	5/05 15:06:41	30		46	5/05 12:17:21	62	5/05 12:21:02
15	1/01 0:01:04	31		47	5/05 12:19:15	63	5/05 13:41:29
16	5/05 15:07:53	32		48		64	

SELECT INDIVIDUAL DATE & TIME TO RESET TO CURRENT TIME

6. TROUBLESHOOTING

6.2. LCN Cable Diagnosis

This display shows three types of large buffer status information for each node on the LCN. Select the SMALL BUF target to call up a similar display that shows the same information for small buffer status for each node.

11

08 May 90 13:44:48 4

\$LNLBUFS

LCN CABLE DIAGNOSTICS

NETWORK SMALL BUFFER STATUS

MAIN MENU

N#	BC	OVRUN	NOMEM	N#	BC	OVRUN	NOMEM	N#	BC	OVRUN	NOMEM	N#	BC	OVRUN	NOMEM
01				17				33				49	17	0	0
02	16	0	0	18				34				50	17	0	0
03				19				35				51			
04				20				36				52	16	1	0
05				21				37				53	12	3	0
06				22				38				54			
07				23				39				55			
08				24				40				56			
09				25				41				57			
10				26				42				58			
11				27				43				59			
12				28				44				60			
13				29				45				61			
14	16	0	0	30				46				62	12	4	0
15	16	0	0	31				47				63	17	0	0
16	28	0	0	32				48				64			

BC = NUMBER OF BUFFERS OVRUN = OVERRUN NOMEM = NO MEMORY AVAILABLE

For each node on the LCN, this display shows the crossed cable message totals. The count for the cable master node is incremented once each minute, then each node is sent the message count held in the parameter. The lowest numbered US is the cable master. In this example node # 1 is the cable master. Each node that receives the count from the cable master, displays the count. A node that has crossed cables will develop an incorrect message count that is "out of sync" with the other nodes. A node that is not present or powered up will not display a message count at all.

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29 Nov 94 12:58:47 4

\$LNACCT

LCN CABLE DIAGNOSTICS

RECEIVED - MESSAGE STATISTICS

CROSSED CABLE MESSAGE TOTALS

MAIN MENU

PAGE 1 OF 2
PAGE FWD

N#	COUNT	N#	COUNT	N#	COUNT	N#	COUNT
01	412	17	411	33	411	49	411
02	411	18	411	34	411	50	411
03	411	19		35	411	51	411
04	411	20	411	36	411	52	411
05	411	21	411	37	411	53	411
06	411	22	411	38	411	54	411
07	411	23	411	39	411	55	
08	411	24		40	411	56	411
09	411	25	411	41	411	57	411
10	411	26	411	42	411	58	411
11	411	27	411	43	411	59	411
12	411	28	411	44	411	60	411
13	411	29	411	45	411	61	
14	411	30	411	46	411	62	411
15	411	31	411	47	411	63	411
16	411	32	411	48	411	64	411

52562

The Cable/Node Marginal Threshold Values display allows you to set the cable/node marginal threshold parameter value. The cable threshold value set for the cable master node is propagated to the US/GUS/EST nodes of the network. The node threshold value set for the cable master node is propagated to all the nodes in the network. The display also indicates the number of errors in Cable A and B.

Note: Honeywell recommends that the threshold parameter value is set in the cable master node and propagated to all the other nodes.

1

LCN CABLE DIAGNOSTICS

CABLE MASTER NODE : 1

MAIN MENU

CABLE/NODE MARGINAL THRESHOLD VALUES	
CABLE MARGINAL THRESHOLD	: 1
NODE MARGINAL THRESHOLD FOR 1 MIN	: 1
NODE MARGINAL THRESHOLD FOR 15 MIN	: 2

PROPAGATE

CABLE ERROR COUNTER FOR CURRENT CYCLE	
CABLE A :	0
CABLE B :	0

14

LCN CABLE SWAP INHIBIT

CABLE A: OK CABLE B: OK

CABLE SWAPPING: PERMIT
INHIBIT TIME REMAINING: 0:00

INHIBIT SWAPPING PERMIT SWAPPING

6. TROUBLESHOOTING

6.2. LCN Cable Diagnosis

The LCN Cable Overview display can be used for diagnosing the LCN cables. When the display is invoked, an algorithm associated with the display collects the data from all the nodes in the network. The algorithm analyses the collected data and displays the following information in the LCN Cable Overview display:

- Cable mastership
- Clock mastership and slave
- Multiple clock mastership
- Crossed cable existence
- Multiple cable master possibility
- Cable status

The LCN node statistics display provides the statistical information about the LCN status for a selected node.

29 Jul 08 12:11:49 2

\$LNDCBL

LCN Cable Overview display
LCN Diagnostics

15

Crossed Cable Identified	NO
Multiple Cable Master Exists	NO
Number of Clock Masters	1
Any Node in Marginal state	NO

Clock Node	20	Master
Clock Node	7	Slave

Cable A Status	OK
Cable B Status	OK
Cable master Node	1

LCN NODE STATISTICS LCN FAULT DIAGNOSIS PING NODE

ENTER NODE NUMBER (01-96): 7 29 Jul 08 12:15:52 2

\$LNDGNST LCN STATISTICS FOR NODE 2

SELECT NODE NUMBER

	CABLE A	CABLE B
TRANSMIT ERROR COUNT	0	
RECEIVE IOCB PACKET ERROR COUNT	0	
TOTAL FCS ERROR COUNT FOR 15 MINS	0	0
FCS ERROR RATE FOR 15 MINS	0	0
TOTAL FRAMING ERRORS FOR 15 MINS	0	0
FRAMING ERROR RATE FOR 15 MINS	0	0
TOTAL OVERSIZE ERRORS FOR 15 MINS	0	0
OVERSIZE ERROR RATE FOR 15 MINS	0	0
TOTAL OVERRUN ERRORS FOR 15 MINS	0	0
OVERRUN ERROR RATE FOR 15 MINS	0	0

LCN CBL OVERVIEW LCN FAULT DIAGNOSIS PING NODE

ATTENTION

While diagnosing, if multiple clock nodes are identified in the network, the following must be performed.

- If the network is loaded with pre 682 release, one of the nodes must be reloaded based on the user's preference.
- For the network that is loaded with R682 or later releases, the online feature can be used for relinquishing / assigning the control to the selected node.

The parameter to be used for online clock mode changes is "Comm (1192)". For more information about Comm (1192), refer to "Appendix C – Using the PSDP Comm Parameter" section of the *LCN Guidelines Implementation, Troubleshooting, and Service Guide*.

6.2.3 LCN Cable Diagnostic Procedures

Both an off-line procedure, for use when the system is not controlling the process, and an on-line procedure, are provided below. One or the other may be required to locate LCN problems. Loss of all nodes while using the on-line procedure is unlikely, but it is possible to cause nodes to become isolated and to be unable to communicate with each other. For such systems, if it is necessary to remain on-line while isolating and repairing LCN problems, the on-line procedure may be used, subject to the restrictions and precautions cited in the procedure.

6.2.3.1 Off-Line LCN Test Procedure

A very effective approach for diagnosing LCN communications difficulties is to take the system off-line. With the system off-line, you can run powerful exerciser and diagnostic programs that cannot be run on line because you risk the loss of viewing the process.

Once the system is off-line, run the tests described in the *LCNI Network Communications Test* manual in the *LCN Service* binder. Run the test in manual mode on the suspected cable. This test creates a high LCN traffic load and yields basic information about which nodes have difficulty communicating with other nodes. This is similar to the results from a review of the error journal, but the increased LCN load gives a better indication of the magnitude of any problem. The program does not automatically test both cables, so the user must give the commands to switch between the A and B cables.

While the LCN Network Communications Test is running, perform a test for crossed cables (i.e., Cable A connected to the Cable B port or vice versa). With the test running in manual mode on either cable, remove the tee connector of the cable not being tested from one of the nodes. If there are no errors in the next five minutes, there are no crossed cables. If there are errors, the A and B cables are interchanged somewhere in the system. This interchange must be found and corrected. The LCN error recovery mechanisms depend on the cables not being crossed.

If the LCN Network Communications Test runs without excessive errors for long periods (four hours on each cable), the network is solid and further testing is not necessary. If errors do occur when the LCN Network Communications Test is run, these errors should be analyzed and a course of action determined.

If the system contains LCN Extenders, the test may show that errors always involve communication across a fiber optic link. If this is the case, replace the suspect LCN Extender boards (more information is in Section 4) and run the test program to determine if the extender is at fault. Bad or loose fiber optic cable connections also may be involved. The fiber optic cable itself also may have been damaged.

The next step in isolating the cause of LCN errors is a visual and manual inspection of the appropriate cable or cables while the LCN Network Communications Test is running. For this step, it is best to have two people involved—one to do the cable inspection and one to monitor the test program display at a Universal Station to see if errors suddenly occur when a particular cable is manipulated. A 2-way radio can be useful.

- Look for kinks or places where the cable has been crushed.
- Ensure that no improper substitute cables have been used.
- Check each tee connector to ensure that it is the proper type for use in an LCN (not a Data Hiway tee that has components inside). The LCN tee must be Honeywell part number 51190728-105 (acceptable alternatives are Kings KC99-52 and Amphenol 31-208-1051).
- Check that all connections are tight.
- Flex the cables at the connections to check for intermittent connections.

6.2.3.2 Using a Time Domain Reflectometer (TDR)

If the above cable inspection does not disclose any cable problems and the errors still exist, it is time to do a Time Domain Reflectometer (TDR) test of all the cables. This should be done according to the procedure outlined in the *LCN System Checkout Manual* in the *Site Planning and Installation* binder. Follow the procedure in the manual, except use a tee connector from the system rather than a splice to attach the selected terminator to the end of the coax cable under test. Thus if a tee connector is bad, it will be found. Every cable and tee connector in the system should be tested in this manner.

If no bad cables are located and errors persist, the problem should be reported to TAC [call 1(800)822-7673 or in Arizona (602) 863-5558] and the TAC may contact IAC Engineering for assistance.

6.2.3.3 On-Line Test Procedure

We recommend that only individuals approved by the Honeywell Technical Assistance Center (TAC) perform this on-line procedure. The approved person and all others involved need to be thoroughly briefed on the procedures to be followed—including recovery procedures—before testing starts. Some portions of the procedure require the coordinated actions of three people.

CAUTION

Contact the Honeywell Technical Assistance Center before attempting to perform this on-line procedure.

Loss of all nodes while using the on-line procedure is unlikely, but it is possible to cause nodes to become isolated and to be unable to communicate with each other. For such cases, if it is necessary to remain on-line while isolating and repairing LCN problems. The On-line procedure may be used, but are subject to the restrictions and precautions cited in the procedure.

Floppy disks or disk cartridges required to reload must be readily accessible.

6.2.3.4 Preparation for On-Line Testing

Before starting on-line cable diagnostic testing, be sure that plant operations personnel are trained and familiar with the following:

1. Plant operations personnel need to predetermine which console and specifically which US is to be the recovery console in the event of an LCN system outage or a loss of view to the process. In addition, the operator of that console should have a current LCN topology drawing to know where all LCN nodes are, their designation, and their function. The following items should be taken into consideration in console selection:
 - Does the console with the designated recovery US have a configured Area Data Base that expedites a view to the process?
 - Does this Area Data Base support the most critical process units on this LCN System?

- Does the recovery US connect directly to the local removable-media drive (cartridge or floppy), to allow this US to be rebooted?
 - Does this US also have a printer directly connected, so that any system errors continue to be printed as necessary?
2. Once the recovery console and associated US have been designated, ensure that—as a minimum precaution—diskettes or cartridges containing the volumes indicated below are located and physically available at the console before initiation of the test procedure. Please note that the versions of these diskettes or cartridges, including whether or not 68000- or 68020-based hardware is in use, must exactly match the current release software version that is loaded and operational on the LCN System (you can use a user-prepared fast load cartridge for this purpose. For more information, refer to subsection 20.2 in the *Engineers Reference Manual*, found in the *Implementation/Startup and Reconfiguration* — 2 binder).

The volumes are as follows:

- System Startup (&ASY) including current customer NCF
- US Standard Abstracts (&DSY)
- US Operator Personality (&OPR) or Universal Personality (&UNP)
- US Engineering Overlays (&OV1, 2, 4)
- HG or PLCG On-Process Personalities (&HGO) (required only when HGs or PLCGs are present)
- NIM On-Process Personalities (&NMO) (required only when NIMs are present)
- Process Manager Personalities (&PMO)

In addition, cartridges or floppies containing the following data must be available:

- Current set of labeled checkpoint diskettes for all HGs, AMs, NIMs, and PMs. If not currently available, do a demand checkpoint (SAVE DATA) to diskette or cartridge before proceeding.
- Current copy of the designated console's up-to-date Area Data Base on diskette or cartridge (&Dnn).
- Current copy of the System's (&ASY) diskette with the current NCF on it. If not available, one must be created. Please refer to the *Customer Release Guide* for the current release on the LCN.

NOTE

Steps 3 and 4 are a test of the readiness of the people involved to reload and restart the system after the loss of several or all nodes. Though unlikely, such a failure could occur, for example, in the event of loss of both primary and backup power to several or all nodes.

3. After completing the checks above and before testing commences, execute the following quick procedure to ensure that the recovery operations person is familiar with restarting (rebooting) a US from diskette or cartridge.
 - Reset the US by lifting the keyboard cover panel and pressing the <RESET> button.
 - Next, press the <LOAD> key and respond to the LOAD prompts specifying the diskette/cartridge drive to load the Operator Personality in the US. Be sure also to specify the appropriate cartridge or diskette for all other required volumes/directories (i.e. &DSY, &ASY), and Area Data Base (&Daa where “aa” is the area number, 0-10). If the HM that stores the NCF is still available, the NCF will be loaded over the LCN network from the HM. Be certain that the NCF on the diskette/cartridge is the same version as the NCF on the HM.
4. The operator should have available a written restart procedure that explains, step-by-step, how to bring the LCN back on-line. This restart procedure should designate the order in which the various nodes of the system are rebooted or reloaded, including what steps to take if the HM has to be reloaded. Design the procedures so they reinstate the operator's view of the process as quickly as possible. For those systems configured in clusters, each operator needs to be instructed in the restart of their assigned cluster in parallel with other clusters on the LCN. The testing as outlined below now may proceed.

6.2.3.5 On-Line Test Instructions

The first step in on-line cable testing is a visual and manual check of the cables. One person should do the cable inspection while another person monitors system operation through the LCN Overview Status display. If errors occur, the cable inspection should be stopped and the equipment in the vicinity being inspected at the time of the errors should be evaluated for possible faults.

CAUTION

To reduce the possibility of getting into a double-fault situation, during the following steps, always inspect one cable and its connectors one at a time without disturbing the other cable or its connectors.

- Perform a visual check for crossed cables. Cable A is marked with yellow and Cable B with green shrink-fit bands at the connectors.
- Look for kinks or places where the cable has been crushed.
- Ensure that no improper substitute cables have been used.
- Check each tee connector to ensure that it is the proper type for use in an LCN (not a Data Hiway tee that has components inside). The LCN tee must be Honeywell

part number 51190728-105 (acceptable alternatives are Kings KC99-52 and Amphenol 31-208-1051).

- Check that all connections are tight.
- Flex the cables at the connections to check for intermittent connections.

If no bad cables are found with the visual and manual checks, the next troubleshooting effort involves doing a TDR test of each individual cable on the suspect A or B cable subsystem while the system is on-line and using the other cable.

6.2.3.6 On-Line TDR Testing

CAUTION

We advise extreme caution in using this procedure because of the risk of introducing double cable faults and the possibility of subsequent isolation of nodes.

If the system includes LCN Extenders and the system configuration permits it, place the RUN/RESET switches on the extenders for both the A and B cable subsystems to RESET, so that separate LCNs result. This allows each individual coax bus segment to be tested without affecting the others. Unfortunately, some systems are not configured to allow this. If you isolate LCN segments in this manner care must be exercised when reactivating the LCN Extenders. The reconnection scenario in Section 8 of this manual must be followed.

The cables should be tested using the procedure outlined in the LCN System Checkout Manual in the LCN Installation binder. Follow the procedure in that manual except that a tee connector from the system, rather than a splice, is used to attach the selected terminator to the end of the coax cable under test. Thus if a tee connector is bad, it will be found. Every cable and tee connector in the system should be tested in this manner.

CAUTION

For the activity in the next paragraph, use extreme caution to avoid removal of the wrong cable.

Because the system is on-line, the cable subsystem being tested must be kept in a state such that it can be returned to service as rapidly as possible. We recommend that a known good piece of coax and a tee connector be substituted for the cable and tee connector to be tested. This temporary coax cable does not need to be routed in the same way as the cable to be tested, because it will be removed as soon as the test is performed. The concept is that the cable subsystem under test should be in an operational state as much of the time as possible. Note—the substituted cable should be approximately the same length as the original cable so as not to violate the rule that the maximum distance between any two nodes in the system, as measured on the A cable, must not differ from the distance measured on the B cable by more than 300 meters.

This procedure can be cumbersome or impractical when long coax cables are to be tested. Still, we recommend this approach to ensure system integrity. Most cables can be checked in this manner. Removal and replacement of the coax cable and an associated tee connector should be a coordinated action to reduce the number of errors generated. One person monitors the cable status display to know when the system switches to the "good" cable subsystem. The cable and an associated tee connector in the "bad" cable subsystem then should be removed and replaced as quickly as possible. For long cables, this requires two people—one at each end of the cable. Thus three people are involved: one to watch the cable status display at a Universal Station and two to perform the simultaneous cable substitution.

Normally, the system switches between the A and B cables once every minute. After errors have occurred, the interval is changed to once every 10 minutes until the system can remain on the suspect cable for an entire 10-minute swap interval. Note that operator-commanded cable switches do restart the "time to swap" timer.

If the 1-minute time interval is not considered a long enough time to safely complete the cable substitution, an error can be safely induced (as long as the cables are not crossed) that will cause the system to use the opposite cable for 10 minutes before switching back. To accomplish this, observe that the system is operating without error and is switching between cables on a 1-minute interval. Then, while the system is using the cable subsystem where a cable substitution is to be made, remove the tee connector of the cable being used from the CLCN A/B board of any node until the system switches to the other cable. **REMOVE THE TEE CONNECTOR FROM THE CLCN A/B BOARD—DO NOT REMOVE LCN CABLES OR TERMINATING RESISTORS FROM THE TEE CONNECTOR.**

The system will stay on the other cable for 10 minutes, then switch cables and remain on that cable for 10 minutes before resuming periodic 1-minute cable swaps. This technique will take longer, but allow more time for cable substitution. Even though more time is available, the cable substitution should be made as expeditiously as possible because of the critical 2-minute time interval described below.

The personnel performing this test must be aware of the following critical time intervals. Each node sends four status broadcasts in two minutes. If a node does not get at least one of those status broadcasts within two minutes, it "votes" against the node. If three nodes all fail to hear at least one status broadcast message from a specific node within a 2-minute window, that node will be isolated. Also, if a single node does not hear from another node within 3 minutes, it isolates it. This LCN cable diagnosis procedure attempts to avoid accidental loss of communications and thus avoid these situations. If a mistake is made and errors occur, cables should be restored as soon as possible and sufficient time given for the system to stabilize before proceeding further.

If no bad cables are located and errors persist, TAC management should contact IAC Engineering for assistance.

6.2.3.7 Error analysis

The diagnostic display provides an automatic means for analyzing the LCN cable problem using an algorithm. The algorithm uses the communication data between the nodes to identify the nodes that are causing the LCN cable errors. The identified nodes are displayed in the descending order, based on the error rates/messages. In addition, the algorithm collects the communication errors and the message counts between the nodes.

The algorithm is associated with the following two diagnostic displays:

- Error rate selection display (\$LNDGERS)
- Error rate analysis display (\$LNDGERA)

The diagnostic displays represent the two phases of the diagnostic algorithm. The first phase of the algorithm identifies the nodes that have communication difficulties with other nodes. The second phase of the algorithm analyzes the communication data between the nodes to identify and prioritize the nodes causing the LCN cable errors.

The Error Rate Selection display runs the first phase of the diagnostic algorithm. To identify the nodes which are causing problems, the algorithm collects the state of the node and the communication errors. The algorithm filters the nodes that are in Shut-down, Failed, and Isolated state. Based on the error rates, the identified nodes are displayed in the descending order and analysed further. Therefore, the final list includes the nodes that are active and running. You can select the nodes that can be included in the second phase of the diagnostic analysis. The number of nodes for further analysis can vary from 1 through 10.

The Error Rate Selection display can be accessed through the LCN Fault

05 Aug 08 11:48:05 1

\$LNDGERS
LCN CABLE DIAGNOSTICS

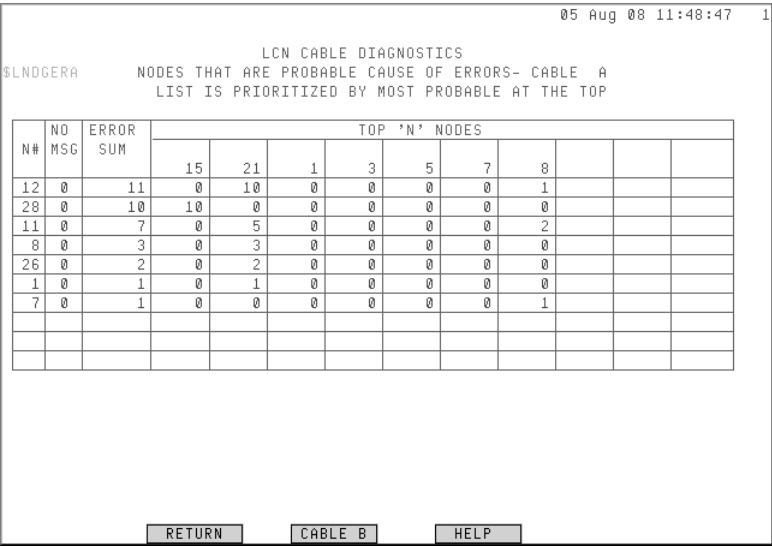
MAIN MENU
ERROR RATES OF ALL NODES ON NETWORK FOR CABLE A

UPDATED - 11:48:03 105 08-05-08

N#	ERROR RATE	INCLUDE Y/N	N#	ERROR RATE	INCLUDE Y/N	N#	ERROR RATE	INCLUDE Y/N	N#	ERROR RATE	INCLUDE Y/N
15	1	YES	27	0	YES						
21	1	YES	28	0	YES						
1	0	YES	29	0	YES						
3	0	YES	30	0	YES						
5	0	YES	41	0	YES						
7	0	YES	54	0	YES						
8	0	YES									
9	0	YES									
11	0	YES									
12	0	YES									
16	0	YES									
20	0	YES									
22	0	YES									
24	0	YES									
25	0	YES									
26	0	YES									

ENTER THE NUMBER OF NODES TO BE CONSIDERED :

UPDATE
CABLE B
ANALYZE

Diagnosis target in the \$LNDGCBL display.	
The Error Rate Analysis display runs the second phase of the diagnostic algorithm. Refer to the screen shot of \$LNDGERA on the right pane for details. The algorithm performs the following functions: Sorts the nodes which are sending no messages and encountering more errors. The primary sorting is done on the messages (NO MSG column) and then on the errors (ERROR SUM column). The errors in the communication between nodes can be read from the matrix following the grid. Example: The errors in the messages sent from Node 28 to Node 15 are 10. If there are no messages and therefore there are no errors, the number zero is indicated in yellow.	
The number of messages received and the error contribution from the other nodes are analyzed for every node in the top N list. - If a node in the network is unable to communicate with a node in the top N list, the node must be monitored. - If a node fails to communicate with multiple nodes in the top list, the node must be prioritized in the list. - If a node can communicate with the other nodes in the top list, the node's error contribution to other nodes is considered for further analysis. - For Trouble shooting, the node selection should be in Top-down order.	

6. TROUBLESHOOTING

6.2. LCN Cable Diagnosis

This analysis can be carried out for Cable A and Cable B separately. The results of the analysis are available even after you navigate to another display and return to this display. The results are available until another analysis is initiated.

The Ping Command display enables you to verify the communication between the selected nodes through a specific cable. You can verify the sending/receiving capability and the cross cable existence for any selected node. Refer to the screen shot of \$LNDGPNG on the right pane for details.

Attention: The crossed cable message shall be sent by the cable master node alone. Hence if the cable master node has K2LCN with Firmware revision P or greater or K4LCN board, the crossed cable identification mechanism shall work irrespective of other nodes platforms.

29 Jul 08 12:24:01 2

\$LNDGPNG LCN Cable Diagnostics : Ping Command

NODE: 2

Node / Cable selection for Ping Command

Source Node Number : 2

Destination Node Number : 7

Selected Cable : A

PING

Ping Command Status

Ping From Node: 2 Ping To Node: 7

7 2

RECEIVED ACK MESSAGE SUCCESSFULLY SENT ACK MESSAGE SUCCESSFULLY

ELAPSED TIME(IN MS): 111

ATTENTION: You may not receive correct results when multiple nodes are trying to connect to the same node, using the Ping functionality.

You can enter the source and destination node number and select the cable in the Ping Node display. The initial value for the source node number is the node number on which the display is invoked.

The ping functionality updates the following parameters:

Parameter...	Is updated with...
For source node	
PingToNode	Destination node number
PingToCable	Selected cable number
For destination node	
PingToNode	0
PingToCable	A

The source node switches to the selected cable and then the pinging is initiated. However, the source node does not switch the cable in the following conditions:

- If the selected cable for pinging is the active cable.
- If the switching interval is more than one second.
- If the Cable Swap Inhibit is ON and the other cable is active. Example: Cable A is active and B is selected for pinging and Swap Inhibit is ON.

After switching to the PingToCable, the source node sends a message to the destination node. The message includes the source node number and the selected cable. The message is transmitted to the destination node through the selected cable. After receiving the message, the destination node updates the following parameters:

Parameter...	Is updated with...
For destination node	
PingToNode	Source node number
PingToCable	Selected cable number

In addition, the destination node sends an acknowledgement to the source node through the active cable. The acknowledgement includes the destination node number and the selected cable.

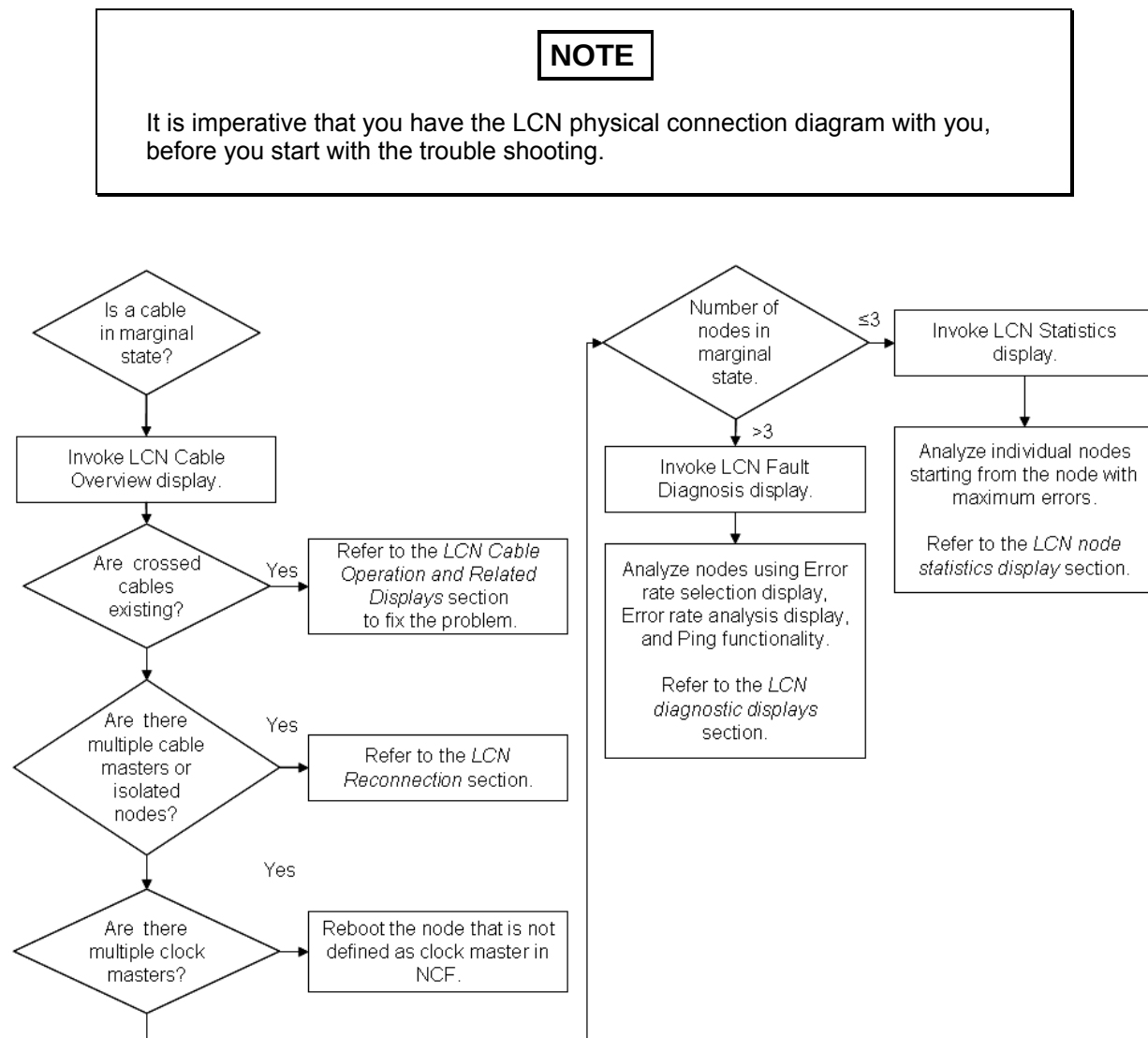
At each stage of the ping functionality, the source and the destination node updates the PingStatus parameter. The ping status is represented as values and is updated, based on the success/failure of the functionality. If the acknowledgement from the destination is not received within 150 milliseconds, the ping status changes to timeout. The following table maps the values with the status of the functionality at each stage and indicates the display behavior of the PingToNode parameter:

PingStatus	Status of each stage	PingToNode display behavior
0	Initial state	Gray
1	Sent Message Successfully	Yellow
2	Received Message Successful	Yellow
3	Sent Ack Message successfully	Yellow
4	Received Ack message Successfully	Green
5	Send Message Failed	Red
6	Send Ack Message Failed	Yellow
7	Send Cable switch Failed	Red
8	Ping timeout	Red

6.2.3.8 Troubleshooting LCN communication problems

The marginal status of the cable and the auxiliary marginal status of the node are indicated in the System Status display. You can analyze the LCN communication problem using the LCN Cable Overview display and the LCN node statistics display.

To analyze the LCN communication problem, follow the steps explained in the flow diagram.





TIP

If a communication problem is detected in a network with bridging devices such as LCN extender/TPN Bridge, ensure to inspect them.

The neighboring nodes that are physically connected to the marginal/suspected node need to be inspected for any communication problem. The inspection should cover the following network components:

- Tee connectors
 - Paddle boards
 - Cables
 - Connectors
-

6.2.4 LCN Diagnostics tool

6.2.4.1 Overview

The LCN Diagnosis tool is a standalone personality image that runs on GUS/EST. The tool helps to investigate the LCN network transaction anomalies. The tool can be loaded on an existing node and its assigned node number can be used for referring the node. If an additional node is added in the network to load the tool, then an unused node number or a number between 97 and 127 is assigned to the new node. The status of the node (numbered 1 through 96) appears as SUSPCT ISOL in the System Status Display.

6.2.4.2 Pre-requisite

The node, on which the tool is running, must have the following:

- Connection to the active LCN
- LCNP4E board with the associated firmware Rev F and above

The LCN Diagnostics Tool R100.1 NodeMonitor Patch 1 works on Windows XP and Windows 7. The following table identifies the platforms that support the tool.

LCN I/F Board	Platform	TPS Node	TPS Release	Experion Node	Experion Release
LCNP4E	T5500	GUS	R401	EST	R3xx
			R410		R400
			R420		

ATTENTION

To invoke the Tool in Windows 7, you should be a member of any one of the Honeywell user group policies.

6.2.4.3 Purpose

After launching, the tool captures the token pass frames and token pass related errors such as media dead, token pass sequence fail, and noise. Online errors are captured and archived. A maximum of 839,680 frames can be stored in the buffer.

The tool helps in finding the cause for the following errors in a node.

- ISOLATION: When there is isolation in the network due to multiple Rings/Tokens, the tool assists in identifying the nodes with errors.
- Media Dead: When the cables are faulty and there are instances of Media Dead, the tool assists in identifying the individual node that is missing the token.

ATTENTION

The tool captures the errors/frames only when it is running.

The node switches to offline when the LCN Diagnostics tool is loaded. Therefore, the node's address is not identified by the other nodes in that particular network. As the node is in offline, it can only receive frames and cannot transmit frames. The node does not participate in token passing.

- The tool may not indicate the cause for the error when there is a noise in the cable. If there are errors in the network, the status of the cable changes to SUSPECT. In such a situation, the tool collects the error information and displays the list of probable error nodes.

6.2.4.4 Token rotation sequence

1. The LCNI, ring initiator initiates the token pass. During the token pass,
 - The ring initiator transmits the data and then attempts to pass the tokenOr
 - If there is no data to transmit, passes the token to the next highest address
2. If the next consecutive address is not able to accept the token,
 - The sender node passes the token to the next highest address until a node accepts the token. The sender node remembers the node that accepted the token.
3. In the next cycle of token pass,
 - The sender node tries to pass the token to the node that did not accept the token in the previous instance.
 - If the node does not accept the token, the sender node passes the token to the node that accepted the token in the previous instance.

6.2.4.5 Installing LCN Diagnostics tool

Use the procedure to install the LCN Diagnostics tool in a GUS node.

Table 6-2 Installing LCN Diagnostics tool

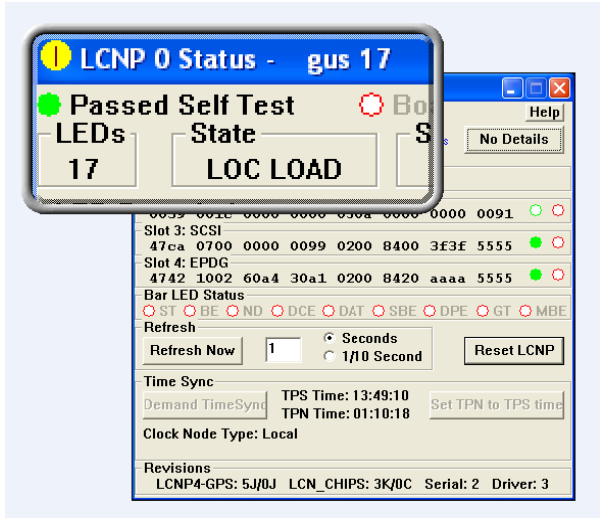
Step	Action
1	Copy the Diagnostics tool (disk_&Z18.LCN) to the hard disk of a GUS machine. Ensure that you have the administrator privilege.
2	Create an emulated disk (disk_&Z18.LCN file). For more information about creating emulated disk, see <i>System Configuration Utility User's Guide</i> .
3	Reset LCNP and wait until the status of the GUS is changed to POWER_ON. After successfully loading the tool, the status of the LCNP changes to LOC LOAD.
	
4	Click the Load button in the status bar of the Native Window.
5	Type the drive number in which the disk_&Z18.LCN is loaded.
	The tool is launched and the menu items are displayed.
	 ATTENTION Verify the TPS and the TPN time in the LCNP status window. If the time displayed in the tool does not match with the real time or the TPS time, the following might be the reason. - When the node is reset, the TPN time in the LCN diagnostic tool is reset. - TPS time corresponds to the local time set in the computer. - Therefore, the TPN time is node specific and differs between the nodes.

Table 6-2 Installing LCN Diagnostics tool

Step	Action	
		
		TIP There are no changes required in the NCF to load this tool.

6.2.5 PC Archiving tool

6.2.5.1 Overview

The PC Archiving tool is part of the LCN Diagnostics tool and helps to investigate the LCN transaction anomalies in detail. The tool can be run on a GUS/EST. The tool archives the token pass frame transactions and the errors occurring in the network. The archived frames are available for later Diagnostics.

The PC Archiving tool has the following tools.

- Data Capture / LCNPMemReader Tool
- Stand Alone Analysis Tool / Node Monitor Tool

6.2.5.2 Installing PC archiving tool

Perform the following procedure to install the PC archiving tool on a GUS node.

Table 6-3 Installing PC archiving tool

Step	Action
1	Insert the installation CD in the CD ROM drive.
2	Follow the instructions on the screen to complete the installation.

6.2.6 LCN Diagnostics tool – Main menu

6.2.6.1 Overview

The LCN Diagnostics tool help in diagnosing the LCN cable problems. However, when there is a noise in the network due to breakage of cable or when there is a termination problem, the tool might process invalid data. The tool identifies the invalid nodes; however the frames may not display any identifiable errors.

ATTENTION

Ensure that you do not press the **Del** key before selecting any of the menu options in the LCN Diagnostics tool. If you are in the main menu and if you press the **Del** key followed by a menu selection, the tool returns to the main menu. In that case, you will need to select the menu option again.

The following table illustrates when and why to use the menu options of the LCN Diagnostics tool.

When...	And to...	Use...
There are multiple Media Dead occurring on the network	Filter the source and the destination address	The Monitor Token Pass frames on Cable A and Cable B menu option to diagnose only the online Media Dead on the active cable. <i>See Monitor Token Pass frames on Cable A and Cable B.</i>
Cable A is in suspect	Capture frames on Cable A irrespective of Cable A/B being active cable	The Monitor Token Pass frames on Cable A menu option to identify the active nodes on Cable A. <i>See Monitor Token Pass frames on Cable A or Cable B.</i>
Cable B is in suspect	Capture frames on Cable B irrespective of Cable A/B being active cable	The Monitor Token Pass frames on Cable B menu option to identify the active nodes on Cable B. <i>See Monitor Token Pass frames on Cable A or Cable B.</i>
	Measure the LCN bandwidth utilization for a selected period of time interval.	The Bandwidth Utilization menu option. <i>See Bandwidth Utilization.</i>

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When...	And to...	Use...
There are multiple masters in the network which causes the swap	View the cable switch and the node that caused the cable switch	The Monitor cable in use menu option to analyse the cable swap. Note: The Media Dead on the active cable is also detected. <i>See Monitor cable in use.</i>
The Token Pass Sequence Fails, there is a noise on the LCN, and the Media Dead error occurs	Record all the token pass frames and to look for errors such as Media Dead	The Monitor errors menu option. <i>See Monitor errors.</i>
	Diagnose the low level problems at an early stage	The Active nodes on Cable A and B menu option. <i>See Active nodes on Cable A/B.</i>

6.2.6.2 Monitor Token Pass frames on Cable A and Cable B

The **Monitor Token Pass frames on Cable A and Cable B** menu option is useful to capture only the online errors on the active cable. If Cable swapping is inhibited (intentional / due to errors), then this menu displays only the errors on the active cable and captures the Media Dead.

To understand the Token pass ring topology, the source and the destination addresses are useful. The Token pass frame sequence is displayed in the **Display Frame** menu, which gives the Token pass ring sequence. The Token pass sequence indicates which node is communicating with the specified known destination address and the unknown address.

Perform the following procedure to monitor all the token pass frames in the network.

Table 6-4 Procedure – Monitor token pass frames on Cable A and Cable B

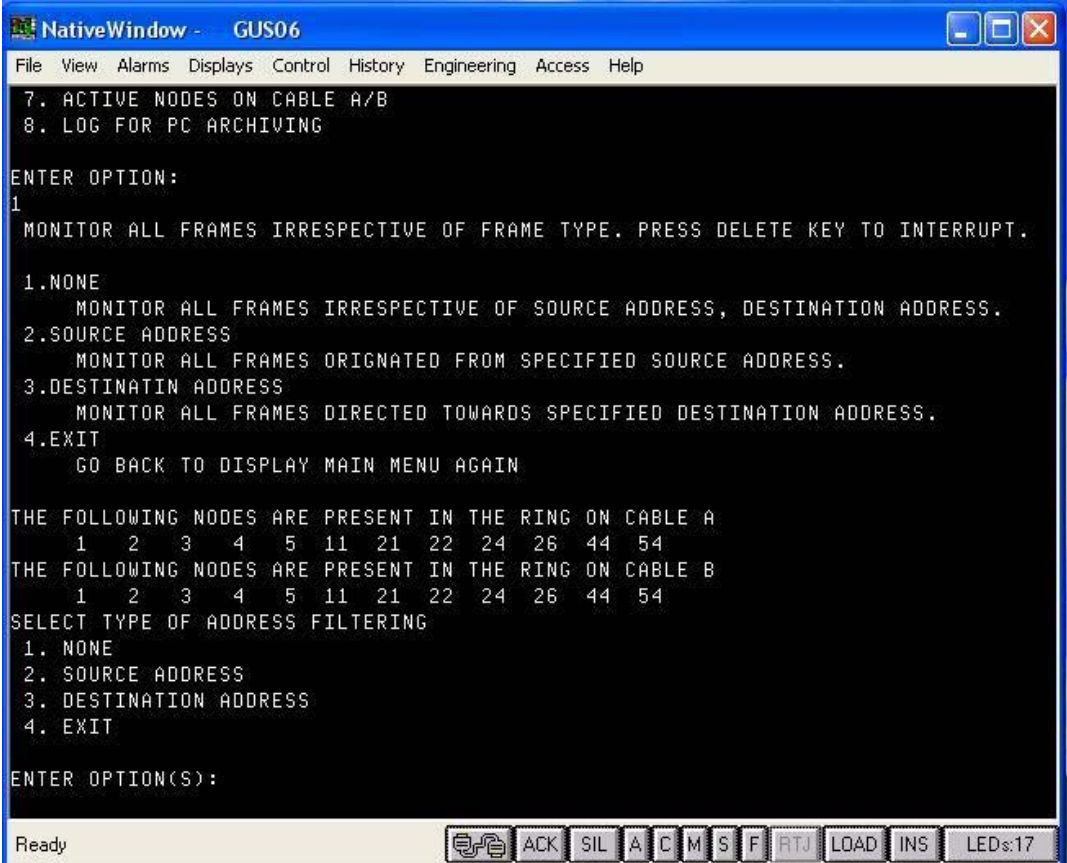
Step	Action
1	Press <1>, to monitor all the token pass frames in the network.
	 The screenshot shows the 'NativeWindow - GUS06' application. The menu bar includes File, View, Alarms, Displays, Control, History, Engineering, Access, and Help. The main window displays the following text: <pre> 7. ACTIVE NODES ON CABLE A/B 8. LOG FOR PC ARCHIVING ENTER OPTION: 1 MONITOR ALL FRAMES IRRESPECTIVE OF FRAME TYPE. PRESS DELETE KEY TO INTERRUPT. 1.NONE MONITOR ALL FRAMES IRRESPECTIVE OF SOURCE ADDRESS, DESTINATION ADDRESS. 2.SOURCE ADDRESS MONITOR ALL FRAMES ORIGINATED FROM SPECIFIED SOURCE ADDRESS. 3.DESTINATION ADDRESS MONITOR ALL FRAMES DIRECTED TOWARDS SPECIFIED DESTINATION ADDRESS. 4.EXIT GO BACK TO DISPLAY MAIN MENU AGAIN THE FOLLOWING NODES ARE PRESENT IN THE RING ON CABLE A 1 2 3 4 5 11 21 22 24 26 44 54 THE FOLLOWING NODES ARE PRESENT IN THE RING ON CABLE B 1 2 3 4 5 11 21 22 24 26 44 54 SELECT TYPE OF ADDRESS FILTERING 1. NONE 2. SOURCE ADDRESS 3. DESTINATION ADDRESS 4. EXIT ENTER OPTION(S): </pre> The status bar at the bottom shows 'Ready' and a row of buttons: ACK, SIL, A, C, M, S, F, RTJ, LOAD, INS, and LEDs:17.
2	Select one of the following types of address filtering.
	None: To view all the Token Pass frames in the network. Source Address: To view all the Token Pass frames originating from a specified source address. Note: You must type the address of the active node available on Cable A and B. Destination Address: To view all the Token Pass frames directed towards a specified destination address.
	The monitored frames are stored in the buffer. The frames are displayed under the following categories. • Frames Stored: Displays the number of frames stored in the local buffer of the GUS. • Error Count: Displays the error count. If an error occurs, the error count is incremented. You must analyze, in the network where the error is originating from.

Table 6-4 Procedure – Monitor token pass frames on Cable A and Cable B

Step	Action
3	Press <4> to exit this option. • Select Terminate to exit. Or • Select Display Frames to analyze the token passing sequence to find which node had the last good token. Note: Press Q to exit the Display Frames.
	 TIP Monitoring stops when the delete key is pressed. The buffer is overwritten by incrementing the error count.

6.2.6.3 Monitor Token Pass frames on Cable A or Cable B

The Monitor Token Pass frames on Cable A and Cable B menu option is used for identifying the active nodes on Cable A and Cable B respectively.

The Display Frame option displays the Token pass ring and the error occurred on a particular node, if any. The Token pass anomaly helps in identifying the physical connection problems, if any. In addition, the Display Token pass ring option displays the current active node list. This list is used for finding the following:

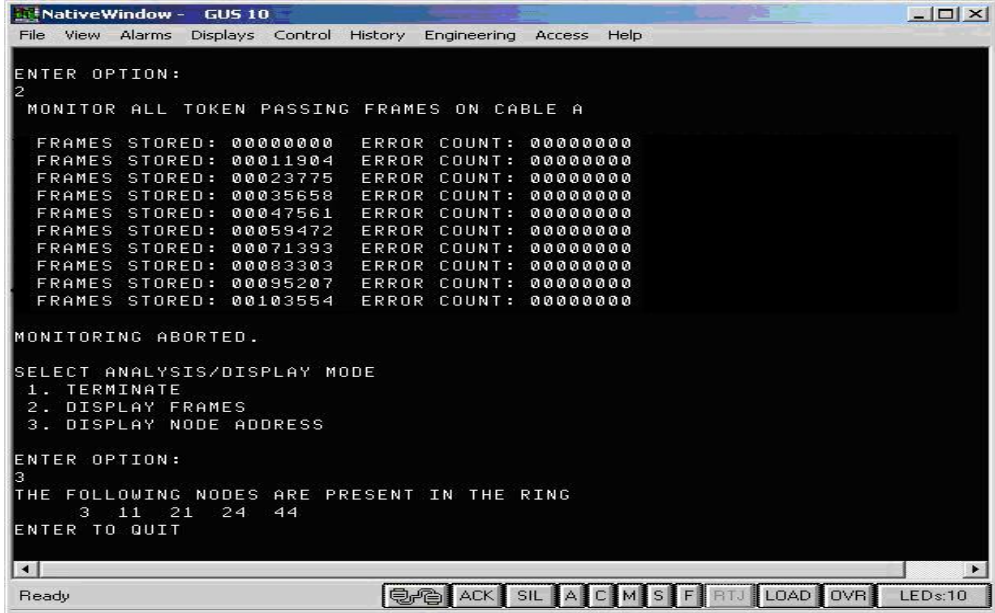
- The node that caused the cable to go to SUSPECT.
- The node that is isolated on Cable A or Cable B.

ATTENTION

If Cable swapping is inhibited on Cable A (intentional / due to errors), then this menu does not report any errors, other than noise. This menu does not display the node which is in ISOLATE state on Cable A or Cable B.

Perform the following procedure to monitor all the token pass frames on Cable A or Cable B.

Table 6-5 Procedure – Monitor token pass frames on Cable A or on Cable B

Step	Action
1	Press <2> to monitor all the token pass frames on Cable A. Or Press <3> to monitor all the token pass frames on Cable B.
	 The screenshot shows the GUS 10 NativeWindow interface. The menu bar includes File, View, Alarms, Displays, Control, History, Engineering, Access, and Help. The main display area shows the following text: <pre> ENTER OPTION: 2 MONITOR ALL TOKEN PASSING FRAMES ON CABLE A FRAMES STORED: 00000000 ERROR COUNT: 00000000 FRAMES STORED: 00011904 ERROR COUNT: 00000000 FRAMES STORED: 00023775 ERROR COUNT: 00000000 FRAMES STORED: 00035658 ERROR COUNT: 00000000 FRAMES STORED: 00047561 ERROR COUNT: 00000000 FRAMES STORED: 00059472 ERROR COUNT: 00000000 FRAMES STORED: 00071393 ERROR COUNT: 00000000 FRAMES STORED: 00083303 ERROR COUNT: 00000000 FRAMES STORED: 00095207 ERROR COUNT: 00000000 FRAMES STORED: 00103554 ERROR COUNT: 00000000 MONITORING ABORTED. SELECT ANALYSIS/DISPLAY MODE 1. TERMINATE 2. DISPLAY FRAMES 3. DISPLAY NODE ADDRESS ENTER OPTION: 3 THE FOLLOWING NODES ARE PRESENT IN THE RING 3 11 21 24 44 ENTER TO QUIT </pre> The status bar at the bottom shows 'Ready' and a row of buttons: ACK, SIL, A, C, M, S, F, RTJ, LOAD, OVR, and LEDs:10.
	The source and the destination address of all the token pass frames received on Cable A/B are stored in the buffer.
	The monitoring stops when the operator intervenes by hitting the Load key
	The token pass frames can be analyzed after the monitoring is stopped.
2	Click one of the following options that appear in the screen after the monitoring stops.
	Terminate: The Frames Received, Frames Stored, and Media Dead are displayed. Display Frames: The sequence in which the token is passed from one node to the other is displayed. If the result is not as expected, you must further investigate with the help of the site's topology map. Note: Press Q to exit the Display Frames. Display Node Address: The node numbers participating in the token pass are displayed.
	The node numbers available in both the cables are displayed. This helps to troubleshoot. If there are any node numbers that are not displayed, analyze those nodes.

6.2.6.4 Bandwidth Utilization

ATTENTION

The Bandwidth Utilization referred here does not relate to the LCN load as determined by the System Load calculation tool.

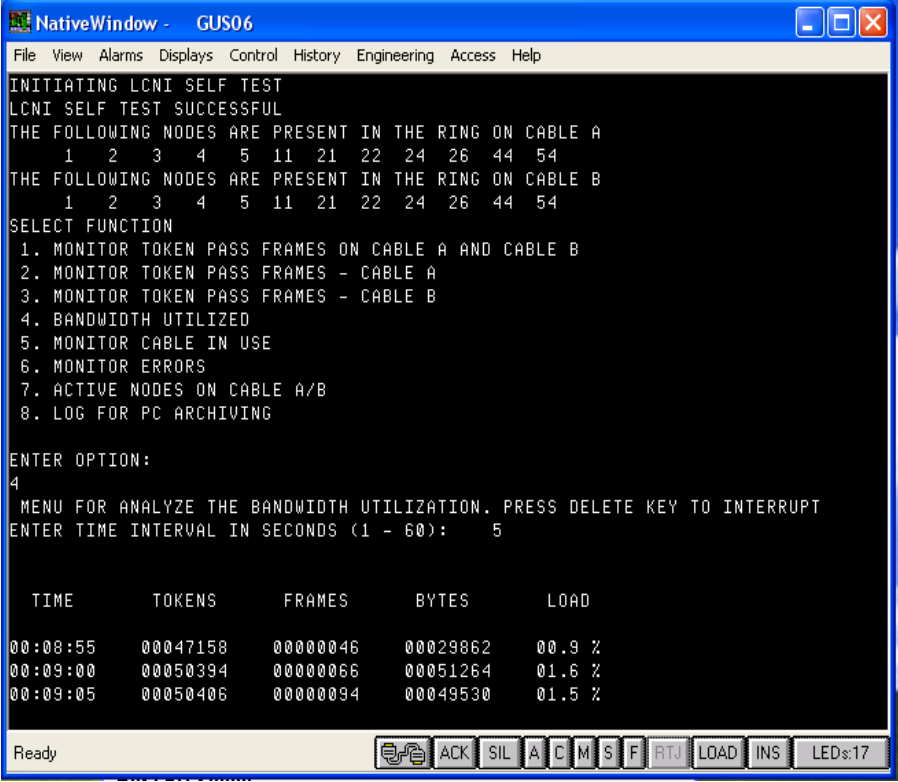
Measuring the bandwidth utilization helps to analyze the overall performance of the entire network. This provides the percentage of data load transmitted for a specified time interval. The tool measures the LCN communication load. At the specified time interval, the number of token pass frames, the number of information frames, total frame bytes and percentage of maximum bandwidth utilized is displayed.

Perform the following procedure to measure the LCN communication load.

Table 6-6 Procedure – To measure the LCN communication load

Step	Action
1	Press <4> to measure the LCN communication load (bandwidth utilization) for a time interval.
2	Enter the time interval in seconds. Range for time interval: 1 – 60 seconds.

Table 6-6 Procedure – To measure the LCN communication load

Step	Action
	 <pre> NativeWindow - GUS06 File View Alarms Displays Control History Engineering Access Help INITIATING LCNI SELF TEST LCNI SELF TEST SUCCESSFUL THE FOLLOWING NODES ARE PRESENT IN THE RING ON CABLE A 1 2 3 4 5 11 21 22 24 26 44 54 THE FOLLOWING NODES ARE PRESENT IN THE RING ON CABLE B 1 2 3 4 5 11 21 22 24 26 44 54 SELECT FUNCTION 1. MONITOR TOKEN PASS FRAMES ON CABLE A AND CABLE B 2. MONITOR TOKEN PASS FRAMES - CABLE A 3. MONITOR TOKEN PASS FRAMES - CABLE B 4. BANDWIDTH UTILIZED 5. MONITOR CABLE IN USE 6. MONITOR ERRORS 7. ACTIVE NODES ON CABLE A/B 8. LOG FOR PC ARCHIVING ENTER OPTION: 4 MENU FOR ANALYZE THE BANDWIDTH UTILIZATION. PRESS DELETE KEY TO INTERRUPT ENTER TIME INTERVAL IN SECONDS (1 - 60): 5 TIME TOKENS FRAMES BYTES LOAD 00:08:55 00047158 00000046 00029862 00.9 % 00:09:00 00050394 00000066 00051264 01.6 % 00:09:05 00050406 00000094 00049530 01.5 % </pre>
	For every time interval specified, the following information is displayed. • Number of Token Pass frames • Number of information frames • Total number of frame bytes • Percentage of maximum load
	Bandwidth Utilization is calculated as follows. LCN communication bandwidth is 5mb/sec. Therefore, 5mbits/8bits = 625000 bytes/sec. Percentage of Bandwidth Utilized = ((Number of bytes received/sec) / 625000 bytes) * 100
	Example: Bandwidth Utilized = ((49530/5)/625000)*100 = 1.5%
3	Press Del Key to exit from this menu.

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6.2.6.5 Monitor cable in use

The **Monitor cable in use** menu option monitors only healthy cables.

This menu option is used for analyzing the cable swap. When there is a media error, the tool displays the probable node with the problem. This helps to troubleshoot the cause of the error. If Cable swapping is inhibited (intentional / due to errors) then this menu does not help in identifying the cable swap.

If there are multiple masters, the tool displays multiple nodes that initiated the cable swap. This helps to initiate actions to identify a single cable master.

Perform the following procedure to monitor the cable that is being used.

Table 6-7 Procedure – Monitor cable in use

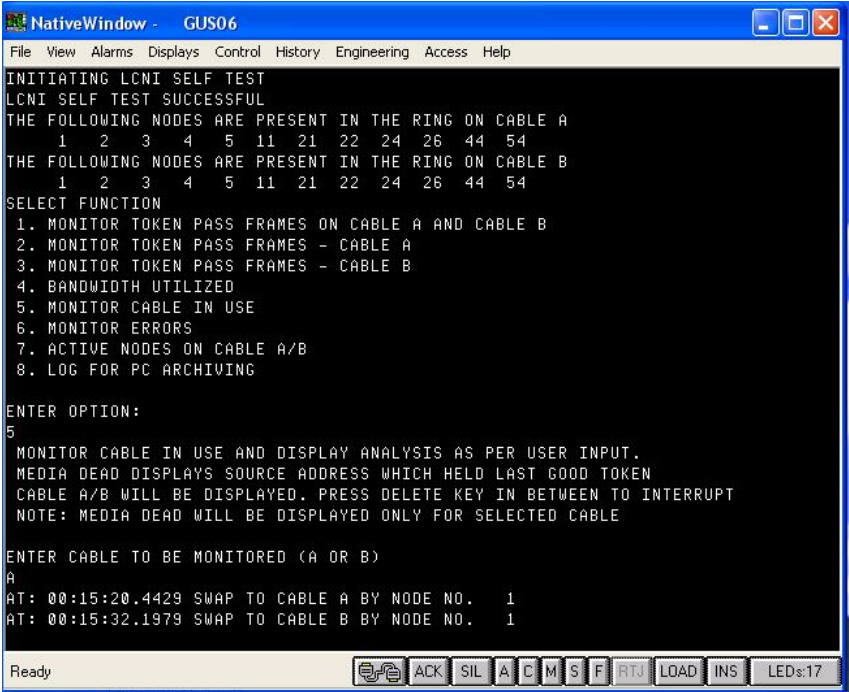
Step	Action
1	Press <5> to monitor the cable in use.
2	Specify the cable to be monitored (A or B).
	
	The cable swap information and the node number that triggered the swap are displayed. This information helps to analyze the Media Dead and the Cable Suspect.
	Media Dead is displayed only for an active cable.

Table 6-7 Procedure – Monitor cable in use

Step	Action
	The US node with the lowest address in the LCN is the cable master node. The master node controls the automatic periodic swapping between the cables A and B. In addition, it declares the status of the cable as GOOD or SUSPECT.

6.2.6.6 Monitor errors

The **Monitor errors** menu option is used for capturing the online errors such as Media Dead, Token pass sequence fail, and noise on the LCN, irrespective of the cable status. This menu displays the first error that occurred on the network. It might display invalid error information on the cable which is already in SUSPECT. When there is a breakage in cable, noise on LCN or cable end terminators, invoking the tool cannot identify the nodes that caused this error.

If Cable swapping is inhibited (intentional / due to errors), then this menu displays errors only on the active cable. It might not display the errors other than noise on SUSPECT (inhibit) Cable. If Cable A is in SUSPECT (inhibit), then this menu might not show any errors. By default, this menu is always monitoring through Cable B. If LCNI does not receive a token pass addressed to itself or any other address for approximately 18 ms, it indicates that the token-passing ring is collapsed and a Media Dead has occurred.

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Perform the following procedure to monitor all the errors.

Table 6-8 Procedure – Monitor errors

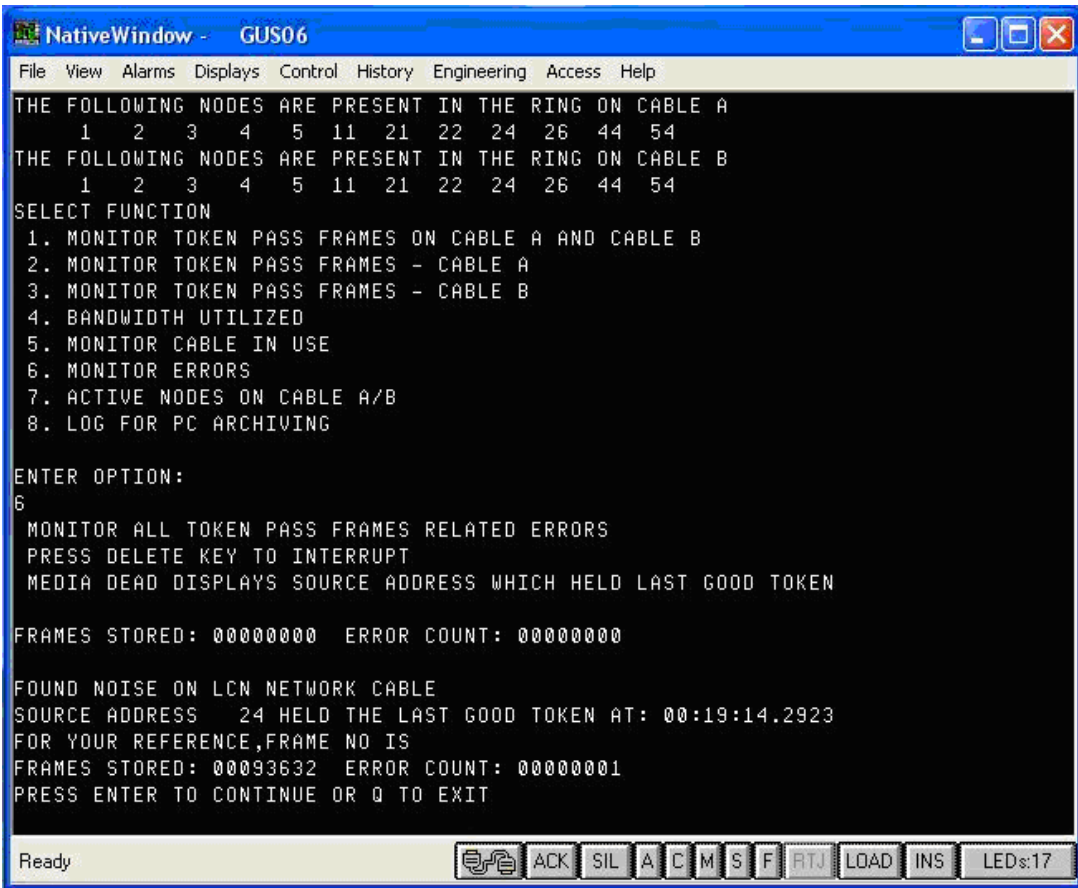
Step	Action
1	Press <6> to monitor all the errors.
	All the Token Pass frames are recorded to identify if the following errors occurred in the frames.
	 <pre> NativeWindow - GUS06 File View Alarms Displays Control History Engineering Access Help THE FOLLOWING NODES ARE PRESENT IN THE RING ON CABLE A 1 2 3 4 5 11 21 22 24 26 44 54 THE FOLLOWING NODES ARE PRESENT IN THE RING ON CABLE B 1 2 3 4 5 11 21 22 24 26 44 54 SELECT FUNCTION 1. MONITOR TOKEN PASS FRAMES ON CABLE A AND CABLE B 2. MONITOR TOKEN PASS FRAMES - CABLE A 3. MONITOR TOKEN PASS FRAMES - CABLE B 4. BANDWIDTH UTILIZED 5. MONITOR CABLE IN USE 6. MONITOR ERRORS 7. ACTIVE NODES ON CABLE A/B 8. LOG FOR PC ARCHIVING ENTER OPTION: 6 MONITOR ALL TOKEN PASS FRAMES RELATED ERRORS PRESS DELETE KEY TO INTERRUPT MEDIA DEAD DISPLAYS SOURCE ADDRESS WHICH HELD LAST GOOD TOKEN FRAMES STORED: 00000000 ERROR COUNT: 00000000 FOUND NOISE ON LCN NETWORK CABLE SOURCE ADDRESS 24 HELD THE LAST GOOD TOKEN AT: 00:19:14.2923 FOR YOUR REFERENCE,FRAME NO IS FRAMES STORED: 00093632 ERROR COUNT: 00000001 PRESS ENTER TO CONTINUE OR Q TO EXIT </pre>
	Token Pass Sequence Fail: The following are the possible causes for the Token Pass Sequence Fail. - Active node on the LCN network is reset - Cable Break - Node isolated - End terminators of LCN cable are removed - Media Dead
	Noise on LCN: The tool detects the noise on the LCN and displays an error message. The noise on the network is due to the invalid data in the received frames. Therefore, an invalid source address (>128) is displayed.

Table 6-8 Procedure – Monitor errors

Step	Action
	Media Dead: The tool identifies the Media Dead and displays the node that held the last good token, irrespective of the cable.
	 TIP The monitoring stops when the Operator intervenes by hitting the Del key.

6.2.6.7 Active nodes on Cable A/B

The **Active nodes on Cable A/B** menu option helps in diagnosing the low level problems at an early stage. It indicates which nodes are communicating on Cable A and Cable B.

Perform the following steps to monitor active nodes on Cable A/B.

Table 6-9 Procedure – Monitor active nodes on Cable A/B

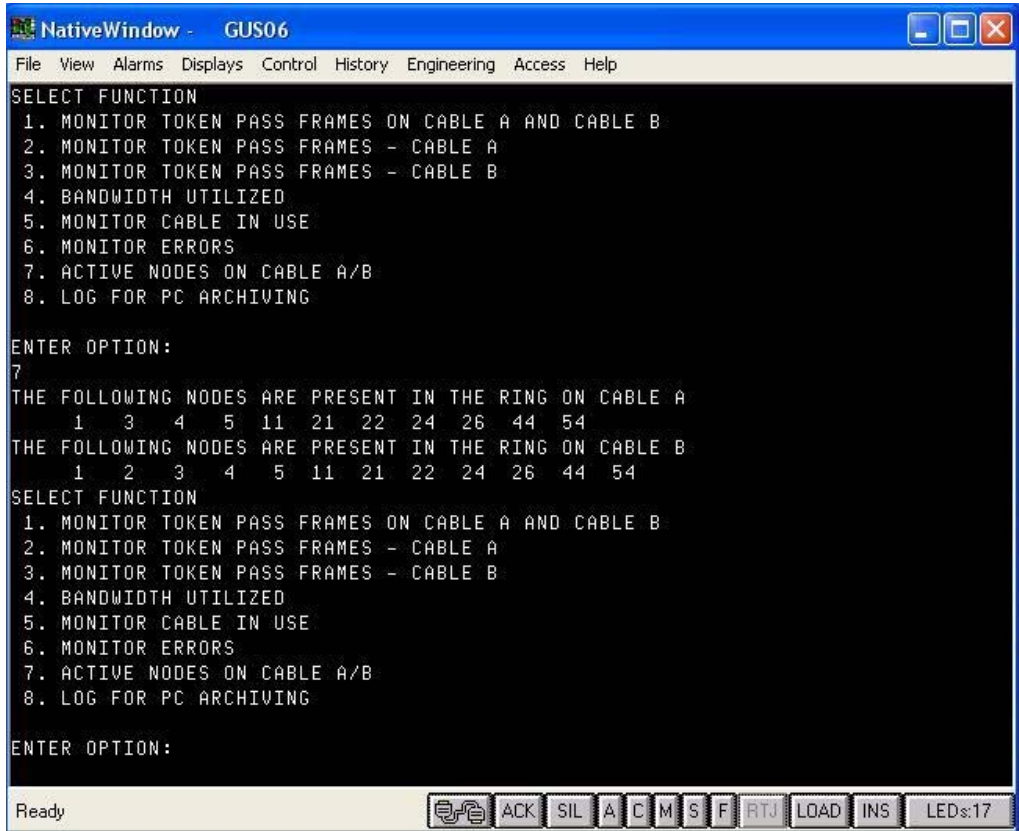
Step	Action
1	Press <7> to monitor active nodes on Cable A/B.
	Lists all the nodes communicating on Cable A and B.
	 The screenshot shows a terminal window titled 'NativeWindow - GUS06'. The menu lists 8 options, with option 7 'ACTIVE NODES ON CABLE A/B' selected. The output shows the following nodes are present in the ring on Cable A: 1, 3, 4, 5, 11, 21, 22, 24, 26, 44, 54. The following nodes are present in the ring on Cable B: 1, 2, 3, 4, 5, 11, 21, 22, 24, 26, 44, 54. The window also shows a status bar at the bottom with 'Ready' and various function keys like ACK, SIL, A, C, M, S, F, RTJ, LOAD, INS, and LEDs:17.

Table 6-9 Procedure – Monitor active nodes on Cable A/B

Step	Action
	The following indicates the error condition. • If the list of nodes on Cable A and B does not match. • If the list does not include all the nodes mentioned in the site's documentation.
	 ATTENTION If there are no active nodes on Cable A or B and the firmware version is not matching with the requirement, the following error message appears. NO FRAMES RECEIVED ON THIS CABLE. EITHER CABLE IS OPEN OR CHECK SETUP. This condition arises due to the improper cable connection to the node on which the tool is loaded. Ensure the hardware and the cable connection is working properly.
	If one or more node numbers of the cable are not displayed, it indicates those nodes are not communicating in the particular cable. The particular node must be analyzed for resolving the issue. If there is a noise in the cable and on repeated usage of this menu option, the tool might display inconsistent node list for the selected cable.

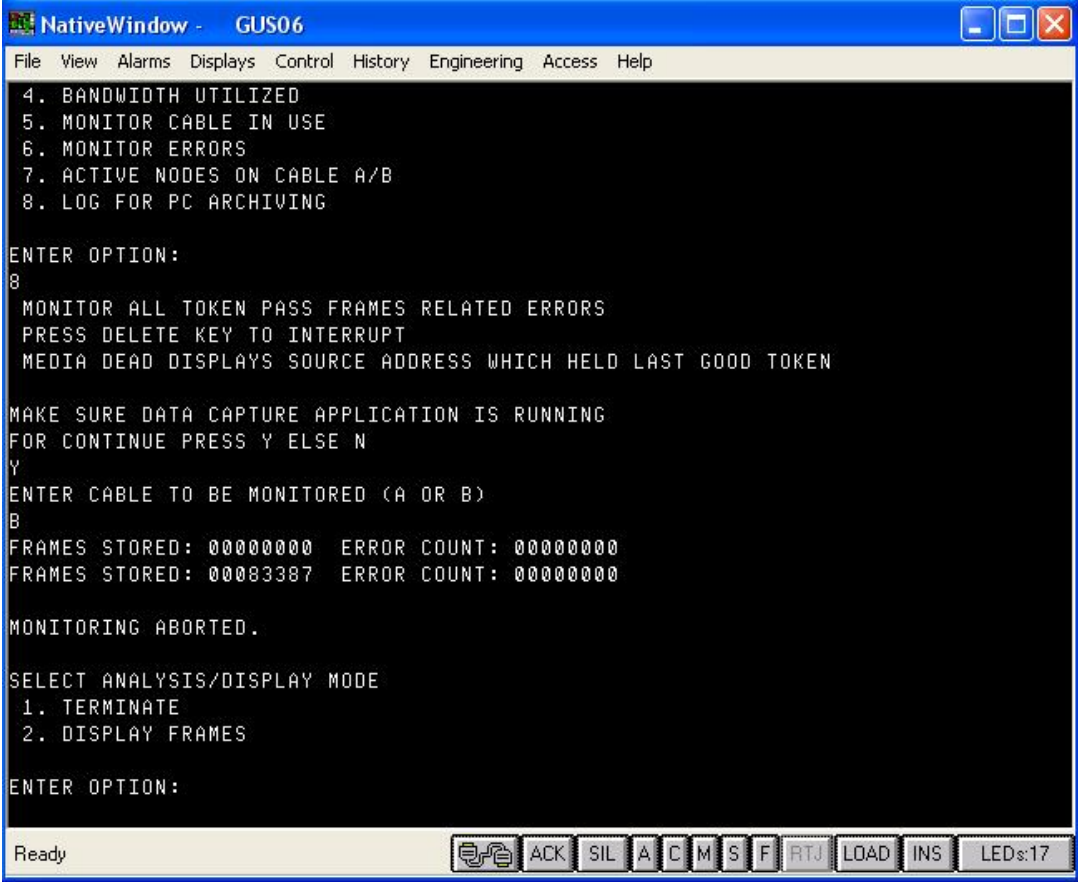
6.2.6.8 Log for PC archiving

Perform the following steps to log on to the PC archiving tool.

Table 6-10 Procedure – Log on to the PC archiving tool

Step	Action
1	Press <8> to log on to the PC archiving tool.
2	• Type Y, if the LCNPMem Reader tool is running. The LCN Diagnostics tool proceeds with the logging for PC archiving. (See Step 3.) Or • Type N, if the LCNPMem Reader tool is not running. The LCN Diagnostics tool displays the main menu (with menu items 1 – 8).
	 ATTENTION If you type Y when the LCNPMem Reader tool is not running, then the LCN Diagnostics tool starts capturing the frames until the buffer is full. When the buffer is full, the following message appears and the tool waits for a confirmation. MAKE SURE DATA CAPTURE APPLICATION IS RUNNING.
3	Type A or B for the cable to be monitored.

Table 6-10 Procedure – Log on to the PC archiving tool

Step	Action				
					
	The following are the criteria for selecting the cable to be monitored.				
	<table> <tr> <th>If...</th><th>Then...</th></tr> <tr> <td>Cable A is OK and Cable B is in SUSPECT</td><td> Select Cable B for monitoring. The tool captures data from the selected cable. If the cable is in SUSPECT after every 10 minutes, the master node swaps the cable from healthy to suspect. When the cable is swapped, the tool captures the information from Cable B too. As there are cable issues, few nodes are not able to communicate and therefore the Token pass fail sequence error occurs on Cable B. The tool displays the nodes with errors. The token ring is re-initialized on Cable A as the token pass failed on Cable B. This causes an error on Cable A too. Hence, when Cable B is selected in this scenario, consider the errors occurring only on Cable B. </td></tr> </table>	If...	Then...	Cable A is OK and Cable B is in SUSPECT	Select Cable B for monitoring. The tool captures data from the selected cable. If the cable is in SUSPECT after every 10 minutes, the master node swaps the cable from healthy to suspect. When the cable is swapped, the tool captures the information from Cable B too. As there are cable issues, few nodes are not able to communicate and therefore the Token pass fail sequence error occurs on Cable B. The tool displays the nodes with errors. The token ring is re-initialized on Cable A as the token pass failed on Cable B. This causes an error on Cable A too. Hence, when Cable B is selected in this scenario, consider the errors occurring only on Cable B.
If...	Then...				
Cable A is OK and Cable B is in SUSPECT	Select Cable B for monitoring. The tool captures data from the selected cable. If the cable is in SUSPECT after every 10 minutes, the master node swaps the cable from healthy to suspect. When the cable is swapped, the tool captures the information from Cable B too. As there are cable issues, few nodes are not able to communicate and therefore the Token pass fail sequence error occurs on Cable B. The tool displays the nodes with errors. The token ring is re-initialized on Cable A as the token pass failed on Cable B. This causes an error on Cable A too. Hence, when Cable B is selected in this scenario, consider the errors occurring only on Cable B.				

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Table 6-10 Procedure – Log on to the PC archiving tool

Step	Action				
	<table><tr><td>Cable A is in SUSPECT and Cable B is OK</td><td>Select Cable A for monitoring. Consider the errors occurring only on Cable A.</td></tr><tr><td>Cable A and B are in SUSPECT</td><td>Select the Active nodes on Cable A/B menu item of the LCN Diagnostics tool to identify the nodes that are not active in the network. Analyze only those nodes if they cause an error.</td></tr></table>	Cable A is in SUSPECT and Cable B is OK	Select Cable A for monitoring. Consider the errors occurring only on Cable A.	Cable A and B are in SUSPECT	Select the Active nodes on Cable A/B menu item of the LCN Diagnostics tool to identify the nodes that are not active in the network. Analyze only those nodes if they cause an error.
Cable A is in SUSPECT and Cable B is OK	Select Cable A for monitoring. Consider the errors occurring only on Cable A.				
Cable A and B are in SUSPECT	Select the Active nodes on Cable A/B menu item of the LCN Diagnostics tool to identify the nodes that are not active in the network. Analyze only those nodes if they cause an error.				
	The token pass frames and all the token pass errors are archived. Later, the archived data can be analyzed.				
4	Press Del to stop the PC Archiving operation. • Select Terminate to exit. Or • Select Display Frames to analyze the token passing sequence. Note: Press Q to exit the Display Frames.				
	For more information about the PC archiving tool, see <i>PC Archiving tool</i> .				

6.2.7 LCNPMem Reader Tool

The LCNPMem Reader tool is a standalone online tool that runs on GUS/EST, to assist with offline Diagnostics of LCN cable transaction anomalies.

The LCNPMem Reader tool can be used for capturing the LCN cable transaction. The custom LCN personality loaded into the system, logs the token pass details into the LCNP4e board memory. This information must be transferred to a computer for further analysis. The LCNPMem Reader tool reads the data from the LCNP4e board memory (logged by the LCN personality). The tool generates a text file that will be used for further analysis.

ATTENTION

Only one instance of the LCNPMem Reader tool must be running. Multiple instances of the tool generate multiple files with the same time stamp and the data in the files might be lost.

Perform the following steps to capture the data using the LCNPMem Reader tool.

Table 6-11 Procedure – LCNPMem Reader tool

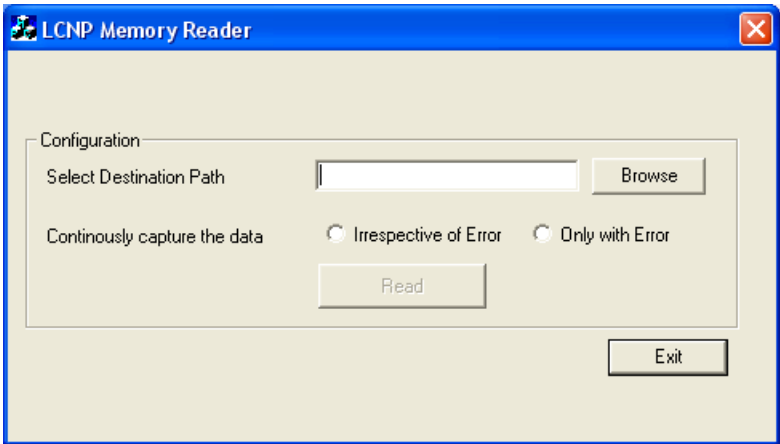
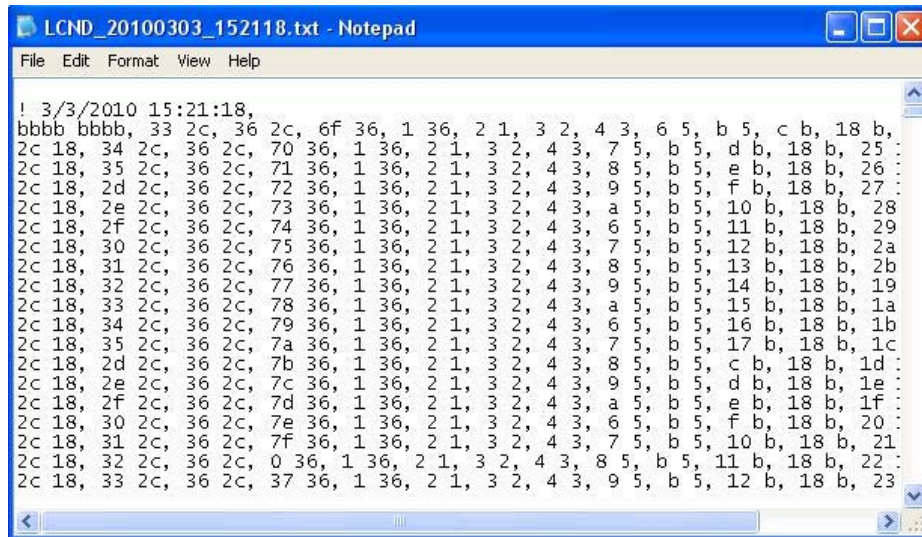
Step	Action
1	Choose Start > All Programs > Honeywell TPS > LCNPMemReader .
	The LCNPMemReader dialog box appears.
	TIP To access the tool on an EST node, choose Start > All Programs > Honeywell Experion PKS>TPS Application> LCNPMemReader.
	
2	Click Browse to select the destination path. Ensure that the destination path is not in the Desktop. The text file generated by the tool is saved in the selected destination path.

Table 6-11 Procedure – LCNPMem Reader tool

Step	Action	
		TIP The tool stops collecting the data if the required space is not available to create the text file. Therefore, select the destination path considering the space requirement.
3	Click one of the following to specify the type of data to be captured.	
	• Irrespective of Error: Captures all data (with zero or more errors) loaded into the LCNP4e board memory by the LCN personality. You will require 7 to 8 GB of disk space to store the data collected in 24 hours. All successful token pass frames are captured along with errors, if any. • Only with Error: Captures data with one or more errors. The size of the data depends on the number of error blocks captured.	
4	Click Read. The tool performs the following: • Creates the text file to store the data (token pass frames and all token pass errors). The name assigned to the text file is in the following format. – LCND_Date_Time.txt (when there are no errors) – LCND_Date_Time_E.txt (when there is at least one error) • Collects the data from the LCNP board memory. • Creates a log file that lists all the filenames that contain the error information.	
		ATTENTION: The data collection is stopped if file handling errors occur. • The following are a few examples of file handling errors. • File is not created • Disk space not available
5	To stop collecting the data from the buffer, perform the following: • Stop the LCN Diagnostics tool on the Native Window. • After 15 seconds, click Exit on the LCNPMemReader tool.	
	You can now analyze the data using the Node Monitor tool.	

The following figure illustrates a sample of the data captured.



The maximum size of the file is approximately 15MB. Each word is separated by a space. Two consecutive words are separated by a comma. Every time the tool is invoked, a new data file and a log file are created irrespective of the size of the last file generated.

6.2.8 Node Monitor tool

The Node Monitor tool processes the text files generated by the LCNPMem Reader tool. The tool then provides a Graphical User Interface (GUI) that allows the user to analyze the data and generate reports. The reports are of different types:

- Node report
- Error report
- Excel report
- Summary report

The reports can be analyzed further to fix the problem in the LCN network. The tool supplements to the existing fault information reported by the LCN Diagnostic Tool.

The tool might display the invalid node numbers due to the noise in the network. The invalid node numbers can be removed from the display using the Filter option in the tool. Therefore, the tool reports only the active nodes.

Perform the following steps to analyze the data using the Node Monitor tool.

Table 6-12 Procedure – Node Monitor tool

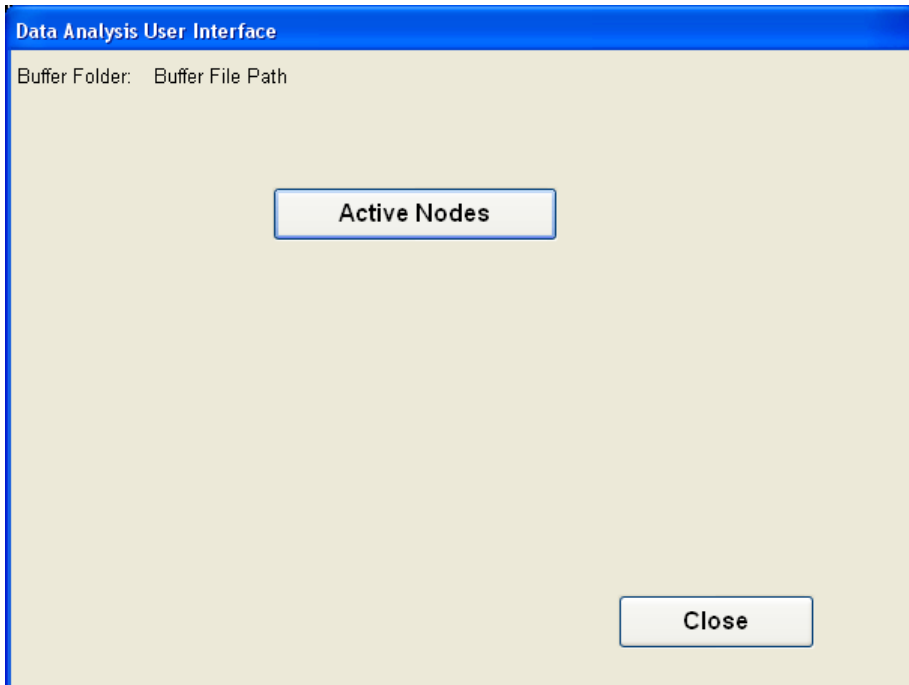
Step	Action
1	Choose Start > All Programs > Honeywell TPS > NodeMonitor.
	The Data Analysis User Interface appears. 

Table 6-12 Procedure – Node Monitor tool

Step	Action
 TIP To access the tool on an EST node, choose Start > All Programs > Honeywell Experion PKS>TPS Application>NodeMonitor.	
2	Click Active Nodes . The Active Nodes Selection Form appears.
	The node numbers (1 through 96) appear in the Active Node List . The node numbers can be re-arranged (based on the plant topology) in the Active Node List using the Up and Down buttons.
3	Select the inactive nodes from the list and then click Remove Selected . The inactive nodes appear in the Non Active Node List . If there are active nodes in the Non Active Node List , select the active nodes and then click Add Selected . The selected active nodes are removed from the Non Active Node List and added in the Active Node List .

Table 6-12 Procedure – Node Monitor tool

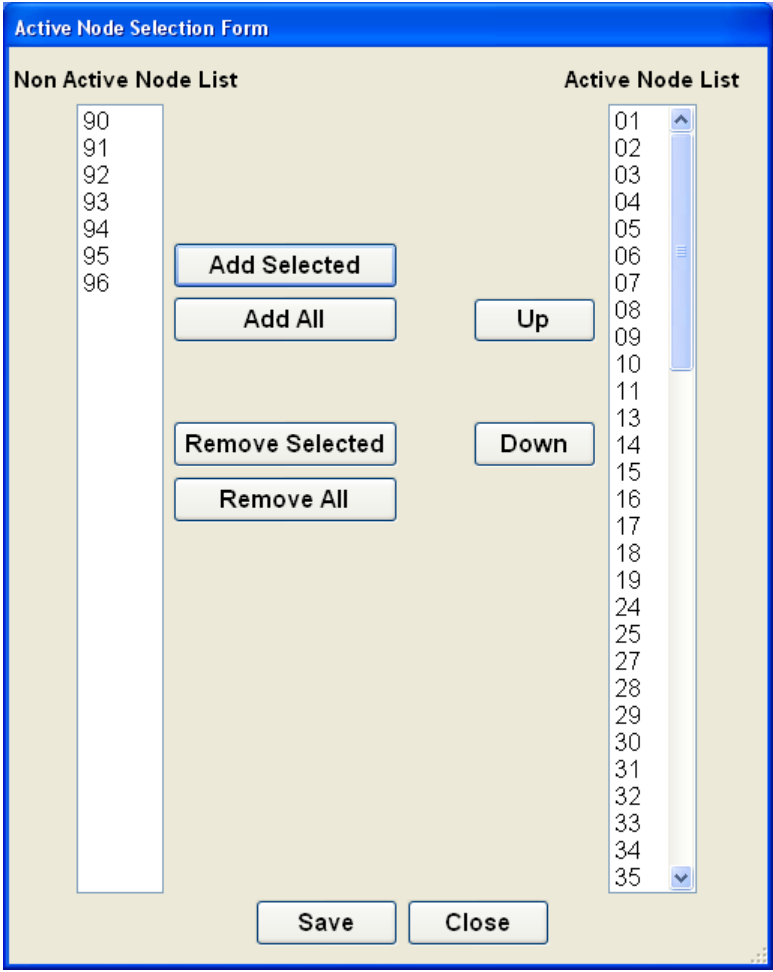
Step	Action
	
4	Click Save to save the active node list.
5	Click Close .
	The Data Analysis User Interface dialog box appears.

Table 6-12 Procedure – Node Monitor tool

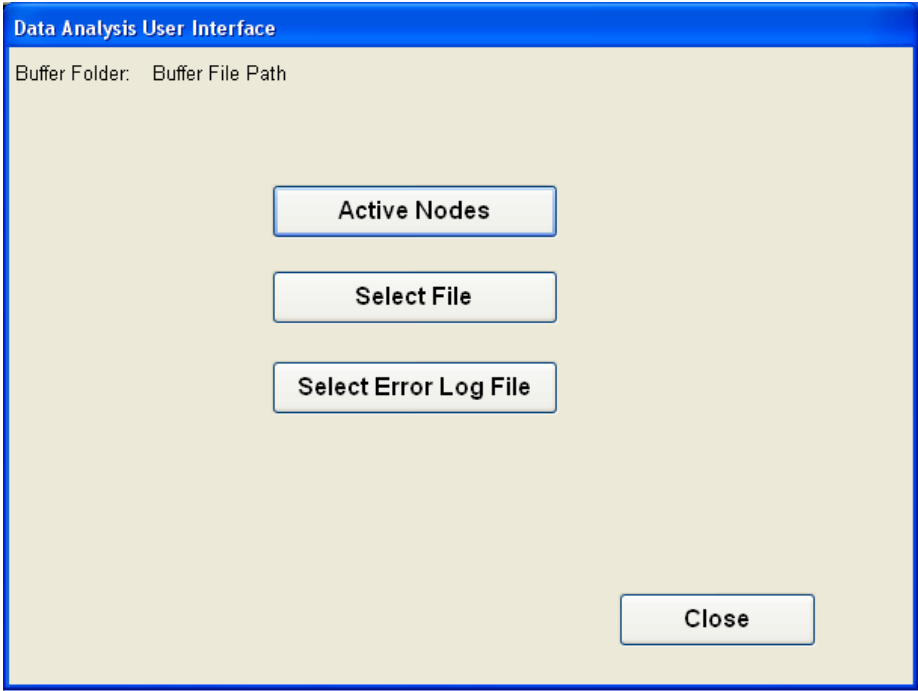
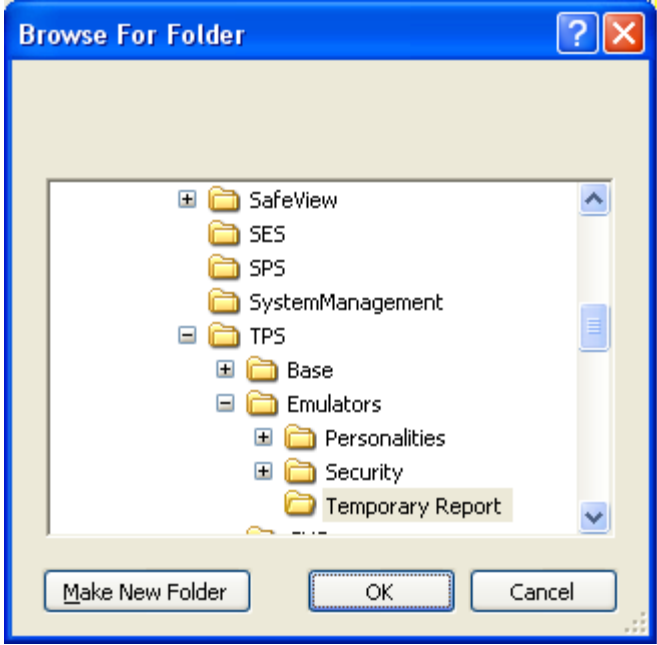
Step	Action
	 The 'Data Analysis User Interface' dialog box is shown. It has a title bar with a blue gradient. Below the title bar, there are two labels: 'Buffer Folder:' and 'Buffer File Path'. In the center, there are three buttons stacked vertically: 'Active Nodes', 'Select File', and 'Select Error Log File'. At the bottom right, there is a 'Close' button.
6	Click Select Files .
7	Go to the folder where the .txt file generated by the LCNPMemReader tool is saved.
	 The 'Browse For Folder' dialog box is shown. It has a title bar with a blue gradient and standard Windows window controls. The main area displays a tree view of folders. The folders are: SafeView, SES, SPS, SystemManagement, TPS, Base, Emulators, Personalities, Security, and Temporary Report. The 'Temporary Report' folder is selected and highlighted. At the bottom, there are three buttons: 'Make New Folder', 'OK', and 'Cancel'.
	The File Select dialog box appears.

Table 6-12 Procedure – Node Monitor tool

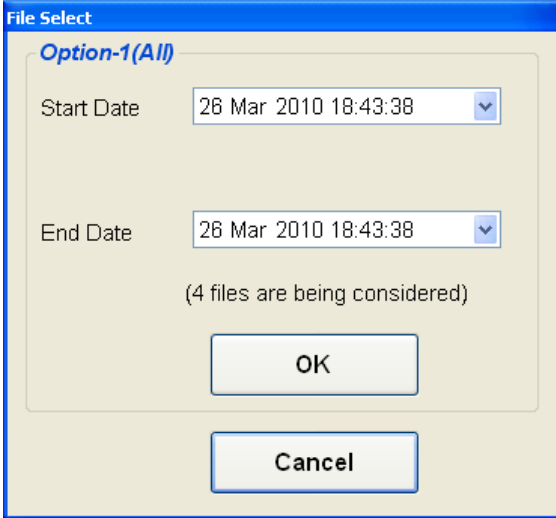
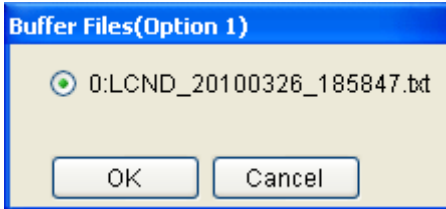
Step	Action
	
8	Select the start date and the end date. Click OK .
	The list of buffer files is displayed in the Buffer Files (Option 1) dialog box.
	
9	Click one of the buffer file names and then click OK .
	The GUI For LCN Node Monitor appears.

Table 6-12 Procedure – Node Monitor tool

Step	Action
	 The screenshot shows the 'GUI for LCN Node Monitor' window. It features a menu bar with 'File', 'Reports', and 'About'. Below the menu, it displays 'Cable selected: A', 'Cable A: Error' (in red), and 'Cable B: Ignored' (in green). It also shows 'Start Date/Time: 3/26/2010 6:58:47 PM' and 'End Date/Time: 3/26/2010 6:59:45 PM'. The main area is a grid of 88 numbered nodes (1-88). Each node is represented by a colored square: Green (OK), Red (SUSPECT), or Yellow (Active but not participating). For example, nodes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 18, 19, 24, 25, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 12, 20, 21, 22, 23, and 26 are Green. Nodes 17, 18, 19, 28, and 41 are Red. Nodes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 18, 19, 24, 25, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 12, 20, 21, 22, 23, and 26 are Yellow. The selected cable and the status of Cable A and B is displayed. The following color code indicates the cable status. - Green: Cable is in OK state. - Red: Cable is in SUSPECT state. All nodes (active and isolated) on the network that has framework entries in the input archive file are displayed. The following color code indicates the node status. - Green: Node is healthy. - Red: Node is in error condition. - Yellow: Node is present in the Active Node list but not participating in Token pass sequence.

Table 6-12 Procedure – Node Monitor tool

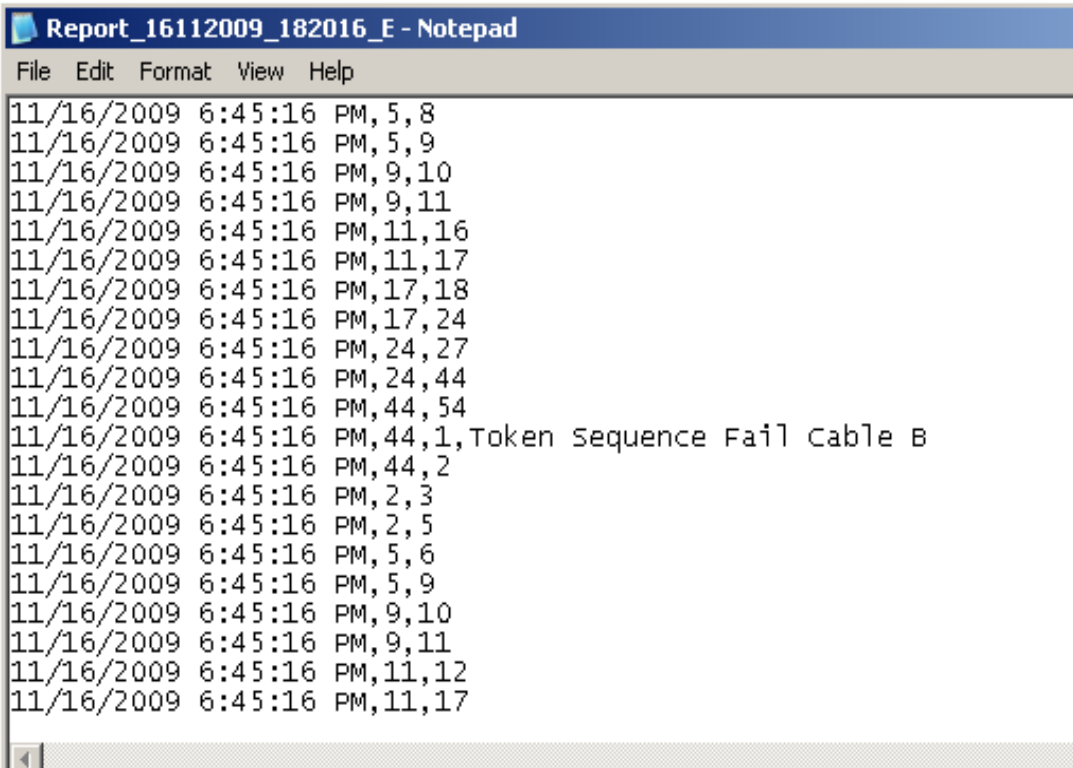
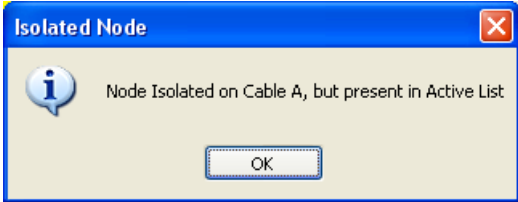
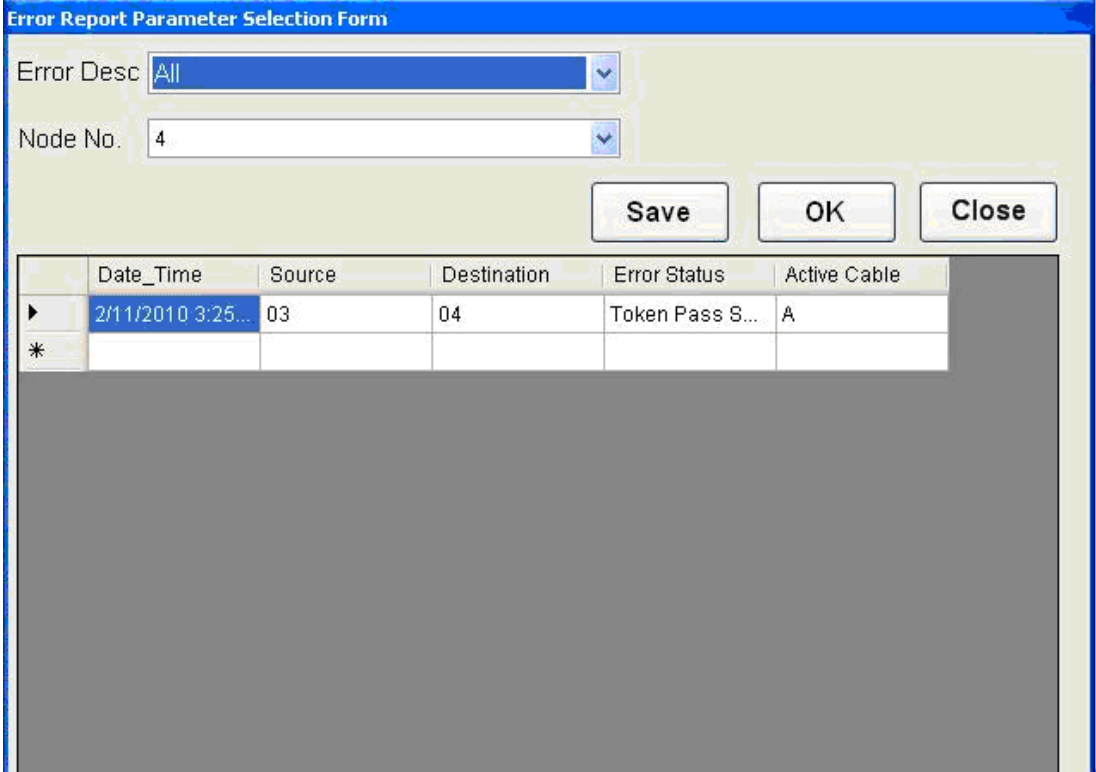
Step	Action
	The captured data is logged in a text file as shown in the following figure.  Note: The date and time stamp in the text file corresponds to the date and time format of the local machine.
10	Click one of the node buttons to view the error status.
	 TIP If the status of the selected node is in Yellow, the following message appears. 
	The Error Report Parameter Selection Form appears.

Table 6-12 Procedure – Node Monitor tool

Step	Action
	
10a	Select one of the error descriptions from the Error Desc drop-down list and then click OK .
	The error status of the node appears in a grid.
10b	Click Save to save the error report in .csv format.
10c	Click Close to close the Error Report Parameter Selection Form .
11	Click Summary .
	The Summary Report window displays a summary of all the errors generated in all the nodes.
	If the user selects Cable A, then all the errors and the nodes of Cable B are ignored in the Summary report and vice versa. While archiving, if both the cables are healthy and an online error occurs on the selected cable, then all the nodes (those were active at least once in the network) are displayed as Active in the summary report irrespective of the change in the status of the node. If a Media Dead occurs and the node is isolated on a specific cable, then - The tool displays Media Dead for that cable - Summary report displays the node as Active
	Note: If a node is present in the Active Node List and while archiving, if that node is not communicating with the network, then that node will appear as Inactive.

6. TROUBLESHOOTING

6.2. LCN Cable Diagnosis

Table 6-12 Procedure – Node Monitor tool

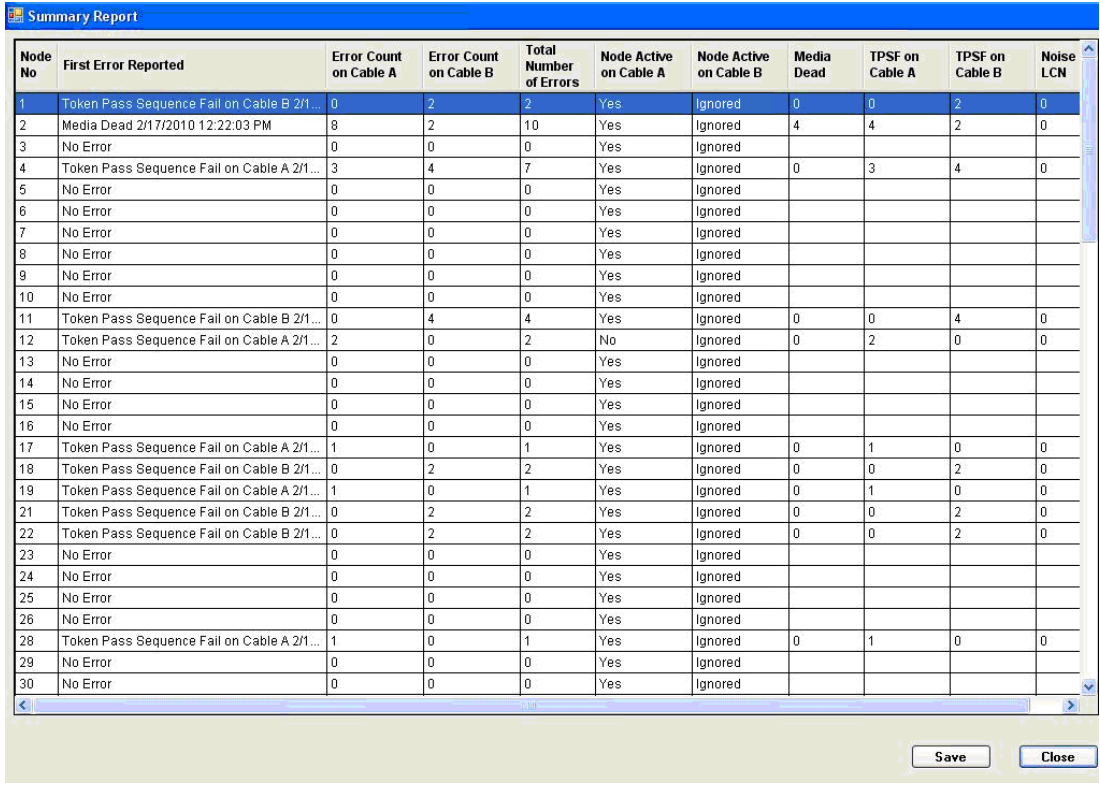
Step	Action
	
11a	Click Save to save the error report in .csv format.
11b	Click Close to close the summary report.
File Menu	
(i)	Choose File > Open to open the already generated reports.

Table 6-12 Procedure – Node Monitor tool

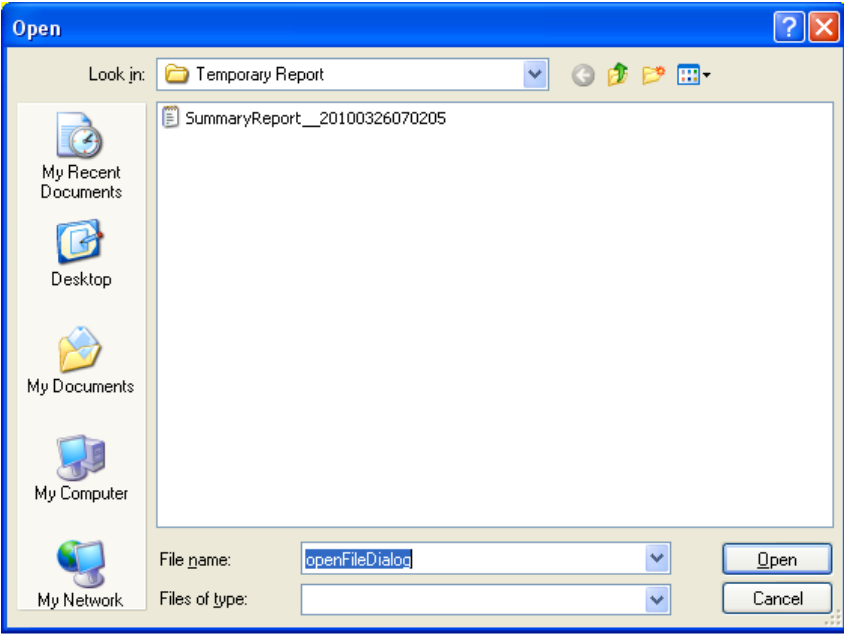
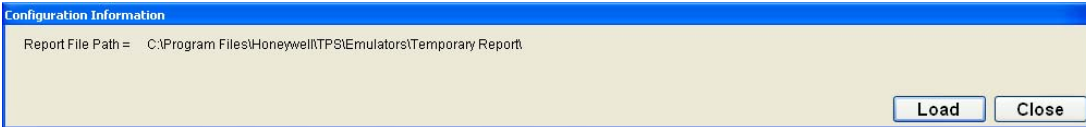
Step	Action
	
(ii)	Choose File > Configuration to change the destination path for the generated reports.
	
	Click Load to choose the new destination folder.

Table 6-12 Procedure – Node Monitor tool

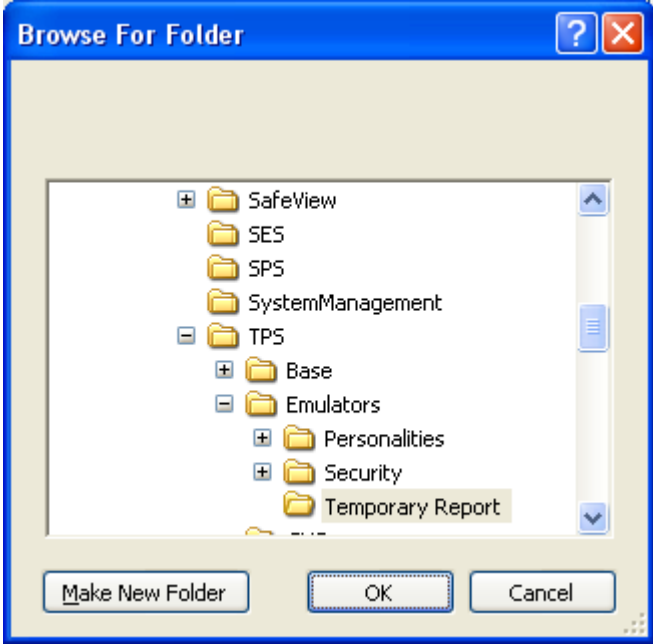
Step	Action
	
(iii)	Choose File > Exit to close the GUI For LCN Node Monitor window.
Reports Menu	
(i)	Choose Reports > Error Report .
	The Error Report Parameter Selection form appears.

Table 6-12 Procedure – Node Monitor tool

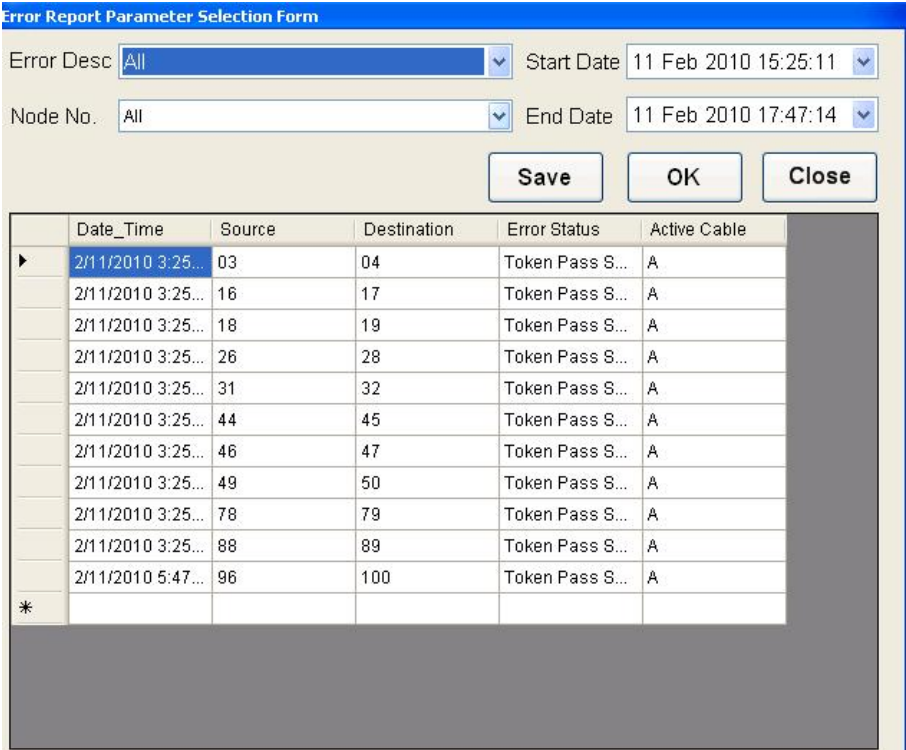
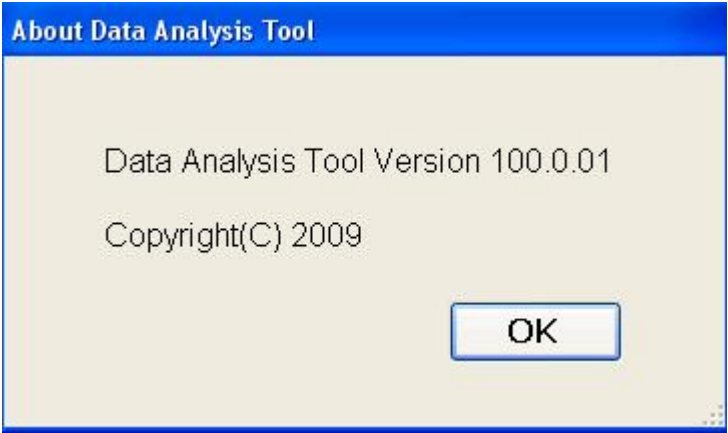
Step	Action
	
	Select the appropriate values for the following options. • Select a specific error description or All. • Select a specific node number or All. • Select the start date and time. • Select the end date and time.
	Click OK .
	The summary of error status is displayed in a grid. The summary contains Date, Time, Source node#, Destination node#, Error status, and Active Cable. If there are no errors, a blank report is generated.
	Click Save to save the summary information in .csv format.
(ii)	Choose Reports > Excel Report .
	The Excel Report Parameter Selection form appears.

Table 6-12 Procedure – Node Monitor tool

Step	Action
	Excel Report Parameter Selection Form Start Date 11 Feb 2010 15:25:11 End Date 11 Feb 2010 17:47:14 OK Close
	• Select the start date and time. • Select the end date and time.
	Click OK .
	The report is generated in .CSV format. The report contains Date, Time, Source node#, Destination node#, Error status, and Cable Type. The report is saved in the user defined path using File > Configuration. Microsoft Excel - TempExcelReport_20112009_161220 File Edit View Insert Format Tools Data Window Help </

Table 6-12 Procedure – Node Monitor tool

Step	Action
	
12	Click Select Error Log File .
	The list of error log files is displayed in the Error Log Files dialog box.
13	Select one of the error log file names and then click OK .
	The list of buffer files is displayed in the Buffer Files (Option 2) dialog box.
14	Click one of the buffer file names and then click OK .
	The GUI For LCN Node Monitor appears.

The LCN Diagnostics tool and the PC Archiving tool must be running to diagnose the errors generated in the following scenarios.

Scenario 1

The status of Cable A or B changes to suspicious fail if there is a communication problem in the LCN network. It is not possible to find out the node causing the problem.

The **Monitor Errors** menu item is used for detecting the node causing the problem. The node that held the last good token is displayed on the screen. This indicates that the node next to the displayed node number has a problem and that must be investigated. The following components of the node must be investigated to rectify the error.

- Node hardware
- Cable connections
- Cable Tees connection
- LCN Tees used for connecting the LCN cable
- LCN cable connections at the node
- Cable end terminators

6. TROUBLESHOOTING

6.3. XLCNE2 And Fiber Optic Link Diagnosis

The cable must be replaced if it has degraded.

Scenario 2

To know the number of active nodes available on the LCN cables A and B.

The LCN network in the plant might be working fine. However, all the nodes on the network might not be present on both the LCN cables. The **Active nodes on Cable A/B** menu item is used for resolving the issue.

This menu item helps in diagnosing the low level problems in the initial stage. The menu lists the nodes that are communicating on Cable A and Cable B. The nodes that are not active must be investigated. After investigation, the menu option is selected again to verify if the nodes are on the network.

Scenario 3

The status of one of the cables changes to SUSPECT and the other cable becomes active.

The **Monitor cable** in use menu item of the LCN Diagnostics tool identifies the node that caused the cable swap.

Note: Select the cable which is in OK state to identify the node that caused the cable swap. If a SUSPECT cable is selected and there is a noise in the network, the tool displays the incorrect information.

Scenario 4

The LCN Diagnostics tool is used for diagnosing the cause for node isolation.

The **Active nodes on Cable A/B** menu item is used for identifying the isolated nodes in the network.

6.3 XLCNE2 And Fiber Optic Link Diagnosis

6.3.1 LED Indicators on Interface Boards

There are up-to-three LED indicators on the boards interfacing the fiber optic cable.

- Red LED — Antijabber Timer Error Flag—Data frame too long, XLCNE2 transmits too long. This flag remains set until reset by cycling the power OFF/ON or by toggling the next switch.
- Yellow LED — Fiber to Coax Transmit—Blinks as data is passed onto the coaxial cable segment.
- Green LED — Indicates the presence of an attached fiber optic cable. The green LED is not an indication of signal quality.

6.3.2 Switches

XLCNE2 boards have a single RUN/RESET switch located on the edge of the board.

- RUN — Normal position that allows data transfer.
- RESET — Stops data transfer and resets Antijabber Error flag (the Antijabber flag must be manually reset before data can be passed again).

6.3.3 Indications of Normal Operation

Under normal system operating conditions the following indicators are seen on the boards that are a part of the XLCNE2.

- Red LEDs is out
- Green LED is on
- Yellow LED either blinks on and off (indicating traffic) or stays on (more traffic)

For an explanation of the other indicators on other boards in a module, see *Five/Ten-Slot Module Service* in the *LCN Service — 1* binder.

6. TROUBLESHOOTING

6.3. XLCNE2 And Fiber Optic Link Diagnosis

6.3.4 Failure Indications

If a remote segment has gone dead, check the following:

1. Red LED on all XLCNE2 boards should be off.
If a red LED is on, then the Antijabber flag is set. The failure that caused this to happen is on the board with the LED on. Replace that board (for more information, refer to *Five/Ten-Slot Module Service* in the *LCN Service — 1* binder.
2. Green LED on XLCNE2, FOC/RCVR, LCNFL boards must be on and steady.

If a Green LED is not on or is flickering then there is not enough light power being received from the fiber optic cable. Check the following:

- a. Cable integrity: Check for cut, damaged, or kinked cable.
- b. Light received: Measure the light power at the receiving end of the cable. A minimum of .8 μw must be measured. Light Power measurement instructions are in Section 7.
- c. Light transmitted: Measure the light power at the transmitter, a minimum of 8 μw must be measured using a one meter length fiber optic cable. Light Power measurement instructions are in Section 7.

Yellow indicators are normally ON, but blink off when no traffic is being passed.

NOTE

If the green LED is on continuously, there may still not be enough light power being received from the fiber optic cable. The green LED indicates the presence of an attached cable.

6.4 Recovery From A System Outage

First, load the nodes needed to reestablish any lost critical paths to the process. We recommend that you use previously prepared fast load cartridges, unless one or more HMs with the appropriate personality files and checkpoints is available as the source for loading. Floppies with the proper personality files and checkpoint data can be used but take more steps and more time. Then use normal node loading procedures to load failed nodes that are not on critical paths to the process (for more information, refer to the *Process Operations Manual*).

To reestablish a critical path to the process using a user-prepared fast load cartridge (for more information, refer to subsection 20.2 in the *Engineer's Reference Manual*), follow these steps:

1. Load a US, by doing a “local boot” of the US, preferably with a fast load cartridge previously prepared to reestablish this critical path. To do so, use the following steps:
 - a. Mount the fast load cartridge in the US to be reloaded.
 - b. Open the cover at the rear of the Operator's Keyboard, press <RESET>, then press <LOAD>. N, 1, 2, 3, 4, X appears.
 - c. Key in the number of the cartridge drive and press <ENTER>.
 - d. When the Console Status display appears, call up the Gateways Status display.
2. To reestablish operation of the NIM or HG on the critical path, use the following steps:
 - a. Select the appropriate node, and then select NODE LOAD. NET LOAD and LOCAL LOAD picks appear.
 - b. Select LOCAL LOAD. The NIM or HG goes busy and loading from the cartridge proceeds.
 - c. When the NIM or HG status goes to OK, use the appropriate Operator Personality or Universal Personality displays to reestablish appropriate control states, data point activity, and modes.
3. Repeat 1 and 2, as necessary to reestablish other critical paths to the process, then reload any other failed nodes. Nodes can be loaded in parallel using multiple USs.

NOTE

If a red LED error occurs on a module, it is necessary to press the reset on that module before that module can be successfully restarted.

6. TROUBLESHOOTING

6.4. Recovery From A System Outage

7. FIBER OPTIC LIGHT POWER MEASUREMENTS

This section gives instructions for power level measurements on fiber optic cable used with LCN Extenders.

7.1 Measuring Light Power In A Fiber Optic Cable

When the installation of the fiber optic cable has been completed, and the XLCNE2 Extender Set has been installed, it is vitally important that power level measurements be made to ensure proper operation of the LCN. Measurements must taken at each transmitter in the system and at the receiving end of each fiber optic transmission line.

To measure light power levels at a transmitter or end of cable, a metering device such as those suggested below must be connected to the end of the cable or to the transmitter by a one foot length of fiber optic cable.

The following are suitable for measurement of the light power level:

Hewlett-Packard model HP 8140A	Meter
Hewlett-Packard model HP 81S11A	Optical Head
Intelco Model 131	Meter
Intelco Model 2511 SMA Adapter	
Laser Precision AM-3500	Meter
Protodyne Model 17XT	Meter
Protodyne Model 2003 SMA Adapter	
Tektronics 2815	Optoscope

7.2 Measurement Procedures

A minimum of 50 μW and a maximum of 200 μW should be measured. Power levels over 200 μW may saturate the receiver and cause malfunctions.

To measure light power levels at a transmitter, a metering device such as those suggested below must be connected to a transmitting board by a 1-meter fiber optic cable.

1. Measure the power output of the transmitter on the power meter.
2. If the measurement is not within the specifications in Table 7-1 the board should be replace. A reading greater than the specification can cause a malfunction.

Table 7-1 — Limits of Optic Power

Cable Size	Minimum Reading	Maximum Reading
62.5 micron	8 μW (-21 dBm)	31.6 μW (-15 dBm)

To measure light power levels at the receiving end of a fiber optic link, disconnect the fiber cable from the receiver board and connect it to the metering head. A maximum of 31.6 μW and a minimum of 8.0 μW must be measured. If this minimum power level is not found:

1. Check the fiber optic cable for kinks, and physical damage. A damaged cable must be replaced or the damaged fiber switched with a spare fiber if any are available in the cable.
2. Check the power level at the transmitter board.
3. If the minimum power level is found, and the green LED still is not lit, replace the receiver board.

If the maximum is exceeded, replace the transmitter board.

8. LCN RECONNECTION

This section provides guidelines and instructions for reconnecting two separated LCN portions, while maintaining control, if that is possible.

CAUTION

You may need the procedures in this section in critical situations, where they must be accomplished quickly and with minimal problems and surprises. We strongly recommend that you carefully review this section and Section 6 before you need the information they contain.

Preparation of people and materials for the handling of unexpected LCN separations can be invaluable if separation ever occurs. People preparation might include operator and engineer training and drill, and a review of who has appropriate keys for US key switches. Material preparation should include local instructions, LCN topology maps (see subsection 6.2), and node-loading removable media (cartridges or floppies) with fresh checkpoints and other data. For systems operating on Release 230 and later, current fast load cartridges (see Section 20 of the *Engineer's Reference Manual*) should be available for each critical path to the process that the system handles.

CAUTION

IF YOU HAVE REDUNDANT NODES:

If the redundant (backup) node of a redundant node pair becomes ISOLATED during an LCN Reconnect, the primary node will fail. This could cause additional loss of critical paths to the process. Backup nodes that will become ISOLATED during LCN Reconnect must be shut down BEFORE executing the LCN Reconnect. Refer to the detailed procedures in subsections 7.7.4 and 7.7.6—they include the necessary steps.

8.1 Definitions

NOTE

The term "LCN portion" is used here to distinguish from the term "LCN segment" as "segment" pertains to segments created by LCN extenders and fiber optic links. While separation of such a segment can occur, so can separation occur between other portions of the LCN that are not established by LCN extenders.

View of the Network—How a node "sees" the other nodes on the network; that is, are they OK, ISOLATED, FAILED, etc. From a US, you can obtain the view of the network from the LCN Overview display (see definition below). You can obtain the view from (as seen by) that US, or you can call up the view from any other node with which you can communicate. You initiate the LCN Reconnect function from the LCN Overview display, but you must use the display that shows the view from the US at which you perform the reconnect (not the view from some other node).

LCN Overview—A display that shows the view of the network as seen from a selected node. Three ways to access the LCN Overview display are:

1. Press the Console Status button on the Operator's keyboard. Select the desired US (it doesn't have to be the US where you are). Then select the LCN OVERVIEW target.
2. Press the System Status button on the Operator's keyboard. Select the desired node type and then select the desired node (it doesn't have to be a US). Then select the LCN OVERVIEW target.
3. From an LCN Overview display, select the LCN OVERVIEW target. In response to the prompt at the top of the screen, enter the desired node number.

View of the Process—The ability of an operator at a US to access an ongoing process. Includes the ability to view or monitor the process, and to control the process as required.

Critical Path—The combination of working elements required to provide an operator with a view of the process. Includes, as a minimum, a US, a gateway device (HG, NIM, or PLCG), and a process control device (MC, PM, APM, or PLC). All devices must be functional and able to communicate over the LCN.

Separation—A change in the LCN that results from a disruption of communications on both LCN cables. After the communication problem is fixed, an LCN Reconnect procedure is used to restore normal LCN function with minimal impact on the process. Typical steps in a separation and reconnection scenario are:

- Communication is interrupted on both LCN cables (see subsection 8.3).
- The LCN is split into two (or more) "portions." The separated portions begin operation independently, functioning as separate LCNs. (If the separated portions are not properly terminated, one or more may not be able to function at all.)
- The fault(s) that caused separation are repaired. The LCN continues to function as independent portions (reconnection is not automatic).
- A US is chosen from which to perform the LCN Reconnect function. (Subsections 7.7.4 and 7.7.6 have information about the criteria used to choose the US.) Nodes that are on a different LCN portion than the chosen US appear as (have a status of) ISOLATED from that US.
- The LCN Reconnect function is initiated from the chosen US. At this time, additional nodes may change state to ISOLATED because they have a different view of the network than that of the US that initiated the reconnect.
- As a part of the LCN Reconnect procedure, critical paths to the process are restored (if any were lost), and remaining nodes are transferred from separated portions of the LCN to the portion on which the chosen US resides.

LCN Reconnect—A process that restores normal operation to a separated (and repaired) LCN. The LCN Reconnect is initiated manually from a US. After initiation, an automatic software process forms a logical group of nodes that consists of the US from which the reconnect was initiated plus all other nodes that have the same, identical view of the network as that US. All nodes that have a different view are isolated from this group, which we refer to as the "good portion." Using a sequence of manual steps, lost critical paths, if any, are restored by shutting down and reloading isolated gateway devices. Finally, using additional manual steps, other nodes are added to the "good portion" of the LCN by shutting down and reloading them one-by-one.

8.2 Goal

To reestablish full operation of all nodes on all LCN portions without breaking any path to the valve," while maintaining the operator's view of the process with a minimum of disruption and alarming.

8.3 How Separation Can Occur

1. Between portions connected by LCN Extenders (caused by switching off both A and B cables at the XLCNE2 or by power loss affecting only one portion).
2. By deliberately disconnecting (and properly terminating) both A and B LCN cables at some point on the network. This can include cases where both portions have been down and were restarted separately.
3. Accidental separation or cutting of both A and B cables by some physical mechanism.

Any situation where communication is at least temporarily lost on both the A and B cables is referred to as a "double cable fault."

8.4 What Happens When Separation Occurs?

If the LCN cables become disconnected as described in section 8.3, a timeout period of approximately 2 minutes begins. If the cables can be repaired or reconnected with proper terminations within the 2-minute period, isolation will not occur, though in a few cases a backup NIM or HG may fail. After the timeout period, nodes will set the status of other nodes with whom they cannot communicate to ISOLATED. At this time, the LCN is separated into two or more portions. After the cables are repaired or reconnected, nodes are physically capable of communication with each other, but they continue to function as separate, independent portions. Nodes ignore messages from nodes they view as ISOLATED, except for "POWER ON" and "QUALIFIED" status messages.

From the discussion so far, you may be tempted to conclude that separation results in two "perfect" groups of nodes—one group on each side of the cable fault, with all of the nodes in each group able to communicate with all other nodes in the same group, but with no communication between groups. A simultaneous break in both cables of a fiber optic link could produce such a "text book" isolation between LCN portions. However, there are many other types of faults that can cause much more complicated and unpredictable results. These include loss of proper cable termination, intermittent opens or shorts, loss of cable ground, as well as various types of hardware failure. When separation occurs under these circumstances, signal reflections or other phenomena can cause nodes on the same side of a fault to have differing views of the network. This is why it may be desirable to examine the view from more

than one node when selecting a US from which to do the reconnect. It is also the main reason for the reconnect function—to obtain an optimum group of nodes, all with an identical view of the network, to use as a base for rebuilding the network.

NOTE

Physically disconnected LCN portions can function separately only if each is terminated at the point of disconnection. Otherwise, communication errors may prevent operation.

8.5 Reconnection Highlights

You must determine which portion will be treated as the "good" portion with which operation of another portion will be reestablished. Usually the good portion will be the portion with the most paths to the process or the most critical paths to the process.

Separated portions can be physically reconnected at any time without any interaction through displays. Nodes that see each other as ISOLATED cannot communicate with each other.

For all releases, physical reconnection is followed by an ordered sequence of node shutdowns and reloads that have the effect of "moving" nodes to the "good" logical portion.

8.6 Reconnection Requirements

- The preferred method to reestablish any lost critical paths to the process is use of a user-prepared fast load cartridge to reestablish each critical path. Instructions for preparing and maintaining such cartridges are provided in subsection 20.2 in the *Engineer's Reference Manual*. Floppies can be used as the source of loadable files and data for restoration of critical paths, but require more time and effort. To restore operation of portions not involved in critical paths to the process, the portion should include a US that has access to current checkpoints on an HM, cartridges, or floppies.
- Up-to-date checkpoints on media (cartridges, floppies, or HMs) accessible by the good portion are necessary to ensure that no changes are lost.
- Once reconnection becomes necessary, the reconnection procedures must be completed before anyone attempts to load any nodes not involved in the reconnection process. No Engineering Personality functions should be performed during the reconnection procedures and control commands should be limited to changes of OP, SP, and Mode.
- Review the key switch levels configured in Network Configuration, especially for operations such as node shutdown, node loading, checkpointing, and history collection. (To review the key switch levels, go the Engineering Main Menu, select SYSTEM WIDE VALUES, select CONSOLE DATA, and press <PAGE FWD> to reach pages 3-6.) The configured levels may dictate that specific people with the appropriate key(s) be present to proceed with the reconnection. Before you actually need to do a reconnection, you should consider reducing key switch levels to assure that necessary key switch access will be available if you ever do a reconnection.

For information about on-line changes to NCF System Wide Values, refer to *Network Data Entry in the Implementation/Startup & Reconfiguration* — 1 binder.

8.7 How To Reestablish Operation On One LCN Portion

8.7.1 Physical Reconnection

For fiber cables—Just repair and reconnect the A and B cables and switch on the XLCNE2s at both ends in no particular sequence. For further information, refer to Section 6 of this manual.

For coax cables—Repair the cable break(s), at least to the point where no unterminated cables remain. Terminators must be at the same points on both cables.

NOTE

When you reconnect coax cables, expect a number of LCN Driver and software errors to be recorded in the LCN statistics accumulated in the Processor Status Data Point COMM parameter in the nodes affected. For more information about LCN cable statistics, refer to subsection 6.2.

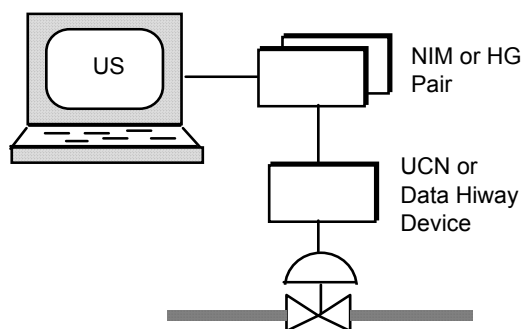
8.7.2 Summary of the Following Procedure

This summary is provided to help you to understand what is to happen in the following procedure and why. You should review the summary and the detailed procedure before you actually need to use this procedure.

- Verify that any LCN extenders involved are now operating properly (refer to Section 6 of this manual). If you proceed with a break still present, you may not be able to reload nodes or back out of the reconnection process.

- If one or more critical paths to the process has been lost, go to a US that is on one of those paths and has a local cartridge drive [floppies can be used for restoring the path(s) but take more steps and more time].

If no critical paths to the process have been lost, choose any US, but choose only one.



- If the chosen US is on a critical path, use it to reload nodes to reestablish the path to the process, and if necessary, reload nodes to reestablish other critical paths.

If this US is not on a critical path, use the LCN Overview display to determine if it should be used to reload nodes to restore them to this portion of the LCN. If so, proceed, if not, continue until you find a US you can use.

- On the chosen US, on its LCN Overview display, select INIT LCN RECNECT but DO NOT press <ENTER> yet. If any redundant (backup) nodes are backlighted in red indicating that they will become ISOLATED, shut down these nodes (see detailed procedures in 7.7.4 and 7.7.6).
- With INIT LCN RECNECT selected on the LCN Overview display of the chosen US, press <ENTER> to initiate the LCN Reconnect procedure. The System Version field on the display shows INIT LCN RECONNECT.
- Load nodes that are isolated or failed, to restore their operation on the LCN portion that the chosen US is on. To do this, you can use any US whose SYSTEM VERS : field on the LCN Overview Display shows INIT LCN RECONNECT.
- Once all nodes have been reloaded and the last node was reloaded because of the INIT LCN RECONNECT, approximately two minutes later the INIT LCN RECONNECT message will disappear from the display.

However, if the message does not disappear and the System Status display still indicates that the SYS VERS is INIT LCN RECONNECT, then this indicates that there is still a node that is not running on the same token as this US/GUS station. There is also the possibility that the message NETWORK VIEW <> (in red) will be displayed indicating that there is a node in the ISOLATED state. When the last node is loaded, SYS VERS will indicate the TPN software version such as R650 that the system is running.

CAUTION

AN ATTEMPT TO LOAD AN ISOLATED NODE MAY RESULT IN THE NODE REMAINING ISOLATED.

Invoke the LCN Reconnect procedure before loading an isolated node.

8. LCN RECONNECTION

8.7. How To Reestablish Operation On One LCN Portion

8.7.3 Prepare to Reestablish Operation on One LCN Portion

NOTE

For TPN R640 (and newer) systems, be sure to disable the **Automatic TPN System Restart** function before initiating the LCN Reconnect. Leave the **Automatic TPN System Restart** function disabled until the LCN Reconnect actions are complete and the LCN is functioning normally.

The LCN is split into two or more separate portions. Each node sees the nodes on the other portion(s) as ISOLATED. Current statistical information useful for analyzing node failures and isolation are recorded in the LCN cable statistics in the PSDP COMM parameters in various nodes (see subsection 6.2.1).

The isolated nodes are indicated as ISOLATED on the System Status displays, the node status displays, and the LCN Overview displays. When a node is isolated from another node, the two cannot communicate with each other. In consoles with USs that see other nodes as isolated, the ISOLATED indicator flashes and the CONSOLE STATUS or SYSTEM STATUS LED blinks until this new status is acknowledged. To acknowledge the isolated status, select the node status display for each node affected and press ACK.

Verify that any LCN extenders involved are now operating properly (refer to Section 6 of this manual), and verify that all cables are in proper states of repair. If you proceed with a break still present, you may not be able to reload nodes or back out of the reconnection process. If the status for both cables is GOOD (green on node status displays), you can proceed. If you cannot afford to wait until the cables are shown as GOOD (this could take as long as 20 minutes), you can proceed with cables indicated as SUSPECT (yellow on node status displays) if you are sure the cables are properly repaired, connected, and terminated.

If critical paths to the process have been lost, continue with the steps in subsection 8.7.4. If critical paths have not been lost, go to subsection 8.7.6 and begin with the steps there.

8.7.4 Determine the Best US to Initiate the Procedure When Critical Paths Have Been Lost

1. If one or more critical paths to the process have been lost, go to one (and only one) US that is responsible for the process affected and has a local cartridge drive [floppies can be used but restoring the path(s) will take longer]. If no critical paths to the process have been lost, choose any US.
2. Call up the LCN Overview display for this US (to call up and LCN Overview display, select the node number for this US on the Console Status display, then select the LCN OVERVIEW pick).
3. Select the INIT LCN RECNCT pick but DO NOT press <ENTER> yet. Boxes for nodes that will become isolated (because they do not have the same view of the network as this US) are backlighted in red. This may indicate additional NIMs, HGs, or HMs that will become isolated if you press <ENTER>, which could break additional paths to the process [here, an LCN topology map (see subsection 6.2) that shows the cable routing, nodes, node types, node numbers, and process points served can be invaluable].
4. Before you press <ENTER> check all of your redundant node pairs on the LCN Overview display. If you have any cases where the primary indicates that it will be OK after reconnect (it is not backlighted in red) and the corresponding backup indicates that it will be ISOLATED after reconnect (it is backlighted in red), you must shut down the backup(s) before executing the reconnect. If you don't do this, the primary will fail during the reconnect procedure. If you have any such cases where the primary indicates that it will be OK but the redundant backup indicates that it will become ISOLATED, use the following procedure (otherwise, go to step 5).
 - Press the <SYSTEM STATUS> key.
 - Select the node type.
 - Select the redundant node that indicated that it will become ISOLATED.
 - Shut down the node.
 - Repeat the above four steps for each redundant node that indicates that it will become ISOLATED.
 - When done, return to the LCN Overview display on the chosen US and select the INIT LCN RECNCT target.

8. LCN RECONNECTION

8.7. How To Reestablish Operation On One LCN Portion

CAUTION

After you press <ENTER> in step 5 below, if a break occurs in both LCN cables while the System Version field on the LCN Overview displays is showing INIT LCN RECONNECT, it will not be possible to determine which of the separated portions is the best one to bring over the nodes on the isolated portion(s), and it may be necessary to shut down and reload all nodes.

5. On the chosen US, press <ENTER>. The System Version field shows INIT LCN RECONNECT and all nodes that have the same view of the network as this US retain their current status. All nodes that do not have the identical view of the network isolate from this LCN portion. Backlighting of the node boxes is discontinued.
6. Continue with the steps under subsection 8.7.5 below.

8.7.5 Restore Critical Paths to the Process

1. Use the Gateways Status display to select and shut down the NIM or HG on the critical path. If this is a pair of redundant NIMs or HGs, shut down both of them.
2. Load one of the NIMs or HGs. We recommend that you use a fast load cartridge that was previously prepared to reestablish this path to the process. You can use floppies that contain the proper personality files and current fresh checkpoints, but this may require that you mount more than one floppy and can take more time. To use a fast load cartridge, do the following:
 - Select NODE LOAD. NET LOAD and LOCAL LOAD picks appear.
 - Select LOCAL LOAD. The NIM or HG goes busy and loading from the cartridge proceeds.
3. Load the nodes needed to reestablish any other lost critical paths to the process. Again, we recommend that you use previously prepared fast load cartridges, unless one or more HMs with the appropriate personality files and checkpoints is available as the source for loading. Floppies with the proper personality files and checkpoint data can be used but may take more steps and more time.

Reload a US, by doing a “local boot” of the US, preferably with a fast load cartridge previously prepared to reestablish the critical path. To do so, use the following steps:

 - Mount the fastload cartridge in the US to be reloaded.
 - Open the cover at the rear of the Operator's Keyboard, press <RESET>, then press <LOAD>. N, 1, 2, 3, 4, X appears.
 - Key in the number of the cartridge drive and press <ENTER>.
 - When the Console Status display appears, call up the Gateways Status display.
 - Select the node number for the NIM or HG to be reloaded. If it has a redundant partner, select one member of the pair to be loaded.
 - Select SHUT DOWN and press <ENTER>. If this is a pair of redundant NIMs or HGs, shut down both of them.
 - Select NODE LOAD. NET LOAD and LOCAL LOAD picks appear.
 - Select LOCAL LOAD. The NIM or HG goes busy and loading from the cartridge proceeds.
4. Go to subsection 8.7.7 and continue with the steps listed there.

8. LCN RECONNECTION

8.7. How To Reestablish Operation On One LCN Portion

8.7.6 Initiate the Procedure When Critical Paths to the Process Have Not Been Lost

1. Determine which US is best to initiate the restoration of operation on one LCN portion. You can do this by calling up the LCN Overview displays on several USs and selecting INIT LCN RECNCT. Any node whose status is ISOLATED or whose box is backlighted in red will have to be reloaded if you press <ENTER>. If any of those nodes are on a path to the process, that path will be broken. If this is the case, check the LCN Overview display on additional USs to see if there is a better choice—a US on which pressing <ENTER> will not cause a path to be lost or will cause fewer paths to be lost.
2. Select the INIT LCN RECNCT pick but DO NOT press <ENTER> yet. Boxes for nodes that will become isolated (because they do not have the same view of the network as this US) are backlighted in red. This may indicate additional NIMs or HGs that will become isolated if you press <ENTER>, which could break additional paths to the process [here, an LCN topology map (see subsection 6.2) that shows the cable routing, nodes, node types, node numbers, and process points served can be invaluable].
3. Before you press <ENTER> check all of your redundant node pairs on the LCN Overview display. If you have any cases where the primary indicates that it will be OK after reconnect (it is not backlighted in red) and the corresponding backup indicates that it will be ISOLATED after reconnect (it is backlighted in red), you must shut down the backup(s) before executing the reconnect. If you don't do this, the primary will fail during the reconnect procedure). If you have any such cases where the primary indicates that it will be OK but the redundant backup indicates that it will become ISOLATED, use the following procedure (otherwise, go to step 4).
 - Press the <SYSTEM STATUS> key.
 - Select the node type.
 - Select the redundant node that indicated that it will become ISOLATED.
 - Shut down the node.
 - Repeat the above four steps for each redundant node that indicates that it will become ISOLATED.
 - When done, return to the LCN Overview display on the chosen US and select the INIT LCN RECNCT target.
4. Press <ENTER> to initiate the LCN Reconnect. The System Version field shows INIT LCN RECONNECT and all nodes that have the same view of the network as "this US" retain their current status. All nodes that do not have the identical view of the network are isolated from this US. Backlighting of the node boxes discontinues.
5. Go to subsection 8.7.7 on the next page and continue.

8.7.7 Continue the Procedure to Reestablish Operation on One LCN Portion with One US

Nodes that show `ISOLATED` status at the chosen US must be shut down and reloaded from that US, to bring them from the isolated LCN portion(s) to the LCN portion on which the chosen US is operating. To do so, use the following steps:

1. Determine if all of the volumes needed to reload nodes are on an HM whose status is `OK` at the US on which you are initiating reconnection (in the Engineering Personality).
 - You can use the Utilities' `LV NET` command to observe and record which volumes are on each HM. The volumes required are as follows:
 - System Volume (&0np) Schematics (usually in &3np)
 - Personality images (&1np) Buttons (usually in &3np)
 - Area data base (&3np) Required checkpoints (&6np, &7np, and &8np)
 - If all of the volumes needed to reload are on an HM (or HMs) whose status is `OK`, you do not have to "autoboot" the HM(s), so go to step 3. Otherwise, continue with step 2.
2. You are to shut down and allowed to restart ("autoboot") the HM(s) with the volumes needed to load nodes on the separated portion(s) of the LCN. If you have a US whose LCN Overview displays do not show `INIT LCN RECONNECT` in the System Version Field and the status of the load-source HM(s) is `OK` at that US, use Approach A. Otherwise, use Approach B. With either approach, this HM (these HMs) is (are) available only on the separated LCN portion.

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8.7. How To Reestablish Operation On One LCN Portion

Approach A (for an HM that shows OK on another US)

The following substeps reload the HM to bring it up on the “good” LCN portion, so that this HM can be used as the source for reloading data to bring isolated nodes over to the “good” LCN portion.

- At the US that shows the HM as OK, use the History Modules status display to initiate a load of the HM (even though it appears as OK). Select OPERATOR PROGRAM, and select the default source (NET) for the program source and the data source. Don't forget to press <ENTER> to start the actual loading. Then go on with step 3.

Approach B (for an HM that does not show OK on any US)

The following substeps force the HM to autoboot to bring it up on the “good” LCN portion, so that this HM can be used as the source for reload data to bring isolated nodes over to the “good” LCN portion.

- Use the History Modules status display on any US to shut down the HM.
 - Shut down the HM a second time. This ensures that the HM goes to the FAILED state.
 - Ensure that no cartridges or floppies are in any drive connected to this US.
 - Use the History Modules status display to initiate a load of the HM. Select OPERATOR PROGRAM, and select any alternate source (cartridge or floppies, even though they are not mounted) for the program source and the data source. Don't forget to press <ENTER> to start the actual loading.
 - If the load request fails, the HM is already autobooting, so go to step 3. Otherwise, continue with the next line.
 - When a prompt requests that you mount the first personality file, select ABORT REQUEST.
 - Clear the red "load failure" message by selecting LOAD NODE and then selecting the LOAD NODE again.
 - Use the status display to shut down the HM again.
 - Ensure that no cartridges or floppies are in any drive connected to this US.
 - Use the status display to initiate a load of the HM. Select OPERATOR PROGRAM, and select any alternate (cartridge or floppies even though they are not mounted) for the program source and the data source. Don't forget to press <ENTER> to start the actual loading.
 - If the load request fails, the HM is already autobooting, so go to step 3. Otherwise continue with the next line.
 - This step should not be necessary, but if it is, locate the HM chassis (be sure it is the HM you have been trying to restart), turn the power switch off for about five seconds, and then back on. The HM should autoboot and after a few minutes and its status should become OK. After a successful autoboot, continue with step 3.
3. Wait for the status of any HMs you have restarted to become OK, then continue with step 4.

4. Use the steps under subsection 8.7.8 to complete the restoration of isolated nodes to the operation on a single LCN portion. In those steps, if both partners of a redundant NIM or HG pair are isolated from all USs, both of them must be shut down and reloaded. If this is necessary, there is no way to avoid the temporary loss of a path to the process.

8.7.8 Complete Restoration of All Nodes on One LCN Portion

NOTE

The SHUTDOWN of any module on the separated portion can be initiated from any Universal Station on the physical network (or by pressing <RESET> at the module). The <LOAD>, however, must always be initiated from a Universal Station on the good portion.

The source for both program and data used in reloading the nodes being moved to the good portion varies. Either can come from a removable medium or from one or more History Modules depending on the system volume assignments.

To attain one or more USs with the appropriate data base for loading HGs and NIMs, move one or more USs to the good portion by shutting down and then reloading them, one at a time. Next, for each remaining HG pair on the separated portion, follow steps 1 through 6 in the sequence shown. Then, for each remaining NIM pair on the separated portion, follow steps 7 through 11 in the sequence shown.

1. For an HG pair, shut down the backup HG (hereafter referred to as HG'). Note that HG redundancy is lost. To start the reload of HG', select the LOAD/DUMP target on the Gateways Status display, then select MANUAL LOAD and then select LOAD SFW UPGRADE. This selection is necessary to enable both HG and HG' to operate as primaries during the transition period. This is the only time that this target is used in the reloading process.
Once the reload of HG' is complete, both HG and HG' act as primaries on separate logical LCN networks. You can now observe operation of Hiway boxes at Universal Stations on both the separated and good LCN portions.
2. Disable alarming on HG from a US on the separated portion (through the HWY ALARM ENB/DSB target of the Gateway Status display), then enable alarming on HG' from a US on the good portion. Event-initiated processing by this HG pair is lost until Step 5 is complete.
3. Prepare for transfer of the AM (steps 12-13) by setting control status for HG to BASIC. (HG' comes up in BASIC control status.)
4. Shut down the original primary HG. If the HG has HTD responsibility, that function is automatically switched to HG'.
5. Select the ENABLE UPGRADE target for HG'. HG' now has full functionality (but no backup).
6. Reload the original primary HG using the AUTOLOAD NET or OPERATOR PROGRAM pick. That HG now comes up as the backup HG and redundancy is restored.
7. For a NIM pair, use the Gateways Status display on a US to shut down the backup (secondary) NIM (hereafter referred to as NIM').

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8. To start the load of NIM' with the new software, select the LOAD/DUMP target on the Gateways Status display, then select MANUAL LOAD, and then select LOAD SFW UPGRADE.

When the load is complete, both NIM and NIM' act as primaries on separate logical LCNs. You can observe operation of UCN nodes at a US on the “good” LCN portion.

After the load is complete, the Gateways Status display shows the status of NIM' as UPGRADE, and the UCN Status Display shows the NIM' status as BACKUP. NIM' is not really a backup NIM, it is running as a primary NIM.

9. Use the Gateways Status display to SHUT DOWN NIM (the original primary)—this must be done to allow completion of the reconnection for this pair.

The Gateways Status display shows NIM as QUALIFIED and NIM' as UPGRADE. The UCN Status display shows NIM as OFFNET and NIM' as BACKUP.

Do not reload NIM until step 10 is completed.

10. At a US, select the ENABLE UPGRADE target for NIM'. NIM' determines if the original primary NIM is still on the UCN. If so, reconnection cannot continue and an error message is issued.

If NIM is still not on the UCN (it should not be after step 10), the status for NIM' goes from UPGRADE to OK, and it becomes the fully functional primary, with NIM as its secondary. NIM' assumes the odd UCN address (previously, it had the even address). The UCN Status display shows NIM' (odd UCN address) as OK.

11. Use the Gateways Status display at a US to load the new software in NIM (select the OPERATOR PROGRAM target). NIM becomes the backup NIM and NIM redundancy is restored.

12. Shut down and reload any separated portion Application Modules, using your normal criteria for selection between COLD, WARM, and FAST restart. For any redundant AM pair, both AMs must be shutdown and reloaded.

13. Set the control status for the Data Hiway or UCN to FULL to enable AM control, then start AM control by placing HG and NIM points into Cascade mode.

14. SHUT DOWN and reload any remaining Universal Stations from the separated portion.

15. Reload any CGs.

16. After the last LCN node has been reloaded because of INIT LCN RECONNECT, wait for approximately two minutes and the System Status display will change from INIT LCN RECONNECT to the software release number, for example "R300" (see Figure 5.3). This step is essential, because you may not be able to recover except by shutting down all nodes, if another separation occurs.

Restoration is now complete. Use the node status displays for all nodes to examine, and if necessary to change, the operating states such as Auto Save, Alarm Enable/Disable, History Collection, and Checkpoint Enable/Disable. There are situations where the restoration procedures can change these states from what the operators expect to see.

8.8 Special Cases

If the LCN portions are not properly terminated, the portions may not be able to establish themselves as rings. Therefore, the first step must be to find the break, then terminate each side at the node closest to the break. You can then begin to decide which further steps are appropriate.

If both portions were successful in establishing rings, reconnection by these instructions should be possible. Return to subsection 8.7.1. Note that there may be one or more noncritical isolated or failed nodes on each portion. When to reload these nodes depends on the circumstances. You may, for example, wish to wait on the availability of a History Module rather than reload from removable media.

If only one (or neither) portion was successful in setting up a ring, you should power down the nodes of the separated portion; reconnect the LCN cables, then reload the nodes in whatever sequence you deem appropriate.

8.9 How To Add A Node Or Nodes To An LCN

A new node can be connected to the cables of a running LCN if care is taken that nothing is inadvertently misconnected and that at least one of the two redundant LCN cables continues to operate through the whole procedure.

Before the new node can function, it must be configured into the network configuration file (NCF) and software must be loaded into it. A node can be added to the NCF anytime before or after it is physically connected to the LCN. It is added to the NCF through the LCN NODES activity on the Engineering Main Menu. A node that had been added to the NCF, but is not physically present or is not operating is represented by an OFF on its Node Status or Gateway Status display.

Do this to connect a new node to operating LCN cables:

1. Inspect both the A and B LCN cables for proper connections and terminations, and for damage. Refer to LCN hookup and termination information in subsection 2.2 of this manual and subsection 3.4 of the *LCN System Installation* manual in this binder.
2. Make sure that power is not applied to the new node. Turn the power switch off and do not plug the power cable into a power outlet.
3. Check the System Status display (see subsection 6.3 and Figure 6-3) and the LCN Cable Statistics (see subsection 6.2.2) to verify that both cables are operating without errors. **If any errors are indicated, do not continue with this procedure before the problem is corrected and both cables are operating without errors.**
4. At the point where you are to insert the new node, disconnect Cable A from the tee connector on next node and wait for the CABLE A pick on the System Status display to flash yellow, indicating an error (this can take up-to-one minute), then put the Cable A connector back on the tee.
5. Use a new length of cable and an new tee to connect Cable A to the new node. Make sure that the A cable is connected to the A tee connector on the CLCN A/B board and that the cable is properly terminated. If the A cable is not properly terminated and a fault occurs on the B cable, the LCN portions separate but continue to operate and the nodes on the two portions appear to the other portion as ISOLATED.

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6. Select the CABLE A pick on the System Status display and press <ENTER>. Make sure that the pick changes to and remains a steady (nonblinking) yellow, or changes to and remains green (it may take up-to-20 minutes for it to turn green). Steady yellow or green means that the reconnected A cable is OK. If the cable is not OK (blinking yellow), a problem has been introduced that must be corrected before you can proceed. If the A cable is not properly terminated and a fault occurs on the B cable, the LCN portions separate but continue to operate and the nodes on the two portions appear to the other portion as ISOLATED.
7. At the point where you are to insert the new node, disconnect Cable B from the tee connector on next node and wait for the CABLE B pick on the System Status display to flash yellow, indicating an error. This can take up-to-one minute.
8. Connect Cable B to the new node. Make sure that the B cable is connected to the B tee connector on the transceiver paddle board and that it is properly terminated. If the B cable is not properly terminated and a fault occurs on the A cable, the LCN portions separate but continue to operate and the nodes on the two portions appear to the other portion as ISOLATED.
9. Select the CABLE B pick on the System Status display. Make sure that the pick changes remain green, which means that the reconnected B cable is OK. If the cable is not OK, a problem has been introduced that must be corrected before you can proceed. If the B cable is not properly terminated and a fault occurs on the A cable, the LCN portions separate but continue to operate and the nodes on the tow portions appear to the other portions as ISOLATED.
10. Apply power to the new node. You can now proceed with the implementation of this node as a participating member of the system.

9. LCN FIBER OPTIC LINK SERVICE

This section provides information about extender board replacement and maintenance of fiber optic cables.

9.1 Board Replacement

When a board that is connected to a fiber optic cable needs to be replaced, the node containing the board to be replaced must be shutdown so power can be turned off. **Before proceeding**, see Section 7 so that you are aware of what must be done, and what is needed to restore the system. An alternative to this procedure is to take down only one cable (A or B) at a time (the other cable is still in communication). This is safe only if the System Status display (Figure 5-3) shows both the A and B cables as a steady green. Do not attempt to run on one cable if the cables have been shown recently as SUSPECT. To shut down one cable at a time, place the Run/Reset switches (on the XLCNE2 board) in the Reset position at both ends of the one (A or B) cable.

After the power is off, disconnect the fiber optic cable connector and replace the board.

Reconnect the fiber optic cable connector, tightening the connector only tight enough to not vibrate loose (finger tight).

After replacing the board and reconnecting the cable, place the Run/Reset switches back to the Run position.

CAUTION

If the fiber optic cable end is hit on a hard object it may be damaged. If the cable is overtightened the lens in the transmitter or receiver can crack. Handle fiber optic cable carefully and do not over-tighten the cable connector.

NOTE

Any time a board or fiber optic cable is replaced an optic power level measurement should be made to determine the viability of the system.

No preventive maintenance is required. Proper installation to avoid stretching, twisting, or kinking should make the cable last the life of the equipment.

If an LCNFL board is replaced, the LCN Node Address must be pinned on the new board just as it was on the old board. See Figure 9-1 for the location of the address pinning module.



10. Appendix A

10.1 Glossary Of LCN Terms

This section defines the terminology used in this manual.

TERM	DESCRIPTION
APP Node	Application Processing Platform – A node on the TPN used to run application programs in connection with the TPS system and components outside of the TPS system.
Cable Fault	A communications malfunction that causes a complete disruption of communication, an excessive error rate, or unrecoverable errors.
Central Segment	Central coax segment to which LCN Extenders can be connected to extend the LCN system.
Coaxial Segment	A continuous connection of TPN nodes by a dual redundant pair of LCN coaxial cables. A coaxial segment is limited to 300 meters maximum and can have a maximum of forty loads including LCN Extenders.
Console	From one to ten Universal Stations configured to function a console. A console is typically used by one process operator.
CS/R	Clock Source Repeater—an I/O size board that transmits a 20 ma current loop signal as a part of the LCN Extender Set.
Double Cable Fault	A cable fault on both Cable A and B at the same time. This is a catastrophic incident.
Fiber Optic Link (segment)	A fiber optic cable with at least two fibers that interconnect LCN Extension Set boards. Each of the two fibers transmits in the opposite direction.
Fiber Optic	Two or more glass fibers wrapped and jacketed for inside, outside Cable aerial, or outside direct burial use.
Five/Ten Slot	A module that has either five or ten card slots that accommodate the boards required to function as a node.
FOC/XMTR	Fiber Optic Clock Transmitter—an I/O-size PWB that transmits through a fiber optic segment.
FOC/RCVR	Fiber Optic Clock Receiver—an I/O-size PWB that terminates a fiber optic segment.
GUS	Global User Station – An operator's window to the process.
History Module	A node that provides mass storage. History Modules (HMs) are principally used to store historical data including continuous history, and various journals and logs. They can also be used to store software, system files, and user files.
HG	Hiway Gateway. A node that interconnects the Data Hiway to the TPN.
LCN	The Local Control Network that interconnects the TPN-based nodes. The LCN employs dual, redundant cables, Cable A and Cable B.
LCNE	LCN Extender board.

TERM	DESCRIPTION
XLCNE2	LCN Extender board.
LCN Extender	Single board (I/O size) that interconnects one coaxial cable in a coaxial segment to a single full-duplex fiber optic cable fiber.
LCN Extension Set	A set of boards that extends the LCN through the use of fiber optic cable. See subsection 4.2.1.
LCNFL	LCN Fiber Link board that terminates two Fiber Optic cables (Cable A and Cable B), and is used when no clock is needed.
LED Module	Light Emitting Diode. A unit containing power supply, fans, and backplane that, with a compliment of boards, performs a specific TotalPlant Solution (TPS) System function.
NIM	Network Interface Module; the node that provides the interconnection between the Universal Control Network (UCN) and the TPN.
Node	An identifiable, addressable entity on the TPN or UCN that participates as a member of the network. TPN node types include GUSs, APP Nodes, USs, UWSs, NIMs, HGs, AM, and CGs. UCN node types include PMs, APMs, HPMs, LMs, and Safety Managers.
Pinning Reconnect	A means to define in hardware the address of a node on a network. To physically reconnect a section of coaxial cable to a T-connector or to logically reconnect two nodes so that they communicate.
Redundant	Two nodes of the same type (one primary and one secondary) form a pair—the secondary backing up the other.
Remote Segment	A coaxial segment that is connected to the central segment only by a single LCN Extender Set.
SAM	Scanner Application Module—Part of the TPS Paper Machine Automation System and executes the scanning and sensing features used in paper-making process activities.
TDR	Time Domain Reflectometer
TPN	TotalPlant Network (formerly referred to as the LCN).
Transceiver	CLCN A/B board which provides the electronic interface to the LCN coaxial node.
UCNE	The Universal Control Network Extender (UCNE) is a fiber optic link that joins two UCN cable segments.
Universal Station	Another TPN-based human interface, the Universal Station (US) is a node on the TPN that provides the window to the process. It includes a color monitor, Operator's Keyboard, optional touchscreen, optional Engineer's Keyboard, optional Removable Media Drives (cartridge or floppy drives), and optional trend pen recorders.
XLCNE2	LCN Extender Board.

11. Appendix B - LCN Topology Map

This section defines topology mapping to be used in analyzing cable problems.

11.1 USE OF LCN TOPOLOGY MAP

A topology map can be used to troubleshoot LCN cable problems and also to plan the addition of new modules to the system. A map is needed that shows:

- Each coax and fiber optic segment
- Cable A and Cable B
- Terminators
- Module types
- LCN Node Address
- LCNEs, if any

11.2 LCN TOPOLOGY MAP ANALYSIS

By recording the data from the error display described in Section 5.2 onto the topology map, a pattern may appear. A cluster of errors in one portion of a coax segment on one cable would indicate a bad coax cable segment that has been damaged and needs to be replaced. If all nodes get a discernibly high rate of errors in messages received from one particular node, the node may have either a bad "T" connector or a bad transceiver board.

11. Appendix B - LCN Topology Map

11.2. LCN TOPOLOGY MAP ANALYSIS

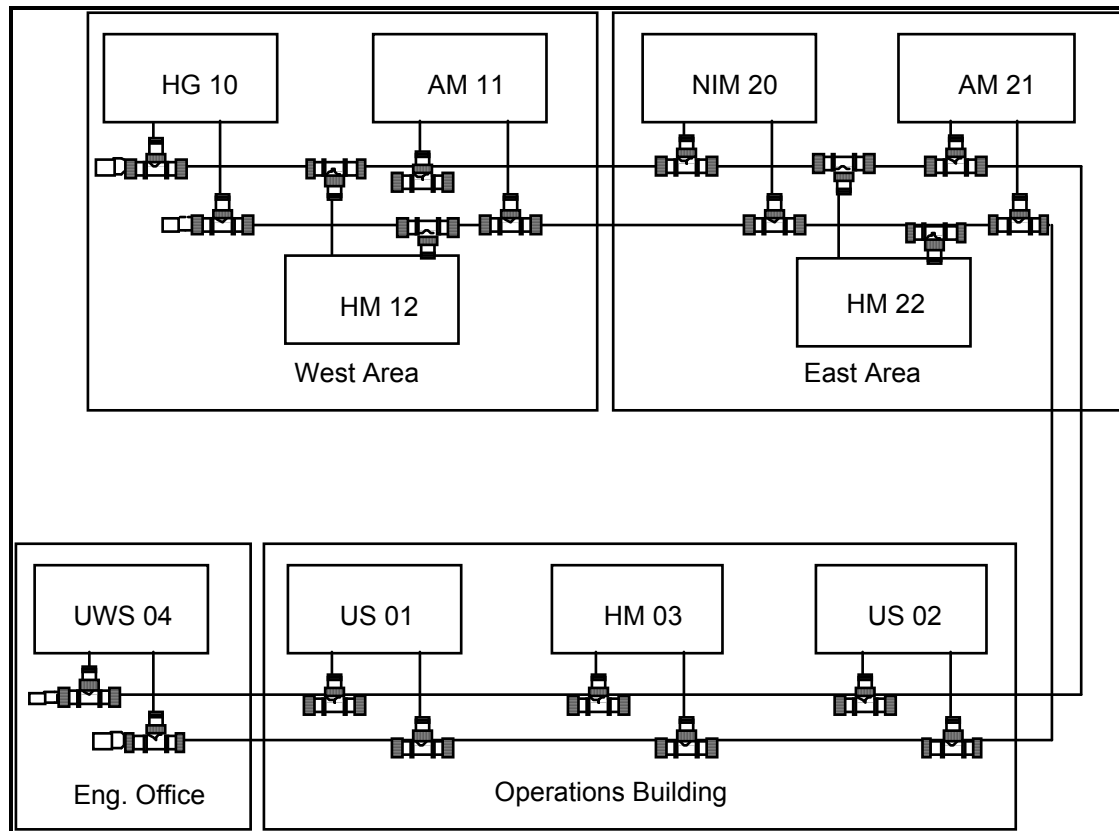


Figure 11-1 — Example of a Topology Map in a Small System

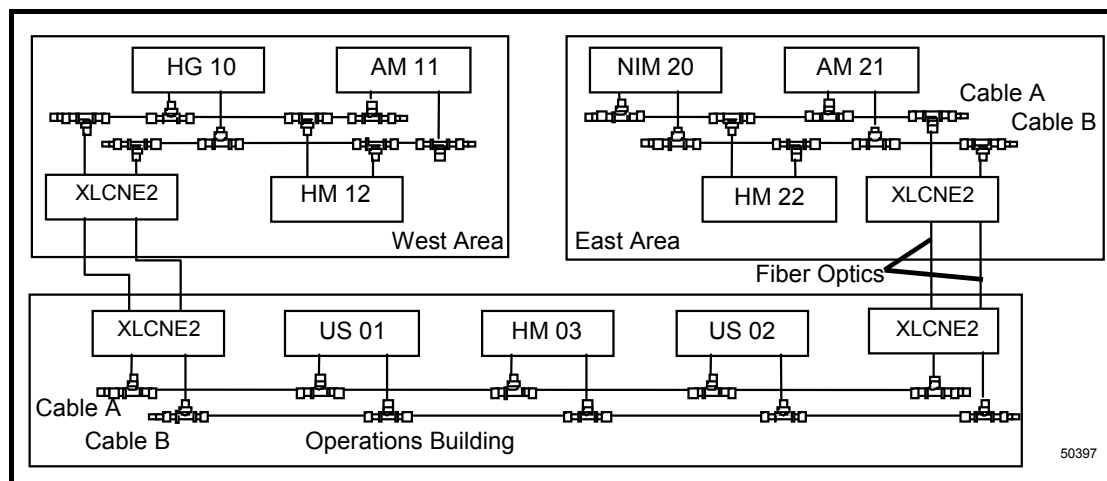


Figure 11-2 — Example of a Topology Map in a System Using XLCNE2s

12. Appendix C - Using The PSDP Comm Parameter

This section contains a tabulation of indexes that may be used to access elements of the data array of the Processor Status Data Point (PSDP) Comm Parameter.

12.1 DIRECT USE OF THE PSDP COMM PARAMETER

A user who is skilled in CL programs or the Picture Editor may use the PSDP Comm Parameter directly to develop custom displays to monitor TPN communications performance or assist in troubleshooting communications problems. This parameter should be used in conjunction with a topology map to develop useful diagnostic statistics.

The PSDP Comm Parameter is defined as follows:

`$PRSTSnn.COMM(y)`

Where “nn” is the node number whose data array is to be accessed, and “y” is the index into the Comm Parameter data array of that node. Tables C-1 and C-2 list the indexes that are used to access the various elements of the data array.

Some examples will illustrate the use of the tables.

`$PRSTS12.COMM(4)`

Will return the system version number for node 12 (see Table C-1 for the index value 4).

`$PRSTS32.COMM(25)`

`$PRSTS32.COMM(26)`

`$PRSTS32.COMM(27)`

Will return the date and time that the statistics counters for node 32 were last reset. As indicated in Table C-2, index 25 returns YYMM, index 26 returns DDHH, and index 27 returns MMSS (year, month, day, hour, minute, and second, respectively).

`$PRSTS15.COMM(271)`

Will return the total number of errors on Cable A on messages received by node 15 from node 25. The “15” in the parameter specifies that node 15’s data array is to be accessed, and Table C-2 shows that the index value 271 corresponds to Total Error data for Cable A from node 25.

12. Appendix C - Using The PSDP Comm Parameter

12.1. DIRECT USE OF THE PSDP COMM PARAMETER

Table 12-1— PSDP Comm Parameter Data for Indexes 1-70

Index	Description
1	PSDP Flag 1 — Counter Reset Flag (<i>Note 1</i>)
2	PSDP Flag 2 — Set Mode Flag (<i>Note 1</i>)
3	PSDP Flag 3 — Reserved for Future Use
4	System Version
5	Communication Management Status (<i>Note 2</i>)
6	Last Driver Error Number (see Section 5, Table 5-1)
7	Cable Master Status (<i>Note 3</i>)
8	Cable Primary Selection (<i>Note 4</i>)
9	Number of Input Buffers Allocated for Channel One (<i>Note 5</i>)
10	Number of Input Buffers Allocated for Channel Two (<i>Note 6</i>)
11	Output Pending Chain Count
12	Output Packet Lost Count
13	Mantissa for Packet Transmission Counter
14	Exponent for Packet Transmission Counter
15	Transmit Error Count
16	Receive IOCB Packet Error Count
17	LCN Recovery Retry Count
18	Indicated Primary Cable (<i>Note 4</i>)
19	Cable After MD Interrupt (<i>Note 4</i>)
20	Last Cable Actually Selected (<i>Note 4</i>)
21	Swap Ticks Remaining
22	YYMM of Failed State Entry Date and Time
23	DDHH of Failed State Entry Date and Time
24	MMSS of Failed State Entry Date and Time
25	YYMM of Date and Time That Counters Were Last Reset
26	DDHH of Date and Time That Counters Were Last Reset
27	MMSS of Date and Time That Counters Were Last Reset
28	Total Frame Check Sequence Errors, Cable A
29	Total Frame Check Sequence Errors, Cable B
30	Frame Check Sequence Error Rate, Cable A
31	Frame Check Sequence Error Rate, Cable B
32	Total Framing Errors, Cable A
33	Total Framing Errors, Cable B
34	Framing Error Rate, Cable A

Table 12-1— PSDP Comm Parameter Data for Indexes 1-70

Index	Description
35	Framing Error Rate, Cable B
36	Total Oversize Errors, Cable A
37	Total Oversize Errors, Cable B
38	Oversize Error Rate, Cable A
39	Oversize Error Rate, Cable B
40	Total Overrun Errors, Cable A
41	Total Overrun Errors, Cable B
42	Overrun Error Rate, Cable A
43	Overrun Error Rate, Cable B
44	Total Manchester Errors, Cable A
45	Total Manchester Errors, Cable B
46	Manchester Error Rate, Cable A
47	Manchester Error Rate, Cable B
48	Mantissa for Total Frames Received, Cable A
49	Exponent for Total Frames Received, Cable A
50	Mantissa for Total Frames Received, Cable B
51	Exponent for Total Frames Received, Cable B
52	Mantissa for Total Errors, Cable A
53	Exponent for Total Errors, Cable A
54	Mantissa for Total Errors, Cable B
55	Exponent for Total Errors, Cable B
56	Error Rate, All Errors Combined, Cable A
57	Error Rate, All Errors Combined, Cable B
58	Alone in Ring, Cable A
59	Alone in Ring, Cable B
60	Media Dead Count, Cable A
61	Media Dead Count, Cable B
62	Stolen Token Count, Cable A
63	Stolen Token Count, Cable B
64	Input Packet Chain Overrun Count for Channel One (<i>Note 7</i>)
65	Input Packet Chain Overrun Count for Channel Two (<i>Note 7</i>)
66	Input Packet Buffer Not Available for Channel One (<i>Note 8</i>)

Table 12-1— PSDP Comm Parameter Data for Indexes 1-70

Index	Description
67	Input Packet Buffer Not Available for Channel Two (<i>Note 8</i>)
68	Buffer Shortage Count for Channel One
69	Buffer Shortage Count for Channel Two
70	Buffer Memory Unavailable (<i>Note 9</i>)

Note 1 Reserved for software use

Note 2 0 = Uninitialized

1 = Initializing

2 = Entering Ring

3 = Waiting

4 = In Recovery

5 = On Network

6 = Failed

Note 3 0 = Cable Slave, 1 = Cable Master

Note 4 0 = Cable A, 1 = Cable B

Note 5 Channel One contains 1000-word buffers

Note 6 Channel Two contains 50-word buffers

Note 7 Ran out of buffers, data was lost.

Note 8 Low on available memory.

Note 9 Long message didn't fit in buffers available.

Table 12-2 — PSDP Comm Parameter Data for Indexes 71-846

Node	Total Errors		Total Frames				Errors/10,000	
	A	B	A-Man	A-Exp	B-Man	B-Exp	A	B
00	71	72	73	74	75	76	77	78
01	79	80	81	82	83	84	85	86
02	87	88	89	90	91	92	93	94
03	95	96	97	98	99	100	101	102
04	103	104	105	106	107	108	109	110
05	111	112	113	114	115	116	117	118
06	119	120	121	122	123	124	125	126
07	127	128	129	130	131	132	133	134
08	135	136	137	138	139	140	141	142
09	143	144	145	146	147	148	149	150
10	151	152	153	154	155	156	157	158
11	159	160	161	162	163	164	165	166
12	167	168	169	170	171	172	173	174
13	175	176	177	178	179	180	181	182
14	183	184	185	186	187	188	189	190
15	191	192	193	194	195	196	197	198
16	199	200	201	202	203	204	205	206
17	207	208	209	210	211	212	213	214
18	215	216	217	218	219	220	221	222
19	223	224	225	226	227	228	229	230
20	231	232	233	234	235	236	237	238
21	239	240	241	242	243	244	245	246
22	247	248	249	250	251	252	253	254
23	255	256	257	258	259	260	261	262
24	263	264	265	266	267	268	269	270
25	271	272	273	274	275	276	277	278
26	279	280	281	282	283	284	285	286
27	287	288	289	290	291	292	293	294
28	295	296	297	298	299	300	301	302

12. Appendix C - Using The PSDP Comm Parameter
12.1. DIRECT USE OF THE PSDP COMM PARAMETER

Table 12-2 — PSDP Comm Parameter Data for Indexes 71-846

Node	Total Errors		Total Frames				Errors/10,000	
	A	B	A-Man	A-Exp	B-Man	B-Exp	A	B
29	303	304	305	306	307	308	309	310
30	311	312	313	314	315	316	317	318
31	319	320	321	322	323	324	325	326
32	327	328	329	330	331	332	333	334
33	335	336	337	338	339	340	341	342
34	343	344	345	346	347	348	349	350
35	351	352	353	354	355	356	357	358
36	359	360	361	362	363	364	365	366
37	367	368	369	370	371	372	373	374
38	375	376	377	378	379	380	381	382
39	383	384	385	386	387	388	389	390
40	391	392	393	394	395	396	397	398
41	399	400	401	402	403	404	405	406
42	407	408	409	410	411	412	413	414
43	415	416	417	418	419	420	421	422
44	423	424	425	426	427	428	429	430
45	431	432	433	434	435	436	437	438
46	439	440	441	442	443	444	445	446
47	447	448	449	450	451	452	453	454
48	455	456	457	458	459	460	461	462
49	463	464	465	466	467	468	469	470
50	471	472	473	474	475	476	477	478
51	479	480	481	482	483	484	485	486
52	487	488	489	490	491	492	493	494
53	495	496	497	498	499	500	501	502
54	503	504	505	506	507	508	509	510
55	511	512	513	514	515	516	517	518
56	519	520	521	522	523	524	525	526
57	527	528	529	530	531	532	533	534
58	535	536	537	538	539	540	541	542

Table 12-2 — PSDP Comm Parameter Data for Indexes 71-846

Node	Total Errors		Total Frames				Errors/10,000	
	A	B	A-Man	A-Exp	B-Man	B-Exp	A	B
59	543	544	545	546	547	548	549	550
60	551	552	553	554	555	556	557	558
61	559	560	561	562	563	564	565	566
62	567	568	569	570	571	572	573	574
63	575	576	577	578	579	580	581	582
64	583	584	585	586	587	588	589	590
65	591	592	593	594	595	596	597	598
66	599	600	601	602	603	604	605	606
67	607	608	609	610	611	612	613	614
68	615	616	617	618	619	620	621	622
69	623	624	625	626	627	628	629	630
70	631	632	633	634	635	636	637	638
71	639	640	641	642	643	644	645	646
72	647	648	649	650	651	652	653	654
73	655	656	657	658	659	660	661	662
74	663	664	665	666	667	668	669	670
75	671	672	673	674	675	676	677	678
76	679	680	681	682	683	684	685	686
77	687	688	689	690	691	692	693	694
78	695	696	697	698	699	700	701	702
79	703	704	705	706	707	708	709	710
80	711	712	713	714	715	716	717	718
81	719	720	721	722	723	724	725	726
82	727	728	729	730	731	732	733	734
83	735	736	737	738	739	740	741	742
84	743	744	745	746	747	748	749	750
85	751	752	753	754	755	756	757	758
86	759	760	761	762	763	764	765	766
87	767	768	769	770	771	772	773	774
88	775	776	777	778	779	780	781	782

12. Appendix C - Using The PSDP Comm Parameter
12.1. DIRECT USE OF THE PSDP COMM PARAMETER

Table 12-2 — PSDP Comm Parameter Data for Indexes 71-846

Node	Total Errors		Total Frames				Errors/10,000	
	A	B	A-Man	A-Exp	B-Man	B-Exp	A	B
89	783	784	785	786	787	788	789	790
90	791	792	793	794	795	796	797	798
91	799	800	801	802	803	804	805	806
92	807	808	809	810	811	812	813	814
93	815	816	817	818	819	820	821	822
94	823	824	825	826	827	828	829	830
95	831	832	833	834	835	836	837	838
96	839	840	841	842	843	844	845	846

Table 12-3 — PSDP Comm Parameter Data for Indexes 1167-1192	
1167	Frame Check Sequence Errors in 15 minutes, Cable A
1168	Frame Check Sequence Errors in 15 minutes, Cable B
1169	Framing Errors in 15 minutes, Cable A
1170	Framing Errors in 15 minutes, Cable B
1171	Oversize Errors in 15 minutes, Cable A
1172	Oversize Errors in 15 minutes, Cable B
1173	Overrun Errors in 15 minutes, Cable A
1174	Overrun Errors in 15 minutes, Cable B
1175	Manchester Errors in 15 minutes, Cable A
1176	Manchester Errors in 15 minutes, Cable B
1177	Mantissa for Total Errors in 15 minutes, Cable A
1178	Exponent for Total Errors in 15 minutes, Cable A
1179	Mantissa for Total Errors in 15 minutes, Cable B
1180	Exponent for Total Errors in 15 minutes, Cable B
1181	Total Ring initializes
1182	Ring initializes in 15 minutes
1183	Total Ring collapses
1184	Total Token initiations
1185	Token initiations in 15 mins
1186	Total errors on Cable A
1187	Total errors on Cable B
1188	PingToNode: The destination node number.
1189	PingToCable: The selected cable for ping
1190	PingStatus: The Status of ping
1191	Ping Time: Turn Around Time
1192	Clock switch (Note 1)

12. Appendix C - Using The PSDP Comm Parameter

12.1. DIRECT USE OF THE PSDP COMM PARAMETER

NOTE 1	
Value Range	Explanation
0	Initial value of '0' is displayed until the operation is complete. The value '0' also indicates that the release/platform does not support the online clock switch functionality.
1	The value of '1' indicates that the online clock switch functionality is supported.
2	To be entered by the user - Instruction to take up the Clock Mastership.
3	To be entered by the user – Instruction to take up the Clock Slave responsibility
4	To be entered by the user - Instruction to relinquish the Clock Mastership
5	To be entered by the user - Instruction to relinquish the Clock Slave responsibility

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